

EXXONMOBIL UPSTREAM

Upstream Best Practices Work Management System (WMS)

Revision 4.0.4

Work Management System (WMS)

1.0 SUMMARY

1.1 BEST PRACTICE

The *Upstream Work Management System (WMS) Best Practice* describes the processes and practices that define the Company's Permit-to-Work management system, and is an Upstream cornerstone used to manage work safely at all operating sites. WMS ensures that work activities at Company-operated or Company-controlled sites are undertaken in an efficient, structured, and controlled manner that reduces incident risk to as low as reasonably possible. The system also provides a framework for identifying hazards, actively managing the risks associated with the work to be performed, and confirming that interfaces with the work activity are considered appropriately.

This Best Practice directly supports compliance achievement to the requirements of OIMS System 6-4, "Work Management."

1.2 KEY PERFORMANCE INDICATORS

Refer to [Section 5.2: "Verification, Measurement, and KPIs"](#).

1.3 SUITCASE TOOLS

- **Supporting Suitcase tool:** Electronic Permit To Work System (ePTWS)

1.3.1 DOCUMENT CATEGORIZATION

This Work Management System Best Practice has the following categorizations:

- Management and Protection of Information (MPI) Classification: Unclassified
- Joint Venture Categorization: Novel

1.3.2 DOCUMENT CONTROL AND DISTRIBUTION

The Primary Record Holder (per ExxonMobil [EM] Records Management Guidelines) for this document is the Best Practice Administrator. An electronic copy is available from the Upstream Best Practices website, WMS SharePoint site, or external references within the ePTWS.

1.4 TRAINING

None

1.5 SUPPORT

Upstream Work Management Network

1.6 CONTINUOUS IMPROVEMENT

This UBP will undergo an annual health review, with updates incorporated on an evergreen basis. Updates will be informed by the health of the overall suite of UBPs.

Global changes to the Manual may be requested by an Operations Unit at any time by using the UBP Work Management Change Request Form.

Reference: [GOTO/WMSFORMS](#), "UBP Forms," UBP Work Management System Manual Change Request Form.

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The request requires approval by the OIMS 6-4 Owner and submitted to the Work Management System Network Owner for review and approval. The Network Lead and Facilitator review the Change Request and accept or reject. Decisions on accepting or rejecting change requests will be communicated back to the requesting Operations Unit. Accepted change requests will be made in the System following the Site BP Principles Process.

1.7 CONTENT DESCRIPTION

The content of UBPs is categorized into the following three components:

- **Critical Expectations, denoted as [CriticalE] throughout**, describe actions that must/shall be met by all Sites. These are complete, single thoughts or actions, worded concisely to communicate something that is necessary, specific, verifiable / actionable and attainable. Generally, Critical Expectations are tied to differentiating capabilities, economies of scale, or needs to close gaps in defensive positions, where deviation results in:
 - Significant personnel safety risk equivalent to multiple fatalities
 - Significant process safety risk equivalent to an HC1 or HC2
 - Significant Controls risks
 - Potential loss of significant value or Corporate Scale advantage
- **Expectations, denoted as [E] throughout**, describe actions that must/shall be met by all Sites. These are complete, single thoughts or actions, worded concisely to communicate something that is necessary, specific, verifiable / actionable and attainable. Generally, Expectations are tied to industry standards, differentiating capabilities or areas where deviation could result in increased personnel, process safety, or controls risk or loss of financial value
- **All else** in this document is generally globally supportive institutional knowledge, tools, and 'how to' guidance. Operations Units have flexibility to adopt this content based on business value.

Content within this Best Practice aligns to common criteria as defined in the UBP Principles document, [Section 2.0: Governance](#). These criteria apply to all Upstream Best Practices.

1.8 IMPLEMENTATION CONSIDERATIONS

Sites should consider the following when developing implementation plans, or changing how this UBP is locally implemented:

- Sites should not expect to be assessed on any content not deemed an Expectation [E] or Critical Expectation [CriticalE], even when adopted by the Site, unless the Site has decided to adopt this content as a means of business control defined within locally-owned Site management systems or business controls catalogues (for example, CIMS catalogues, OIMS protocols).
- Implementation coordinators and UBP practitioners are encouraged to engage local Controllers and OIMS contacts during UBP change management exercises to support identification and resolution of any business controls issues associated with a proposed change.

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1.9 APPROVALS AND ENDORSEMENTS

Refer to the [UBP Principles](#) document, section 2.0 for responsibilities associated with the roles listed below.

Endorsed:	DocuSigned by:  D76B20EE783D48B Tommee Lambert Best Practice Lead Global Operations WMS Supervisor	June 29, 2026 Date
Approved:	DocuSigned by:  50471AA24CD7407... Marcos Azuaje Best Practice Owner Global Process Owner – Control of Work	June 30, 2026 Date

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REVISION RECORD

MANAGEMENT OF CHANGE

Changes made to the Work Management System Best Practice are subject to the standard change management protocols as described in the UBP Principles document. For the purpose of executing change to this document, roles are assigned as follows:

- Best Practice Change Coordinator: WMS Team Lead
- Best Practice Owner: OS/UBP Manager, Upstream

All management of change records are retained by the document Administrator or delegate.

DOCUMENT UPDATE LOG

The latest changes to this document are summarized in the Table below. Detailed change history is available by contacting the UBP Process Admin.

Rev.	Description of Change	Date
1.0	Version 1 original issue	April 2010
1.1	Post implementation lesson capture, input from Business Units and Network Workshop Version 1 Rev 1.0	April 2013
1.1	Best Practice Summary issued as part of 2013 UBP revitalization	June 2013
2.0	Periodic Review Cycle - Input from Business Units and Network Workshop	January 2017
2.1	Learning from incidents, organizational changes, input from Business Units and Network Workshop.	January 2020
3.0	Periodic Review Cycle - Updates to reflect organization changes, updates to rigging and lifting practices, inclusion of handover requirements, deviation process Enhancing Process Safety terminology, and other minor updates driven by the WMS network.	July 2020
3.1	Updated to reflect changes to the THA (WAME & Rigging & Lifting), changes to CSE entry thresholds for Oxygen and VOC, minor updates to complex lift definition, clarifications to hydrocarbon freeing, long term isolation requirements, and other minor clarifications.	January 2021
3.2	Update to adjust assessment requirements and include the Global SWP links.	October 2022

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Document Update Log, Continued.

General

- Updated the document to the new UBP format
- Governance aligned to UBP principles including [E] & [CriticalE] identification
- Removal of redundant information throughout the manual

Section 4: Introduction

- Updated language associated with PSMS components
- Included guidance for Wells applicability of WMS
- Defined Non-Centralized Remote Field Operations & Updated Scalability Matrix
- Aligned Deviation/Exception process to UBP Principles
- Updated Roles & Responsibilities with PSMS linkages & guidance
- AIA title updated to Site Coordinator / Area Interface Authority

Section 5: WMS Framework

- Framework updated to include PSMS safety in the moment
- Updated LSRAs, Start Work Checks, JSA, Pre-task briefing and other guidance to align to PSMS standards
- Added JSA Use Scenarios to align with PSMS
- Simplified the Work Pack Requirement Matrix
- Removed desktop assessment language and requirements

4.0

- Added Non-Centralized Remote Field DWCM Agenda

February 2023

Section 6: Permit to Work

- Remote Field Operations (RFO & NC RFO) Scalability updated
- Added NC RFO Work Planning DWCM alternate endorsement process & workflow
- DWCM meeting frequency change for NC RFO
- Removal of General Permit
- Updated Hot Work on Hydrocarbon Systems

Section 7: Energy Isolation & Temporary Defeat

- Positive Isolation terminology replaced with Blinding, Spading or Disconnection
- Updated flange classes and fluid types on the Minimum Isolation Standard Table
- Updated Isolation Method Standard Table to align to Upstream GPs
- Guidance for butterfly valve usage for isolation aligned with General Practices
- Defined Personal Locks

Section 8

- Added OPITO Training Standard requirements for AGT

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Section 4

- Removed PSMS leadership linkages from WMS roles

Section 5

- Removed reference to JSA only work in DWCM agenda and replaced with non-permitted work
- Added designated supervisor to the pre-task briefing expectations.
- Removed Table 2: JSA Use Scenarios

Section 6

- | | | |
|-------|---|----------------|
| 4.0.3 | <ul style="list-style-type: none">• Removed [E] expectation to utilize JSA for work authorization• Incorporated Hot Work barrier requirements interim directive for onshore activities within 10 feet of a hydrocarbon source. | September 2025 |
|-------|---|----------------|

Section 7

- Removed requirement to use JSA for specific activities in line with PSMS expectations.

Section 10

- Laid down requirement to use Forklift Operations Checklist where Working Around Mobile Equipment Start Work Check is used

• General

- Aligned with Upstream organization title changes
- RAMS 2-1 process for HC1/HC2 approvals
- Business Unit transitioned to Operating Site and when abbreviated (BU) to Site

- | | | |
|-------|--|-----------|
| 4.0.4 | <ul style="list-style-type: none">• Inclusion of Interim Directives<ul style="list-style-type: none">○ Section 7<ul style="list-style-type: none">▪ Updated Section 7.3 to include management of Critical Safeguards utilizing Temporary Defeats▪ Updated Section 7.5.3.4 Break Point Tagging▪ Updated Section 7.5.1.17 Electrical Isolation Plans | July 2026 |
|-------|--|-----------|
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2.0 CORPORATE SEPARATENESS NOTICE

Nothing in this material is intended to override the corporate separateness of local entities. Working relationships discussed in this material do not necessarily represent a reporting connection, but may reflect a functional guidance, stewardship, or service relationship. Where shareholder consideration of a local entity matter is contemplated by this material, responsibility for action remains with the local entity.

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4.0 INTRODUCTION

OVERVIEW

This document provides a detailed description of the Upstream Best Practice (UBP) *Work Management System* (WMS) for use by all Upstream Oil and Gas field facilities. [CriticalE] The Work Management System **must** be used to manage all work activities from the pre-job planning phase through the work execution and work site restoration phases.

This section provides a general overview of the Work Management System.

USE OF THE TERM COMPANY

In this manual, the term "Company" refers to the ExxonMobil Oil and Gas Company (UPSTREAM)

4.1 USING THIS DOCUMENT

4.1.1 PURPOSE

This Work Management System (WMS) describes the key aspects and requirements of the Company's Processes and Practices regarding the Planning, Preparation, Execution and Reinstatement of work activities in UPSTREAM's Operating Areas. The WMS is essential for managing work safely at operational sites. The WMS confirms that the activities at Company-operated or -controlled sites are undertaken in a structured and controlled manner to reduce the risk of incidents. The system also provides a framework for identifying hazards, actively managing the risks associated with the work to be performed, and confirming that interfaces with the work activity are considered appropriately.

This Work Management System addresses the requirements of Operations Integrity Management System (OIMS) System 6-4 Work Management and incorporates relevant aspects of the Company's Personnel Safety Management System (PSMS).

4.1.2 SCOPE AND APPLICATION

The Work Management System applies to the control of work in all UPSTREAM field and plant operations, including contracted activities. A key aspect of the WMS is the use of a Permit to Work (PTW) Process described later in this manual.

The WMS controls the use of all three permits (Hot Work, Critical and Confined Space Entry Permits), referred to as "three (3) permit areas" and:

- Applies to work in all areas except those areas that are designated by the Operating Site.
- Is designed primarily for use in areas containing hydrocarbon-bearing equipment and supporting or ancillary systems necessary to operate and control producing operations.

4.1.3 TYPICAL ACTIVITIES INCLUDE

Work on or in close proximity to vessels, tanks, piping, pig launchers and receivers, pipelines, wells, flow lines, pumps, compressors, generators, switchgear, control rooms, electrical systems, control systems, protective systems, safety equipment, fuel gas, and pneumatic and hydraulic systems.

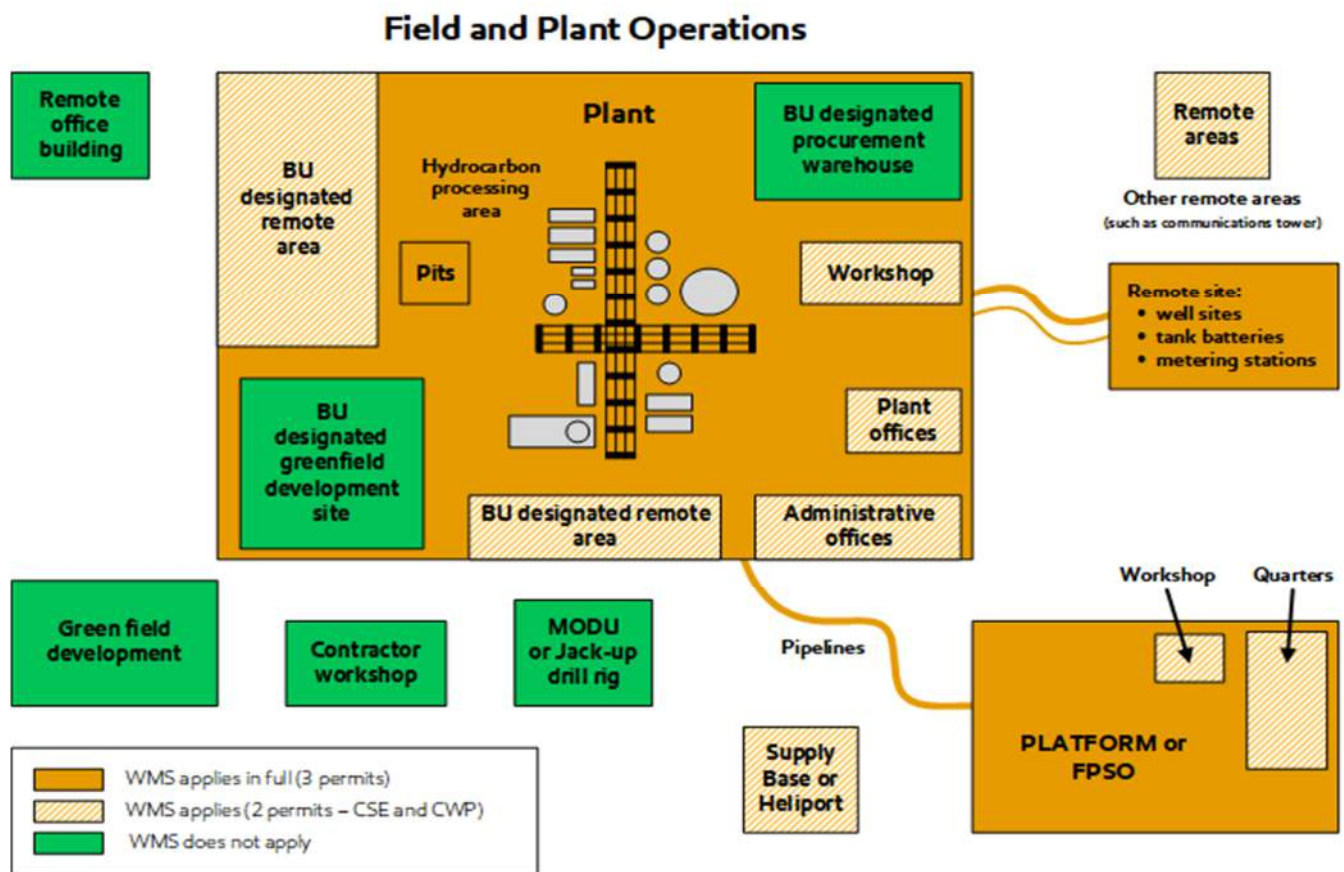
The Site may designate specific areas, remote from hydrocarbon-bearing equipment, where only the Confined Space Entry and Critical Work Permits apply, referred to as two (2) permit areas, such as:

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- Buildings
- Field offices
- Quarters
- Safe welding areas
- Procurement warehouses
- Radio towers
- Logistics (supply bases, heliports and transport)
- Open areas

[CriticalE] Operating Sites **must** clearly identify areas for each of their facilities where three and two permit areas apply and any areas designated to be managed by another function; these designations **must** be approved by the Site Manager or equivalent. This information should be readily available to reference by all people involved in work activities at the facility. Refer to [Figure 1: Work Management System Applies to Field and Plant Operation](#).

Figure 1: Work Management System Applies to Field and Plant Operation



4.1.4 DOES NOT APPLY

Work Management System (WMS)

The Upstream Oil and Gas (UPSTREAM) WMS is not required for activities and sites controlled by other functions (for example, Global Services Company [GSC] managed office or a Development fabrication site) outside of UPSTREAM field and plant operations and areas.

NOTE: Consideration can be given for utilization by other functions.

Within UPSTREAM field and plant areas, provided that SIMOPS and interface are assessed, and the Site Manager or equivalent approves, an area can be designated to be managed by another function using a separate WMS. SIMOPS guidelines and interfaces should be defined in conjunction with the other function in the bridging documents and approved by the Site prior to implementation. Examples where this might be appropriate are GSC-managed warehouses within a plant site or designated Greenfield construction area, such as, additional process train within a plant complex.

[CriticalE] Greenfield construction areas managed by third parties within a Production Controlled area **must** have an equivalent system for controlling work activities. The UPSTREAM Work Management System takes precedence where interfaces exist with Operations.

[CriticalE] The UPSTREAM Work Management System **must** be applied to manage activities on equipment or systems when hydrocarbons are introduced and to manage any adjacent activities that can impact these areas (SIMOPS). Projects, system completions and operations teams need to define the process to manage the introduction of hydrocarbons; consideration should also be given to energizing electrical and other ancillary systems.

While this manual does not specifically address work management at Company-owned or -controlled fabrication or construction yards or new facility (Greenfield) construction sites, many of the work management controls in this system should be applied through:

- OIMS System 3-1 Project Execution
- OIMS System 8-1 Third-Party Services

[CriticalE] Wells activities may be managed by a Wells or equivalent third party work management system where Wells has care, custody and control of a location and a handover of the location to Wells from Operations has been documented. Scenarios may include:

- Onshore field single or multi well pads well work or completions activities
- Onshore field new drills (handover only required where other producing wells may be on site i.e. multi well pad)
- Offshore drilling independent of producing facility (e.g. MODU development / exploration well)

[CriticalE] Where interface with Operations are required and care, custody, control of location has **not** been handed over to Wells, work coordination and authorization through WMS must be applied and scenarios documented in the Site local Appendix A. Scenarios may include:

- Multi-Well locations (e.g. well work / rig up on a producing facility)
- Offshore platforms where operations have care, custody and control of the entire facility

4.1.5 WMS SCALABILITY

Work Management System (WMS)

The Work Management System is documented to take into account the different UPSTREAM operations around the globe while maintaining the ability to manage work safely and practically.

In order to enable practical implementation of the WMS to large plants, offshore platforms, remote platforms, pipelines, remote field operations (such as, well pads or pump stations), workshops, and so forth, the controls documented are scalable.

[CriticalE] Non-centralized Remote Field Operations (RFO) using a supporting suitcase tool other than ePTWS must consult the UBP SME.

The scalable application introduced by each Site for their Field and Plant Operations is consistent with the scalable components defined in the WMS and summarized in the following table. The decisions made by the Site are risk based and traceable (documented) through their Management of Change (MOC) system. Refer to [Figure 2: Work Management System – Scalability Matrix](#)

Figure 2: Work Management System – Scalability Matrix

SCALABLE COMPONENTS \ LOCATION	FIELD AND PLANT OPERATIONS				Non WMS Site - Green Field site, remote Development Fabrication site, or GSC Managed Office
	Three (3) Permit Area	BU Designated Two (2) Permit Area	Remote Site	Non-centralized Remote Site	
Roles and Responsibilities	Roles and Responsibilities				Not Applicable. Function to Manage
Permit Approvals	Documented		Can be wet ink, verbal, eMail, text etc.		
Isolations verified by second person	YES		YES, if available otherwise follow RFO guidance		
Personal Isolations	BU to determine activities allowed				
Non-Permitted Work	BU to determine activities allowed				
JSA	As required by Personnel Safety Management System (PSMS)				
ePTWS (electronic Permit To Work System)	YES (Consider if required when EOFL* is < 5 yrs)	Consider implementation if permits can be centrally administered / issued		Consider paper or alternate tool that fufills WMS requirements	
Exceptions	As per UBP Principles				

*EOFL – End of Field Life

4.1.6 REMOTE FIELD OPERATIONS (RFO)

There are unique circumstances when managing work on remote fields or facilities such as satellite platforms, field wellheads, tank batteries, pipelines, and valve stations cannot be executed with the same segregation of duties as those on fully staffed platforms or plants. In these circumstances, the intent of this WMS to conduct activities safely must be achieved, (for example, verification of isolations and demonstration of zero energy and distinct steps are taken by the Permit Holder).

The Work Management System recognizes two modes of Remote Field Operations. These two operations require additional scalability to effectively execute work.

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Remote Field Operations (RFO) - Workers operate out of a central office, often beginning and ending their day in that location. ePTWS software and tools are readily accessible.

Non-centralized Remote Field Operations (NC RFO) - Workers travel directly to worksites and do not start their shift at a central/field office. Non-centralized RFOs typically:

- Have extended travel times between locations and cover large geographic locations
- Do not have access to ePTWS, printers, data connectivity, etc.
- Have individuals or small independent teams executing work with limited interface with other functions
- Have low complexity facilities (i.e. simple well pads, tank batteries, etc.)
- Have minimal potential for SIMOPS
- Develop & review permits & non-permitted activities the day of work execution

Further explanation for handling works in remote fields and facilities is provided in relevant sections of this System

References:

- [Section 5.0 Work Management Framework](#)
- [Section 6.1.3.4: RFO Scalability.](#)
- [Section 7.0: General Energy Isolation Process.](#)

4.1.7 OBJECTIVES

The objectives of the Work Management System are to confirm that all activities necessary to produce, process, and export oil and gas are performed in accordance with Company and government regulations and industry systems, and to fulfill the OIMS System 6-4 objectives.

To achieve these objectives:

- All work activities are planned with due consideration to physical interfaces at the work site, potential simultaneous activities, hours of work required without a rest break, and the impact of shift and crew changes on the work activity.
- All work activities with potential higher risk are managed.
- Equipment and facilities are prepared for work activities to be performed, including safe and environmentally sound draining, venting, purging, and isolation.
- Controls are applied to the temporary disarming or deactivating of the following:
 - Critical alarms
 - Process control, shutdown equipment and systems
 - Other critical safety equipment (such as, fire pumps)
- A permit-to-work program is implemented that incorporates checks and authorizations that are consistent with the risks likely to be encountered.
- Work activities are performed in accordance with Company safety expectations.
- Equipment and facilities are reinstated to a safe and environmentally sound working condition after the work activity is completed and appropriate personnel are informed of the revised equipment status.

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- Lessons learned from planning and executing the work activities are captured for similar future work activities.
- Personnel at an operational facility are trained in the purpose and application of work management processes.

4.1.8 USERS

The users of this System are personnel who are responsible for managing, planning, supervising, or executing work on the Company facilities. This includes the following:

- Site Manager, Operations Manager and Operations SLS
- Superintendents (Operations, Maintenance, Construction, and Workover)
- Persons in Charge (PICs), Production Supervisors, Field Foreman and Operations Supervisors
- Discipline (Maintenance, Construction, and Workover) Supervisors
- Safety, Health, and Environment (SHE) and OIMS Advisors
- Operators and work execution personnel

4.1.9 ACTION HIGHLIGHTS

In this System, WARNING, CAUTION, IMPORTANT, and NOTE are presented throughout the text and used to highlight various actions as follows:

WARNINGS indicate situations in which bodily injury or death *could* occur through negligence or failure to follow the proper procedure.

- **Example:**

! WARNING: Under normal operation, all operating machinery and electrical equipment **must** have functional safety guards, switches, and alarms in place. Follow the proper operating procedures.

CAUTIONS are used to inform users of undesirable consequences of actions or non-actions. Cautions are not as serious as warnings.

- **Example:**

▲ CAUTION: Do not attempt to stop a closing elevator door with your hands or feet.

IMPORTANT information is more critical for the user than a note, but it does not pertain to actions that can result in serious consequences.

- **Example:**

! IMPORTANT: Welding, burning, or hot work on painted surfaces or stainless steel or other alloys should be evaluated for appropriate exposure control methods.

NOTES express incidental information that is helpful in addition to the regular text information.

- **Example:**

NOTE: Always plan an escape route in case of emergencies.

4.1.10 LOCAL OPERATING SITE CONTENT

[CriticalE] Operating Sites **must not** alter the content of a published Best Practice. When Site-specific information is needed (for example, to accommodate local regulatory requirements) an individual

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addendum should be written by the site, and attached to Best Practice. This manual provides a [Section 12.0 Appendix A](#) that may be used to capture these items.

4.1.11 WMS FORMS

There are a number of forms that support execution of the Work Management Best Practice.

Examples: PTW Forms, Isolation Certificates, TD Forms, Daily Work Coordination Meeting Agenda, Assessment Forms, SWP Checklists, Change Request Forms, etc.

All forms referenced in this manual can be found at [GOTO/WMIFORMS](#).

4.2 REGULATORY REQUIREMENTS

The Company will comply with all applicable laws and regulations when applying this System including application of permissible mechanisms wherein the Company is granted written approval to operate under specific design codes, specifications, or protocols. In addition to the regulatory requirements, the Company may also place additional requirements on operations that will meet or exceed regulatory requirements.

These and other regulations, Company-specific approvals, and obligations deriving from them are maintained in accordance with OIMS System 4-2 Compliance with Laws, Regulations, and Permits along with the Site's *Regulatory Compliance Manual*.

[E] Each Site **must** review applicable regulatory requirements pertaining to Work Management, and reference or apply the regulatory requirements in addition to this System to meet or exceed regulatory compliance.

4.3 DEVIATIONS AND EXCEPTIONS

4.3.1 APPLICATION OF THE SYSTEM AND DEVIATIONS

Variance or deviations from the requirement(s) in this System (Critical Expectation [CriticalE] or Expectations [E]) by a Operating Site should be carefully considered. [CriticalE] "Deviations" from CE and E requirements in this System **must** be handled as follows:

- Deviations **must** be reviewed by a multidisciplinary team of individuals knowledgeable with the issues associated with the proposed deviation
- Knowledgeable personnel on the Variance/Deviation multidisciplinary should consist of personnel such as:
 - The developer of the Variance/Deviation;
 - A technical process representative (e.g. process engineer for the applicable unit);
 - An hourly operation representative (e.g. experienced operator in the applicable unit), and
 - A health and safety representative.
- Deviations from requirements must follow the deviation process described in the UBP Principles Document (**Reference:** "UBP Principles" [Section 3.0 Control and Sustainment](#))
- No deviations are allowed for **Prohibited** activities in the SIMOPS Matrix (**Reference:** [Section 9.0 Simultaneous Operations](#)).

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[CriticalE] Approval for deviations is only granted if other safeguards will be introduced that maintain the safety of personnel, the environment, and facilities.

Example: If physical disconnect or blinding for confined space entry cannot be achieved, then a deviation is required.

NOTE: This deviation would only apply to the specific job scope and duration.

4.3.2 EMERGENCY DEVIATIONS FROM THE SYSTEM

[CriticalE] Emergency deviations are permitted, but the safety of personnel and protection of the environment and property **must not** be jeopardized. Once the emergency is controlled, the requirements contained in this manual will prevail. Any change implemented during the emergency **must** be re-assessed and approved as a deviation.

4.3.3 EXCEPTION TO THE BEST PRACTICE

Exceptions to the System are defined as a **continuing or permanent deviation** from the expectation(s) (Critical Expectations [CriticalE] or Expectations [E]).

The Work Management Upstream Best Practice content and tools that are not implemented will require an exception. [CriticalE] All Exceptions follow the exception process described in the UBP Principles Document (**Reference:** “UBP Principles” [Section 3.0 Control and Sustainment](#))

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4.4 ROLES AND RESPONSIBILITIES

INTRODUCTION

This section contains the following information:

- Positions referenced in the system
- Combined roles

4.4.1 POSITION REFERENCED IN THE SYSTEM

INTRODUCTION

The following outlines the core functional roles and primary responsibilities associated with the WMS.

4.4.1.1 PERSON IN CHARGE (PIC)

The facility or site senior Person in Charge of the day-to-day operations. This is typically the Offshore Installation Manager (OIM), Facility Supervisor, Vessel Master, or designated Person in Charge by management. The PIC has primary responsibility for:

- Ensuring that competent personnel are assigned to plan, review, authorize and execute work at their facilities.
- Endorsement of work planning activities, including simultaneous operations (SIMOPS).
- Conducting Daily Work Coordination meetings and *Endorse* permits.
- Ensuring that trained and competent personnel are assigned to plan, approve, and execute work at their facilities.
- Oversight and compliance to the WMS.
- Assigning personnel to conduct V&V (verification and validation) activities (consistent with PSMS) and WMS assessments as well as conducting periodic V&V / assessments to understand WMS status and effectiveness.

4.4.1.2 AREA AUTHORITY (AA)

The person designated by the Site (such as Production/Operations Supervisor/Foreman, Area Operator, or Functional Supervisor/Lead) who is responsible for:

- Reviewing Permit Packs for quality and completeness (e.g., job scope details, identification of LSRAs and life-saving safeguards, required SWP Checklists, JSA content (where required), procedures, Temporary Defeats).
- Ensuring competent individuals are assigned to execute tasks and appropriate tools are available.
- Verify isolation plans for their functional area of expertise (e.g. electrical, process, etc.)
- Safe work coordination (including SIMOPS management) within the designated area of authority.
- Authorizing permits to allow PH to commence work activity; withhold authorization until satisfied on readiness for safe execution
- Conducting periodic WMS V&V activities and assessments, reviewing results and validating lessons learned.

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4.4.1.3 AREA AUTHORITY FUNCTIONAL (AAF)

The person designated by the Site (such as Area Operator, or Functional Supervisor/Lead) who is responsible for:

- Reviewing Permit Packs for quality and completeness (e.g. job scope details, identification of LSRAs and life-saving safeguards, required SWP Checklists, JSA content (where required), procedures, Temporary Defeats).
- Ensuring competent individuals are assigned to execute tasks and appropriate tools are available.
- Verify isolation plans for their functional area of expertise (e.g. electrical, process, etc.)
- Safe work coordination (including SIMOPS management) within the designated area of authority.
- Conducting periodic WMS V&V activities and assessments, reviewing results and validating lessons learned.

4.4.1.4 AREA OPERATOR (AO)

The person designated by the Site (such as Unit, System, and Equipment Operators or Central Control Room Operators) who is responsible for:

- The operating equipment and systems in their area.
- The field review and issuance of the work permit to the PH.
- Conducting or assigning isolation and temporary defeat requirements.
- Acting as second verifier for isolations (if required).
- Reviewing and communicating work site controls and safe working processes to the PH.
- In a remote field, consider if more than one work scope on the same location could present SIMOPS concerns, and consult AA if required.
- Verifying work status and restoring the equipment or system.
- Identifying how Zero Energy will be demonstrated to the Permit Holder.
- Demonstrating Zero Energy.

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4.4.1.5 PERMIT HOLDER (PH)

Discipline Supervisors, Foreman, Front Line work crew Supervisor (or designee), crew lead or single individual performing work under a permit. The PH has primary responsibility for:

- Preparing and/or reviewing the Permit Pack documents to confirm accurate work scope, location, identification of hazards, applicable LSRAs, life-saving safeguards, controls for work and completing / facilitating the JSA when required
- Interfacing with the Area Authority
- In a remote field, consider if other work on the same location could present SIMOPS concerns, and consult AA if required
- Leading or arranging the Pre-Task Briefing(s), including Start Work Check discussion(s) and reviewing Permit Pack documentation with the work team
- Ensuring START work readiness and reinforcing STOP Work Expectations with the team/individuals, and promote usage of LMRA
- Verifying and validating identified life-saving safeguards and work site controls are in place and effective
- Verifying and securing (lock and tag) the energy Isolation Control Point (ICP) (if applicable)
- Witnessing the demonstration of zero energy by the Isolating Authority before accepting the permit
- Completing work and restoring the work site to a clean and hazard-free condition
- Lead After Action Reviews (AARs) when planned / warranted

4.4.1.6 ISOLATING AUTHORITY (IA)

Personnel who have the skills, knowledge, and competency to perform energy isolation on equipment and systems necessary for the safe execution of work. Typically, IAs are Area Operators, discipline trade personnel (electrical, mechanical, instrument, etc.), or persons deemed competent by site management. The IA has primary responsibility for:

- Identifying isolation requirements in association with the work scope and the PH.
- Applying all isolations in accordance with the WMS and safe work practices.
- Conducting and verifying isolations to ensure Zero Energy state is achieved and maintained.
- When requested, assist AO to demonstrate Zero Energy state to the PH.
- Removing isolations and restoring equipment and systems to normal.

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4.4.1.7 SITE COORDINATOR (SC) / AREA INTERFACE AUTHORITY (AIA)

Personnel designated to be in charge of a specific location where there are multiple work activities happening concurrently which have the potential to conflict (SIMOPS or Concurrent Work). This individual is typically designated by the PIC. The SC/AIA is required (where applicable) to:

- Is designated as responsible for the work location where above activities are taking place
- Is typically the person responsible for the highest risk work at the location
- Understands what the higher risk activities are for the location which could lead to life-altering injuries or process safety events.
- Ensures work authorization process are followed
- Evaluates the proposed permit activities, location, and hazard controls, in relation to their existing work to verify awareness and provide endorsement for the work to proceed
- Identifies any barriers, scope changes, or competing priorities impacting ability to ensure work is executed safely (e.g., conflicting work interfaces, SIMOPS conflicts, significant scope changes).
- Notifies AA/AAF as needed and align on appropriate mitigations before proceeding
- Countersigns or signs the Permit to Work when managing activities that require coordination or interface with other work teams.
- May be a PH for one of the activities at site
- Has ultimate Stop Work Authority if work conflicts arise

4.4.1.8 AUTHORIZED GAS TESTER (AGT)

Personnel who have performed the required training and are deemed competent to perform atmospheric testing of work areas, equipment, and systems to enable the safe execution of work. They are primarily responsible for:

- Assessing the work location and identifying all atmospheric testing requirements for the safe execution of work.
- Verifying that acceptable conditions exist to perform the proposed activity by performing atmospheric gas tests and documenting the information on the work permit as required.
- Ensuring testing equipment is fit-for-purpose, calibrated, and functional.
- Communicating abnormal atmospheric test information to the AA or AO.

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4.4.1.9 FIRE WATCH

A person deemed competent by site management to perform Fire Watch duties and is identified by the AA or AO to monitor the work area during hot work or when requested for potential spark throwing and spark generating activities. The Fire Watch is primarily responsible for:

- Ensuring the following:
 - Fire-fighting equipment is available and ready for immediate use.
 - Flammable materials are cleared away from the work site.
 - Drains are covered and sealed.
 - Sparks and welding spatter are contained.
- Initiating emergency alarms and response if a fire or gas release occur in the area.
- Monitoring the work area after work completion until a safe state is established.

4.4.1.10 CONFINED SPACE ENTRY ATTENDANT

A person deemed competent by site management to act as the Confined Space Entry Attendant to ensure safe work control during all entries. The Attendant is primarily responsible for:

- Understands hazards and consequences of working in the confined space
- Reviewing the Permit Pack entry requirements and confirming all identified controls and safeguards are in place to allow safe entry
- Confirming all entry and rescue equipment is in a ready state and available on site
- Ensuring a rescue plan is available and understood by involved personnel and rescue team
- Confirming atmospheric testing is completed, documented, and monitored
- Monitoring and documenting entry access and egress on the Confined Space Entry Permit, and maintaining communication with entrants
- Monitoring entry area and evacuating personnel if an unsafe condition occurs

4.4.1.11 CONFINED SPACE ENTRY SUPERVISOR

The person who ensures the Confined Space Entrants and Confined Space Entry Attendant are trained to recognize the hazards of working in confined spaces, provides entrants with proper PPE, ensures a rescue team and alarm system is in place, and rescue team personnel are trained in first aid and cardiopulmonary resuscitation (CPR).

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4.4.1.12 STAND-BY / SAFETY WATCH

The person assigned to monitor work activity who is familiar with the work being done, safety requirements of the job, and their role and responsibilities in regards to emergency response. Some of these roles and responsibilities can include:

- Maintaining a watch of ongoing work activities and safeguards to ensure controls are in place.
- Maintaining a watch of personnel during their execution of work to mitigate risk exposures that may occur.
- Initiate emergency response requirements and maintain communication to gain assistance where required.

4.4.1.13 PERMIT COORDINATOR

The person that helps support the application of the Work Management System and the day-to-day coordination of permitted work. The PTW Coordinator supports the Site Management team by assisting with planning, coordinating, and safe execution of work. The Coordinator must be competent in the Work Management Best Practice, familiar with work activities, and proficient in the ePTWS where applicable.

4.4.2 COMBINED ROLES

Due to the organizational framework and scalable nature of operations within Company assets, roles and responsibilities can be combined to enable safe planning and execution using this System. When establishing combined roles and responsibilities, the following guidelines should be considered:

- Persons assigned are deemed competent to fulfill the roles and are capable of executing their responsibilities outlined in the WMS.
- Site Management identifies and assigns roles to individuals to ensure that adequate safeguards exist that allow personnel to perform multiple roles.
- Additional focus should be given to segregation of responsibilities for critical activities such as Hot Work, Confined Space Entry, and Critical Work.

▲ CAUTION: [E] A person **must not** assume all three roles (PIC, AA, and AO) associated with a single work activity (insufficient segregation of responsibilities.)

The PIC approvals may be verbal under some circumstances, such as remote fields. Approval is given after discussion and confirmation that the job or change of scope can be done safely.

Reference: Section 6.1.3: "PTWS Principles," specifically "Remote Operations Scalability."

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4.5 DEFINITIONS

4.5.1 INTRODUCTION

The following definitions are referenced throughout this System. These definitions do not cover all terms, processes, and activities in the System. These definitions are intended to provide understanding of some key processes and terminology. Users of the Work Management System should seek understanding of any terminology, processes, and activities before executing work. Refer to [Table 1: Definitions](#)

Table 1: Definitions

Description	Definition
Acceptance testing	A planned testing process to confirm that the equipment is in ready-state and functioning correctly.
After Action Review (AAR)	Proactive learning process designed to informally learn from work with higher potential consequences and improve the ability to apply learnings to future work.
Air freeing	Removing oxygen from process equipment to prevent flammable mixtures occurring when hydrocarbons are reintroduced.
Atmospheric testing	Describes the activities and equipment used to test an areas' atmosphere to determine what contaminants are present at what quantifiable levels to ensure a <i>hazardous atmosphere</i> does not exist. Also known as "gas testing".
Blinding	The process of installing pressure-rated blinds or spades into process piping and equipment.
Breaking containment	Breaking the containment on equipment or systems such as hydrocarbons, chemicals, air, steam, fluids and gases. Examples of breaking containment activities are: ▪ Unbolting flanges and manways or removing screwed fittings on systems, such as, pipes, vessels, exchangers, valves, turbines, compressors, and pumps. ▪ Opening enclosures such as pig launchers or receivers and filter housings. ▪ Opening vents or drains to atmosphere ▪ Removing orifice plates ▪ Taking samples (when venting or draining to atmosphere)
Cleaning	Cleaning systems of residual products and substances so work can be conducted.
Competent Person	Person with knowledge and skills required to safely perform job-related tasks to achieve desired business outcomes.

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Table 1: Definitions, Continued

Description	Definition
Confined Space Authorized Entrant(s)	Personnel entering and working inside the confined space who know the hazardous conditions they are working in, use the proper personal protective equipment (PPE), communicate with the Confined Space Entry Attendant, recognize any warning signs, and exit from the permit space when ordered to do so, or when a hazardous atmosphere or condition is detected.
Basic Lift	The Basic lift classification covers lower risk lifting activities that are performed on a regular basis (a simple common, frequent operation) and that do not meet the Complex Lift Definition.
Complex Lift	Constitutes personnel lifts (unless risk assessed and approved by the Site Manager or equivalent), blind lifts, lifts that exceed 80% of the crane's current certified capacity or design operating limits, lifts over or near pressurized hydrocarbon filled equipment, lifts over or near well work equipment, lifts near overhead power lines, lifting hazardous materials, lifting through hatches.
Critical Safeguards	Designated safeguards to be managed as part of Scenario Management for a Site's highest consequence scenario(s). Critical safeguards are specific pieces of equipment or specific tasks that are critical to managing higher consequence scenarios. Critical safeguards are shown on the Scenario Management bow-ties and Site Scenario Register. Independent Verification: Tasks related to one or more Critical Safeguards may require additional risk control measures such as independent verification. Independent Verification means inserting hold points into the procedures for these critical tasks and having a co-worker or supervisor separately, and independently, confirm it is safe to proceed.
De-isolations	Unlocking and moving isolation valves, reconnecting systems, removal of blinds, and unlocking electrical switches to their normal operating state.
Designated welding area	A designated welding area is an area that meets at least one of the following criteria: ▪ An area functionally and physically isolated (horizontally and vertically) from sources of potentially flammable or combustible materials. ▪ A designated area that has no flammable or combustible material. NOTE: Designated welding areas may be pre-approved for hot work, allowing welding or similar activities to be performed without a hot work permit.

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Table 1: Definitions, Continued

Description	Definition
Draining and depressurizing	The process of draining fluids from vessels and pipework and removing pressure from isolated systems and equipment.
Energy isolation (lock-out/tag-out)	Energy isolation (lock-out/tag-out) is a process used to facilitate and maintain equipment and systems at a zero energy state. This could involve positioning isolation valves, physical disconnection, blinding, securing electrical breakers and switches, and other approved isolation methods.
Equipment testing	The process of temporarily de-isolating equipment and energizing to a live-state for operational testing or fault finding.
Error-Prone Situation	A work condition in which there is greater chance for error when performing a specific action.
Gas-free areas: toxicity	An area is considered to be gas free with respect to toxicity if the concentration of gas is less than the levels stipulated by regulatory agencies or by Company guidelines. Reference: <i>Upstream Safety Manual</i> , "Hazardous Materials"
Habitat (Hot Work)	Pressurized structure installed over and enclosing the work area where hot work is conducted without shutting-in and depressurizing the hydrocarbon process system in the vicinity.
Hazardous atmosphere	An atmosphere that could expose workers to the risk of death, incapacitation, impairment of ability to self-rescue, injury, or acute illness.
Higher Potential Consequences	Life-altering injuries and/or significant process safety events
Higher-risk electrical work	Higher-risk electric work involves activities with the potential (accidental or otherwise) for individuals to come in contact with energized electrical components of 300 volts or greater, or to be exposed to high temperatures or flashes resulting from electrical arcing or electrical fault conditions. Work on live electrical systems above Extra Low Voltage (ELV) levels. Extra Low Voltage circuits have a voltage not exceeding 50 Volts RMS A.C. between conductors or earth. E.g. 24 VDC instrument loops, CCTV, data networks etc. Reference: <i>Upstream Safety Manual</i> , "Electrical Safety"
Hold Point	A planned interrupter used to move personnel from Fast to Slow Thinking and drive us to pause work.
Hydrocarbon freeing	Removing residual hydrocarbon liquids and gases from vessels, pipework, and equipment to prevent flammable mixtures when opened to atmosphere.

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Table 1: Definitions, Continued

Description	Definition
Hydrocarbon source activity	Hydrocarbon source activities are activities that can result in a hydrocarbon release. These include: ▪ Breaking containment ▪ Hydrocarbon bunkering (for example, transfer from supply boat to platform). Pressuring up process equipment that has been previously opened.
Immediately dangerous to life and health (IDLH)	An atmosphere that poses an immediate threat to life, would cause irreversible adverse health effects and/or would impair an individual's ability to escape from a dangerous atmosphere. Includes oxygen deficient atmospheres, and atmospheres where the hazardous contaminant(s) cannot be measured or reasonably estimated.
Inerting	Inerting is the displacement of an atmosphere by a nonreactive gas (commonly nitrogen) to such an extent that the resulting atmosphere is no longer within the flammable limits. ! WARNING: Inerting creates an oxygen-deficient atmosphere that is immediately dangerous to life and health (IDLH).
Isolation planning	The process in which competent individuals identify energy isolation requirements, assess relevant hazards, document equipment and system isolation points on an Isolation Control Certificate (ICC), develop isolating plans (where required), and ensure energy isolation verification and demonstration is performed prior to work execution.
Job closeout	At the completion of a job, the process of closing out Isolation and Temporary Defeat Certificates, closing out work permits, and updating blind lists, status boards, and other documentation associated with the Permit Pack.
Leak testing	Introducing pressure to the system to confirm that integrity has been restored and there are no leaks.
LEL	Lower explosive limit (LEL) is the lowest concentration of a flammable gas or vapor in air that ignites when an ignition source is present.
Permit Pack	Compilation of documents that accompany a Permit to Work for the safe execution of work. Permit Pack content is determined by using the Permit Pack Requirement Matrix. Permit Packs may include procedures, checklists, safe work practices, piping and instrumentation diagrams (P&IDs), Isolation Control Certificates, Temporary Defeat Certificates, and any other document necessary to undertake the work.

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Table 1: Definitions, Continued

Description	Definition
Permit system	A written process for preparing and issuing permits for work to be performed (that is, hot work, critical work, and confined space entry) and for returning the permit space to service following the termination of the work performed.
Permit to work	A work control document required for the planning, communication, authorization of work execution, and reinstatement as identified in the Work Management System. Reference: Section 6.0 Permit to Work
Personal isolation	Reference: Section 7.5.1: Energy Isolation and Isolation Control Certificate , "Personal Isolation (Delegated Isolation)".
Purging	Purging is an operation in which a gas or liquid is displaced from a vessel or piping by another gas or liquid. Reference: <i>Upstream Safety Manual</i> , "Operating Equipment and Facilities," and "Piping Connections"
Recommissioning	At the conclusion of work, the process of reinstating equipment components (for example, pipework, flanges, circuit breakers and control systems), removing blinds and spades, and confirming that the system integrity is restored.

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Table 1: Definitions, Continued

Description	Definition
Remote Field Operations (RFO)	Remote Field Operations can be designated by the Site in instances where access to the work location and ePTWS are limited and resources are not readily available, such as: ▪ Normally unstaffed platforms ▪ Well locations outside of plant fence lines ▪ Tank Batteries ▪ Pipeline valve stations ▪ Metering Stations ▪ Standalone compressor stations outside of plant operating areas. ▪ Remote work areas where considerable travel distance or time is required. The Work Management System recognizes two modes of Remote Field Operations: Remote Field Operations (RFO) - Workers operate out of a central office, often beginning and ending their day in that location. ePTWS software and tools are readily accessible. Non-centralized Remote Field Operations (NC RFO) - Workers travel directly to worksites and do not start their shift at a central/field office. NC RFOs typically: ▪ Have extended travel times between locations and cover large geographic locations ▪ Do not have access to ePTWS, printers, data connectivity, etc. ▪ Have individuals or small independent teams executing work with limited interface with other functions ▪ Have low complexity facilities (i.e. simple well pads, tank batteries, etc.) ▪ Have minimal potential for SIMOPS Develop & Review permits & non-permitted work the day of work execution
Risk Management / Risk Assessment / Risk Screening	During work planning, job execution, or any other time that it is not clear that the relevant hazards have been sufficiently controlled per WMS expectations, additional evaluation of the risk must occur prior to task execution. This evaluation follows the risk assessment process outlined in the Risk Assessment Management System (RAMS) manual (OIMS 2-1). If the hazards cannot be adequately managed, the task must not be performed.

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Table 1: Definitions, Continued

Description	Definition
Start/Stop Work Expectations	Expectation that personnel delay START of work or STOP work if there is uncertainty, life-saving action safeguards are not effective, or it is unsafe to do so.
Temporary Defeat (TD)	A Temporary Defeat is the isolation or removal of a critical device or a critical system while the equipment or facility being protected can remain in service because adequate alternative controls that maintain safety and integrity can be provided. Reference: Section 7.3: "Temporary Defeat of Critical Devices"
Trigger	A prompted interrupter used to move us from Fast to Slow Thinking and drive us to pause or stop work (e.g., unexpected outcome during a task).
Zero Energy	Zero Energy state is achieved when sources are isolated and a safe condition is established where: ▪ There is no potential for inadvertent or accidental de-isolation. ▪ There is no potential energy release that would be hazardous to people. A safe working condition is maintained while breaking containment or working on the equipment.

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5.0 WORK MANAGEMENT FRAMEWORK

INTRODUCTION

This section provides a general overview of the Work Management System framework for work planning, site preparation, work execution, and closeout / reinstatement phases.

5.1 WORK MANAGEMENT PROCESS FLOW

INTRODUCTION

Use of the Work Management System Framework is key to strengthening barriers in areas associated with people, environment, processes, and equipment. The following work elements describe a series of work flow process steps that enable work on the facilities to be planned, coordinated, controlled, and executed in a safe and efficient manner. These elements form the basis of the Work Management System (WMS) and fall into the general categories listed in this section. Work activities must be planned, prepared, executed and closed out in a manner consistent with the Work Management Process flow described in this section.

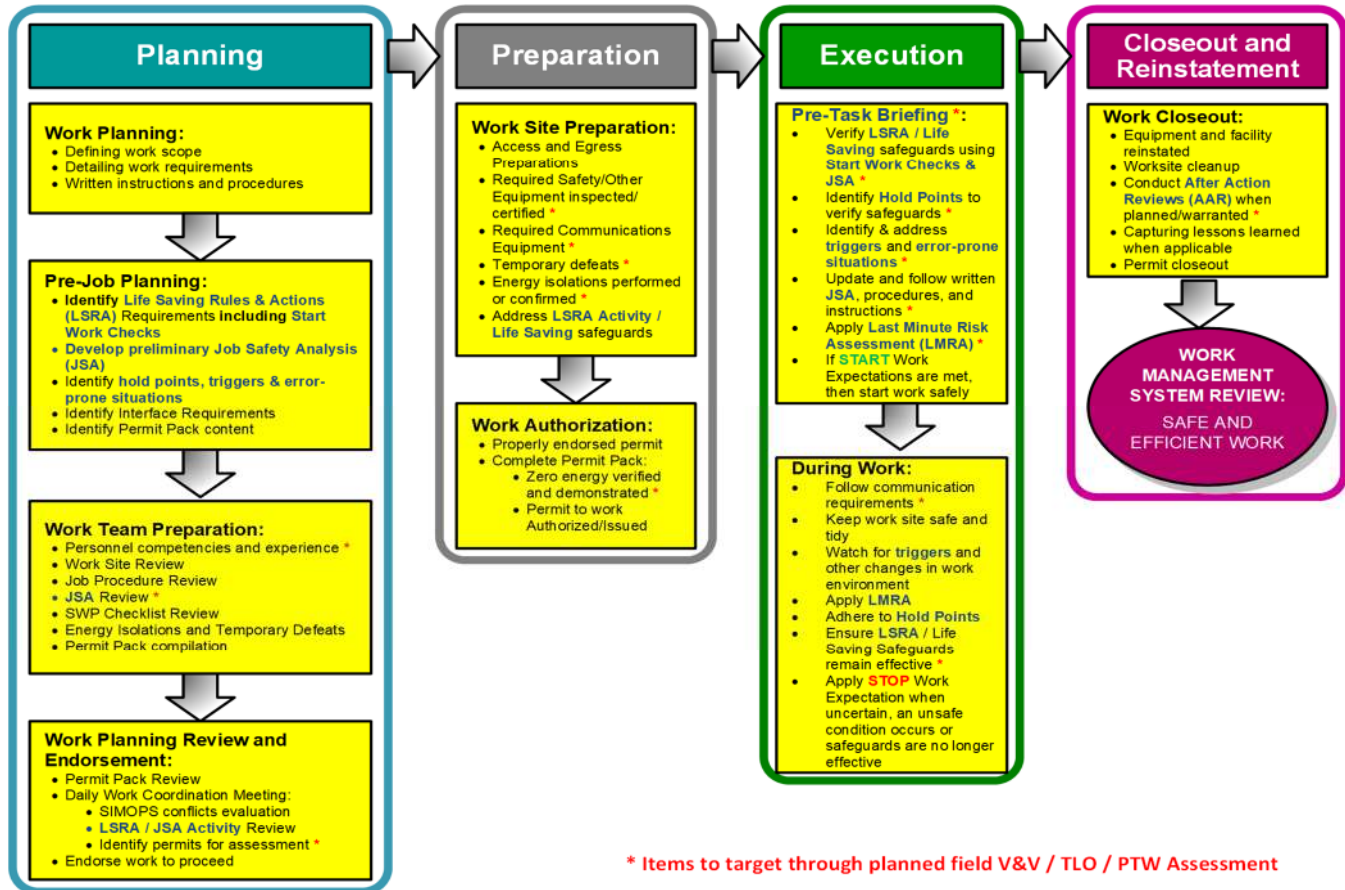
WMS FLOW CHART

The following flowchart summarizes the Work Management framework and process flow: Refer to [Figure 3: Work Management Framework and Process Flow](#)

Work Management System (WMS)

Figure 3: Work Management Framework and Process Flow

Work Management System Framework



PERSONNEL SAFETY MANAGEMENT SYSTEM

The WMS Framework has been designed to work in conjunction with the Personnel Safety Management System (PSMS). PSMS is a corporate consistent approach used to deliver Safety in the Moment for work with higher potential consequences and to build / validate Safety Capacity with the objective of eliminating life altering injuries and fatalities.

Refer to [GOTO/PSMS](#) for more information on the Personnel Safety Management System (PSMS) and the following PSMS components referenced throughout this manual:

- Pre-Task Briefing
- Life Saving Rules & Actions (LSRAs) & Start Work Checks
- Job Safety Analysis (JSA)
- Hold Points / Triggers / Error-Prone Situations
- START** / **STOP** Work Expectations
- Last Minute Risk Assessment (LMRA)
- Verification & Validation (V&V)
- Targeted Learning Observation (TLO)
- After Action Reviews (AAR)

5.1.1 WORK PLANNING

Work Management System (WMS)

INTRODUCTION

This section provides details on the key activities that fall under the Work Planning element of the Work Management process flow. The activities described in this section are intended to confirm that jobs on Company facilities are planned appropriately.

5.1.1.1 WORK SCOPE

It is important that the work scope is clearly and accurately documented so there is no ambiguity regarding the proposed work, the equipment, or location involved.

The work scope detail should include information regarding Preparation, Execution, and Closeout & Reinstatement aspects of the work activity.

Consider the following while defining the work scope:

- Precise tasks to be performed and clearly document those tasks on the permit.
- Order for performing the tasks.
- Identify specific portions of work that have higher potential consequences (life-altering injuries and/or significant process safety events)
- Identify where HC1/HC2 scenario management and critical task execution apply or could be impacted by the task
- Tools, materials, and equipment required.
- Specific location and system or equipment to be worked (with consideration for hazardous area classification where appropriate).
- Identification and documentation of required energy isolations and temporary defeats.
- Division and communication of responsibilities for functional groups (such as, operations, maintenance, and construction) are well defined, documented, and understood.

Work Management System (WMS)

5.1.1.2 WORK REQUIREMENTS

When determining work requirements, the following factors should be considered and documented (when required) to ensure work is performed safely and efficiently:

- Required resources and skills
- Time required to do the work
- Allowable weather conditions
- Potential conflicting activities (SIMOPS)
- Equipment and tool inspection/certification and preparation
- Materials and consumables requirements
- Existence of relevant standards, written procedures, instructions, Safe Work Practices and Checklists or Work Aids.
- Development of new procedures (if required)
- Safe access and egress for personnel and equipment
- Specific personal protective equipment (PPE)

5.1.1.3 WRITTEN INSTRUCTIONS AND PRACTICES

Work performed on facilities should be covered by documentation. These can come from several sources, depending on the:

- Complexity of the work
- Frequency with which the work is performed
- Risk inherent in the work
- Potential consequences of executing work improperly

The sources of these written practices or instructions can be:

- Personnel Safety Management System (PSMS)
- Upstream Safety Manual
- System industry practices
- Regulatory requirements
- Manufacturer procedures
- Facility operating procedures and checklists
- Facility maintenance procedures
- Safe Work Practices and associated Checklists
- Work aids
- Work orders
- Specific job procedures, precautions, and communication requirements.

Work Management System (WMS)

5.1.2 PRE-JOB PLANNING

INTRODUCTION

This section provides details on the key activities that fall under the Pre-Job Planning element of the Work Management process flow.

The pre-job planning activities confirm that the safety aspects of work on the company facilities are appropriately planned and will assist in identifying the required processes and documentation needed in the Permit/Work Pack. The level of detail for each activity depends on the complexity of the work activity or task and the risks involved.

5.1.2.1 LIFE SAVING RULES AND ACTIONS (LSRA) OVERVIEW

Life Saving Rules and Actions (LSRA) define the most critical and life-saving actions controlled by a worker. These worker actions become key safeguards in eliminating higher potential consequences (i.e., life altering injuries and significant process safety events). LSRAs provide a corporately consistent approach to identifying and managing activities with higher potential consequences.

Start Work Checks (SWC) for each LSRA help workers understand and verify that LSRA safeguards are in place and effective prior to starting work and throughout the task.

5.1.2.2 LSRA APPLICATION

[CriticalE] LSRA safeguards **must** be discussed / verified with the work team during the Pre-Task Briefing using the associated Start Work Checks. This discussion is led by the Permit Holder or designated supervisor of the work activity. [CriticalE] Verification that all life-saving safeguards, including those identified using the Start Work Check(s), are in place and effective prior to starting work is a requirement (Start Work Expectations).

Life-saving safeguards are monitored during the work activity to ensure that they remain effective. [CriticalE] If a safeguard is no longer in place or becomes ineffective, workers **must** stop work (Stop Work Expectations).

During the Daily Work Coordination Meeting, all activities with higher potential consequences, including those with Life Saving Rule & Actions, are identified with consideration given to the following:

- Job Scope Verification
- On site Supervision
- Assess resource and competency requirements

More information on the Life Saving Rules & Actions and associated Start Work Checks can be found at [GOTO/LSRA](#).

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5.1.2.3 IDENTIFYING LSRAS

The Company has LSRAs for work activities involving higher potential consequences. The LSRAs provide workers with a specific set of actions they must do to validate that safeguards are in place and effective prior to the start of work. Refer to [Figure 4: Life Saving Rules & Actions](#)

Figure 4: Life Saving Rules & Actions



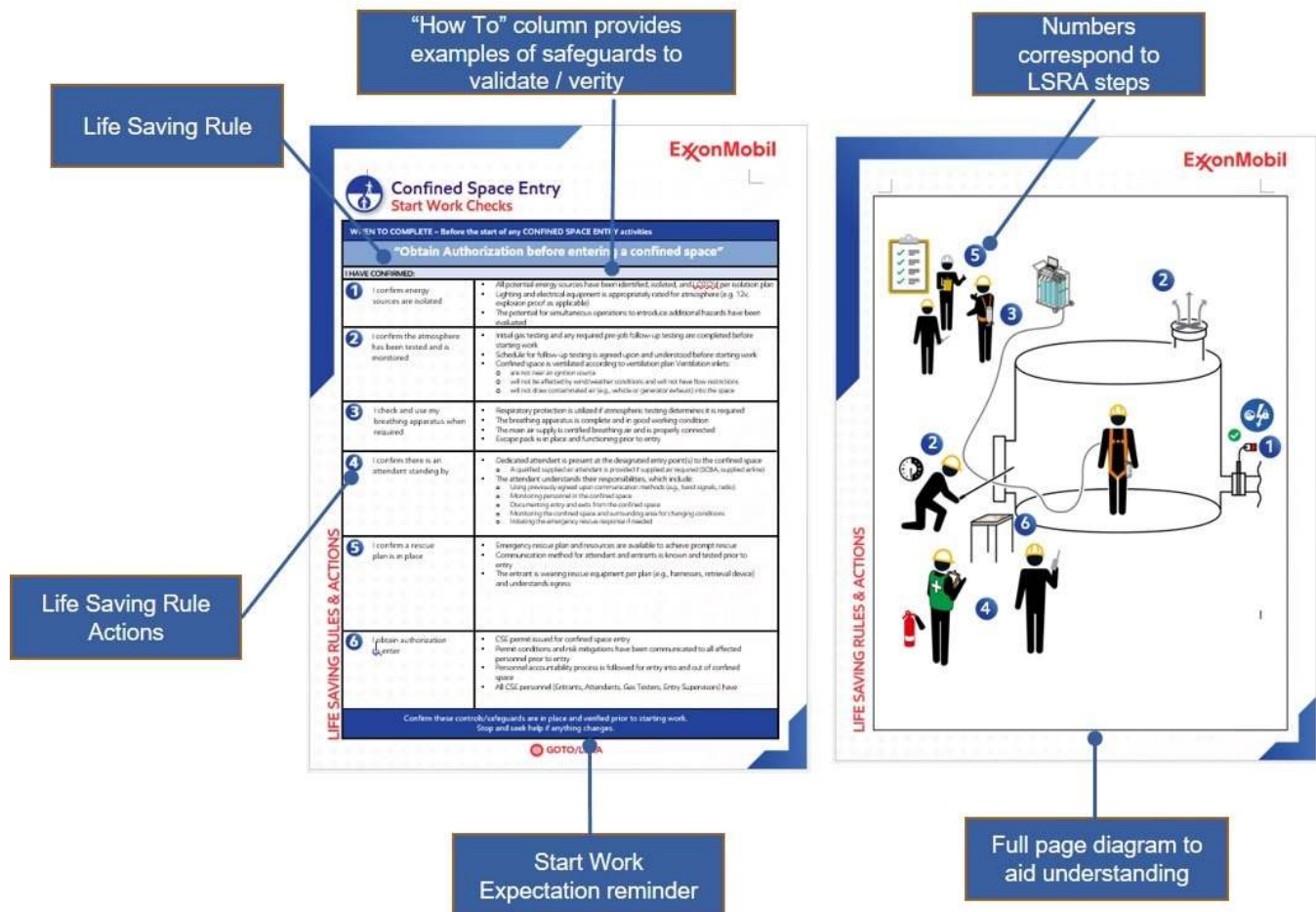
Work Management System (WMS)

5.1.2.4 START WORK CHECKS (SWC)

Start Work Checks (SWCs) are used by Permit Holders or designated supervisors during Pre-Task Briefings to lead a two-way discussion with workers using the “How to” section of the SWC to confirm safeguard effectiveness prior to starting work. SWCs are also used to monitor and verify safeguard effectiveness during work activities. In the Job Planning phase they are used to help identify hold points, triggers, and error-prone situations that may occur during the course of the work activity.

SWCs will also be used by other ‘verifiers’ (e.g., supervisors, field leadership, SMEs) to periodically verify safeguards remain in place and effective during work. Refer to [Figure 5: Start Work Checks](#)

Figure 5: Start Work Checks



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5.1.2.5 JSA APPLICATION

A Job Safety Analysis (JSA) is required for permitted and non-permitted work activities when:

- LSRAs and Start Work Checks do not adequately address the higher potential consequences, or
- Work activities do not involve an LSRA but still include higher potential consequences.

The JSA should be developed during the planning phase and document task specific safeguards and information to prevent higher potential consequences (i.e., life-altering injuries and/or significant process safety events).

Where JSAs are used for tasks with no higher potential consequences, the JSA should focus on the worst case scenario(s) for the task, most significant hazards, and associated safeguards. If there is doubt regarding the use of a JSA, consult the Area Authority or designated Supervisor.

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5.1.2.6 JSA FORM

[E] If an existing JSA (reference JSA) is not available or is not adequate because there are any of the following:

- additional life-saving safeguards (beyond LSRA SWCs),
- multiple hold points,
- multiple triggers, and/or
- error-prone situations,

It must be updated or a new JSA must be developed by the Permit Holder or designated supervisor.

Refer to Figure 6: JSA Form Page 1 and Figure 7: JSA Form Page 2

Figure 6: JSA Form Page 1

JSA No:	JOB SAFETY ANALYSIS (JSA) FORM			Rev Oct 24, 2022
Unit/Area:	Specific Location:	JSA Issuer:	JSA Receiver:	Date:
Job Description for Tasks with Higher Potential Consequence (i.e., life altering injury, significant process safety event) (include Equipment Numbers as Practical):				
Check LSRA(s) applicable to the tasks and review Start Work Check(s). If there are additional task steps with higher potential consequence or no applicable LSRA, complete Page 2 of the JSA.				
 Confined Space Entry Obtain authorization before entering a confined space ☐ 1. I confirm energy sources are isolated 2. I confirm the atmosphere has been tested and is monitored 3. I check and use my breathing apparatus when required 4. I confirm there is an attendant standing by 5. I confirm a rescue plan is in place 6. I obtain authorization to enter	 Bypassing Safety Controls Obtain authorization before overriding or disabling safety controls ☐ 1. I understand and use safety-critical equipment and procedures which apply to my task 2. I obtain authorization before: • Disabling or overriding safety equipment • Deviating from procedures • Crossing a barrier	 Energy Isolation Verify isolation and zero energy before work begins ☐ 1. I have identified all energy sources 2. I confirm that hazardous energy sources have been isolated, locked, and tagged 3. I have checked there is zero energy and tested for residual or stored energy 4. I never remove or tamper with energy control devices without authorization 5. I confirm equipment is back to safe operating condition prior to re-energizing	 Excavation Obtain authorization before digging or entering excavations ☐ 1. I obtain authorization prior to performing excavation / trenching / ground disturbance activities 2. I confirm underground utilities and structures are located, de-energized or protected 3. I confirm hazard controls and confined space entry as required 4. I monitor for changing conditions and exit as appropriate	
 Safe Mechanical Lifting Plan lifting operations and control the area ☐ 1. I confirm that the equipment and load have been inspected and are fit for purpose 2. I only operate equipment that I am qualified to use 3. I establish and obey barriers and exclusions zones 4. I never walk under a suspended load 5. I understand the lift plan prior to starting work and keep hands off load unless authorized 6. I establish and maintain communication with lifting and rigging personnel	 Hot Work Control flammables and ignition sources ☐ 1. I identify and control ignition sources 2. Before starting any hot work: • I confirm flammable material has been removed or isolated • I obtain authorization 3. Before starting hot work in a hazardous area I confirm: • A gas test has been completed • Gas will be monitored continually	 Working Around Mobile Equipment See and be seen; maintain a safe distance ☐ 1. I plan work to separate workers and mobile equipment 2. I only operate mobile equipment if qualified and use spotters where required 3. I verify safety devices are functional on mobile equipment 4. I establish clear means of communication with equipment operator 5. I stay out of the equipment's exclusion zone while in operation 6. I confirm controls are in place to prevent unintentional movement of mobile equipment and/or loads	 Working at Height Protect yourself against a fall when working at height ☐ 1. I inspect my fall protection equipment before use 2. I secure tools and work materials to prevent dropped objects 3. I tie-off 100% to approved anchor points while outside protected area 4. I have validated prompt rescue capability when wearing fall protection 5. I will follow safe work practices when climbing, descending or working from ladders	
 Breaking Containment/OPE Manage ignition sources and exposure to energy & hazardous substances ☐ 1. I verify opening/break location(s) 2. I confirm energy isolation is complete 3. I manage ignition sources and confirm gas testing is completed when required 4. I use appropriate PPE and defensive behaviors when opening process equipment	 Driving Wear a seat-belt and do not speed; no distractions ☐ 1. I always wear a seatbelt 2. I do not exceed the speed limit, and reduce my speed for road conditions 3. I do not use phones or operate devices while driving 4. I am fit, rested and fully alert while driving 5. I follow energy journey management requirements	 Work Authorization Work with a valid work permit when required ☐ 1. I have confirmed if a permit is required 2. I am authorized to perform the work 3. I understand the permit 4. I have confirmed that hazards are controlled and it is safe to start 5. I stop and reassess if conditions change	 Line of Fire Keep yourself and others out of the line of fire ☐ 1. I position myself to avoid: • Moving objects • Vehicles • Pressure releases • Dropped objects 2. I establish and obey barriers and exclusion zones 3. I take action to secure loose objects and report potential dropped objects	
Confirm these safeguards are in place and effective prior to starting work using the applicable SWC(s). During the task, if safeguards are no longer in place or effective, STOP Work.				

Work Management System (WMS)

Figure 7: JSA Form Page 2

Are there any additional higher potential consequence hazards present?							
Electrical Energy		Moving Objects		Radiation		Simultaneous Operations (SIMOPS) or Stacked Work	
Stored Energy		Muscular		Lighting & Visibility			
Thermal Energy		Hazardous Substances		Noise & Vibration			
Higher Potential Consequence Task Steps (Description)		Check off to indicate Hold Point	POTENTIAL HAZARDS & CONSEQUENCES Include applicable Error Prone Situations and Triggers		SAFEGUARDS (Actions & Physical Safeguards) Eliminate or Mitigate the hazards		
Example: Isolate and de-pressure line		Hold Point 	Example: <u>Hazard:</u> Line pressure > 2000 psi <u>Consequence:</u> Fatality and Significant Pressure Release <u>Error Prone Situation:</u> 2 identical valves in same area, and bleeder valve not clearly identified <u>Trigger:</u> Residual line pressure is encountered		Example: Clearly identify (e.g. label) bleeder valve to reduce potential for error <u>Hold Point:</u> verify zero energy before proceeding to the next step. <u>Trigger:</u> If residual line pressure is encountered, STOP work, inform supervisor and assist in resolving issue		
		Hold Point 					
		Hold Point 					
		Hold Point 					
		Hold Point 					
Confirm these safeguards are in place and effective prior to starting work using this JSA. During the task, if safeguards are no longer in place or effective, STOP Work.							
Lessons Learned: What worked well? What improvements can be made? Discussion leader to provide feedback to update planning package and inform JSA owner if needed.							

JSA FORM (CONTINUED)

The JSA Form contains the following components:

- JSA Heading – identifies work scope, location, JSA issuer & JSA receiver
- LSRA listing – for ease of reference for work teams to ensure all applicable SWCs are applied
- Hazard Identification Table – to assist in planning and identifying any additional site specific hazards that could lead to a higher potential consequence event
- List of only Higher Potential Consequence Task Steps with additional safeguards (beyond those addressed by the LSRA SWC(s)), hold points, triggers, and/or error-prone situations
- Hold Point Indicator – identifies steps that require a planned hold point
- Potential Hazards and Consequences – focus on only higher potential consequence hazards. This section should also identify any known error-prone situations and triggers.
- Safeguards – Actions and Physical Safeguards to eliminate or mitigate hazards. This section can also include hold point detail and actions responding to identified triggers and/or error-prone situations
- Start/Stop Work Confirmations
- Lessons Learned (when applicable)

Work Management System (WMS)

5.1.2.7 IDENTIFY HOLD POINTS, TRIGGERS & ERROR-PRONE SITUATIONS

Understanding Human Performance in relation to task or work scope is important to help prevent serious injuries and process safety events. When planning work activities, consider / identify the following:

- Hold Points – Identify these planned interrupters at key times in the task where the worker is the last line of defense or during key task transition points when specific safeguards need to be verified.
- Triggers – Identify these prompted interrupters or prompted events / situations / conditions that would result in specific action to STOP work (e.g., “IF this happens, THEN I will STOP work until resolved”).
- Error-Prone Situations – Identify conditions / factors in which there is greater chance for error when performing specific actions (ex. similar valves in congested areas, equipment tags not aligned with equipment order, unclear procedures, time pressures, fatigued or distracted personnel, etc.).

For more information on these and other Human Performance concepts, visit the [GO&S Human Performance site](#).

5.1.2.8 INTERFACE REQUIREMENTS

A thorough understanding of the work to be performed and its interfaces with routine operations or other planned work is required. Although practices and policies exist to guide the process of interface identification, parties involved in the work and in the work permit preparation and approval should identify all interfaces that could be impacted adversely.

These interfaces can be:

- A requirement to communicate with others during the work.
- A requirement to coordinate activities with others during the work.
- A physical conflict with other work.
- A safety conflict with other work.
- Too many activities taking place in one area.

The process of identifying interfaces should be taking place during the entire work planning, scheduling, and job safety planning process, and included in the JSA for tasks with higher potential consequences.

Work or activities that cross shifts or crews require that work status and precautions are communicated across crew and shift changes. This communication can be accomplished in several ways:

- Detailed handover notes and meetings
- Pre-Task Briefing
- Permit Pack content
- SIMOPS Deviation
- Assigning an Area Interface Authority (AIA)/ Site Coordinator (SC) to the work scope

Work Management System (WMS)

5.1.2.9 IDENTIFY PERMIT PACK CONTENT

The scope of work for the permit can be used with the Permit Pack Matrix to identify the documentation required for the safe execution of the work, for example:

- Permit type
- Life Saving Rules & Actions (LSRAs) and associated Start Work Checks (SWCs)
- Job Safety Analysis (JSA)
- Isolation Control Certificate
- Relevant SWP Checklists
- Procedures
- P&IDs
- Rescue plan
- Lift plan

The Permit Pack Matrix identifies documents required for specific work activities along with other typical documentation that could be considered for use during work execution. The PH in consultation with the AA or AAF determines the required documentation needed in the Permit Pack during the Planning phase.

5.1.3 WORK TEAM PREPARATION

INTRODUCTION

This section provides details on key activities that fall under the Work Team Preparation element of the Work Management process flow. Once the job is planned appropriately and the relevant safety elements are addressed, the work team prepares for the job. These activities culminate in the preparation of the Permit Pack.

5.1.3.1 PERSONNEL COMPETENCIES AND EXPERIENCE

Personnel who work on UPSTREAM sites **must** have the required competencies for the activities they will perform and the equipment they will use.

Work Management System (WMS)

5.1.3.2 WORK SITE REVIEW

The need for an inspection of the job site should be determined at the planning stage and may include the following:

- Potential access or egress problems for equipment or personnel
- Requirement for scaffolding or rope access
- Required crane access
- Required lifting devices, including a lifting point review
- Location of safety equipment such as fire extinguishers, fire hoses, chemical showers, eyewash stations, and flame and smoke detectors
- Location of communications equipment such as telephones, call boxes, alarm speakers, and lights.
- Location of isolation points, drain points, pressure relief points, and purge points
- Location of potential gas sources and ignition sources
- Location of equipment or control devices to be protected
- Location of drains to be stopped/plugged
- Location of appropriate ventilation intakes
- Areas to be access-restricted during the operation, including the location of barriers and signs.
- Requirements for restricting access due to the following:
 - Potentially falling objects
 - Potentially blowing debris or sparks
 - Potential hydrocarbon escapes
 - Exposure to high voltage electricity
 - Exposure to high pressure leaks
 - Exposure to dangerous chemicals
 - Exposure to hydrogen sulfide (H₂S)
 - Exposure to asbestos or fibrous materials
 - Exposure to noise
 - Exposure to radiation or naturally occurring radioactive material (NORM)
 - Exposure to explosives

5.1.3.3 JOB PROCEDURE REVIEW

Any specific procedures prepared to assist in executing the job safely, including specified Hold Points and associated actions, are reviewed with the work team to ensure that everyone has a clear understanding of how the job will be executed.

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5.1.3.4 JSA REVIEW

The JSA is reviewed prior to the work execution. This review should include:

- Hazards identified in the JSA (focus only on hazards with potential for life altering injuries and/or significant process safety events)
- Identified safeguards (including LSRA SWCs and other life-saving safeguards)
- Any applicable Hold Points, Triggers, and/or Error-Prone Situations and the expected action to take if and when those situations / conditions are encountered during the work
- Potential interfaces that may be required to manage SIMOPS

5.1.3.5 SWP CHECKLIST REVIEW

Safe Work Practice (SWP) Checklists included in the Permit Pack are completed as the job progresses through each phase. SWP Checklists are added to the Permit Pack for review by the AA/AAF. The PH will confirm that any checklist items that are required before the job commences are in place and will ensure that all required checklist items are checked as the job progresses. The PH completes the SWP Checklist items at the Job Site prior to execution.

By signing the permit each shift, the PH acknowledges that all Permit Pack requirements are being followed and implemented when required, including the Checklist items.

For non-permitted work, the task leader will ensure that any checklist items are being followed and implemented when required.

Reference: [GOTO/WMSEFORMS](#), "Safe Work Practices" for the available SWP Checklists.

Checklists are tools to facilitate compliance with Safe Work Practices from job planning through to closeout.

Sites can have additional SWP Checklists where required in their Appendix A. [E] Line items must not be deleted from the SWP Checklists contained in [GOTO/WMSEFORMS](#), "Safe Work Practices." Justifiable items can be added to a checklist by the Site e.g. regulatory checks.

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5.1.3.6 ENERGY ISOLATION AND TEMPORARY DEFEATS

[E] A review of the requirements for energy isolations and Temporary Defeats required to perform the job **must** take place and may include:

- A documentation review of relevant piping and instrumentation diagrams (P&IDs), process flow diagrams (PFDs), safe charts, electrical drawings, and other facility documentation.
- A physical inspection of the affected facilities so that potential energy sources are identified in the documentation review.
- Identification of drain points, purge points, hydrocarbon test points, and pressure test points.
- The physical marking of isolation, drain, purge, vent and test points on the plan and on copies of the appropriate documentation. This documentation should be included in the Permit Pack when available.
- Determination of critical devices requiring temporary defeat.

5.1.3.7 PERMIT PACK COMPILATION

[E] In preparation for the review by the Area Authority, the Work/Permit pack **must** have all associated certificates, applicable LSRA Start Work Checks, SWP Checklists, Procedures and other required documents. [CriticalE] Process Safety Risks and the health of Critical Safeguards identified with HC1/HC2 scenarios **must** be considered during work preparation and the Team should consider if the planned work can impact the Critical Safeguards. The completed work planning activities serve as inputs for the preparation of the Permit Pack. Based on the work planning activities, typically the Permit Holder commences preparation of the Permit Pack.

The Permit Pack Requirements Matrix (See [Figure 8: Work Management System Permit Pack Requirements Matrix](#)) are attached and the Permit Pack is ready to be reviewed.

Reference: [GOTO/WMSEFORMS](#), "Permit to Work," for a stand-alone version of the Permit Pack Requirements Matrix.

Work Management System (WMS)

Figure 8: Work Management System Permit Pack Requirements Matrix

Work Management System Work Pack Requirements Matrix

Permit Category	Permit Pack Requirements		
M Mandatory for permit pack C Consider if relevant to the task/work team – to be determined by PH & AA/AAF	LSRA Start Work Check(s)*	Global Safe Work Practice Checklist	Atmospheric Test Record
Hot Work	M		M
Confined Space Entry	M		M
Critical Work:			
• Breaking Containment	M		C
• Complex Lift	M	M	C
• Working at Heights	M		C
• Diving Operations	C*	M	C
• Ground Disturbance/Excavation	M		C
• High Risk Electrical	C*	M	C
• Abrasive Blasting			C
• Spark Potential	C*		M
• Working on Pressurized Systems	C*		C
• Over-the-Side Work	C*		C
• Working in Exposed Locations	C*		C

* Note: In addition to the LSRA Start Work Check available for a specific Permit Category, all LSRA Start Work Checks need to be evaluated for applicability to the work, including Energy Isolation, Bypassing Safety Controls, Line of Fire, Work Authorization, Work Around Mobile Equipment, and Driving.

Other Documents to Consider:

- | | | |
|--|---------------------------------------|---|
| ☐ LSRA SWCs | ☐ JSA | ☐ Other Lifting Checklists |
| ☐ Atmospheric Test Record | ☐ Rescue Plan | ☐ Other SWP Checklist |
| ☐ Temporary Defeat | ☐ P&ID/Drawing | ☐ SIMOP Deviation |
| ☐ Isolation Certificate | ☐ Procedures | ☐ Safety Data Sheet |

Cross Referenced Certificates:

Type	Certificate/Permit Number(s)	
Isolation:		
Temporary Defeat:		
Work Permit(s):		

☐ List Required Checklists: _____

☐ Other (describe): _____

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5.1.4 WORK PLANNING REVIEW AND ENDORSEMENT

INTRODUCTION

This section provides details on the key activities that fall under the Work Planning Review and Endorsement element of the Work Management process flow. The intent of these activities is to confirm that the appropriate degree of job and safety planning has taken place, and that all appropriate preparations and precautions are specified and planned for scheduled work.

For break-in or non-scheduled work, the Permit Pack is reviewed by the Area Authority and Area Operator. The Area Authority is responsible for notifying the PIC of this work and ensuring interfaces have been informed. The Permit Holder can assume the role of the Area Authority and Area Operator in a remote location and discuss the proposed activities with the PIC before commencing.

5.1.4.1 PERMIT PACK REVIEW

A review of the Permit Pack is conducted by the Area Authority (AA/AAF) to ensure that it is properly prepared and the necessary documentation is included for safe execution of work. The Area Authority for the Permit Pack review can be the Operations or Functional Supervisors.

This review by the Area Authority should include but not be limited to the following:

- Is the correct Work Permit used?
- Have specific portions of the work that have higher potential consequences (life-altering injury or significant process safety event) been defined, including consideration of mobilization and demobilization activities?
- Have all relevant LSRAs and associated SWCs been determined?
- Is a JSA required; if so, is it adequate?
- Has the size and complexity of the task been considered?
- Is the work description adequate and scope clearly defined?
- Are hazards clearly identified and managed with appropriate controls?
- Does this work relate to one of the Site's highest consequence scenario(s) and associated Critical Safeguard(s) as designated by Scenario Management?
- Are the specified preparations and precautions adequate?
- Are the required mechanical, instrument, and electrical isolations specified?
- Are the required Temporary Defeats specified?
- Are the required resources and skills available?
- Are any physical or organizational interfaces addressed?
- Are there any impacts due to SIMOPS?
- Are weather conditions taken into account?
- Is the timing of work execution acceptable?
- Are any specific equipment preparations, inspections or certifications required?
- Are applicable Safe Work Practice Checklists attached?
- Are relevant procedures identified and attached?
- Does the Permit Holder or designated supervisor plan to conduct an After Action Review?

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5.1.4.2 DAILY WORK COORDINATION MEETING AND SIMOPS CONFLICTS

Daily Work Coordination Meetings (DWCMs) should be attended by the key asset personnel, specifically, the PIC, AA, and functional supervisors.

Reference: Section 5.3: "Tools," Typical Daily Work Coordination Meeting agenda

The objective of the meeting is to review and coordinate interfacing activities and manage the risks involved with concurrent activities and simultaneous operations (SIMOPS) by evaluating the proposed jobs prior to endorsement. New and suspended permits are reviewed. Non-permitted work that could conflict with planned activities are also discussed at this meeting.

Reviewing the next day's work by location and job type enables Site Management to identify any potential work conflicts and potential interface issues.

NOTE: Cross-referencing conflicting work permits on the PTW can help identify/manage work conflicts.

[E] The PIC **must** determine and communicate to functional supervisors the level of oversight that is required for the High Risk Activities (HRAs) that were identified. Oversight needs will vary and may include but are not limited to the following:

- Company oversight for the duration of the execution
- Contractor management oversight for the duration of the execution (including specialty contractor)
- Company FLS or Contractor Supervision during higher risk portions of the work

Physical and organizational interfaces are managed so work proceeds safely and efficiently. Physical interfaces can exist when facilities are connected by either hardware (e.g. pipelines) or software (e.g. communications systems or shutdown systems). Organizational interfaces can be within groups (e.g. crews or shifts), between functions (e.g. operations and drilling), or between two facilities (e.g. a diving support vessel working near an offshore platform or between two platforms).

[E] Any change of scope or conditions that can impact the job at any affected location **must** be communicated to other affected facilities as soon as practical. If the change affects the safety of the work being performed, work will cease immediately and the Permit will be suspended. Work will not recommence until the permit is either re-authorized by both facility Area Authorities or a new permit is processed.

In assessing work site conflicts, the SIMOPS Matrix and Restriction Table will be consulted.

Reference: Section 9.0 "Simultaneous Operations"

At the Daily Work Coordination Meeting, the PIC should also identify those permits, ICCs and Temporary Defeats that should be assessed as part of a V&V activity or a Work Permit Assessment. Assessments are assigned to the Functional AA (AAF) for coordination and completion.

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5.1.4.3 PERMIT PACK ENDORSEMENT

In order for work to progress, the permit will be endorsed by the PIC indicating:

- The work is coordinated and approved for implementation as planned.
- There will not be any concurrent activity, SIMOPS, and schedule or resource conflicts.
- Equipment required for isolation can be made available.

Endorsement at this stage grants permission to progress the permit.

5.1.4.4 NON-CENTRALIZED REMOTE FIELD OPERATIONS (NC RFO)

In NC RFO operations, daily work coordination and endorsement may be achieved through alternate processes (e.g., PIC Review of job board or worklist).

[E] The alternate process **must** fulfill the same objectives of the DWCM, which are to review and coordinate interfacing activities and manage risks involved with concurrent activities and SIMOPS by evaluating the proposed jobs prior to endorsement.

In some non-centralized remote field operations (NC RFO), permit packs and JSAs (where required) are prepared on the day of work execution, and are therefore unavailable for review and endorsement the day prior.

[E] In place of complete permit packs, the AAs/AAFs and PHs **must** submit all proposed work scopes to the PIC for endorsement the day before execution. The detailed list of work scopes will be used to accomplish the DWCM objectives. Each work scope submission must include the:

- Type of work (permit type),
- Job scope description,
- Location,
- LSRA activities,
- Required isolations including type (e.g., personal, ICC, SVI),
- Required TDs,
- Any other information relevant to SIMOPS and work coordination.

The NC RFO DWCM agenda can be followed by the PIC independently, without meeting participants in attendance. The PIC must confirm with the AAs prior to endorsement if any further clarifications are required. In this scenario the PIC will follow the DWCM agenda while reviewing the lists. [E] A formal DWCM meeting with all relevant parties in attendance is held at least weekly to evaluate planned work for the working period.

Reference: [Section 5.3.1 agenda "Tools," Daily Work Coordination Process Objectives](#)

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5.1.5 WORK SITE PREPARATION

INTRODUCTION

This section provides details on the key activities that fall under the Work Site Preparation element of the Work Management process flow. This section addresses the equipment and facilities handover process and Work Permit approval process whereby the Area Operator, Area Authority, and Permit Holder confirm that all requirements for safe and efficient work are met.

5.1.5.1 ACCESS AND EGRESS PREPARATION

Permit Holder confirms that the site preparations are completed, such as:

- Access (for example, scaffolding).
- Egress (for example, rerouted emergency routes are established and signed).

5.1.5.2 REQUIRED SAFETY EQUIPMENT

Permit Holder confirms that all required safety equipment is on location as described in the work pack documentation. Permit

5.1.5.3 REQUIRED COMMUNICATION EQUIPMENT

Many locations have communication equipment readily available around the facility but special equipment is required at times for high noise activities or confined space entries. The work team needs to be familiar with the operation of these systems and contacts for normal or emergency situations.

5.1.5.4 EQUIPMENT INSPECTION AND/OR CERTIFICATION

Verify equipment has been inspected and/or certified, where appropriate.

5.1.5.5 TEMPORARY DEFEATS

Required temporary defeats are in place as dictated by the scope of the work. These are identified and implemented in cooperation with the relevant people e.g. Area Operator, Control Room Operator and Area Authority.

5.1.5.6 ENERGY ISOLATIONS

All required isolations of energy sources are conducted with [Section 7.0](#) of this manual.

5.1.5.7 LSRA / LIFE SAVING SAFEGUARD PREPARATIONS

Required LSRA or other life-saving safeguards are put in place and made effective prior to starting work (e.g., fall protection anchor points, barricades, ventilation).

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5.1.6 WORK AUTHORIZATION

INTRODUCTION

On the day of the proposed work, the permits are authorized, issued and accepted. Authorization will not be given until the Area Authority conducts a final check of activities already underway or proposed to confirm that the job can proceed safely.

5.1.6.1 PERMIT PACK AUTHORIZED

Area Authority *Authorizes* the Permit Pack if the work can proceed as planned. The Area Authority role for authorizing the work is assumed by Operations except for remote field operations where Operations personnel may not be present.

5.1.6.2 ZERO ENERGY VERIFIED AND DEMONSTRATED

A completed isolation plan will be reviewed by the Area Operator with the Permit Holder and Zero Energy is demonstrated to the Permit Holder prior to signing the permit. Each individual working on the job has the right to request Zero Energy be demonstrated.

When demonstration of Zero Energy to the Permit Holder is impractical (for example, diving operations or remote field locations), the deliberate step of confirming Zero Energy will be specifically delegated to the Permit Holder.

5.1.6.3 WORK PERMIT ISSUE AND ACCEPTANCE

Permit Issue by the Area Operator is formal permission for the work to proceed.

The permit can then be accepted by the Permit Holder. The permit is not considered "Live" in the field until the Area Operator and Permit Holder sign the permit. [CriticalE] Worksite controls and any LSRA and/or life-saving safeguards required for the job to commence **must** be in place and effective.

Work Management System (WMS)

5.1.7 WORK EXECUTION

The Work Execution element of the Work Management process flow involves accomplishment of the work activities.

5.1.7.1 PRE-TASK BRIEFING

[CriticalE] Prior to work execution, the Permit Holder **must** arrange a Pre-Task Briefing with the Work Team. The purpose of the Pre-Task Briefing is to lead a two-way discussion with the work team at the work location, immediately prior to commencing a higher potential consequence task. Pre-Task Briefings can also be used when conditions/scope significantly change to focus the work team on items important to safe work execution.

Prior to the Pre-Task Briefing, Permit Holders should:

- Review permit/Work pack documentation to identify critical steps and life-saving safeguards
- Confirm resources are sufficient for the job (e.g., number of workers, appropriate tools/equipment)
- Confirm the work team has competencies necessary to execute the work safely

[CriticalE] Permit Holder or designated supervisor expectations during the Pre-Task Briefing:

- Confirm worker readiness (i.e., head in the game, fit for duty) and their understanding of safeguards through two-way engagement
- Share learnings to be applied to the job/task
- Identify START & STOP Work situations / conditions and reinforce expectations
- Communicate planned field engagements (e.g., targeted learning observations, V&V, after action reviews, work permit assessments, other audits/assessments)

[CriticalE] The Pre-Task Briefing includes discussion on:

- Work Authorization is in place for all work including consideration for adjacent work activities
- Emergency response and muster considerations have been identified
- Independent verifications required (e.g. safeguards, energy isolation)
- Procedures to be used for the task including steps associated with Hold Points where the person executing the step is a last line of defense in preventing a higher potential consequence event.
- LSRA Start Work Checks to verify safeguards are in place and effective & how safeguards will be maintained
- JSA (where applicable) and verification that associated life-saving safeguards are in place and effective, as well as:
 - Error-prone situations and necessary corrective actions
 - Additional Hold Points and specific Triggers (if / then situations) to STOP work
- Other specific hazards and mitigations not already addressed through LSRAs or JSA
- Last Minute Risk Assessment (LMRA) discussion focusing the group on identifying error-prone situations, hold points, or specific triggers to STOP work
- Reinforcing individual use of LMRA and STOP Work expectations

Work Management System (WMS)

5.1.7.2 DURING WORK

While executing work activities, work team members are expected to:

- Follow communication requirements
- Keep the work site safe and organized
- Apply the LMRA process during task/job step changes
- Watch for triggers and other changes in the work environment
- Adhere to Hold Points discussed during the Pre-Task Briefing
- Ensure LSRA and other life-saving safeguards are in place and remain effective
- Apply STOP Work expectations when uncertain, an unsafe condition occurs, or safeguards are no longer effective

5.1.8 WORK CLOSEOUT

OVERVIEW

The Closeout element of the Work Management process flow ends the work execution activities and initiates documentation activities that are critical to the Work Management System.

A key aspect of properly closing out the work is the equipment and facility reinstatement activities.

The general sequence of reinstatement can include:

- Re-commissioning
- Purging
- Leak testing
- Verification of isolation status
- Removing energy isolations (process, mechanical, and electrical)
- Acceptance testing
- Removing temporary defeats of critical devices
- Completing [After Action Reviews](#) when planned or when warranted based on the learnings from the work (e.g., went better than expected, was more difficult than expected, learning opportunities for next time)
- Capturing lessons learned when applicable
- Work site cleanup
- Permit closeout

Work Management System (WMS)

5.2 VERIFICATION & VALIDATION (V&V), MEASUREMENT, AND KPI'S

INTRODUCTION

[E] Periodic Verification & Validation activities and WMS Assessments **must** be conducted to verify that all Work Management System activities (i.e., PTW, ICC, TD, JSA) are in compliance with the Work Management System (WMS) and performance improvement opportunities are identified and addressed.

The Work Management review process involves management oversight of permit to- work- activities leading to a continuous improvement effort. The Work Management review process is intended to confirm that:

- Work Permits are being used consistently and correctly on a daily basis for work on company facilities.
- Isolations are compliant with the WMS and Zero Energy demonstration is occurring.
- Temporary Defeats are being managed safely and adequate alternative controls are implemented when equipment will remain in service. Risk Assessments are requested, when necessary.
- LSRA and other life-saving safeguards are in place and effective as defined by the LSRA SWCs and the JSA.
- Work without a Permit and other low risk activities that are not part of the Permit Pack documents are also reviewed and assessed.
- The Work Permitting process is effective at achieving safe and efficient work in accordance with the Life Saving Rules & Actions.

5.2.1 VERIFICATION AND MEASURE

The Operating Site is responsible for managing and conducting verification & validation (V&V) activities, including the collection and reporting of key performance indicators (KPIs) to the Operating Site Leadership Team.

Site's are encouraged to utilize Permit Pack assessments and Work Site inspections to maintain focus on higher risk activities, non-routine tasks and complex isolations. Site's should establish practical targets, specific to their operations that will deliver meaningful metrics for the assessment of WMS compliance and effectiveness.

Additional indicators may be developed and reported for specific purposes or periods of time; however, standard key performance indicators to be reported are described in OIMS System 6-4.

5.2.2 WORK SITE INSPECTION

Work site inspections are the periodic, on-site verification & validation (V&V) of work execution by Site Management to assess compliance with WMS and PSMS. It consists primarily of supervisors, leads, or designates (including contractors) proactively visiting job sites to verify & validate that appropriate safeguards are in place, personnel working on the job are implementing these safeguards and coaching personnel on system components.

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5.2.3 PERMIT PACK ASSESSMENTS

Permit Pack assessment tools are built into the ePTWS to facilitate reviews and reports. The tool assesses Permits (including the JSA where required), Isolations, and Temporary Defeats. For Site's that do not have ePTWS access, assessments can also be accessed through [GOTO/WMSEFORMS](#), "Assessments,"

Supervisors, leads, or designates can conduct an assessment. Assessments have been built for in Field application.

- **Field Assessments** are conducted at the work site to verify and validate compliance with the WMS expectations during work execution and to ensure life-saving safeguards are in place/effective during tasks. This includes reviews of Permits, TDs, ICCs as well as conversations with Permit Holders and work team members.

Focus of assessment activities is recommended for higher risk activities such as confined space entry, hot work, LSRA activities, and non-routine activities and to support in field engagements with work teams. Additional focus areas may include newer work teams, work with complex isolations and as follow-ups to ensure learning from events.

No minimum thresholds are in place for assessment activity completion. Sites are encouraged to conduct assessment activities to the extent needed to ensure strong compliance and effectiveness of WMS processes is being achieved.

Reference: [GOTO/WMSEFORMS](#), "Assessments," for the Work Management System Assessment Tool.

5.2.4 CAPTURING IMPROVEMENT OPPORTUNITIES

Results of WMS assessments that identify work permitting compliance issues or improvement opportunities should be shared with the local work teams and issues closed out in a timely manner. Significant learnings or improvement opportunities should be shared with the OIMS System 6-4 Owner, Administrator, and subject matter experts (SMEs) and stewarded to closeout.

Assessments are *Reviewed* by the AA or PIC to confirm the quality of the assessments, add comments, and agree or reject any action items identified, and then Management Reviewed by the PIC who adds comments.

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5.3 TOOLS

FORMS AND TOOLS

These forms are referenced in this section and listed on following pages:

5.3.1 DAILY WORK COORDINATION MEETING AGENDA

Table 2: Daily Work Coordination Meeting (DWCM) – Agenda

Description	Guidance
<u>Establish Status of Activities</u> Review ongoing live and suspended permits & non-permitted work. Identify those jobs that continue the next day that must be considered for SIMOPS conflicts or congestion.	All in attendance should be knowledgeable about their ongoing and requested jobs and also be somewhat familiar with other jobs that are active and/or requested in the nearby vicinity and be ready to discuss.
<u>Review new work permits and Non-permitted work</u> Discuss Reviewed Permit Packs, and Non-permitted work with Functional Supervisor to discuss readiness: - Parts available, cranes organized, scaffolds in place, weather suitable, resources organized, and so forth. - Discuss availability of equipment from operations. - Identify isolation and/or TD requirements. - Discuss Non-permitted work. - Review of activities involving Life Saving Rules & Actions and/or other higher potential consequences - Review procedures (including contractor) for readiness of execution	Review and understand the ongoing and proposed work for the next day. Things to consider: - What is the job and associated activities? – LSRA? Consider all activities taking place within the job scope. - Who is executing? – What is their competency? Do they have experience with this activity? - Where is the activity taking place – high consequence area? H₂S Area? - Remote location? - How will this activity be executed? – Will this be a normal or complex execution? Will there be machinery involved? - Does this activity involve or affect one of the Site's identified High Consequence Scenario(s) Critical Safeguard

Work Management System (WMS)

Table 2: Daily Work Coordination Meeting (DWCM) – Agenda, Continued

Description	Guidance
<u>Higher Risk Activities and Oversight:</u> Identify the Permits and/or non-permitted work to be considered as the High-Risk Activities for the day. Discuss/establish oversight expectations for all High-Risk Activities identified.	From the guidance above, determine which activities would be considered higher risk for the day. Ensure everyone fully understands the activities and why they are considered higher risk. The PIC must determine and communicate the level of oversight that is required for the HRA's that were identified. Oversight needs will vary and may include but are not limited to the following: ▪ Company oversight for the duration of the execution ▪ Contractor management oversight for the duration of the execution (including specialty contractor) ▪ Company FLS or Contractor Supervision during higher risk portions of the work NOTE: Postpone activities if resources are not available to ensure safe execution.
<u>Temporary Defeats:</u> Review requested and applied TDs.	▪ Steward overdue and aging defeats (including required approvals) ▪ Review defeats and how they may affect other activities in the same area.
<u>Assess SIMOPS and concurrent work interfaces by location</u> Discussion/review of permitted work or other non-permitted activities.	Assess permits for the following: • SIMOPS matrix issues • Congestion • What could happen as it applies to adjacent or concurrent activities

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Table : Daily Work Coordination Meeting (DWCM) – Agenda, Continued

Description	Guidance
<u>Endorse new Permits</u> Endorse new permits that do not have SIMOPS or concurrent work conflicts.	PIC is to ensure that good communication is taking place between the functional supervisors to understand requested work and also ensures permits are linked if/when necessary. Permits are only endorsed if no conflicts exist. Confirm that Life Saving Rules & Actions have been addressed in Work Plan. Consider whether the jobs can go ahead before endorsing the permits. Function Supervisors to support their endorsement discussing the LSRA activity, safety and production impacts. Conduct spot checks of Permit Packs (for example, permit details, LSRAs, JSA quality, required SWP Checklists attached, required IC procedures attached, etc.). Provide timely feedback and direction to improve performance.
<u>Assessments:</u> Review activities to identify where assessments may provide value and assign where applicable.	PICs should look for higher risk activities, non-routine tasks, activities with newer work crews, complex isolations etc. when determining when to request assessments Engage the DWCM Team in a conversation on previous day assessment findings, ensure Team members are carrying the message down to the Workforce.
<u>Long Term Isolations:</u> Review Long Term Isolation (Monthly).	Conduct monthly review to determine if the LTI can be reinstated, is an assessment required or is a MOC required.

Attendees: PIC, AA, AAF (EM and Contract), others as necessary (for example, SSHE, Permit Coordinators, PH for specific input on unique work).

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5.3.2 NC RFO DAILY WORK COORDINATION MEETING

Table 3: NC RFO Daily Work Coordination Process (NC RFO DWCM)

Description	Guidance
<u>Establish Status of Current Activities</u> Review ongoing live and suspended permits & non-permitted work. Identify those jobs that continue the next day that must be considered for SIMOPS conflicts or congestion.	Functional Supervisors/AAFs have responsibility to provide updated statuses and expected duration of work to the PIC for review. PIC to review and understand current activities and permits.
<u>Review New Work Scopes, Permits and Non-Permitted Work</u> Review Work Scopes (including permitted work, and Non-permitted work with SIMOPS potential) to be conducted the next day. Each Functional Supervisor to supply information on readiness, such as: ▪ Availability of equipment from operations. ▪ Isolation and/or TD requirements. ▪ Non-permitted work. ▪ Identify Life Saving Rules & Actions activities.	AAFs/Functional Supervisors must provide list of proposed work for the following day to the PIC. PIC to review the ongoing and proposed work for the next day. Things to consider: ▪ What is the job and associated activities? – LSRA? Consider all activities taking place within the job scope ▪ Who is executing? – What is their competency? Do they have experience with this activity? ▪ Where is the activity taking place – high consequence area? H₂S Area? ▪ Remote location? ▪ How will this activity be executed? – Will this be a normal or complex execution? ▪ Does this activity involve or affect one of the Site's identified High Consequence Scenario(s) Critical Safeguard.
<u>Higher Risk Activities and Oversight:</u> Identify the work scopes to be considered as the High-Risk Activities for the day. Establish oversight expectations for all High-Risk Activities identified.	From the guidance above, determine which activities would be considered higher risk for the day. The PIC must determine and communicate to functional supervisors the level of oversight that is required for the HRA's that were identified. Oversight needs will vary and may include but are not limited to the following: ▪ Company oversight for the duration of the execution ▪ Contractor management oversight for the duration of the execution (including specialty contractor) ▪ Company FLS or Contractor Supervision during higher risk portions of the work NOTE: Postpone activities if resources are not available to ensure safe execution.

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Table 3: NC RFO Daily Work Coordination Meeting (DWCM) Continued

Description	Guidance
<u>Temporary Defeats:</u> Review requested and applied TDs.	▪ Steward overdue and aging defeats (including required approvals) ▪ Review defeats and how they may affect other activities in the same area.
<u>Assess SIMOPS and concurrent work interfaces by location</u> Review location of current and proposed work. Review of permitted work and non-permitted activities.	Assess work scopes and permits for the following: • Location • Planned job scopes • Consult SIMOPS matrix and restriction tables when more than one activity on a location • Consider assigning an AIA to manage activities
<u>Endorse Work Scopes and Permits</u> Endorse new work scopes, permits that do not have SIMOPS or concurrent work conflicts.	PIC is to ensure that good communication is taking place between the functional supervisors to understand requested work and also ensure concurrent work is communicated. Confirm AIA requirements. Work is only endorsed if no conflicts exist. Confirm that Life Saving Rules & Actions have been identified in the work scope. Conduct spot checks of work list and work packs (for example, well defined work scope and location, LSRA identification, TD/isolation requirements identified.). Provide timely feedback and direction to improve performance.
<u>Assessments:</u> Review activities to identify where assessments may provide value and assign where applicable.	PICs should look for higher risk activities, non-routine tasks, activities with newer work crews, complex isolations etc., when determining when to request assessments
<u>Long Term Isolations:</u> Review Long Term Isolation (Monthly).	Conduct monthly review to determine if the LTI can be reinstated, is an assessment required or is a MOC required.

Contributors: PIC, AA, AAF (EM and Contract), others as necessary (for example, SSHE, Permit Coordinators, PH for specific input on unique work).

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6.0 PERMIT TO WORK

INTRODUCTION

This section provides the process and responsibilities for the Permit to Work System (PTWS) used by the Company.

6.1 PERMIT TO WORK OVERVIEW

6.1.1 PURPOSE AND SCOPE

The purpose of the Permit to Work System (PTWS) is to accomplish the following:

- Control the work activities and functional interfaces on Company facilities in order to prevent incidents resulting from uncontrolled work.
- Communicate information clearly to avoid an inadequate understanding of work task hazards.
- Gain necessary approvals and authorizations to execute work.
- Manage and control:
 - Risks relating to individual operations and work tasks
 - The process relating to the safety of individuals and facilities

The PTWS addresses the following work activities:

- Hot work
- Confined space entry
- Critical work
- Energy isolation
- Temporary defeats of critical devices
- Atmospheric Testing
- Simultaneous Operations

6.1.2 GOALS AND OBJECTIVES

Specific goals of the PTWS are to accomplish the following:

- Eliminate near misses and incidents on Company facilities.
- Consistently identify appropriate safeguards and safety precautions prior to and during work on Company facilities.
- Coordinate and approve work activities requiring work permits.
- Control the condition of the work site prior to beginning work.
- Control the commencement and completion of work.
- Maintain safety of the work site, operable systems, and personnel during work on Company facilities.

The PTWS goals are supported by the following:

- Clearly describe and document the work to be carried out and the time when it can be done.
- Assign authority and accountability for all aspects of the work.

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- Provide written confirmation of the steps already taken to confirm that the work site is prepared and made safe before work commences.
- Describe outstanding hazards and controls.
- Clearly indicate the personnel who issue and receive the permit..
- Confirm that the facilities and equipment are handed back to the operating function in a safe condition for reinstatement as per the closeout process.

6.1.3 PTWS PRINCIPLES

INTRODUCTION

Several independent parties may be involved in controlling, coordinating, and managing activities.

The following parties work together during the planning and execution phases of work:

- **Operations** is responsible for the care, custody, and control of the equipment and facilities.
- The **Work Team** performs the work on the equipment and facilities.
- **Site Management** has the responsibility to confirm that the PTWS is properly executed and achieves safe and effective work.

6.1.3.1 OPERATIONS

Operations are the owner of the Company facilities and have the expertise and skills to operate them.

Operations have the following duties and expectations:

- Maintain monitoring and control of the operable systems and areas.
- Manage all required energy isolations.
- Prepare equipment and surrounding area prior to work commencement.
- Reinstate systems and isolations after completion of work.
- Maintain control of interfacing activities related to the operable systems and areas.

6.1.3.2 WORK TEAM

The Work Team may be a single worker or a group of workers. In either case, the person responsible for safely performing the work is the Permit Holder (PH). The PH confirms that personnel in the team possess the correct expertise and skills for performing the work activity. The Work Team:

- Performs required safeguards related to the work activity.
- Confirms that work preparations and appropriate materials, tools, and equipment are in place.
- Confirms that the work site is safe in cooperation with the Area Operator (AO).
- Confirms that the work is executed according to the requirements of the Permit Pack.
- Recognize any changes in work scope that require work to STOP and changes to be addressed.

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6.1.3.3 SITE MANAGEMENT

Site Management is responsible for overseeing the execution of the PTWS and has the following duties and expectations:

- Coordinates and supervises work on Company facilities.
- Reviews the proposed work activity to confirm that potential interfaces are identified and appropriately managed.
- Review of work permits for completeness of job and safety planning efforts, and endorsement of work permits as appropriate.
- Assesses and stewards the effectiveness of the PTWS in controlling work and in achieving the goal of safe and efficient work on Company facilities.

6.1.3.4 REMOTE FIELD OPERATIONS (RFO & NC RFO) SCALABILITY

It is recognized that work is also performed at remote field locations where a single person (or a limited number of personnel) is responsible for the safe execution of work.

In these cases, it is the expectation that a Operating Site identify what work can be performed by a single person in these locations. Consideration should be given to critical activities and the segregation of responsibilities to ensure that adequate safeguards are in place and work execution is managed.

[E] Where a person in a remote location either:

- Identifies additional work not already approved ("break-in"),
- Or the scope of the permitted work that was approved changes,

The individual **must** notify the Person in Charge (PIC) to gain approval to proceed and document this conversation and approval on the permit/work pack.

[E] Where a single person in remote locations is performing work, that person **must** be deemed competent by the Site to perform duties and fulfill the requirements of the Work Management System (WMS). In these cases, the individual will be the Permit Holder (PH) and may fulfill the requirements of one or both of the Area Authority (AA) and Area Operator (AO) and sign the appropriate sections.

In cases where a person holds more than one role and is executing work requiring a permit, each step of the permit process **must** be followed to ensure that work is planned, hazards assessed, controls established (including isolations and temporary defeats), work site safety checks are performed, (including zero energy confirmation and demonstration to other personnel assisting), work is safely executed, and equipment and systems safely reinstated.

In cases where authorization is obtained verbally or through another means (text, email etc.), the requirement for signatures on the permit or JSA can be satisfied by the permit holder documenting the name of the individual, role and means of authorization (for example verbal AO issuance, or Jane Smith - AA authorization via email). [E] Authorization **must** be received before commencing work.

Permits and JSAs continuing the next day/shift, should be suspended through notification to the AO or AA. [E] Re-authorization of the permit at the start of the next shift **must** be received before work commences and required signatures, or individuals' information, be documented on the permit.

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6.1.4 PTWS REQUIREMENTS

INTRODUCTION

PTWS supports the Work Management System by:

- Requiring planning and preparation
- Requiring pre-job checks
- Approving the work activity
- Maintaining control during the work permitting process
- Requiring closeout checks

6.1.4.1 BASICS FOR USING A WORK PERMIT

Using a work permit typically involves the following:

- [E] Only WMS trained individuals are allowed to perform permitted work. The training required depends on their roles and responsibilities.
- [E] An individual can be exempted from the training by the PIC for short duration specialist jobs because of their specialized expertise required at short notice; however, a dedicated WMS trained Permit Holder will be in attendance.
- With the exception of breakdown or break-in type work, a work permit is typically progressed to the Endorsed stage the day before the work is planned so that it can be properly reviewed at the Daily Work Coordination Meeting (DWCM).
- The type of work permit cannot be changed e.g. a Critical Work Permit cannot be used as a Hot Work Permit.
- If the work scope or conditions change during the term of the permit the work stops and the work permit is suspended. [E] The Operations AA will determine whether a new permit will be required or whether the changes should be reflected in the documents of the Permit Pack (e.g. JSA) and communicated to affected personnel. [E] If the changes are not adequately controlled or mitigated, a new permit **must** be developed.
- All work permits can be cross-referenced on all associated documentation required for the work to be performed.
- The permit is not valid if the required endorsements, approvals, and signatures are not in place.
- The Central Control Point (CCP) is the dedicated location where the work permit is controlled.
- The permit is not considered "Live" until it is signed by the Area Operator (AO) and Permit Holder (PH).
- The Permit Pack in the field is the controlled copy, this includes all updates and required signatures. Minor changes to the Permit Pack documents can be made in the field.

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6.1.4.2 PLANNING AND PREPARATION

Planning and preparation activities include identifying tasks, reviewing, and specifying the following:

- Type of work, location, and conditions to be managed with work permits.
- Type of work permit required for each job, based on the risks involved.
- Description of the work based on a work order (or equivalent).
- Detailed and specific precautions and preparations required to protect the following from potential hazards:
 - Personnel
 - Environment
 - Equipment
 - Materials

6.1.4.3 PRE-JOB CHECKS

The work permit process contains pre-job checks so that:

- The work site is safe.
- The necessary defeats or energy isolations are performed, verified, and demonstrated.
- The necessary and appropriate atmospheric testing is performed.
- The work does not unsafely interfere with other work planned or in progress.
- Those affected by the work are informed of the work and of its possible effects on their work.
- Appropriate communication and coordination meetings are held, and key decisions are documented.

6.1.4.4 WORK PERMIT APPROVALS

The work permit approval is typically performed in two stages:

- **Work Planning:** Daily Work Coordination Meetings (DWCMs) should be attended by the key Work Management stakeholders, specifically, the PIC, AA, and functional supervisors. The objective of the meeting is to review and coordinate interfacing activities and manage the risks involved with concurrent activities and simultaneous operations (SIMOPS) by evaluating the proposed jobs prior to endorsement. New and suspended permits are reviewed. Non-permitted work that could conflict with planned activities are also discussed at this meeting.
Endorsement at this stage grants permission to progress the permit and proceed with the work activities as planned.
- **Reference:** [Section 5.3: "Tools,"](#) for a recommended Daily Work Coordination Meeting agenda.
- **NC RFO Work Planning:** In non-centralized remote field operations, daily work coordination and endorsement may be achieved through alternate processes (ex. PIC Review of job board or worklist). [CriticalE] The alternate process must fulfill the same objectives of the DWCM, which are to review and coordinate interfacing activities and manage risks involved with concurrent activities and SIMOPS by evaluating the proposed jobs prior to endorsement. The NC RFO DWCM agenda can be followed by the PIC independently, without meeting participants in attendance. [CriticalE] In this scenario, the AA/AAFs must submit a work list to the PIC for review, and the PIC will follow the DWCM agenda while reviewing the lists. A formal DWCM meeting with all parties in attendance should be held at least weekly.

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Reference: Section 5.3: "Tools," for a recommended NC RFO Daily Work Coordination Meeting agenda.

- **Work Execution:** Operations (Area Operator, Area Authority) confirms that prerequisites for the work specified on the work permit are in place. The Permit Pack is then Authorized and Issued.

6.1.4.5 VALID PERIOD OF WORK PERMITS

A work permit is valid only under the following conditions:

- Permits are valid for the duration of the work or until the end of shift (whichever comes first) up to a maximum of 16 hours with an approved extension.

! IMPORTANT: Extended permits typically remain Live during a shift change if the PH does not change. However, it is expected that the incoming AA and AO will review and sign the Permit Pack. [E] The AO **must** also visit the work site and sign the work site copy of the Permit pack documents.

- [CriticalE] When the PH changes out during the shift, the new PH **must** review the Permit Pack documents to confirm that controls remain in place, the scope of work has not changed, all isolation control points (lock box, bag, etc.) have the required functional locks, zero energy has been demonstrated, and signs the work site copy of the permit.

NOTE: The AO will be notified whenever the PH changes out.

- [CriticalE] A work permit may be issued for up to seven days, but the permit **must** be progressed with the appropriate signatures each shift (Authorized by the AA, Issued by the AO and Accepted by the PH), after conducting all necessary verifications, briefings and demonstration of zero energy.
- Controls verifications and safety checks **must** be performed by the oncoming work team before work starts. Pre-Task Briefings are required at each shift change and when new personnel join the Work Team.
- The scope of work does not change from the job description on the permit.
- A new permit is developed after a maximum of seven days for the longer-term work.
- Work should typically start within one hour of the permit going live at the work site (AO and PH signatures).
- [E] Work **must** stop before the permit expires each shift unless an extension is requested and approved.

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6.1.4.6 SUSPENSION AND WITHDRAWAL

A work permit can be suspended or withdrawn in the following situations:

- If anyone is seen working in noncompliance or an unsafe manner.
- Urgent conflicting work develops.
- Work scope changes that do not align with the permit conditions and controls.
- If the work will continue on the following shift or day.
- An emergency occurs or an emergency alarm sound.

! IMPORTANT: [E] In an emergency situation or if an emergency alarm sounds, all work **STOPS**. Once the emergency is controlled, the PIC (or designate) assesses the situation, and determines any work (permitted and non-permitted work) that will be suspended or whether it is safe to approve work to resume. Pre-work safety checks by the PH and Work Teams are required before work is recommenced.

6.1.4.7 CLOSEOUT CHECKS

Permit closeout checks confirm that:

- Work was completed or suspended satisfactorily.
- Isolations and Temporary Defeats are returned to service
- Equipment is returned to a safe condition and the work site is organized.
- Completion of the work is communicated to the affected personnel.

6.1.5 CRITERIA FOR REQUIRING A PERMIT

INTRODUCTION

Work permits are required to manage activities by planning, evaluation, interface coordination, preparation, controls, communication, approval, handover, and feedback. [CriticalE] Sites are expected to follow the Permit Guidance and types provided in [Section 6.0: Permit to Work](#) of WMS.

6.1.5.1 WORK THAT REQUIRES A PERMIT

This section includes a description of each permit type and provides examples of work activities that fall into each type category. The examples provided are based on typical Operating Site work activities but may not be exhaustive to the types of work being performed at a facility.

- Hot Work
- Confined Space Entry
- Critical Work

Examples of work activities that fall into each category of permit can be found in:

- [Section 6.3: "Hot work permit activities"](#)
- [Section 6.4: "Confined space examples"](#)
- [Section 6.5: "Critical work permit activities"](#)

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! IMPORTANT: Each Operating Site should assess work activities at their facilities to determine work permit requirements for those activities not listed above.

6.1.6 GUIDANCE FOR WORK WITHOUT A PERMIT

OVERVIEW

Some activities included in the permit categories may be considered lower risk and not requiring a permit by the Operating Site.

! IMPORTANT: For these non-permitted activities, controls must be identified to manage any hazards. (ex. Work Authorization, SWC, JSA, LMRA, Procedure, etc.)

[CriticalE] Non-permitted work activities **must** be assessed and mitigating controls documented and approved by the OIMS 6-4 Owner.

6.1.6.1 WORK AUTHORIZATION FOR WORK WITHOUT A PERMIT

[E] When evaluating work not requiring a permit, Sites **must** determine if work authorization is required for the task. [E] When work authorization is required, Sites **must** establish a means of work authorization. Some examples may include:

- Verbal communication (call to supervisor, CCR, etc.)
- Email, text or other written authorization
- JSA issuer/receiver
- Work Authorization log / list

6.1.6.2 WORK NOT REQUIRING A PERMIT

The following are some examples of work activities/criteria that could be considered by the Site:

- Routine operations (for example, production or marine) covered by approved procedures
- Work inside workshops or in the living quarters that do not involve critical work activities or confined space entry
- Cleaning and housekeeping
- Sampling from an engineered (purposely designed) sample point by a competent person
- Checking and refilling lube oil levels
- Checking for leakages
- Visual inspections and monitoring of pressure or temperatures
- Operation (i.e. start, stop, operate) production/processing facilities (for example, wells, process equipment, utility systems, bilge and ballast systems) that are managed by procedures
- Surveillance and checking of production/processing facilities
- Routine operator activities (for example, equipment preparation for maintenance or construction, returning equipment to service, and housekeeping)
- Helicopter operations (landing, takeoff, and re-fueling)
- Loading or unloading containers from supply boats (refer to *complex lift* definition)

6.1.7 INTERFACE MANAGEMENT

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INTRODUCTION

Many work permits require managing interfaces between facilities, groups or individuals in order for the work to be executed safely. This section describes the types of interfaces that need to be effectively managed, as follows:

- Organizational interfaces
- Shift or crew interfaces
- Physical interfaces

6.1.7.1 ORGANIZATIONAL INTERFACES

Organizational interfaces occur when more than one company or organization can be affected by a work activity. Examples include drilling, diving and construction/projects activities.

The management of organizational interfaces are typically addressed in a bridging document, or other approved document that aligns functional oversight of the work activities. A handover checklist or other tool may be used when change of custody will occur. The work permits can be signed/countersigned by the designated **Area Interface Authority (AIA) / Site Coordinator (SC)** where required.

For potential interface issues, refer to the SIMOPS Matrix and Restriction Tables to identify and address them.

Reference: Section 9.4.4: "SIMOPS Matrix".

6.1.7.2 SHIFT OR CREW INTERFACES

[CriticalE] Work or activities that cross shifts or crews require a handover with appropriate documentation to support the transition by all impacted personnel. This is to ensure critical information is transferred from one responsible person/group to another and limit SSH&E and Process Safety exposure as a result of lacking information. (For details on shift handover practices see [Section 6.1.8: Handover Practices](#)).

This is intended:

1. To ensure an effective interface and communication of critical operating information and issues between Operating Shift Teams.
2. To provide a consistent and thorough communication process for operating shift teams and the day organization (Operations Team & support staff).
3. To ensure shift personnel returning from normal days off or an extended absence are aware of critical changes, non-routine and abnormal operating conditions or maintenance activities on the unit/site.
4. To establish clear work priorities.

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6.1.7.3 PHYSICAL INTERFACES

Physical interfaces are work or activities that can cross physical areas or locations. The management of physical interfaces can be accomplished in several ways:

- Identifying secondary task areas associated with the permit.
- Requiring work permits to be signed/countersigned by additional users (e.g. the Area Interface Authority / Site Coordinator).
- Preparing a separate work permit for the affected site and cross-referencing it to the work permit at the controlling site.

Examples of physical interfaces include but are not limited to:

- Cutting into a deck or firewall on an offshore platform can cause sparks in the adjacent area.
- Pressure testing pipework can extend beyond the area where the pressure is being applied.
- Pigging a pipeline or flow line between two locations requires coordinated activities at both locations.
- Work on communications systems between locations can affect automatic shutdown functions on the location.

The Daily Work Coordination Meeting should identify physical interfaces and confirm that the work permit requirements are appropriately reflected. **Reference:** [Section 5.3: "Tools,"](#) for a recommended Daily Work Coordination Meeting agenda.

6.1.8 HANDOVER PRACTICES

OVERVIEW

For the purpose of this Best Practice, the term "shift" is intended to represent the time frame between the responsibility of one person/group for a given equipment/system and another person/group. It can vary depending on facility, organization structure, and span of control. It could be 12-hour, daily, weekly, or rotational.

[CriticalE] When shift changes are required, the Sites **must** establish handover processes to ensure sufficient knowledge is transferred to limit operational and SSH&E exposure as a result of lacking information. This handover and interface should be managed in a controlled manner with appropriate documentation to support the transition.

This section is intended to enhance existing Operating Site handover practices to ensure all pertinent Safety, Work Management and key site-specific information requirements relating to equipment or process, are passed on by key personnel and covered during shift and crew changes, including:

- Work status
- Structured round completion
- Equipment health
- Process or equipment deviations

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6.1.8.1 OBJECTIVE

The objective of shift handover is to ensure that Key personnel discuss all relevant information with oncoming personnel. Handover information **must** be documented by **key personnel** and be maintained as required per the site retention guidance. The information will be kept current and accurate during the shift. This information should be retained in either an electronic or hard copy format for future reference and shall be readily accessible to on coming key personnel returning after an extended absence.

Key personnel include:

- PIC
- Area Authorities
- Functional Supervisors/Leads
- Operators
- Console Operators
- Other Site identified key personnel

6.1.8.2 SHIFT OR CREW HANDOVER PRACTICE

[CriticalE] Handover practices vary from facility to facility; however, the following information **must** be included as part of the handover process (where applicable):

- Incidents
- Critical safeguards that are impaired or not meeting performance criteria
- Temporary Defeats that are in place or in progress (active/in-place)
- Suppressed alarms/system controller issue
- Process or equipment deviations or abnormal operating conditions
- New and existing active critical operating parameters (COPs)
- Live work that is extended or continuing into the oncoming shift
- Isolations that have been applied, removed or in progress on critical equipment
- Car Seal Register changes
- New Operational or Safety directives

The following information should be considered for inclusion where relevant:

- Lock-out/tag-out (LOTO) custody transfer of isolation keys
- Critical equipment status and readiness
- Planned work activities for the oncoming shift.

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6.1.8.3 SHIFT HANDOVER METHODS

Shift Handover methods provide a consistent approach to allow discussions and the transfer of information related to Safety and Operational activities and conditions. Shift Handover methods consist of:

- Structured Log (written or electronic)
- Person to Person Handover
- Shift Team Meeting

6.1.8.4 STRUCTURED LOG

A structured log is the first component in a good shift handover. The structured log is a standardized document, used for Shift Handover. It will be kept current and accurate during the shift and kept on file for the Site specified period. It can be a hard copy or electronic and retained for future reference. It shall be readily accessible to employees returning after an extended absence. This standardization makes the recording and retrieving of information efficient for the users of the logs. A shift log should be generated for each shift position.

Structured Logs should be formatted to accommodate effective communication of the Operating Principles. These operating principles include Safety, Health, Environmental, Reliability and Safe Operating Conditions. Following are examples of information that can be included in these operating principles:

- Safety and Health that include all incidents that affect personnel, facility and equipment or the community in which we operate. Elevated areas of risk (personnel safety or process hazards), potential external impacts, e.g., community complaints or weather impacts, losses and near losses.
- Environmental items that include losses to the sewer or atmosphere, flaring events, spills, releases and the steps taken.
- Reliability items that impact equipment dependability or operations, impaired or not meeting performance criteria of a safeguard, non-availability of critical equipment, and critical maintenance work.
- Safe Operating conditions and problems including actions taken, product rate change, quality, constraints, operating envelope exceptions, abnormal non-routine operating (ANRO), events including any loss of containment, deviations from normal operations, major changes, and safety critical devices out of service or deactivated.

6.1.8.5 PERSON TO PERSON

Person to Person (Individual) Handovers should be conducted by shift personnel (i.e. field/plant, console, and supervisory) where practical. The failure to conduct a person to person handover between individuals exchanging responsibilities requires management oversight and contingency management by shift supervision. Person to person methodology:

- The person being relieved **must** give their report to the person coming on duty before leaving the work area. Shift Relief takes place at the designated location.

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- Handover participants:
 - Are responsible for taking the appropriate amount of time to accurately discuss pertinent points to avoid miscommunication
 - Ask questions to ensure an accurate understanding of conditions, appropriate actions and other information. Go to the worksite if necessary to enable understanding of the details of the information being exchanged.
 - Use the Structured log to guide the discussion and enable a consistent, complete and accurate information transfer.
- An employee coming to work after scheduled off days or an extended absence has the responsibility to make sure they are adequately updated.

6.1.8.6 SHIFT TEAM MEETING

Shift Team meetings prepare teams for the days' tasks and always address the four principle areas: Safety (personal and process) and Health, Environmental (air, water, and soil), Reliability (equipment, processes and production) and Operations Optimization (efficiency and cost). Shift Team Leaders are accountable for the quality and content of the Shift Team Meeting. The Shift Team Leader will:

- Communicate issues and incident information
- Test the quality of the individual shift handovers by asking for each person to report out the key information from their part of the unit/area.
- Make sure that team members are working priorities consistent with the long-term objectives and immediate day's demands.
- Focus on the status of on-going Operations as well as preparatory activities, such as process adjustments and equipment isolations.
- Checks that the Team is aligned on priorities for work execution.
- Reviews specific topics during special operations handovers. (E.g. hydrocarbon freeing, flushing, draining, steam-out procedures especially during transient operations such as unit transitions or shutdowns or startups)

6.1.8.7 VERIFICATION AND MEASURE

The verification and measurement of an effective shift handover practice ensures personnel conduct safe and effective communications. Periodic reviews or assessments should be conducted at the Site to evaluate the health and effectiveness of the handover practices.

6.2 COMPLETING THE WORK PERMIT

INTRODUCTION

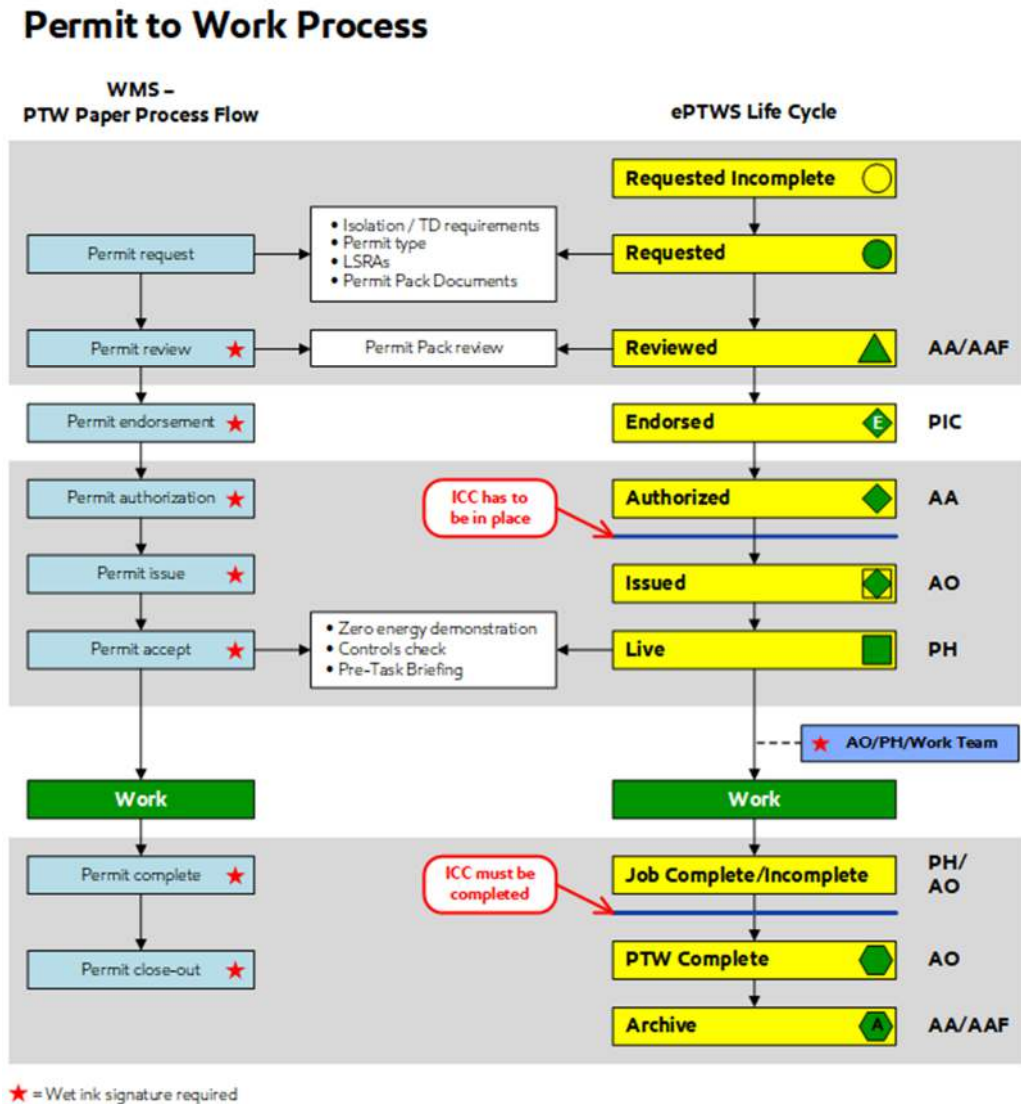
This section provides details of the steps in preparing and completing a permit. This section also describes the process flow in developing the Permit Pack with the defined responsibilities for each step of the process. This section closely ties with the Work Management process described in [Section 5.0: Work Management Framework](#).

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6.2.1 FLOW CHART

The following flowchart summarizes the Work Management process referenced during the work permitting process. Refer to [Figure 9: Permit to Work Process Flow 1](#)

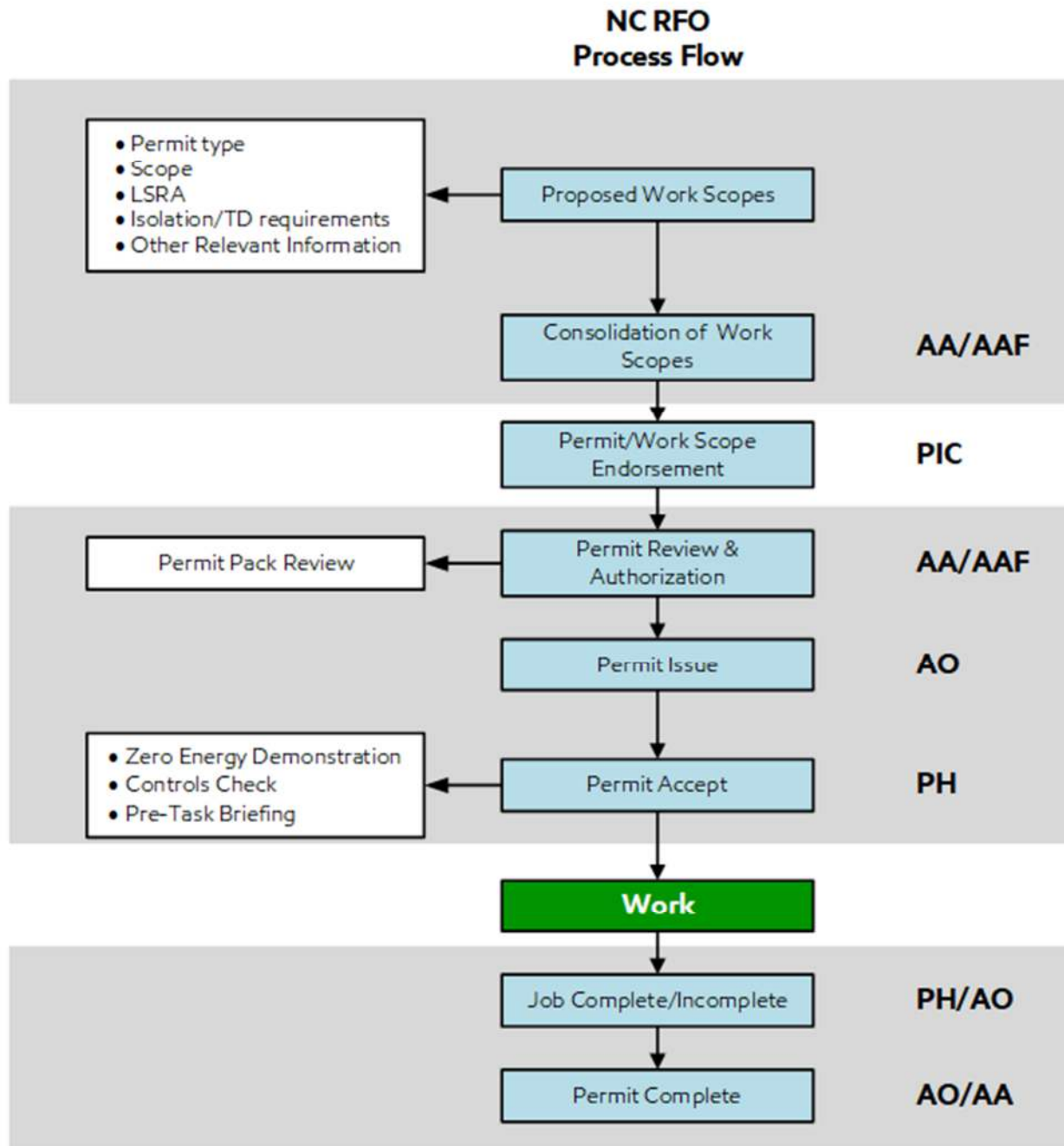
Figure 9: Permit to Work Process Flow 1



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Figure 10: Permit to Work Process Flow 2

Permit to Work Process



*For additional information on the NCO RFO lifecycle and flexibility. Refer to [Section 6.1.6](#) and [Section 6.3.3](#)

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6.2.2 WORK PERMITTING PROCESS

The following table details each of the Work Permit Process flow phases, typical steps required at each of the phases, reference to the applicable sections of the permit(s), and the responsible person(s) required to execute the requirements at the specific phase. Refer to

Table 4: Work Permit Process Details

In non-centralized remote field operations (NC RFO), permit packs and JSAs (where required) are prepared on the day of work execution, therefore the order in which the typical steps of the permitting process are executed may vary. [E] In NC RFO all steps of the permitting process must still be completed.

Table 4: Work Permit Process Details

Task	Activity	Responsible Person
PTW Request NC RFO List of Work Scopes	- Obtain work order, assigned work, or work request from Site Management representative. - Determine permit type or types required (Hot Work, Confined Space Entry or Critical). - Complete the required fields on the permit, include: - Date of work. - Time of proposed work. - Specific location of work. - Equipment/system tag or ID number. - Work order number (if applicable). - Permit Holder and company name assigned to the job. - Provide details in the Work/Task Description. Ensure alignment with the work order or work assignment detail. Ensure that the work scope and details are clear and sufficient to communicate to any person the location, activities, and equipment required to complete the job. - Identify LSRA activities - The permit requestor selects the isolation type required (none, personal or Isolation Control Certificate [ICC]/ Remote Field Isolation Certificate [RFIC]) and indicates the need for a Temporary Defeat (TD) in the Permit Pack Document (PPD) Other.	Permit Requestor (PH, Planner, Supervisor or other person)
Permit Preparation	The requestor or PH visits the worksite to assess potential hazards associated with the planned work.	Permit Requestor

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Table 5: Work Permit Process Details, Continued

Task	Activity	Responsible Person
Permit Preparation, Continued	- The Permit Holder then initiates the Job Safety Analysis (JSA) (if required). A Reference JSA can be used and updated to accurately address the job scope. The PH engages the Work Team and other relevant functional group representatives in the JSA development, review and updating. - The PH reviews the Permit Pack Requirements Matrix to identify all required Permit Pack Documents (PPDs). The PH can identify additional external reference documents (Safe Work Practice Checklists, Procedures, Safety Data Sheets [SDSs], and others) that are also required to ensure the job progresses safely. The Permit Document requirements are documented in the Permit Section and attached/linked with Permit for Review and use during work execution.	Permit Holder (PH)
Permit Review	The Permit Holder (PH) reviews the Permit Pack with the Area Authority (AA). The AA for the Permit Pack review can be the functional supervisor (or lead) Area Authority Functional (AAF) who has the knowledge and understanding of the proposed job.	PH and AA/AAF
	The AA/AAF, in consultation with the PH, reviews the Permit Pack to ensure all hazards and controls are identified and Permit Pack required documents are correct. If isolations or temporary defeats are required, the AAF consults the AA (Operations) to verify that all isolation and TD requirements are accurate before progressing the permit to Reviewed.	
Permit Endorsement	The Permit is endorsed by the PIC at the Daily Work Coordination Meeting (DWCM). [CriticalE] A comprehensive check for any SIMOPS or conflicting work issues will be conducted before the permit is moved to the <i>Endorsed</i> state. In some cases, the PIC may verbally endorse permits for Remote Field Operations and Non-centralized Remote Field Operations (RFOs & NC RFOs) allowing the work to progress.	Person in Charge (PIC)
	Necessary isolations and/or TDs will be considered at the DWCM. It is the responsibility of the Operations AA to assign duties in performing Isolations and Temporary Defeats.	AA

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Table 5: Work Permit Process Details, Continued

Task	Activity	Responsible Person
Permit Endorsement, Continued	3. The Isolating Authority (IA) develops the isolation and TD plans and submits it to the Operations AA for verification. The IA in consultation with the Area Operator (AO) and Control Room Operator, as required, performs the isolations and defeats when required. Isolation Certificate and Temporary Defeat Certificates can be cross-referenced to the permit(s).	IA
Authorization	On the day the permit is required, [CriticalE] the AA (Operations) who is responsible for managing the permitted activities <i>Authorizes</i> the permit if there are no conflicting activities.	AA
Work site Verification/ Permit Issue	Work site verifications must occur before the final permit signatures by the AO and PH and commencement of work. [CriticalE] Prior to issuing (signing) a permit in the field, the AO must familiarize themselves with the work scope and ensure the following: ▪ Required isolations and/or TDs are in place ▪ Isolation integrity is verified and zero energy is demonstrated to the PH ▪ Controls identified on the permit/JSA are in place ▪ SWP checklist & SWC items have been confirmed up to the point prior to work execution ▪ Atmospheric gas tests are conducted by an AGT when required Reference: GOTO/WMSEFORMS, "Atmospheric Test Record," for an Atmospheric Test Record for situations where work is conducted without a permit or when additional space is required.	AO
Permit Acceptance	[CriticalE] Once the Permit is signed by the AO, the PH must arrange the Pre-Task Briefing with the Work Team before work can commence. [CriticalE] The PH and Work Team must verify that each control identified in the Permit Pack is in place at the time it is required. The PH signs the permit and the permit goes Live at this point. Work can commence!	PH and Work Team

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Table 5: Work Permit Process Details, Continued

Task	Activity	Responsible Person
Work execution	1. All personnel involved with the job are to make sure that all required controls from the Permit, JSAs, SWCs, and SWP Checklist, procedures are in place when required. [CriticalE] The team members must perform a last-minute Risk Assessment (LMRA) before, during and after the job.	PH and Work Team
	2. <u>Shift Requirements:</u> The AO and PH sign the permit each shift after all required demonstration and verification has taken place.	AO, PH and Work Team
	3. <u>Permit Extension:</u> Should there be a need to extend the duration of the permit to work beyond the shift to complete the job or activity. The PH requests the extension and a same shift extension can be approved by agreement between the PH, AO, and AA to a maximum of 16 hours total from the start of the shift. The AA can approve the extension for up to 4 hours after shift change provided the work is being completed by the same crew. The AO must be notified when this occurs. The work can continue through the Operations shift changes; however, the incoming AO must visit the work site and sign the permit.	AA, AO and PH

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Table 4: Work Permit Process Details, Continued

Task	Activity	Responsible Person
Work execution, Continued	<u>PH or Work Team Change out:</u> When the PH changes out during the shift, the oncoming PH signs the permit after reviewing the Permit Pack, confirming the isolation control point (ICP) functional lock is in place. The AO must demonstrate zero energy when required. NOTE: The AO will be notified when the PH changes out. When the Work Team changes out the PH must review the Permit Pack, confirm controls are in place and arrange a Pre-Task Briefing with the Work Team prior to the work recommencing."	PH and Work Team
Permit Close-out	1. When the work has been completed or when the work has stopped under the conditions and duration of the permit, the permit must be closed out.	PH
	2. The PH is responsible for accounting for all personnel involved in the work, and the work site is restored to a safe and tidy condition. The PH is also responsible for communicating if the job has been completed or, if not, the status of the work.	
	3. Once the PH completes the job, the PH notifies the AO and returns the Permit Pack to the agreed location or person. The PH confirms <i>Job Complete</i> .	PH and AO
	4. For work that involves isolations or TDs, the AO confirms that the job is complete, and then notifies the AA of the job status.	
	5. The AA authorizes the controls to be removed if there are no other permits linked to the ICC or TDs.	
	6. Once the isolations or TDs (or both) are removed and the work site inspected, the AO completes the permit.	

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6.3 HOT WORK

INTRODUCTION

This section discusses hot work activities and the precautions to be taken while performing hot work. These precautions are intended to protect personnel and equipment when performing hot work. This section also discusses the responsibilities for implementing protective measures during hot work.

[E] Procedures, including welding or hot tapping, and Safe Work Practices Checklists **must** be part of the Permit Pack. Site procedures and safe work practices should align with the *Upstream Safety Manual*.

Reference: [GOTO/WMSEFORMS](#), "Safe Work Practices," for SWP Checklists.

6.3.1 HOT WORK PERMIT ACTIVITIES

The Hot Work Permit covers any open flame or spark-throwing operations.

Reference: [GOTO/WMSEFORMS](#), "Permit to Work," for the Work Permit.

Work involving non-spark-throwing activities is covered by the Critical Work Permit category of "Spark Potential."

Examples of open flame or spark-throwing work activities requiring a Hot Work Permit include the following activities:

- Open flame
- Welding, burning, brazing, and grinding
- Use of a de-burring tool
- CADWELD® welding
- Stress relieving equipment that could develop a surface temperature higher than 400°F (200°C)
- Cutting and gouging
- Propane heating equipment
- Any activity that could generate sparks or slag that could be thrown out from the point of work.

6.3.1.1 POTENTIAL HYDROCARBON SOURCES

The following are examples of potential hydrocarbon sources that should be considered for hot work location, distance, and barrier requirements: Refer to [Table 5: Potential Hydrocarbon Sources](#)

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Table 5: Potential Hydrocarbon Sources

Production Operations	Workover / Drilling Operations
▪ Wellheads and wellbays ▪ Production areas and process areas ▪ Separators and pressure vessels ▪ Flammables and hydrocarbon tanks ▪ Flammables and hydrocarbon vents ▪ Header systems ▪ Flanges ▪ Valves ▪ Controllers and instrumentation ▪ Pig traps ▪ Hydrocarbon sump systems ▪ Enclosed or inadequately ventilated buildings (where there is a potential for an explosive atmosphere) ▪ Coatings or painting operations ▪ Chemical, waste transfer, and storage locations ▪ Sewers and deck drains ▪ Flammable dusts	▪ Rig floor ▪ Sub-rig floor area ▪ Mud manifold ▪ Mud tanks, ditch, or trench ▪ Mud pump in enclosed location ▪ Shale shaker ▪ De-sander, de-silter, de-gasser, or vent ▪ Diverter line vent opening

6.3.1.2 HOT WORK LOCATION

A hot work location is one that the Site has classified as containing or having the potential to contain hydrocarbons or other flammable or combustible materials.

6.3.1.3 HOT WORK SAFETY REQUIREMENTS

[CriticalE] The Hot Work Barrier Requirements table **must** be used to determine requirements for establishing fire-resistant barriers when there are points of potential hydrocarbon discharge on piping or equipment when conducting hot work activities. Refer to [Table 6: Hot Work Barrier Requirements](#)

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Table 6: Hot Work Barrier Requirements

Potential hydrocarbon source on piping or equipment located...	THEN...
Within 10 ft. (3 m) of the hot work location	- For offshore operations: Shut down and depressurize the equipment or effectively shield it from the hot work activity using a pressurized habitat. - For onshore operations: Shut down and depressurize the equipment or effectively shield it from the hot work activity. Shielding can be achieved by using a pressurized habitat or providing 100% containment of sparks and slag using fire-resistant materials (ex. welding curtains, fire blankets etc.). !IMPORTANT: When preparing shielding for hot work consider the direction of sparks and slag and their ability to impact areas above and below or pass through cracks or openings in the walls, doorways, or windows. !IMPORTANT: Consider the use of a pressurized habitat in areas with known integrity issues or where equipment operation may result in the release of flammable substances !IMPORTANT: The fire watch is responsible to monitor the effectiveness of the shielding. - Any potential hydrocarbon or flammable sources or materials within 10-35 ft. (3-11 m) must be protected. - Other hot work control and precautions identified in this section must be in place.
Between 10 ft. (3 m) and 35 ft. (11 m) , or within 85 ft. (26 m) for high vapor pressure liquids (such as, propane or butane) of the hot work location	- A fire-resistant wall or barrier must be erected between the hot work and the piping, or equipment must be shut in. - Any potential hydrocarbon or flammable sources or materials within the applicable distances must be protected. - Other hot work controls and precautions identified in this section must be in place. ! IMPORTANT: Hot work controls must address hazards located at higher and lower elevations.
Beyond 35 ft. (11 m) or 85 ft. (26 m) for high vapor pressure liquids (such as, propane or butane) of the hot work location	Use of barriers or shutting down equipment is discretionary with reliance on the JSA to determine any controls and precautions required, taking into account the conditions of the day (for example, wind speed and direction, elevation, adjacent activities, exposures above or below the hot work site, and so forth).

6.3.2 HOT WORK PERMIT CONSIDERATIONS

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INTRODUCTION

[CriticalE] Hot work activities within a classified area **must** be conducted under a Hot Work Permit unless the work is performed in a *designated welding* area or workshop setting. Hot work should be performed in a designated hot work area where possible.

6.3.2.1 DESIGNATED HOT WORK AREAS

Hot work can proceed without a Hot Work Permit if the following conditions are met:

- The area is a designated hot work area.
- No flammable or combustible liquids, gases, or materials are present.
- The area is defined by fences, ropes, or barriers.
- Access is restricted, when practicable.
- A protective screen around electric arc welding is provided to protect other personnel from eye damage.
- The area is clearly communicated and identified by signage.

6.3.2.2 DESIGNATED ONSHORE WORKSHOP

Hot work can proceed without a Hot Work Permit in an onshore designated workshop setting that is outside of the process area if the following conditions exist:

- The work is on equipment, pipe, or a vessel that has been rendered completely free of hydrocarbon or combustible substances.
- Safeguards or checks for the presence of naturally occurring radioactive material (NORM), lead-based paint, and pyrophoric scale are implemented if the work is on process equipment removed to the workshop.
- No flammable materials are within 10 ft. (3 m) of the hot work activity.
- Shielding is erected to contain all sparks or hot metal particles, if there are any flammables within 35 ft. (11 m).
- Shielding is provided to protect others entering the workshop from arc flash, or to exclude others from entering.
- Fire extinguishers are readily available at the hot work site.

▲ CAUTION: An appropriate intake and exhaust ventilation system is required for a workshop to be considered as a designated onshore workshop.

▲ CAUTION: Personnel should always assume that hot work is in progress in the workshop and take the proper precautions upon entering.

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6.3.2.3 DESIGNATED OFFSHORE WORKSHOP

In *addition* to the conditions for onshore workshops, hot work may proceed without a Hot Work Permit in an offshore workshop setting if the following additional conditions exist:

- The workshop is pressurized and has gas detectors in service.
- Oxygen and acetylene cylinders are stored outside the building.
- Acetylene, oxygen, and service air lines have automatic shutoff valves that close upon loss of power to the workshop.
- A manual fire pull station is near the exit.

6.3.2.4 TEMPORARY DEFEAT OF FIRE PROTECTION SYSTEMS

For hot work outside the living quarters or an approved workshop, the following restrictions apply:

- Temporary Defeat of the passive and active fire and gas protection systems is not allowed where hot work is taking place unless adequate mitigating measures are initiated. For example:
 - Direct monitoring performed by the Operator or Standby Person in the affected area.
 - Availability of direct communication with the Central Control Room (CCR) in the event of an incident.
 - Availability of fire extinguishers or fire hoses.
 - Approval for the temporary defeat is obtained.

6.3.2.5 RESTRICTIONS

Specific activity restrictions and safety precautions associated with a Hot Work Permit depend on the exact nature and location of the work. The specific activity restrictions should be identified by the following:

- Job planning and scheduling (work description, interface identification)
- Job safety planning
- SIMOPS Matrix and Restriction Tables

References:

- Section 9.4.4 "SIMOPS Matrix"
- Section 9.4.5: "SIMOPS Restrictions Tables".

6.3.3 HOT WORK RISK MANAGEMENT

INTRODUCTION

Risk management is the process of identifying potential hazards, undesirable consequences, and mitigation steps for hot work so appropriate precautions can be taken to minimize these undesirable consequences as low as reasonably practicable (ALARP).

6.3.3.1 HAZARDS

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The potential hazards of hot work are the generation of:

- Sparks or temperatures high enough to ignite flammables and combustibles
- Eye-damaging light during electric arc welding
- High amount of electrical energy during electrical arc welding
- Noxious or poisonous fumes

6.3.3.2 POTENTIAL CONSEQUENCES

The potential consequences of hot work are:

- Ignition of hydrocarbons or flammable materials if present. It is noted that:
 - Hot work can throw sparks or hot metal some distance from the source if not contained and can cause burns to personnel and damage to equipment if hydrocarbons or flammable materials are ignited.
 - Some hot work activities require time for the equipment or metal to cool to a temperature that is below the ignition temperature of hydrocarbons.
 - Some hot work activities have the ignition source present for a long time (for example, use of resistance mats).
- Looking at electric welding arcs can cause permanent damage to vision.
- The fumes generated by hot work may be hazardous or poisonous if inhaled.
- Electrocution or electrical burns.
- Heat burns from hot surfaces.

6.3.3.3 MITIGATION STEPS

Potential risk mitigation steps for hot work are:

- Remove the item that requires hot work to a nonhazardous area.
- Use cold cut methods when possible.
- Schedule hot work to coincide with planned facility or equipment shutdowns.
- Conduct a JSA and apply the Risk Management process, if required.
- Restrict conflicting work in adjacent areas.
- Conduct atmospheric testing.
- Use specific job procedures.
- Control health hazards.
- Use appropriate personal protective equipment (PPE).
- Use barriers and spark containment (habitats, fire-resistant blankets, and so on).
- Emergency preparedness.

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6.3.4 CONTROL OF HOT WORK HEALTH HAZARD

6.3.4.1 HAZARDOUS FUMES AND VAPORS

Welding or burning operations can generate metal fumes, gases, and dusts. The contaminants can include lead, nickel, carbon monoxide, iron oxide, oxides of nitrogen, ozone, NORM, chromium, and zinc. The sources of hazardous fumes, the various welding and cutting processes, and the hazards they represent are as follows:

- Resins or coatings on electrodes.
- Surface coatings on the metal-like galvanizing or paint (especially lead-based paint).
! IMPORTANT: For work that will disturb existing paint or coatings containing greater than 0.5% lead, it is necessary to remove the paint or coating before welding or cutting proceeds.
- Gases that are added to the process such as carbon dioxide, argon, and helium.
- Gases formed by electric arcs such as ozone or oxides of nitrogen.
- Manual metal arc welding creates metal fumes and the resulting ozone levels often exceed acceptable toxic exposure limits and thresholds.
- Gas metal arc welding creates metal fumes that exceed acceptable toxic exposure limits and thresholds.
- Gas tungsten arc welding creates welding fumes that are usually below the toxic exposure limits and thresholds, but the resulting ozone levels can be well above acceptable toxic exposure limits and thresholds.
- Cutting, particularly plasma cutting, can generate large quantities of fumes and dust and may also be very noisy.
- NORM residue and dust can release fumes during welding.
- Leaking or damaged oxygen or acetylene hoses and fittings can present both a health and explosion hazard.
- Grinding, burning, and abrasive blasting generate large quantities of fumes and dust and may also be very noisy.

6.3.4.2 VENTILATION OR RESPIRATORY PROTECTION

Monitoring of welding, cutting and burning processes has consistently shown the potential for unacceptable exposure to metal fumes and particulates. The following recommendations are made in relation to ventilation of hot work areas:

- Within a habitat or enclosure there should be extraction ventilation and makeup air.
Note: Testing has shown that even habitats or enclosures without roofs are prone to the buildup of unacceptable concentrations of heavier metal fumes if they are not ventilated.
- Local exhaust ventilation should be considered in each individual case, particularly if the plume can be seen rising into the breathing zone.

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Note: In an open environment, there is less risk of unacceptable levels of metal fumes building up. However, the position of the welder's face over the job may mean that exposures to welding fumes are unacceptably high.

The only exception to the use of ventilation may be when the proper respiratory protection is worn. In certain circumstances a powered air purifying respirator or supplied-air breathing apparatus may be used as this can combine a welding helmet with respiratory protection and offers a higher degree of protection. A Safety Advisor or Industrial Hygienist should be consulted on the choice of a suitable respirator.

6.3.4.3 HOT WORK IN A CONFINED SPACE

[E] For hot work inside pressure vessels or tanks the use of a habitat is not required; however, all other hot work and confined space entry requirements **must** be met to protect personnel and equipment from any hazards.

6.3.5 HOT WORK PRECAUTIONS

INTRODUCTION

This section describes the general precautions to be followed when performing hot work. Prior to performing hot work, alternatives to hot work in the process area should be considered to minimize the hazards to workers and the company facilities.

6.3.5.1 JOB PLANNING

Hot work jobs should be assessed for hazards and complexity to determine if job-specific procedures should be prepared. The following should be considered in planning a hot work job:

- Determine if hot work can be eliminated.
- Identify and document any job-specific procedures that are required to address:
 - Technical job requirements
 - Welder certification requirements
 - Hazard and risk mitigation assessment
 - Job preparation requirements
 - Emergency procedures
 - Barrier requirements
- Obtain a Hot Work Permit.
- Confirm that the appropriate PPE is available on the job site and is operational.
- Confirm the appropriate barriers, spark containment, and site protection.
- Confirm that an appropriately equipped and competent Fire Watch is present (if required).
- Identify requirements for atmospheric testing (initial, periodic and continuous).

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6.3.5.2 SITE PRECAUTIONS

[E] Use the following general site precautions:

- All equipment, hoses, and cables **must** be adequately rigged and protected to prevent the blocking of evacuation routes, damage, or becoming a trip or fire hazard.
- Equipment **must** be grounded (earthed).
- All drains at a minimum distance of 35 ft. (11 m) from the work location **must** be plugged and water-filled or properly covered.
- Before welding, burning, or grinding, all drums, barrels, or other closed containers that contain or may contain combustible materials **must** be:
 - Removed from the work area
 - Thoroughly steamed
 - Filled with water
 - Made safe by other approved methods

6.3.5.3 HOT WORK ON HYDROCARBON SYSTEMS

[CriticalE] Welding and burning, other than associated with approved hot tapping, **must not** be conducted on pipes, vessels, or systems containing hydrocarbons or other flammable substances unless the following actions are taken:

- Shut down the equipment.
- Isolate the equipment from hydrocarbon sources and other potential energy sources.
- Depressurize the equipment.
- Clean the equipment of any hydrocarbon deposits or debris (see section 7.5.6 Cleaning)
- Confirm the equipment is hydrocarbon freed (see section 7.5.5 for hydrocarbon freeing methods) to 0% LEL
- If 0% LEL cannot be achieved after completing the above steps, inert gas purge the equipment to an **oxygen** content of less than 1%.

▲ **CAUTION:** Systems with complex internals, dead legs, significant elevation changes, below grade jumpers, etc. are at increased risk for containing trapped and/or residual hydrocarbon. In these scenarios, inert gas purging is recommended.

▲ **CAUTION:** If N₂ or other inert gas purging occurs and gas testing is performed, then gas testing must be completed using a detector capable of measuring in an inert atmosphere (example: Infrared detector)

NOTE: Special precautions **must** be used when welding on certain systems. For example, some chemicals when subjected to heat can create poisonous gases, or oil vapors in some systems can create an explosion hazard.

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6.3.5.4 PERSONAL PROTECTIVE EQUIPMENT (PPE)

[E] Personnel performing hot work **must** wear personal protective equipment (PPE) appropriate for the job they are performing. The PPE requirements should be specified on the JSA when required. Typical hot work PPE includes the following:

- Protective clothing for protection from arc burns, sparks, slag, and hot surfaces such as face shields, hoods and gloves, and so forth.
- Hard hats (where there is potential for head injury)
! IMPORTANT: Welding hoods do not provide sufficient head protection.
- Eye protection
- Glare protection
- Breathing apparatus in any of the following situations:
 - During welding, burning, or grinding on painted or coated surfaces or on pipes or vessels containing materials that may produce noxious gases.
 - If there is no proper ventilation.
 - Welding in narrow or poorly ventilated spaces.

6.3.5.5 ATMOSPHERIC TESTING AND MONITORING

[CriticalE] The following gas testing and monitoring requirements must be followed during hot work:

- An Authorized Gas Tester (AGT) **must** take gas readings with a portable gas detector and the area declared gas-free (0%LEL) before the work commences. Initial and periodic test results **must** be recorded on the gas test record.
- A portable gas detector **must** be used throughout the job.
! IMPORTANT: Portable gas detectors **must** have current calibrations.

Reference: Section 8.1: "Gas Testing," for additional information.

6.3.5.6 FIRE WATCH

[E] A Fire Watch is required when conducting hot work except for locations or activities designated by the Operating Site.

[E] The Fire Watch must:

- Know his or her duties as a Fire Watch.
- Know the hazards that may be encountered during hot work.
- Know the appropriate emergency responses in the event of an incident.
- Be competent in use of the portable gas detector to be used during the job.
- Continuously monitor the hot work activity and surrounding area.
- Continue monitoring for at least 30 minutes after the work is completed to allow hot surfaces to cool and the work area deemed safe.

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6.3.5.7 FIRE-FIGHTING EQUIPMENT

[E] At the minimum, one fire extinguisher **must** be positioned at the Fire Watch location and be ready for immediate use before starting the hot work as follows:

- The fire extinguisher **must** be designated for Fire Watch use only and appropriate for the potential fire type and classification.
- Fire extinguishers **must not** be removed from a facility fire extinguisher station for Fire Watch use if there is no redundancy.

Before hot work begins, the following precautions should be considered:

- A charged fire hose available for use by the Fire Watch.
- A charged deluge system.
- Additional backup fire extinguishers may be positioned at the Fire Watch location.

6.3.5.8 RADIO SILENCE

[E] In the event of an activity that requires radio silence:

- Electric welding **must not** be performed during the radio silence; permits will be suspended.
- Other hot work activities may be allowed if the Area Authority approves them.

6.3.5.9 ELECTRICAL HOT WORK EQUIPMENT

[E] When performing duties with electrical hot work equipment that is not in an approved workshop or designated welding area the following precautions **must** be taken:

- The electrical equipment **must** be disconnected in the following situations:
 - Alarm situations
 - Exercises, such as fire and gas drills
 - Work breaks
- Engine-driven welding machines **must**:
 - Have controls and accessories that are in good operating condition.
 - Be installed, operated, and maintained consistent with the manufacturer's recommendations.
 - Have properly insulated cables and electrode holders.
 - Be shut down or turned off when not in use.
- Follow these rules when using arc welding cable:
 - The cable insulation **must** be in good condition to avoid arcing.
 - Use only enough cable to reach the job.
 - Keep the remainder of the cable rolled on racks.

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6.3.5.10 TRAINING

The following training requirements apply to hot work:

- [CriticalE] Welders **must** be certified on the type of welding they perform. Certification documentation should be available.
- Every person in the work party is familiar with the operation and location of all safety equipment provided in the hot work area (fire extinguisher, facility status lights, telephones, surface process shutdown buttons, and the methods of operating the deluge system, etc.) and is aware of all emergency procedures and notifications.

6.4 CONFINED SPACE ENTRY

INTRODUCTION

This section discusses Confined Space Entry (CSE) Permits and precautions to be taken while performing activities within confined spaces. This section also discusses the responsibilities of personnel and the protective measures to be taken during confined space activities.

[E] Procedures and Safe Work Practice Checklists **must** be part of the Permit Pack. Site procedures and safe work practices should align with the *Upstream Safety Manual*.

Reference: [GOTO/WMSEFORMS](#), "Safe Work Practices," for SWP Checklists

CONFINED SPACE DEFINITION

A confined space is defined as an area that meets the following requirements:

- Is not designed for continuous occupancy.
- Large enough and configured so that a person's head can enter the space.
- Has limited or restricted entry or exit.

And one or more of the following:

- A substance that has the ability to engulf or asphyxiate the entrant.
- A potentially hazardous atmosphere.
- Has an internal configuration that may cause an entrant to be trapped or asphyxiated by inwardly converging walls or by a floor that slopes downward, and tapers to a smaller cross-section.

CONFINED SPACE ENTRY

The Confined Space Entry Permit covers any entry into a confined space. Entry occurs when the breathing zone of the worker breaks the plane of the confined space opening. The entire body does not have to be inside the space for it to be considered an entry.

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! IMPORTANT: Hazardous atmospheres may still be present near openings even when the breathing zone of the worker does not break the plane of the opening or space. Proper PPE may be required prior to entry.

! IMPORTANT: [E] A confined space within a confined space needs to be treated as a separate CSE (for example, pontoons in a floating roof tank).

CONFINED SPACE EXAMPLES

[E] Each Site **must** identify the confined spaces in their facilities. It is recommended that facilities Confined Spaces have signs or identifiers. In addition, the safety planning done on every job evaluates the possibility that the work site meets the confined space definition criteria and determine if a confined space entry procedure is required. Confined spaces may include:

- Storage tanks, process vessels, steam generators, boilers, pressure vessels, skirted areas under vessels, vaults, any tank-like compartments (such as, helicopter fuel tanks).
- Transformers
- Open-topped spaces and diked areas (dikes over 4 ft. [1.2 m] high), such as pits, excavations, bunds, wellhead cellars, basements (turbine or compressors) and tanks.
- Pipes, piping raceways, sewers, sumps, shafts, ducts, and similar structures.
- Any shipboard space entered through a small hatchway or access point, cargo tanks, cellular double-bottom tanks, ballast tanks, compartments, oil tanks, helicopter fuel tanks, and void spaces.

! WARNING: A person whose upper body or head is within the confined space or whose head can be periodically within the confined space (for example, while digging a trench) is considered to have entered a confined space. For a trench or excavation, a person is considered to have entered a confined space if the person's head is *below ground level*.

CONFINED SPACE ENTRY PERMIT

[CriticalE] The Confined Space Entry Permit is required for all confined space entry operations. The permit allows entry for inspection and for performing atmospheric testing; the permit does not allow any specific work to take place within a confined space. [CriticalE] Any work within a confined space **must** be carried out under a separate work permit.

Reference: [GOTO/WMSEFORMS](#), "Permit to Work," Work Permit.

! IMPORTANT: If any doubt exists on what constitutes confined space entry or what requires a Confined Space Entry Permit, the Area Authority provides the necessary guidance on the minimal requirements for the work to be carried out.

! IMPORTANT: A separate confined space entry permit is required for each confined space to be entered.

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6.4.1 ASSESSING CONFINED SPACE ENTRY HAZARD

INTRODUCTION

This topic discusses the requirement to assess the condition of a confined space before starting work, and the required precautions to be followed during work based on the entry class determination.

6.4.1.1 ASSESSING CONFINED SPACE CONDITION

[CriticalE] Before issuing a Confined Space Entry Permit, the condition of the confined space or vessel **must** be assessed to determine its entry class consistent with [Table 7: Confined Space Entry Class Determination](#). Checks should be made for:

- Atmospheric content, including in the following order:
 - Oxygen levels
 - Flammable gases and vapors
 - Potential air contaminants or toxic substances

! IMPORTANT: Initial gas testing done from outside the vessel can become unrepresentative when residual material inside the confined space is disturbed, or the ambient temperature rises.

- Cleanliness of the confined space interior (to determine whether further cleaning of the interior is required before vessel is entered).
- Radioactivity levels (determines if radioactive protection suits are required).
- Any obstructions or voids within the space that could pose a risk of injury.
- Verify energy isolation, including any moving parts are isolated and secured.

6.4.1.2 IMPROVING CONFINED SPACE CONDITION

Once the condition of a confined space is assessed, the following steps should be considered to improve the condition of the confined space, if necessary:

- Additional ventilation
- Additional Purging
- Additional drainage and removal of residual materials (i.e. scale, sludge, fluids)
- Cleaning of interior

6.4.1.3 CONFINED SPACE ENTRY CLASS DETERMINATION

[CriticalE] Gas tests **must** be conducted to determine the entry class of all entries into confined spaces.

[CriticalE] Oxygen, flammability and benzene tests are required for all entries into confined spaces that are known to be exposed to a hydrocarbon stream and atmospheric testing must be conducted in accordance with [Table 7: Confined Space Entry Class Determination](#)

[CriticalE] Other known potential hazardous substances specific to the operation and production stream, for example, mercury, NORM, asbestos, refractory linings, and catalysts require atmospheric tests. Consult with local Site MOH and SSHE to determine other atmospheric testing requirements.

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[CriticalE] Entries into Immediately Dangerous to Life and Health (IDLH) atmosphere are not permitted unless:

- A risk assessment is conducted and approved by the appropriate level of management.
- Hazards and controls have been clearly identified and residual risk are brought to ALARP.
- All precautions for personnel safety have been documented and implemented.
- All IDLH entries **must** be performed by competent personnel.

Table 7: Confined Space Entry Class Determination

NOTE: Limits are subject to change periodically. Confirm local regulatory requirements and apply highest level of control.

Type of Gas	Class 1 Entry Allowed Without Respiratory Protection [1]	Class 2 Entry Allowed With Respiratory Protection [2]
	Atmospheric Concentration	Atmospheric Concentration
Oxygen	≥ 19.5% to ≤ 23.5%	≥19.5% to ≤23.5%
Flammability: Non-Hot Work	< 1% of LEL (HC)	< 10% of LEL (HC)
Flammability: Hot Work	0% of LEL (HC)	0% of LEL (HC)
Toxic Contaminants		
Volatile Organic Compound (VOC) *	< 100 ppm	< 1000 ppm (IDLH) Or unknown
Benzene **	< 0.2 ppm	< 500 ppm [IDLH]
H ₂ S	< 5 ppm	< 100 ppm [IDLH]
SO ₂ ***	< 0.5 ppm	< 100 ppm [IDLH]
CO ₂	< 5,000 ppm	< 40,000 ppm [IDLH]
CO	< 25 ppm	< 1,200 ppm [IDLH]
Organic Lead	< 0.1 mg/m ³	≥ 0.1 mg/m ³
IMPORTANT: Values can change periodically, check local Operating Site class limits or with Medical & Occupational Health (MOH). * Consult local Operating Site MOH/SSHE for latest testing requirements and thresholds ** Benzene: ½ face APR = ≥0.2 - < 2 ppm; FF APR = ≥ 2 - <10 ppm; SCBA >10 ppm *** SO ₂ : ½ face APR = >0.5 - < 5 ppm; FF APR = 5 - < 25 ppm; SCBA = >25 – 100 ppm		

! WARNING: [CriticalE] Entry into atmospheres above the Class 2 limits **must not** be conducted until Risk Assessed and approved.

! IMPORTANT: Values can change periodically. The Medical & Occupational Health (MOH) intranet site has the latest limits.

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NOTES:

1. The values equate to the levels to which personnel may be exposed safely for up to eight hours/day.
2. [CriticalE] For confined spaces where the levels of one or more contaminants are within the ranges represented in the Class 2 column, respiratory protection **must** be worn for entry to be allowed. [CriticalE] Monitoring of the atmosphere **must** be conducted at intervals appropriate to the conditions that might affect the concentration of gases, hydrocarbons, or toxins in the air. [CriticalE] If air monitoring or gas testing indicates that the level of contaminant(s) is increasing and appears likely to exceed those within the Class 2 range, personnel **must** be evacuated immediately.

! IMPORTANT: The Confined Space Entry Class Determination table does not consider any possibility of NORM within the confined space. Presence of NORM may require the use of respiratory protection, particularly if disturbed.

Benzene and VOC tests may be discontinued after two consecutive days' tests indicate that exposure levels for the entry classes are not exceeded. [E] If the vessel contains sludge or walls that are impregnated, then testing **must** be continued and monitored daily regardless of previous readings.

6.4.2 CONFINED SPACE ENTRY RISK MANAGEMENT

INTRODUCTION

Risk management is the process of identifying potential hazards, undesirable consequences, and mitigation steps for performing confined space entry jobs so appropriate precautions can be taken to minimize these undesirable consequences as much as is reasonably practicable.

6.4.2.1 HAZARDS

The potential hazards of work in a confined space are:

- Explosive, toxic, and oxygen-deficient or excessive atmosphere.
- Limited access into and egress from a confined space.
- Improper isolation of energy or chemical sources into a confined space.
- Existence of moving parts, obstructions, and voids within a confined space.
- A confined space within a confined space.
- Corroded internal structures may collapse.
- Live equipment within confined space (such as, asbestos removal enclosure).
- Existence or generation of residual hydrocarbons, benzene, or hydrogen sulfide (H₂S) pyrophoric scale (iron sulfide [FeS]), SO₂, steam, CO₂, N₂.
- Existence of naturally occurring radioactive material (NORM) or low specific activity (LSA) within process vessels.
- Concentration of dangerous chemicals or fumes within a confined space.
- Presence or generation of explosive or flammable gases.
- Flammable or toxic materials introduced when conducting the work (for example, pressurized canisters, leaking oxygen, or acetylene hoses or fittings and sprays).

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- Disturbing residual material or increase in ambient temperature can generate toxic or flammable vapors (or both).
- Existence of asbestos gaskets or fiber particles.
- Electrocution.
- Falls, trips, and slips.
- Entrapment.
- Claustrophobia.
- Reduced visibility.
- Exhaustion.
- Drowning.
- Dropped objects from overhead.
- Heat and cold stress.
- Unstable physical containment of surroundings (for example, trenches and excavations, steep slopes, soil conditions).
- High levels of noise within a confined space.
- Difficulty in communicating or seeing alarms while in a confined space.
- Difficulty in conducting rescue operations (restricting personnel in the space).

6.4.2.2 POTENTIAL CONSEQUENCES

The potential consequences of work in a confined space are:

- Inhalation of or exposure to dangerous chemicals, gases, or fumes
- Exposure to hazardous atmosphere due to leak from live equipment
- Asphyxiation due to lack of oxygen
- Exposure to NORM or LSA
- Ignition or explosion
- Engulfment
- Injury from moving parts, obstructions, or voids
- Fall injury due to corroded structure collapse
- Hearing damage from high noise exposure
- Not responding appropriately to external emergency situations
- Injury due to slips, trips, and falls
- Difficulty in exiting confined space in an emergency situation
- Heat stress
- Serious or fatal injury due to asphyxiation

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6.4.2.3 MITIGATION STEPS

The potential risk mitigation steps for work in a confined space are:

- Consider alternative methods for doing the work (avoid confined space entry).
- Refer to Safety Data Sheets, (SDSs), to determine required personnel protection.
- Standby person oversight of the confined space entry.
- Design enclosures and excavations to exclude live equipment or flanges where possible. If not possible, address risk in JSA.
- Creation of a confined space entry plan, procedure, permits, and work practices.
- Confined space preparation (isolation, cleaning, purging, and ventilation).
- Initial and continuous atmospheric testing and monitoring.
- Personal protective equipment relative to the hazard.
- Breathing apparatus.
- Communications.
- Emergency response plan and rescue equipment.
- Lighting equipment.
- Personnel training.
- Good housekeeping—removal of sludge, debris, and tripping hazards.
- Barricade or cover voids and identify or eliminate obstructions.
- Inspect corroded internal structures for integrity.
- Appropriate signage.
- Provide multiple entry and exit points if practical.

6.4.3 CONFINED SPACE ENTRY PRECAUTIONS

INTRODUCTION

This section describes the general practices and precautions to be followed during confined space entries. A Confined Space Entry Supervisor should be appointed to oversee all aspects of planning and executing confined space entry work. The Permit Holder may fulfill this role.

6.4.3.1 JOB PLANNING

The complexity and risks associated with the confined space entry should be used to determine if a job-specific procedure should be prepared. In addition, the following general practice should be followed:

- Before starting a confined space entry plan, personnel must complete the following requirements:
 - Visit the work site.
 - Plan the work.
 - Prepare a JSA if required.

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- Request a Confined Space Entry Permit. The Confined Space Entry Permit authorizes entry into the confined space. A Critical Work Permit, Hot Work Permit or JSA will be required for any work to be conducted in the confined space.
- Prepare and execute an energy isolation plan to remove energy sources that are potentially hazardous to the workers who will enter the confined space. Energy sources must be isolated and made safe (See Section [7.0 Energy Isolation & Temporary Defeat](#) for Energy Isolation requirements).
- Evaluate the status of other activities e.g. drilling nearing the pay zone or having downhole difficulties at the time.
- Write job-specific procedure or checklist (or both) appropriate to the complexity of the job that should include:
 - Work scope
 - Job preparation requirements
 - Emergency, communication, and rescue procedures/process
- Prepare emergency and safety equipment:
 - PPE
 - Safety and rescue equipment
 - Atmospheric testing equipment
- Make an initial assessment of the condition of the confined space or vessel and determine the entry class for the confined space entry. [E] This initial entry **must** be performed wearing appropriate PPE and taking precautions for the worst expected conditions.
Reference: Section 8.2: "General Gas Testing Practices," under the heading "Gas testing in confined spaces".
- Once the condition of a confined space is assessed, risk mitigation steps must be considered to improve the condition of the confined space if necessary.
- Prepare a Permit Pack with all safety precautions detailed and obtain approval of the CSE Permit.
- Establish the confined space for work according to the requirements including the regulations specific to the work.
- The atmosphere must be tested according to the requirements.
 - [CriticalE] Ventilation, if required, **must** be maintained and monitored and be alert to any short circuiting that may prevent ventilation of the entire space.
- [CriticalE] Rescue personnel **must** be available at the facility, adequately trained and equipped for the rescue, pre-briefed on the rescue plan, and notified of when confined space entry starts and ends.
- [E] Access points created for CSE entry must be protected from entry when work is not ongoing.

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6.4.3.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

All required PPE including respiratory protection must be used as defined by the permit and conditions. Depending on the conditions inside the confined space, this equipment can include the following:

- Respiratory protection:
 - Approved positive pressure breathing apparatus
 - Positive pressure airline respirator equipped with an escape set
 - Air-purifying respirator equipped with the correct filters or cartridges for the air contaminants
- Hearing protection
- Communications equipment
- Personal detector or monitor as required by the hazard
- Rescue harness and lines

6.4.3.3 ATMOSPHERIC TESTING

Atmospheric testing, as prescribed in the job plan, includes testing prior to initial entry, and continuous thereafter while personnel are occupying the confined space in accordance with [Section 8](#).

[E] Atmospheric test must be conducted by an AGT **and** be recorded on the Confined Space Entry Permit, "Atmospheric Test Record." [CriticalE] If any gas concentrations are not within the limits for the type of confined space entry as defined in the Confined Space Entry Class Determination table, confined space activities **must** cease and the appropriate precautions **must** be taken for the new entry class.

[CriticalE] The atmosphere within a confined space **must** be tested:

- As close as possible, but no longer than one hour, before the initial entry occurs. Ventilation **must** be shut down at least 30 minutes prior to the initial test.
- If entry is required for initial testing (insufficient external points to gain representative samples) the initial entry is with respiratory protection and all other CSE controls until the safety of the space is confirmed.
- After any cleaning or additional ventilation is performed to improve conditions.
- Continuously with a portable gas detector while personnel are wearing respiratory protection
! IMPORTANT: Portable gas detectors **must** have current calibrations.
- Atmospheric monitoring within the confined space **must** be performed during the course of work because external activities can result in gas or toxic substances entering a confined space, such as:
 - Flange leak
 - Vessel blowdown
 - Activation of an atmospheric pressure safety valve (PSV)
- Work activities (hot work, chemical cleaning, use of dye penetrates, movement of sludge) may contaminate the atmosphere.

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[CriticalE] Continuous atmospheric monitoring:

- **Must** be done while personnel are inside of the confined space.
- **Must** be done by means of a portable gas detector (with an alarm) that is placed inside the confined space for the types of hazards expected.
- May be done with the mechanical ventilation system operating.

[CriticalE] Workers **must** vacate the confined space:

- If the gas detector alarm sounds.
- If conditions make it hazardous for them to work without the use of breathing apparatus or respirators.
- In the event of a facility emergency alarm.

6.4.3.4 VENTILATION AND LIGHTING

In order to establish and maintain a safe working environment when personnel enter a confined space that poses an actual or potential atmospheric hazard, the [CriticalE] confined space **must** be ventilated by natural air draft, forced air ventilation, or mechanical ventilation.

NOTE: The use of continuous forced air ventilation is preferred for all confined space entries. For hot work within a confined space, additional ventilation requirements may be required to ensure safe atmosphere.

[E] Adequate lighting **must** be provided at all times when workers are within a confined space. The following types of artificial lighting may be considered:

- Air driven
- Portable electric
- Self-contained (light sticks)

Should any electric cables for the use of lighting or other electrical equipment be required, an attempt should be made to route them into the confined space by a route other than the one being used for entry into or exit from the confined space. They should also be placed or protected in a manner that abrasion or movement cannot damage other pieces of equipment. [E] Power **must** be isolated or switched off whenever the cables are being moved into or out of the confined space.

6.4.3.5 SAFETY WATCH / ATTENDANT

[CriticalE] One or more personnel **must** act as a Confined Space Entry (CSE) Safety Watch/Attendant whenever work is performed for each Confined Space Entry.

[E] The CSE Safety Watch/Attendant **must:**

- Review the CSE Permit before anyone enter the confined space.
- Be trained in their duties and in how to use communications and rescue equipment.
- Perform no other duty that might interfere with their primary responsibility during the course of the confined space entry activities.

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6.4.3.6 COMMUNICATIONS

[CriticalE] It is critical that the following communications links be established and maintained while personnel are inside a confined space:

- Communication between all personnel inside the confined space and the Safety Watch/Attendant.
- Communication between the Safety Watch/Attendant, the Control Room or Central Control Point, and rescue team. In areas where there is no Control Room or Central Control Point, the Safety Watch/Attendant must have communication with the Area Operator in all instances.

These communication links must be:

- Effective (able to be clearly understood by all parties regardless of background noise levels within and outside the confined space).
- Reliable (not subject to failure, such as battery failure, without having backup equipment immediately available).
- Instantaneous (able to be used immediately at any time regardless of the position of the relevant personnel).

6.4.3.7 RESCUE PROVISIONS

[CriticalE] Rescue provisions must be provided for the specific hazards and conditions of the entry. These can include:

- Contains or has the potential to contain:
 - A hazardous atmosphere.
 - An oxygen-deficient atmosphere.
- Has the potential for engulfing the worker (e.g. a collapsed trench).
- Has an internal configuration such that the worker could be trapped or asphyxiated by inwardly converging walls or a floor that slopes downward and tapers to a small cross section.
- Contains any other recognized safety or health hazard.

▲ **CAUTION:** The Confined Space Entry Safety Watch/Attendant may enter a confined space to attempt rescue if he or she is trained and equipped for rescue operations and if another qualified Standby Person is assigned the duty of attending the confined space with backup breathing apparatus before rescue operations begin.

6.4.3.8 ENTRY RECORDING

[E] All entrants entering and exiting a confined space **must** inform the CSE Safety Watch/Attendant and confirm that their names are recorded on the Confined Space Entry – Personnel Log-In/Log-Out Sheet.

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6.4.3.9 CONFINED SPACE ENTRY PROTECTION

[E] Voids within the confined space area **must** be covered to prevent a worker from dropping into the void and remain covered until completion of the activities. Obstructions that may be hazardous to personnel should be identified and eliminated or managed.

After completion of work in a confined space, entry into the confined space can still pose a potential hazard. The following precautions are taken:

- Cordon off the entry to the confined space and place appropriate signage

Example: "Danger – Entry Permit Required – Do Not Enter"

NOTE: Wording of danger signs can vary as long as the hazard is identified.

- Close the entry into the confined space as soon as practical.

6.4.3.10 POTENTIALLY HAZARDOUS SCALE

In some processes naturally occurring radioactive scale might be present. This is referred to as naturally occurring radioactive material (NORM or low specific activity [LSA] scale. The main danger of NORM is breathing in airborne radioactive dust particles. Whenever grinding, scraping, or needle-gunning activities are performed within the confined space, personnel within the space should use dust masks, suitable respirators, or breathing apparatus. [E] Tests for radioactive scale **must** be conducted during the initial condition assessment of the confined space.

Iron Sulfide is a black, powdery, scaly compound that can be formed over a long period of time in an oxygen-free atmosphere by a chemical reaction between a sulfur compound such as hydrogen sulfide (H_2S) and iron oxide. Iron sulfide is a pyrophoric scale and should be kept wet. If the scale dries out in the presence of oxygen, it can oxidize rapidly and the resultant heat can cause it to ignite spontaneously. If a flammable mixture is present, it will ignite.

Disposal of pyrophoric scale into open containers is not recommended because this can result in fires either at the facility or during transportation. It is recommended that all scale and other materials removed from the vessel be immersed in water and that a suitable means of disposal be arranged.

6.4.3.11 JOB CLOSEOUT

After all work is completed, the following activities should be performed (where applicable):

- Confirm that all space internals are reinstated to their normal state. Verify that all internal residue and waste has been removed, and openings are clear of obstructions. These could include welding rods, demister pad fixing wire, rags, and so forth.
- Check to confirm that all tools and equipment are removed.
- Remove void covers and lighting equipment.
- Properly close all hatches used for ventilation and for routing hoses and electrical cables.
- [E] The Area Operator and/or Area Authority **must** make a final inspection of any confined space entered before the hatch(es) are closed.

! WARNING: [CriticalE] There **must** be no doubt that all people have exited the space, do not rely on the log-in/log-out register before the hatch(es) are closed.

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- The space **must** then be re-commissioned according to the process for reinstating facilities and equipment.

6.4.3.12 TRAINING

[CriticalE] Personnel who perform work in, or associated with, confined spaces **must** have training for confined space entry hazards, mitigations and their associated specific, designated duties. They should be trained in:

- Possible hazards within the confined space.
- Symptoms of toxic effects due to exposure to hazardous materials that they could encounter in the confined space.
- Emergency procedures/process.
- Operation of simple fire-fighting equipment (for example, fire hose and dry chemical extinguisher).
- Operation of fall restraint or fall arrest equipment, if required.
- Self-rescue techniques.
- Use of the following:
 - All required personal protective equipment (respirators, SCBAs etc.)
 - Portable gas testers and personal monitors
 - Communications equipment
 - Ventilating equipment
 - Lighting equipment
- Duties of the CSE Standby Person (if assigned).

6.5 CRITICAL WORK PERMIT

INTRODUCTION

[CriticalE] The Critical Work Permit is used to manage permit-required work activities where there are higher risks or special considerations. The Critical Work Permit is required for the critical work activities listed in this section.

Reference: [GOTO/WMSEFORMS](#), "Permit to Work," Critical Work Permit.

[E] Procedures and Safe Work Practice Checklists **must** be part of the Permit Pack. Site procedures and safe work practices should align with the Upstream Safety Manual.

Reference: [GOTO/WMSEFORMS](#), "Safe Work Practices," for SWP Checklists

6.5.1 CRITICAL WORK PERMIT ACTIVITIES

Critical work activities follow the same permitting process described in [Section 6.2: "Completing the Work Permit."](#) Examples of critical work activities requiring a Critical Work Permit include (but are not limited to) the following activities. Refer to [Table 8: Critical Work Permit Activities](#)

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Table 8: Critical Work Permit Activities

Critical Work	Example Activities
Abrasive blasting	▪ Sand or grit blasting ▪ Any high pressure blasting that could generate sparks or static electricity ▪ Blasting equipment that could cause serious injury
Breaking containment	Breaking the containment on equipment or systems such as hydrocarbons, chemicals, air, steam, fluids and gases. Examples of breaking containment activities are: ▪ Unbolting flanges and manways or removing screwed fittings on systems, such as, pipes, vessels, exchangers, valves, turbines, compressors, and pumps ▪ Opening enclosures such as pig launchers or receivers and filter housings ▪ Opening vents or drains to atmosphere ▪ Removing orifice plates ▪ Taking samples (when venting or draining to atmosphere). ! IMPORTANT: A Critical Work Permit is not required (unless specified by the Site due to inherent risks such as H₂S, Benzene and NORM) for venting, purging, or draining equipment to closed systems or to atmosphere conducted by a competent person during their normal duties, provided there are no interfaces or simultaneous operations (SIMOPS) issues to manage. However, a JSA or procedure may be required (refer below). ! IMPORTANT: A JSA should be considered when draining or venting to atmosphere unless the residual pressure being vented is below flare or closed drain pressures. ! IMPORTANT: [E] If continuous venting to atmosphere at operating pressure is required for operational reasons, an approved procedure and JSA must be obtained.
Complex lift	A lift is classified as <i>complex</i> when one or more of the following applies ^{1, 2, 3}: ▪ Using taglines, push poles, or hands on load on vessel deck ▪ Personnel transfer lifts ▪ Personnel work basket lifts ▪ Lifting Fast Rescue Crafts and Line Boats ▪ Lifts exceeding 80% of the cranes current capacity or environmental limitations during the lift planning activity ▪ FSO/FPSO lifts where a sudden change of heading may affect the lifting operation

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Table 8: Critical Work Permit Activities, Continued

Critical Work	Example Activities
Complex lift, Continued	▪ Lift over or near pressurized or hydrocarbon filled equipment ▪ Lifts over or near occupied buildings and accommodations ▪ Lifts over or near well work equipment ▪ Blind lifts ▪ Lifts through hatches ▪ SIMOPS restricted activities (per SIMOPS Matrix) ▪ Lifts in to or out of a confined space ▪ Mobile crane/tele handler lifts, supported on tires or when traveling with a suspended load ▪ Lifts with potential for sudden load transfer or change of tension ▪ Lifting hazardous materials ▪ Scaffold as suspension point for rigging/lifting Load- and pull-test ▪ Tandem lifts ▪ Lifts using both aux and main crane hoists simultaneously ▪ Lifts near overhead power lines ▪ 50 ton/tonne load weight NOTES: 1. [E] Where country or local regulatory requirements are more stringent, the Regulations must always prevail. 2. Criteria for defining the planned lifting activity as Complex may not be limited to the scenarios in this section. The lifting activity may also be defined as Complex due to factors such as installation specific considerations, landing areas and lift path or type/design of lifting equipment used. 3. Likewise, shape and weight of the load, rigging configuration, change in environmental conditions, weather, lightning and potential of dropping the whole or parts of the load or rigging should be evaluated. Additional information available in the Rigging and Lifting Safeguards SWP.
Diving operations	All manned diving operations on, or adjacent to an operating facility, including activities on operating pipelines.

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Table 8: Critical Work Permit Activities, Continued

Critical Work	Example Activities
Excavation or ground disturbance	▪ Excavation: Any fabricated cavity or depression in the earth's surface formed by earth removal that produces unsupported earth conditions as a result of the removal. ▪ Cavity or trench: A narrow excavation with a width no greater than 15 ft. (4.6 m). ▪ Ground disturbance: Any digging, drilling, or other activity that penetrates the earth which could present a hazard.
High risk electrical	High risk electric work involves work on live electrical systems above Extra Low Voltage (ELV) levels. Extra Low Voltage circuits have a voltage not exceeding 50 Volts RMS A.C. between conductors or earth. E.g. 24VDC instrument loops, CCTV, data networks etc. High Risk Electrical Work also includes activities with the potential (accidental or otherwise) for individuals to come into contact with energized electrical components of 300 volts or greater, or to be exposed to high temperatures or flashes resulting from electrical arcing or electrical fault conditions. Safety Watch/Standby Person The HRE Safety/Standby person should (in addition to the Safety Watch responsibilities): ▪ Be familiar with job scope and hazards associated with the task. ▪ Be trained in the use of required electrical PPE. ▪ Be trained to perform basic first-aid, including CPR and use of an AED (if available). ▪ Know where and how to de-energized the electrical circuit or equipment. ▪ Keep non-essential personnel out of the work area. ▪ Be present during the safety critical task steps with no other work execution roles assigned (observe only). ▪ Be empowered to intervene or stop work if unsafe acts or conditions occur. ▪ In an emergency, know how to communicate and request additional help. ▪ Know when and how to remove the worker from an electrical hazard without endangering themselves. ! WARNING: The main duty of the Standby Person is to monitor the activity, stop work if an unsafe condition develops, and alert the area authority (AA) of any incidents so that appropriate emergency team can respond. [E] The standby person must not enter the area if there is any potential for injury.

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Table 8: Critical Work Permit Activities, Continued

Critical Work	Example Activities
Work at heights (WAH)	- Working at heights above 6 ft. (1.8 m) without handrails or barriers. - Work requiring the person's center of gravity to extend beyond the vertical plane of an approved scaffold or guardrail. - Working within 6 ft. (1.8 m) of an unguarded or unprotected edge on an elevated (exceeding 6 ft. [1.8 m]) structure. - In areas exposed to impaling hazards where the potential for injury exists by falling on them, even when the height is less than 6 ft. (1.8 m). - Rope access work (abseiling or rappelling). NOTE: [E] Personnel must have an approved rescue plan and not work alone when working at heights Reference: Work at Heights Safe Work Practice, for more detailed information.
Over-the-side work	Work outside an approved deck, or structure that could result in a fall to sea. NOTE: [E] Personnel must have an approved rescue plan and not work alone when working at heights. Reference: Work at Heights Safe Work Practice, for more detailed information.
Work in exposed or congested locations	Any work in congested or tight areas that can result in injury due to moving equipment, energy release, or risk associated with the location (for example, hot or cold surfaces, sharp edges).
Spark potential (spark generating) in hydrocarbon processing areas or areas containing flammables	- Meggering - Engine-driven equipment (gas or diesel including vehicles and mowers). - Non-classified battery-operated equipment (e.g. cameras and laptops). - Use of a heat gun or soldering iron at a temperature less than 400°F (200°C)*. - Chipping (use of needle gun). - Work on live, non-intrinsically safe electrical equipment in classified hazardous locations. - Use of electrical tools or lights not classified for hazardous locations. - Opening and working on live explosion-proof or flameproof enclosures. - Controlled post-weld heat treating (stress relief) (heat blankets) less than 400°F (200°C). *NOTE: A Hot Work Permit and associated controls are required if greater than 400°F (200°C).

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Table 8: Critical Work Permit Activities, Continued

Critical Work	Example Activities
Working on or with pressurized systems	▪ Hot tapping activities ▪ Hot bolting activities ▪ Hydro cutting and hydro blasting (for example, tube cleaning) ▪ High pressure testing (for example, hydro testing) ▪ Any other work that could result in a high-pressure release from the operating system or equipment used.
Working with explosives or detonation devices	▪ Explosive fasteners ▪ Wellwork activities, such as: – Tubing perforations – Wireline operations NOTE: A radio-safe detonating system uses a radio-safe blasting cap or electro-explosive that does not require radio silence. If radio-safe detonation systems are used, the Person in Charge (PIC) should determine if this mitigates the hazard.

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7.0 ENERGY ISOLATION AND TEMPORARY DEFEAT

INTRODUCTION

This section provides guidance on the practices to be followed when isolating facility systems and equipment for maintenance or other activities, and subsequently reinstating them to their normal safe operating condition. It also describes the lock out/tag out (LOTO) process which is integral to the safe execution of work activities on process equipment.

This section also includes the Management of Temporary Defeat as it applies to work management and facility critical devices and systems.

7.1 PROCESS SUMMARY

7.1.1 PURPOSE

This section documents the practices followed to enable the safe execution of activities on or in process equipment and ancillary systems and in particular:

- Safe isolation of equipment
- Establishing a zero-energy condition
- Safe reinstatement of facilities to normal operating conditions

Adhering to these processes and practices enables the work to be conducted safely.

[CriticalE] All isolations **must** meet or exceed the requirements of the Work Management System.

7.1.2 SCOPE

This isolation and reinstatement process is intended to manage the safety and environmental aspects of facility operating energy sources. Hazards associated with energy sources must be managed at all times through the use of Isolation Control Certificates, Temporary Defeat Certificates, and Work Permits as described in WMS.

7.1.3 SYSTEM ISOLATION AND INTRUSIVE WORK

System: A process system is a section of the facility that can include a number of equipment items, such as vessels, pumps, pipework, valves, and instrumentation.

A system to be isolated and prepared for maintenance or other activities:

- [CriticalE] **Must** have appropriate locked and tagged (or appropriately identified) isolation points at all boundaries. The isolation points should be marked on a piping and instrumentation drawing (P&ID) or an electrical diagram.
- [E] When displacing with fluid, must have sufficient high point vents open and the fill rate adjusted to suit venting capacity.
- [E] Must have Vents open at the high points before draining liquids to confirm that a vacuum is not created.
- Should have adequate means to test and monitor the integrity of the isolation(s).

Intrusive work: Any activity that requires working on and/or the breaking of containment of any type of energy, such as mechanical, hydraulic, chemical, electrical, or thermal, and can include:

- Breaking open a flange to repair or replace equipment

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- Removing a pipe spool or valve
- Entry into a process vessel (confined space entry)
- Shutting off power to rotating equipment
- Removing protection from electrically energized equipment

7.1.4 ISOLATION AND REINSTATEMENT ACTIVITIES

The following activities can be performed to isolate facility equipment and systems for maintenance, and to subsequently reinstate equipment and facilities to normal operating conditions safely:

- Isolation planning
- Temporary defeats application/removal
- Energy isolations / De-isolation
- Draining and depressurizing
- Breaking containment / Blinding / De-blinding
- Hydrocarbon freeing
- Purging
- Cleaning
- Job closeout
- Equipment testing
- Air freeing
- Leak testing / Pressure Testing
- Acceptance testing
- Re-commissioning

7.2 SAFETY

7.2.1 ISOLATION RISK MANAGEMENT

7.2.1.1 INTRODUCTION

Risk Management is the process of identifying, assessing, and controlling the risk (consequence and frequency) associated with work activities such as energy isolation, work execution, and equipment reinstatement. Potential hazards are identified and mitigation or preventative controls are implemented to reduce undesirable consequences to as low as reasonably practicable (ALARP). If the level of residual risk for the proposed scope of work is unacceptable, then the work scope or execution methods must be modified to reduce the risk to an acceptable level.

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Various tools and methods are utilized to identify and control hazards such as:

- Last Minute Risk Assessment (LMRA)
- Start Work Checks (SWC)
- Job Safety Analysis (JSA)
- Risk Screening
- Structured Risk Assessment when required
- Safe Work Practice Checklists (SWP)

7.2.1.2 HAZARDS

Some potential hazards when isolating or reinstating facility systems are:

- Incorrectly installing, removing, or reinstating isolations
- Incorrectly isolating control devices, including distributed control systems (DCS)
- Hazardous products or materials trapped or held-up in standpipes, weirs, loops, or low points
- Unexpected pressure while opening lines or vessels
- Hydrostatic pressure within tall vessels or vertical piping
- Sudden, unexpected liquid release caused by:
 - System cooling and causing vacuums, releasing liquids that are in pockets.
 - Plugs (particularly solidified process materials or hydrates) moving.

! IMPORTANT: All systems are considered as having the potential to discharge hazardous liquid from the open ends of lines or broken flanges at any time, even after the system is drained and depressurized.

- Equipment becoming over-pressured due to thermal expansion of trapped liquids
- Chemical reactions between cleaning materials and a vessel or its contents
- Exposure to gaskets containing asbestos or other fibrous materials
- Loss of pressure containment while pressure testing
- Use of off-spec parts, materials or equipment
- Incorrectly reinstating control devices, including distributed control systems (DCS)
- Equipment overpressure or under pressure when isolating and reinstating pressure protection devices

7.2.1.3 CONSEQUENCES

Potential consequences of incorrectly isolating or reinstating facility systems are:

- Loss of containment
- Process fluid or gas release to the atmosphere, which can be under pressure and cause:
 - Injury to personnel
 - Damage to the environment
 - Damage to the facility

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- Fire or explosion
- Electrocution
- Personnel exposure to hydrocarbon/process when entering vessels that are not successfully hydrocarbon-freed or cleaned

7.2.1.4 RISK MITIGATION STEPS

Potential risk mitigation steps for isolating or reinstating equipment are:

- Compliance with the requirements of this Work Management System
- Job-specific written procedures for isolating and reinstating activities
- Ensuring that personnel have the required competencies
- Follow the risk management process if needed

Reference: *Safety Management Manual*, "Risk Management" and the *Risk Assessment and Management Manual (RAM)*

- Compliance with QA/QC and positive material identification (PMI) processes to assure use of appropriately specified parts, materials and equipment
- Appropriate personal protective equipment (PPE)
- Standby personnel
- Restricted site access

7.2.2 SAFETY PRECAUTIONS

7.2.2.1 INTRODUCTION

This section describes precautions to be taken during equipment isolating and reinstatement activities.

7.2.2.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

Although every effort is made to depressurize and hydrocarbon-free the system before breaking containment, additional PPE may be required until the effectiveness of the equipment preparation can be confirmed. In determining the additional PPE required, workers should consider the hazards that could be present (for example, fluid, gas, pressure, temperature, toxicity, and flammability).

Additional PPE will address the potential hazards and can include goggles, face shield, respirators, or breathing apparatus.

7.2.2.3 BARRICADES AND BARRIERS

The work permit and/or JSA (if required) should specify any barriers that will be erected before performing work and the activities that might present hazards to personnel in the area. The entire work area, as well as adjacent areas that could potentially be exposed, should be barricaded. The impact area due to potential residual inventory, pressure, or energy of the system and activities to be conducted is to be considered when placing barricades.

[E] All personnel not directly involved with these activities are to remain outside the barricades. [E] If they are required to be in the barricaded area, they **must** be protected by the same level of PPE as the people performing the work.

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7.2.2.4 EMERGENCY RESPONSE

Before commencing work activities, personnel should familiarize themselves with escape routes and the location of safety equipment, such as call points, telephones, chemical showers, eye-wash stations, and other emergency equipment. Access to safety equipment and egress/escape routes shall be maintained at all times or alternate equipment and escape routes will be provided.

7.2.2.5 STANDBY PERSON

During certain facility isolation and reinstatement activities (such as breaking containment, blinding, hydrocarbon freeing and leak testing), consideration should be made for designating a Standby Person.

! IMPORTANT: The main duty of the Standby Person is to monitor the activities, stop work if an unsafe condition develops, and alert the Area Authority (AA) of any incidents so that appropriate emergency teams can respond. [E] The Standby Person **must not** enter the area if there is any potential for injury.

7.3 TEMPORARY DEFEAT OF CRITICAL DEVICES

7.3.1 INTRODUCTION

This section describes the precautions used to maintain safety for people, equipment, and the environment when critical devices are temporarily defeated.

Temporary defeats are an integral part of Permit to Work. They control the removal and reinstatement of the functionality of *critical protective devices*.

[CriticalE] Temporary defeats **must** be put in place as required by the scope of the work, and **must** be identified, approved and implemented in cooperation with the relevant people e.g. Area Operator, Control Room Operator and Area Authority.

7.3.1.1 OVERVIEW

The process for temporarily defeating critical devices exist to allow the required work to be performed while maintaining acceptable levels of safety on the facilities.

The requirements for defeating critical devices should be identified during the Job Planning phase.

Reference: [Section 5.1.1: "Work Planning"](#)

7.3.1.2 DEFINITIONS

Temporary defeat: Occurs when a critical device or a critical system is bypassed, inhibited, impaired, over-ridden or taken out of service while the equipment or facility being protected remains in service.

Critical devices: Individual components, sections, or the whole of a critical system. Examples of critical devices are:

- A lifeboat, fast rescue craft, or escape craft
- A component or section of the fire main, fire door, fire wall penetrations
- A fire, gas, heat, or smoke detector
- Vibration monitors
- An emergency shutdown (ESD) activation point

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- Public address (PA) systems (if part of a facility emergency response system)
- Critical Operating Parameters (COP) related alarms
- Critical alarms e.g. H₂S, LEL, fire, or abandon facility lights and horns
- Well surface and subsurface valves and controls
- Limit switches (for example, crane limits or drill rig crown saver)
- Pressure safety valves (PSVs) and rupture disks where no redundancy exists
- Critical process safety devices such as:
 - Level shutdown high-high (LSHH)
 - Level shutdown low-low (LSLL)
- Isolation valves on control or safety systems
- Car Seal Valves (e.g. PSVs, BDVs, Thermal)
- Devices designated as Critical Safeguards on one of the Site's highest consequence scenario(s).

References:

- Local Facilities Integrity Management System (FIMS) critical device list to ensure understanding of devices that could require a temporary defeat. Defeats include "A" Critical equipment per FIMS (SSHE Consequence I & II).
- Alarm Management and Critical Operating Parameter, Upstream Best Practice for the expectations regarding the temporary suppression (defeat) of noncritical alarms.
- Scenario Management and Bow-Tie information related to the Site's highest consequence scenario(s).

Critical systems: Systems required to protect personnel and equipment during normal and emergency situations. The following are examples of critical systems:

- Emergency shutdown (ESD) systems (such as sensors, valves, and controllers)
- Critical programmable logic controller (PLC) or DCS shutdowns per FIMS
- Fire and gas detection system or annunciator panels
- Well blowout preventers (BOPs) and their controls
- Well kick detection systems (such as pit alarms)
- Evacuation system
- Fire-fighting systems
- Public address systems (when part of the facility emergency response system)
- Communications systems (such as radios, telephones, and voice-powered phones)

NOTE: Radio silence requires the use of a Temporary Defeat Certificate.

- Transportation aids (such as navigation lights and beacons, helideck and runway lights)
- Emergency power or uninterruptible power supply (UPS) on safety systems

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7.3.1.3 PURPOSE OF TEMPORARY DEFEATS

The purpose of the temporary defeat process and certificate is to control the defeating and reinstatement of critical devices on equipment that remain in service, and to confirm that the following activities occur:

- Approvals are obtained prior to critical devices being isolated.
- Affected personnel are aware of the defeats.
- Only the required defeats are allowed.
- No other work is underway that requires the critical devices or systems to be operational (for example, fire-fighting systems cannot be isolated during hot work in the area).
- A THA specific to temporary defeats is performed.
- Alternative measures are put in place, when required, while critical devices or systems are defeated.
- Critical devices or systems are reinstated as soon as possible after the work.
- Personnel are prevented from starting machinery or equipment that could cause injury because critical devices are isolated.
- Temporary defeats of critical devices are documented to ensure that they are returned to service and not overlooked.

7.3.1.4 TEMPORARY DEFEAT PLANNING

When planning the temporary defeat of a critical device or system, the following should be considered:

- Critical protective devices that will be defeated.
- Existing TDs in the same area or on the facility.
- Equipment or systems that will be unprotected.
- Method by which the critical device will be defeated.
- Expected duration of the defeat(s).
- Completing the Temporary Defeat THA to identify the hazards and establishing alternative (mitigative or preventative) controls.
- Evaluation as to whether a situation could be created that results in the need for manual intervention.
- Level of approval required.

! WARNING: [CriticalE] If an adequate level of protection cannot be provided and maintained, then the work requiring the temporary defeat(s) will be deferred or the protected equipment will be shut down for the duration of the defeat.

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7.3.1.5 ALTERNATIVE CONTROLS

Alternative (mitigative or preventative) controls that may be used during temporary defeating of critical devices include:

- Continuous surveillance by a competent person, such as:
 - Stationing personnel to overlook areas where fire, heat, or smoke detectors have been overridden.
 - Stationing personnel at hose reels or portable fire monitors while deluge branches are out of service.
- Closely monitoring process trends.
- Lowering throughput to lower the likelihood of upset conditions.
- Reliance on detectors in close proximity or provision of alternate portable detection devices.
- Implementing an interim directive as the result of creating a critical operating parameter (COP).

NOTE: Although alternative controls may be in place, every attempt should be made to expedite the work requiring the temporary defeat.

7.3.1.6 AWARENESS

[CriticalE] Temporary defeats of critical devices, systems or critical safeguards **must** be communicated to affected personnel and included in shift handover. Emergency Response Team(s) should be advised of temporary defeats of these systems, when required.

7.3.1.7 TROUBLE-SHOOTING AND FUNCTION TESTING

[E] When troubleshooting or function testing, certain defeats (such as hardwired bridges, disconnections and software defeats) are not required to be itemized on the TD certificate, however the TD must cover the defeat of the system being worked on.

The worker performing the troubleshooting or function testing should not leave the job while any defeats are in place unless the protected equipment is shut down or alternative controls are in place.

7.3.1.8 TEMPORARY DEFEAT APPROVAL

[E] Temporary defeats of critical devices systems or critical safeguards **must** be approved by the Area Authority, and higher management approval will be sought, if necessary, due to the magnitude or duration of the proposed defeat.

[E] When a system is to be defeated, a higher level of approval is required, depending on the risk and mitigation implemented, consistent with the risk matrix. Refer to [Table 9: \[E\] Approval Level for Temporary Defeats](#) and

[Table 10: \[E\] Approval Level for Temporary Defeat Duration](#).

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NOTE: Sites with extra management layer, Department Head (DH, Ops TLS) and Operations Manager (Ops 4th Line), Operating Sites have flexibility to assign approval authority to persons deemed competent to fulfill the role and are capable of executing their responsibilities outlined in WMS.

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Table 9: [E] Approval Level for Temporary Defeats

Approval Level	Temporary Defeat
Area Authority (AA)	Individual device(s), such as gas heads
Person In Charge (PIC)	Section of plant or platform
Operations Manager	Large section of plant (train) or the whole platform
Site Manager	Plant or several platforms

Table 10: [E] Approval Level for Temporary Defeat Duration

Position	Temporary Defeat Duration
Area Authority (AA)	End of shift (up to 12 hours)
Person In Charge (PIC)	12 hours to 7 days
Operations Manager	7 to 30 days
Site Manager	> 30 days and every subsequent 30 days

NOTE: A temporary defeat extending beyond a shift leading to a critical safeguard associated with a Higher Consequence scenario not meeting performance criteria requires additional notifications per the Risk Assessment and Management Manual Section Manage Critical Safeguards.

Table 11: [E] Approval Level for Critical Safeguards

Position	CSG Defeat Duration
Person In Charge (PIC)	<24 hours
Operations Manager	24 hours to 7 days
Site Manager	Greater than 7 days and every 30 days after

NOTE: [E] Temporary defeats of Critical Safeguards **must** be approved by the PIC, and higher management approval will be sought, if necessary, due to the magnitude or duration of the proposed defeat per the Risk Assessment and Management Manual section on Critical Safeguards.

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7.3.1.9 DEFEAT EXTENSION

Temporary defeats that require extension will be approved by appropriate levels of management using the table above.

Temporary defeats extending beyond 60 days should be risk assessed if necessary and the repair/resolution plan captured. [E] The status of temporary defeats >30 days **must** be communicated to the Site Manager or delegate. [E] Aging defeats **must** be stewarded at PM level every subsequent 30 days.

If a permanent change is to be introduced, then the MOC process will be followed and the MOC number will be recorded on the Temporary Defeat.

[E] The Temporary Defeat will remain in place until the critical device or critical safeguard is repaired, replaced, taken "out of service," or deemed noncritical by the MOC process and closed out.

Temporary Defeats resulting in the creation of a Critical Operating Parameter (COP) that extend beyond 60 days must follow the MOC and risk assessment requirements as outlined in the Alarm Management and COP Management Upstream Best Practice. Some examples of temporary defeats that may result in the creation of a COP include but are not limited to:

- Temporarily defeating a hydrocarbon tank level high-high shutdown, potentially resulting in overfill.
- Temporarily defeating a pipeline pressure high-high shutdown, potentially resulting in rupture.
- Temporarily defeating a reciprocating compressor suction drum level high-high shutdown, potentially resulting in liquid carry-over and loss of containment.

7.3.1.10 SHIFT OR CREW CHANGE

[CriticalE] Active temporary defeats of critical devices, systems or critical safeguards must be a part of the written and verbal handover between shifts or crews.

Reference: Section 6.1.8.2: "Shift or crew interfaces and handover"

7.3.2 TEMPORARY DEFEAT CERTIFICATE

7.3.2.1 TEMPORARY DEFEAT CERTIFICATE

[CriticalE] A Temporary Defeat Certificate **must** be used for isolation of critical devices or systems protecting equipment that will remain in service, or the degradation or defeat of critical safeguards.

NOTE: A Temporary Defeat Certificate is not required for facility Start-Up for devices, equipment, or systems if specified in an approved Procedures. In these cases, a Temporary Defeat Certificate shall be processed if they are to remain in place.

Protective devices isolated on out-of-service equipment do not require a Temporary Defeat Certificate as their functionality is not required. Any protective devices isolated will be documented on the Isolation Control Certificate and reinstated as part of the re-commissioning.

Reference: [GOTO/WMSFORMS](#), "ICC & TD," Temporary Defeat Certificate

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7.3.2.2 TEMPORARY DEFEAT RECORD

[CriticalE] All temporary defeats of critical devices, systems or critical safeguards **must** be recorded and stewarded within the electronic system (ePTWS), board, or log.

7.3.2.3 CLOSING TEMPORARY DEFEATS

Temporary defeats should be closed as soon as possible after the work requiring the defeats is complete. This includes:

- Removing the isolation and restoring the functionality of the device(s). If the defeat was implemented because the device or system was faulty, the device or system will be tested to confirm its functionality following repair or replacement.
- Closing the Temporary Defeat Certificate.
- Removing the temporary defeat from the ongoing stewardship tool. (Electronic system, board, or log)

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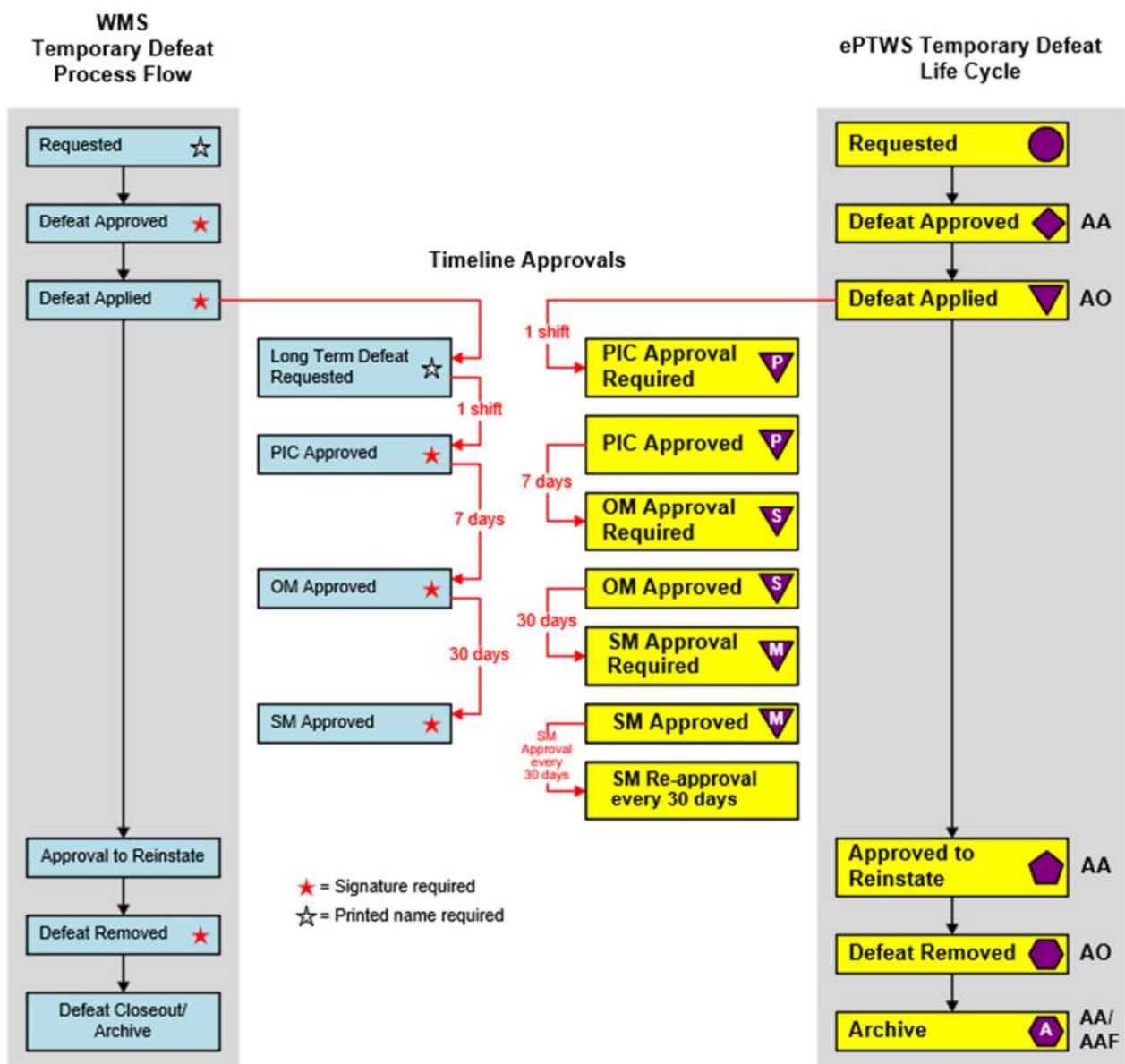
7.3.3 TEMPORARY DEFEAT PROCESS

7.3.3.1 TEMPORARY DEFEAT PROCESS

The following table details each of the Temporary Defeat process flow phases, typical steps required at each phase, reference to the applicable sections of the permit(s), and the responsible person(s) required to execute the requirements at the specific phase. Refer to [Figure 11: Temporary Defeat Process Flow](#) and [Table 12: Temporary Defeat Process](#)

Figure 11: Temporary Defeat Process Flow

Temporary Defeat Process Flow



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Table 12: Temporary Defeat Process

Task	Activity
1.	Plan and prepare a Temporary Defeat (TD) by completing a Temporary Defeat certificate. Review the local FIMS "Critical Devices" list to identify devices that may require a temporary defeat. Defeats are typically related to "A" Critical equipment per FIMS (SSHE Consequence 1 and 2). For complex Temporary Defeats, review the appropriate drawings or diagrams to identify the required device to be defeated, inhibited, overridden, and/or bypassed. A copy of these marked-up diagrams should be attached to the Temporary Defeat. When multiple devices are required to be defeated (for example, fire and gas preventive maintenance [PMs]), the AA can approve devices to be sequentially defeated.
2.	The Requestor selects the TD category (Electrical, Mechanical, Safety, or Instrument/Process).
3.	List the devices to be defeated, and the reason why the defeat is required.
4.	If necessary (e.g. complexity, types of device and magnitude), consider performing a Risk Assessment of the TD plan and provide details for approval.
5.	The Requestor should indicate whether defeating the device will result in a critical operating parameter (COP). If yes, provide details in the TD and have the Site Manager or equivalent approve.
6.	If there is an approved device bypass procedure, attach the procedure to the Temporary Defeat.
7.	Complete a Temporary Defeat THA to identify the hazards and establish alternative (mitigative or preventative) controls. Identify specific hazards and define specific mitigation controls to be implemented during the defeat.
8.	A temporary defeat can be cross-referenced to a permit. The permit can be "Issued" without the TD "In Place." Prerequisite controls should be used when a TD must be in place before the job commences (for example, to avoid a shutdown). The defeat can be identified as a prerequisite control on the temporary defeat .
9.	[E] Temporary defeat of critical devices or systems must be approved by the Area Authority up to the end of the shift. Higher management approval will be sought by the Area Authority, if necessary, due to the magnitude or duration of the proposed defeat. [CriticalE] The approver must indicate whether defeating the device will result in a critical operating parameter (COP); if yes, provide details in the TD and have the Site Manager (or equivalent) approve. The approver should consult technical/engineering resources if needed to support the determination of a COP.
10.	Temporary defeats should be returned to service as soon as possible after the work requiring the defeats is complete. The Area Authority is required to approve reinstatement prior to the Area Operator removing the defeat.

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7.4 ENERGY ISOLATION METHODS AND SYSTEMS

PURPOSE

The purpose of the energy isolation process is to enable work to proceed safely by:

- Ensuring correct isolation of all energy sources and removal of all hazardous substances or conditions when work is to be performed on equipment and facilities.
- Controlling the de-isolation process to ensure that equipment and facilities are not re-energized while work is ongoing.
- Controlling the reinstatement of equipment and facilities to ensure that all protective devices are functional and the integrity of the system (for example, flanges, vents, drains, and sample points) is restored.

This control is maintained by the use of Isolation Control Certificates (ICCs) and through Permit to Work (PTW) controls.

Reference: [GOTO/WMSEFORMS](#), "ICC & TD," Isolation Control Certificate.

7.4.1 ISOLATION METHODS

7.4.1.1 PROCESS ISOLATION TECHNIQUES

The isolation and removal of all potential sources of pressurized fluids and gases from a vessel or section of piping will be achieved by:

- Closing and securing a series of valves.
- Opening vent lines, bleed points, or drains between the isolated valves in a safe and environmentally sound manner to drain and depressurize the equipment.

7.4.1.2 BLINDING, SPADING OR DISCONNECTION

Blinds, spades, plates, clamped flanges, or other terms describe the devices used to positively segregate the energy, substances, and materials from the proposed activity. [CriticalE] These devices **must** be rated for the design pressure of the system if they are part of the isolations and could be exposed to system pressure.

[CriticalE] The Isolating Authority (IA) **must** visually verify that all blinds or spades have been installed.

Disconnection also includes electrical equipment whereby the power delivery pathway is interrupted by an air gap that is created by a removal of the power or signal conducting cables.

[CriticalE] The Isolating Authority (IA) **must** visually verify that an air gap has been created.

7.4.1.3 APPROVED STANDARD ISOLATION METHODS

The following are approved isolation methods:

- Blinding, spading or disconnection
- Double Isolation and Bleed (DIB)
- Double Isolation and Bleed Valve (DIBV)
- Single Valve Isolations (SVI) with additional mitigations

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NOTE: [CriticalE] Non-return valves, modulating control valves, and other valves that do not provide tight shutoff **must** not be used as isolation points. (e.g. concentric (non-offset) butterfly valves, flow control valves, pressure control valves, backflow / check valves).

High performance, lug style butterfly valves such as triple offset valves, as defined as GP 03-12-02, may be suitable for certain isolations. Consideration for use must include unidirectional sealing nature of the valves and field labeling. Field labeling should differentiate from typical butterfly valves and note directionality limitations. Where these specific cases have been approved for use, it shall be documented in the Site's local [Section 12.0 Appendix A](#).

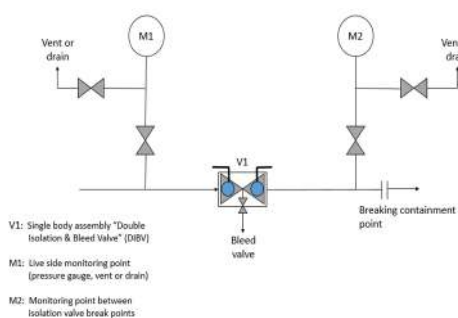
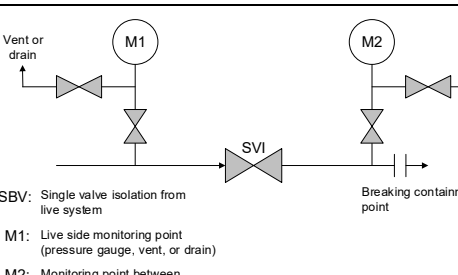
For more information on isolation valve types, refer to Upstream GPs 03-01-01U & 03-12-02U

Table 13: Isolation Method Standards

Isolation Method Standards			
Isolation Type	Definition	Illustrations	
Blinding Spading or Disconnection	Installation of a physical barrier (blind or spade) to separate energy from an activity Or Physical disconnect	F1: Blind/Spade installed M2: Monitoring point between isolation valve break points F2: Physical disconnection / Air gapped	
Double Isolation and Bleed	Two independent block valves in series with a bleed valve between them (i.e. 2 independent gate, globe, ball, plug, or needle valves)	V1: First upstream isolation from line system M1: Live side monitoring point (pressure gauge, vent or drain) V2: Second downstream isolation valve from line system M2: Monitoring point between valve break points	

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Table 13: Isolation Method Standards, Continued

Isolation Method Standards		
Isolation Type	Definition	Illustrations
Double Isolation and Bleed Valves	A single valve assembly with: 2 independent obturators in a single common valve body with a bleed between or Expanding gate or expanding plug valve with a single obturator with two mechanical seats that do not require pressure to seal (Refer to Type C Valve in GP 03-12-02U & 03-01-01U for use limitations)	 V1: Single body assembly "Double Isolation & Bleed Valve" (DIBV) M1: Live side monitoring point (pressure gauge, vent or drain) M2: Monitoring point between isolation valve break points
Single Valve Isolation	A Single Valve Isolation (SVI) consists of the closure of a single isolation valve.	 SBV: Single valve isolation from live system M1: Live side monitoring point (pressure gauge, vent, or drain) M2: Monitoring point between valves and break point

7.4.1.4 MINIMUM ISOLATION REQUIREMENT STANDARD

[CriticalE] The minimum isolation requirements (Table 14: Minimum Isolation Standard Table) **must** be used for isolation planning and provides guidance for establishing safe conditions to perform work on equipment in an isolated state. The guidance table takes into consideration product or process systems, temperature and pressure ratings.

! IMPORTANT: [CriticalE] Confined Space Entry requires blinding, spading or physical disconnection; if isolation standard cannot be met then a risk assessment **must** be conducted and approved by the appropriate management based on the residual risk, and requires a deviation.

[E] Condition A: If isolation standard cannot be met, and single valve isolation is planned then the isolation certificate **must** be countersigned by the PIC prior to endorsement of permit. The Operations AA or competent delegate (independent verifier to the AO) **must** be at the site to witness the demonstration of zero energy.

[E] Condition B: If isolation standard cannot be met, and single valve isolation is planned then the isolation certificate **must** be countersigned by the PIC prior to endorsement of permit. The Operations

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AO **must** be at the site to witness the demonstration of zero energy, it cannot be delegated to the permit holder.

NOTE: The intent of the Operations AA acting as a verifier of the zero-energy demonstration is to confirm that zero energy has been achieved (no false indicators: i.e. plugged valve, incorrect bleed/isolation point, etc.) and to confirm that it is safe for work activities to commence. The PIC signature indicates support of the isolation plan, need to execute work under a single valve isolation and the plan to mitigate potential risk in conducting associated activities using a single valve isolation, including the zero-energy demonstration method.

Table 14: Minimum Isolation Standard Table

Fluid	Flange Class	Min isolation standards	Conditions if Double Isolation Standard is Not Met
Flammable (e.g. crude oil, methanol, diluent, dilbit, emulsion, solvent and diluted froth) Includes combustibles above flash point or <8°C/15°F from flash point	≥ 300	Double isolation & bleed, or Blind and/or physical disconnect	Condition A
	< 300	Double isolation & bleed, or Blind and/or physical disconnect	Condition B
Toxic or harmful (e.g. H ₂ S, acids, caustics and amine)	All	Double isolation & bleed, or Blind and/or physical disconnect	Condition A
Non-flammable and operating at > 66°C /150°F (e.g. steam, boiler feed water)	≥ 300	Double isolation & bleed, or Blind and/or physical disconnect	Condition B
	< 300	Single isolation	N/A
Non-flammable and operating at ≤ 66°C/150°F (e.g. lube oil, water, glycol, air)	≥ 900	Double isolation & bleed, or Blind and/or physical disconnect	Condition B
	< 900	Single isolation	N/A

7.4.1.5 ACTUATED VALVES

[CriticalE] The following additional barriers are required for actuated valves such as emergency shutdown (ESD) valves, remotely operated valves (ROVs), or motor operated valves (MOVs):

- The power (hydraulic, pneumatic, or electric) supply **must** be isolated.
- Control fluid accumulators **must** be depressurized or disconnected when not required to maintain the isolated state of the valve.
- The control cabinet or power supply **must** be secured and locked.

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- Control system isolation points **must** be appropriately tagged.
- Where possible, mechanical restrictions (lock-out caps) should be used on valves so equipped.
 - ▲ **CAUTION:** Consider blocking in or disconnecting exhausts to the flare from gas-powered actuators so that they cannot be driven open by backflow from the flare header at times of high flaring and pressure buildup in the flare header.

7.4.1.6 SPECIAL ISOLATION TECHNIQUES

In instances where the facility/equipment design prevents an approved standard isolation method from being used, devices or methods such as mechanically expandable plugs, vented plugs, stopples, frozen plugs, pipe freeze or water seals can be used. [CriticalE] Use of special isolation methods **must** be Risk Assessed with required approvals based on risk category.

- ▲ **CAUTION:** There is a risk that these devices or materials (plugs, stopples, or frozen plugs) can be ejected. Personnel must not have any parts of their body in the potential ejection path (line-of-fire).

Vented plugs may be used for vapor or liquid seals in drained and depressurized systems to contain and direct any small amounts of residual vapor or liquid to a safe location. Atmospheric monitoring must be used to continuously monitor for the presence of vapor that could create a hazard.

7.4.1.7 DOUBLE ISOLATION AND BLEED (INDEPENDENT)

Double isolation and bleed (DIB) isolations are established by the closure of two independent block valves in series with an open bleed valve between them to eliminate the potential for pressure buildup between them, causing leakage through the closed valves to the work site.

Emergency shutdown (ESD) valves may be used for a DIB isolation, provided they can be reliably immobilized; for example, disconnecting the driver or power source.

7.4.1.8 DOUBLE ISOLATION AND BLEED VALVE (SINGLE ASSEMBLY)

A Double Isolation and Bleed Valve (DIBV) as defined in the context of this manual is a single assembly valve with 2 independent obturators (i.e. balls, expandable gates, globes, or plugs) in a single common valve body. The valve assembly is constructed of two separate seating surfaces, one per obturator with a cavity bleed. These valves are equipped with two independent handles, one for each obturator, and provide the equivalent protection as double isolation and bleed isolations.

- ▲ **CAUTION:** Some single assembly double isolation valves require line pressure to energize the upstream or downstream seals. These valves are constructed of a single obturator and bleed off cavity. The two seating surfaces in these instances are not considered acceptable for double block and bleed isolations by the company and should be treated as **“Single Valves”**.

- ▲ **CAUTION:** [CriticalE] When venting hydrocarbons, flammable, or toxic substances to atmosphere as part of an energy isolation, all vent points **must** be directed to a safe location. For these purposes a safe location is defined as an area where venting can occur that minimizes exposure to personnel and potential ignition sources. Single valve isolation planning should consider potential system back pressure risk when leaving vent lines open to systems above ambient pressure.

Safe locations include but are not limited to;

- Flare systems

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- Atmospheric vents and drains
- Sumps and tanks
- Outside of buildings, or congested / restricted areas

[CriticalE] A method **must** be in place to monitor vent points for the presence of liquids or vapors that could create a hazardous condition.

[CriticalE] Open-ended Hydrocarbon, Flammable, or Toxic Vents **must** be closed if unattended or not being monitored.

▲ CAUTION: [E] Valve cavity bleeds are not to be attempted if there is only a grease nipple fitting installed into the cavity bleed port without an isolation valve.

7.4.1.9 SINGLE VALVE ISOLATIONS

A Single Valve Isolation (SVI) isolation consists of the closure of a single isolation valve. [CriticalE] Any valve used for SVI isolation **must** provide a reliable seal and valve integrity **must** be confirmed before commencing work.

[CriticalE] If the relevant hazards have not been sufficiently controlled and zero energy not achieved, the work **must** stop, the site made safe, and a risk screening **must** be conducted. (Refer to [Section 7.4.2: Process Isolation Integrity Tests](#)).

NOTE: Single valve isolations are allowed for some lower risk or procedural activities and equipment engineered for specific purposes. Examples of such instances include, installing and/or removal of blinds and spades, isolating instrument components, replacing pressure gauges, opening pig launchers or receivers, and the removal of orifice and turbine meters. Operating Sites should assess and identify lower Risk SVI activities within their assets and document in the [Section 12.0: Appendix A](#) of their WMS Manual.

NOTE: Due to operational design it is not always possible to prove SVI integrity where there is no upstream pressure or there is not a downstream testing point. In these cases, a Risk Assessment will be conducted with required based on the level of risk categorization.

[CriticalE] Single Valve Isolations are not permitted if:

- The single valve isolation is Unable to isolate (UTI) (i.e. inoperable, defective or has integrity concerns).
- Zero energy cannot be demonstrated at a point close to the single valve isolation that is free of possible obstruction or plugging.

! IMPORTANT: [CriticalE] Should these conditions not be met, then a Risk Assessment is required and an acceptable means of isolation must be identified, documented, and approved before Breaking Containment.

7.4.1.10 RISK ASSESSMENT OF NON-COMPLIANT ISOLATIONS

[CriticalE] In the event that isolations compliant with the requirements of the Work Management System cannot be achieved, the risk management process **must** be followed. Consult a Risk Screener to establish the appropriate type of Risk Assessment to be conducted.

NOTE: In these cases, a deviation is required in addition to the risk assessment.

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Examples may include but aren't limited to:

- Isolating storage tanks where only concentric (non-offset) butterfly valves exists
- Isolating firewater or utility water systems where only concentric (non-offset) butterfly valves exist

7.4.2 ISOLATION INTEGRITY

INTRODUCTION

This topic defines the systems for executing and testing isolations.

7.4.2.1 ISOLATION INTEGRITY SYSTEM

All isolations must achieve and maintain an effective containment of the relevant hazards. Blinding, spading or physical disconnection is regarded as the most secure method and must always be considered first when planning work.

The isolation methods described earlier in this section should be referred to when planning process isolations.

This system also identifies the mandatory safeguards that the Isolating Authority (IA) is responsible for implementing to manage risks.

7.4.2.2 LOCK-OUT/TAG-OUT

[CriticalE] For isolations, valves **must** be secured and locked in their required position to prevent unauthorized or inadvertent operation. [CriticalE] Where valves are required to be locked (open or closed) as part of an isolation, only the following methods are acceptable:

- Lock and a securing method (metal/steel chains, metal/steel cable, lockable covers/fittings) that takes a deliberate action to remove.
- Specially designed valve locking systems.
- Actuated valves that fail in the *isolation-safe* position (would not compromise the isolation) **must** have their control lines isolated and either locked in the vented position or physically disconnected, tagged, and recorded on the ICC. Electrically actuated valves **must** have their control circuit electrically secured to ensure isolation.
- Actuated valves that fail in the *isolation-unsafe* position (would compromise the isolation) normally should not be used for isolation purposes. If no alternative is reasonably practicable, the valve **must** be prevented from moving.
- For all isolation points, an isolation tag **must** be provided at each point with the relevant details.

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7.4.2.3 ISOLATION CONTROL POINT (ICP)

Devices used to secure lock keys on equipment in an isolated state. These devices can be stored at the worksite or a central location and must be secured with a lock. Some examples of these are lock bags or lock boxes.

7.4.2.4 LOCK SETS

Lock sets are used to isolate equipment using one or more sets (Refer to [Figure 12: Isolation Locks](#)):

- Lock sets are keyed alike (one key opens all locks in the set). Lock sets can be small or large; for example, 5 or 100 locks for small and large isolations.
- Keys used for isolating the equipment are placed in the Isolation Control Point (ICP) box and secured using the uniquely keyed Functional and/or Personal locks.
- [CriticalE] Each Permit Holder performing work on isolated equipment **must** apply their Functional and/or Personal lock to the ICP before signing acceptance of the permit.

7.4.2.5 FUNCTIONAL LOCKS

Functional locks are applied as follows (Refer to: [Figure 13: Function Locks](#)):

- [E] Functional locks are uniquely keyed and identifiable for each Function (for example, by color, code, or type).
- The Functional Supervisor for each group is responsible for issuing Functional locks. [E] Each lock should have two keys; one key is issued with the lock, and any spare keys will be kept by a functional supervisor in a secure location.
- [E] Provision must be allowed for the application of personal locks by work team members upon request or where required by regulation.

Reference: Multiple Isolation Point Locking Process. Refer to [Section 7.0: Energy Isolation & Temporary Defeat](#)

7.4.2.6 PERSONAL LOCKS

Personal locks are applied as follows:

- [E] A Personal lock and key are controlled by a single individual and have either a unique mark or identification traceable to the owner of the lock (for example, name engraved or label applied).
- [E] A personal lock may only be applied and removed by the same worker.
- Any person working under an isolation may request to use a personal lock.

! WARNING: [CriticalE] If at any time functional or personal locks are removed from the ICP, resulting in the ICP box being left unsecured without a lock; STOP work expectations must be applied. [CriticalE] Before recommencing work, the ICP box must be secured, the ICC must be re-verified and zero energy demonstrated again.

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7.4.2.7 TAGGING

[E] All isolations **must** be accompanied by an Isolation Tag to align with the Isolation Control Certificate.

Tags should identify:

- Isolation points such as valves, breakers, flanges, start/stop switches, instrument disconnects etc.
- Open bleeds, vents, sample valves, or removed plugs that need to be highlighted and recorded on the Isolation Control Certificate so that their reinstatement is not overlooked.
- Blinds, spades, skillets, and other mechanical isolating devices.

The following information should be printed legibly on Isolation Tags.

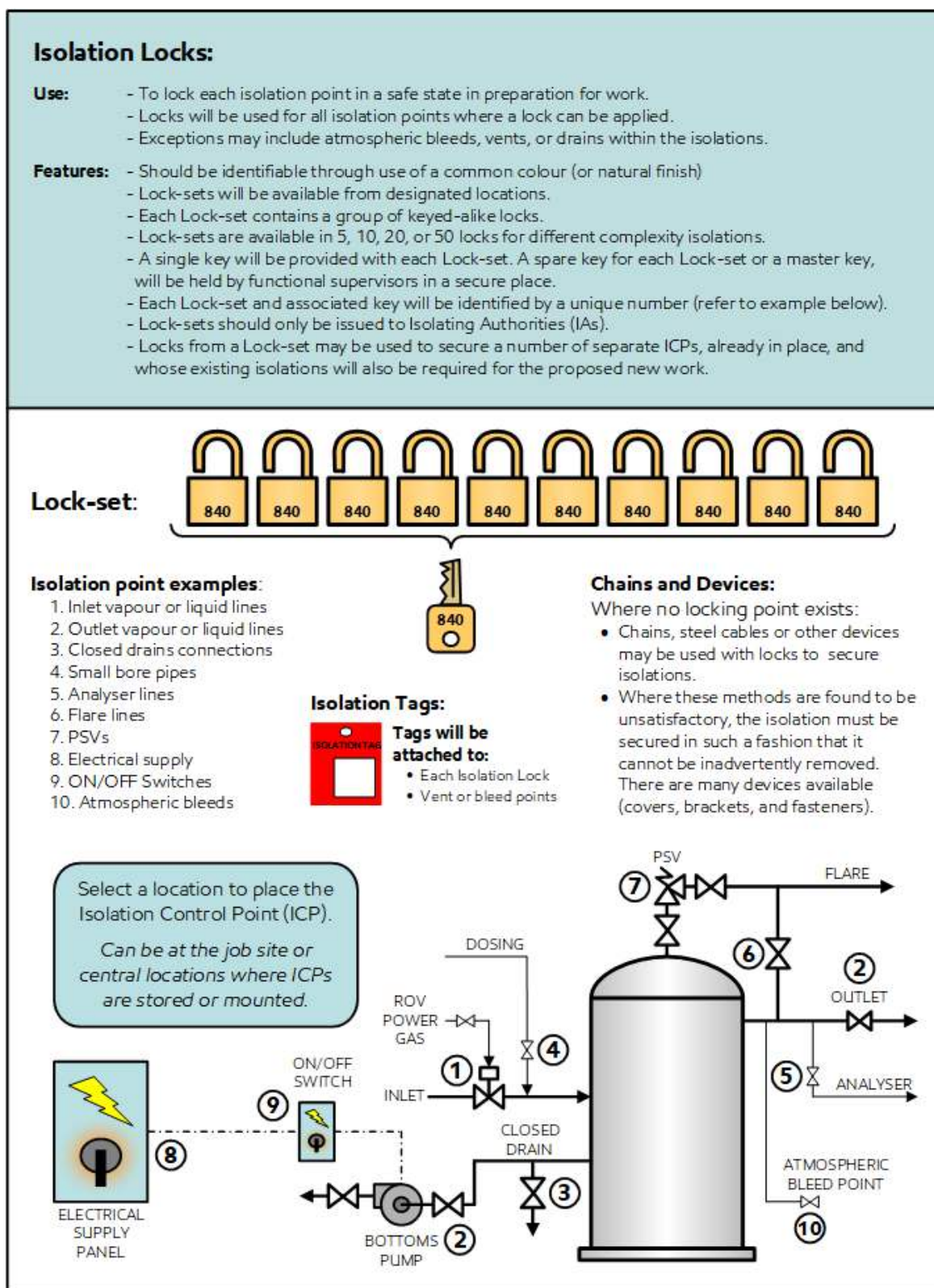
- Isolation point number (for example, from P&ID or electrical diagrams)
- Isolation Control Certificate (ICC) number
- Isolated state
- Work permit number (if applicable)
- Person performing isolation

The Isolation Control Certificate will maintain a list of all locked and/or tagged points to serve as a record and a checklist when reinstating equipment. Tags remain in place until the tagged isolation point is safely restored to its normal operating position. The Isolating Authority who removes the locks should also remove the corresponding tags. [Refer Figure 12: Isolation Locks](#) and [Figure 13: Function Locks](#)

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LOCK-OUT, TAG-OUT PROCESS

Figure 12: Isolation Locks



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Figure 13: Function Locks

Function Locks:

Use:

- To secure the Isolation Control Point (ICP) and ensure workgroup control over the isolations.
- To identify, which workgroup function(s) is/are involved.

Features:

- Will be identifiable and matched to specific workgroup functions, such as Operators, Electricians, Instrument Technicians, Mechanics, and others such as contractors.
- Available individually from designated points of issue.
- Each lock is uniquely keyed and a single key will be issued with each lock. A spare key for each lock or a master key will be held by functional supervisors in a secure place.
- Each lock and associated key will be marked with a unique identification number.

Variation 1 Area Operator – Own Isolations (personal lockouts):

- Operators may use several Operator Function Locks to lock out equipment for their own work and retain the keys in their possession. Alternatively, they can use a keyed-alike lock-set and keep the key in their possession.
- **Other workgroups must NEVER rely on these isolations alone to conduct their own work.**

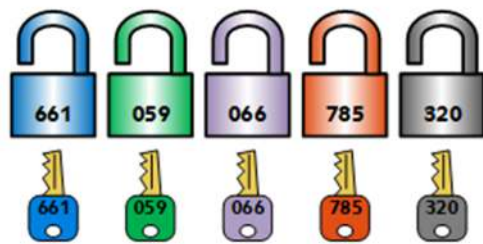
Variation 2 Remote Locations – Own Isolations (personal lockouts):

- Maintenance Technicians or contractors may use several Function Locks to lock out equipment for their own work and retain the keys in their possession. Alternatively, they can use a keyed-alike lock-set and keep the key in their possession.
- **Other workgroups must NEVER rely on these isolations alone to conduct their own work.**

Function Locks:

Colors may be used by the PU but should be distinguishable from the keyed-alike isolation lock-sets. For example:

- Operators (blue)
- Electricians (green)
- Instrument Techs (purple)
- Mechanics (orange)
- Other, such as Contractors (grey)



Isolation Control Point – example:

Isolation Control Bag/Box has a unique number and may be secured:

- To an Isolation Point
- To a nearby structure
- Within a control room
- Other designated location



Identification Tags should be attached to each Function Lock

Use of a tag may be waived only on designated Remote Low Complexity sites where the work is undertaken by the same person who has placed the isolations, or their successor.

Example shows an Isolation Control Point (ICP), used to capture the Isolation Lock-set key(s), that is first secured shut by an Operators Function Lock (blue in the example).

Additional Function Locks securing the ICP indicate that two permitted jobs are relying on these isolations: one "Electrical" workgroup (green) and one "Mechanical" workgroup (orange). Lock colors are selected by the PU, but should be different for the various functions.

Relevant documentation for each job is documented on a tag attached to each Function Lock.



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7.4.2.8 ISOLATION INTEGRITY TESTS

[CriticalE] After the locking devices are installed, the Area Operator **must** test the integrity of the lock-out (for example, attempt to turn the valve handle or close the circuit breaker to prove the lock-out effectively prevents manipulation).

All isolations should be tested and shown to be effective before containment is broken. For more complex isolations, a job specific isolation procedure may be required, specifying the methods by which the isolations will be verified and how zero energy will be demonstrated.

▲ **CAUTION:** A reasonable time period (pipework and valves should reach ambient temperature) should be allowed to confirm the integrity of any isolation before stating that the isolation has been proven.

Valve isolation integrity tests are described in [Section 7.7: "Tools."](#) It is important that all bleeds (vents and drains) are checked and free from obstruction prior to performing any tests.

Reference: [Section 7.7: "Tools."](#) for:

- Double Isolation and Bleed (DIB) Isolation Integrity Test
- Double Isolation and Bleed Valve (DIBV) Isolation Integrity Test
- Single Isolation Valve (SIV) Isolation Integrity Test

After completing integrity testing of isolations, the bleed and vent valves should normally be left in the open position and unlocked to enable periodic monitoring.

▲ **CAUTION:** [CriticalE] When venting hydrocarbons, flammable, or toxic substances to atmosphere as part of an energy isolation, all vent points **must** be directed to a safe location. For these purposes a safe location is defined as an area where venting can occur that minimizes exposure to personnel and potential ignition sources. Single valve isolation planning should consider potential system back pressure risk when leaving vent lines open to systems above ambient pressure.

Safe locations include but are not limited to;

- Flare systems
- Atmospheric vents and drains
- Sumps and tanks
- Outside of buildings, or congested / restricted areas

[E] A method **must** be in place to monitor vent points for the presence of liquids or vapors that could create a hazardous condition.

[CriticalE] Open-ended Hydrocarbon, Flammable, or Toxic Vents **must** be closed if unattended or not being monitored.

If zero energy is not achievable when checking the integrity of valve isolation, attention should be given to:

- Checking all valves for tight shutoff
- Closing additional isolation valves
- Alternative isolation through physical disconnection or blinding

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[CriticalE] If "zero energy" is still not achieved and relevant hazards have not been sufficiently controlled during work preparation or execution, the work **must** stop, the site made safe and a Risk Screening **must** be conducted to determine whether it is acceptable to proceed. The Risk Screening should consider:

- The risk of proceeding with the work activity
- Effective mitigations and alternate methods.
- The nature of the fluids and gases
- The pressure and inventory of the process fluids and gases.
- The anticipated length of time the isolation will be required.
- The job to be performed
- Potential mitigation measures, including:
 - Acceptability of any leak rates
 - Standby Person to monitor the isolation and take appropriate action if the leak rate becomes unacceptable.
 - Bunding or containment
 - Toxicity, impact on personnel, and required PPE.
 - The use of blinds or plugs.
 - Deferral of the work until the work can be done safely (e.g. shut down).

! WARNING: There may be certain instances where demonstrating zero energy to the Permit Holder is impractical (for example, diving operations or remote field isolations). Verbal communication can be used in these cases, providing that PIC approval has been obtained.

! WARNING: [CriticalE] When using blinds or plugs to contain fluids or vapors from leaking valves, the provision for depressurizing and draining before removal of the blind or plug must be included. Note that the exposure time to install and remove blinds can, at times, present a higher risk than conducting a short duration job with mitigations in place.

7.4.2.9 ISOLATION INTEGRITY RETESTING

[CriticalE] A retesting and reconfirmation of energy isolations by demonstrating zero energy **must** be performed at the following times:

- At each shift change before signing the permit.
- At the request of the work party.
- At the discretion of the Area Authority (AA) and Area Operator (AO).
- After any temporary discontinuation of energy isolation for testing (STT).

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7.4.3 VALVE SECURITY

INTRODUCTION

The correct position of valves in the process system is critical to efficient and safe operation of the facility. In many cases, the correct position of fully open or fully closed valves is also critical to the integrity of the facility. Steps must be taken to help confirm that critical valves are not inadvertently moved from their normal operating state without the proper administrative controls.

7.4.3.1 VALVE SECURITY LEVELS

The appropriate position for fully open and fully closed valves is normally shown on the P&IDs for the facility. They are usually one of the following: Refer to [Table 15: Valve Security Levels](#)

Table 15: Valve Security Levels

Valve Position	Security and Allowed Operator Actions
Normally open (NO)	The valve is unsecured and the Area Operator or designate may temporarily close the valve as part of their normal operating duties. The valve should be reopened as soon as possible.
Normally closed (NC)	The valve is unsecured and the Area Operator or designate may temporarily open the valve as part of their normal operating duties. The valve should be reclosed as soon as possible.
Car sealed open (CSO)	The valve is secured with a cable tie to prevent inadvertent operation. The Area Operator or designate may close the valve when an Isolation Control Certificate (ICC) is in the correct state. The valve should be locked when not in its normal position. The valve may be reopened during de-isolation.
Car sealed closed (CSC)	The valve is secured with a cable tie to prevent inadvertent operation. The Area Operator or designate may open the valve when an ICC is in the correct state. The valve should be locked when not in its normal position. The valve may be reclosed during de-isolation.
Locked open (LO)	The valve is secured with a metal chain or metallic cable tie and lock to prevent unauthorized operation. The Area Operator or designate may close the valve when an ICC is in the correct state. The valve should be locked when not in its normal position. The valve may be reopened during de-isolation.
Locked closed (LC)	The valve is secured with a metal chain or metallic cable tie and lock to prevent unauthorized operation. The Area Operator or designate may open the valve when an ICC is in the correct state. The valve should be locked when not in its normal position. The valve may be reclosed during de-isolation.

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NOTE: Valves are car-sealed and tagged to identify their correct status (open or closed) when their inadvertent operation could result in a higher risk of a safety, health, or environmental incident. A Structured Risk Assessment (such as a Hazard and Operability Review [HAZOP]) usually identifies the need for a valve to be car-sealed in a certain position due to potential line rating or relief issues if the valve was moved to a different status.

7.4.3.2 DOCUMENTATION

Critical valves should be marked on facility P&IDs. [CriticalE] A Car Seal Register **must** be kept of all CSO/CSC and LO/LC valves, which states the reason each valve is so designated. All CSO/CSC and LO/LC valves **must** have a field tag that indicates the normal valve position and gives a unique identification (tag) number that can be referenced to the Car Seal Register.

7.4.3.3 PROCESS

[CriticalE] When a CSO/CSC or LO/LC valve is moved from its normal position:

- The Car Seal Register **must** be referenced to determine whether alternative controls (identified from HAZOP) must be implemented prior to moving the valve from its normal position.
- This action **must** be authorized by an Area Authority.
- If the valve is part of an isolation, it **must** be secured, locked, and tagged.
- This action should be managed with a Temporary Defeat certificate, if it is not part of an isolation.

[CriticalE] When a CSO/CSC or LO/LC valve is returned to its normal position:

- This action **must** be authorized by an Area Authority and the valve **must** be locked (for LO/LC valves) or car sealed (for CSO/CSC valves).
- Close out any Temporary Defeat certificates.

NOTE: If a Car Seal or Locked valve is manipulated as part of a Critical Procedure and returned to its normal state at the completion of the procedure there is no requirement to create an isolation certificate or temporary defeat. Any removal of a car seal **must** be captured on the car seal register.

7.4.3.4 CHANGE MANAGEMENT

[CriticalE] A permanent change to a critical valve's normal position or security level **must** be approved through the Management of Change (MOC) process. This will initiate an appropriate level of technical and risk review, and will confirm that the change, once approved, is documented in the following:

- P&IDs
- Site Valve Register (if applicable)

7.4.3.5 EFFECTIVENESS CONFIRMATION

[E] All CSO/CSC and LO/LC valves must be field-checked against the latest P&IDs as per operating site car seal program

- The valve is in the correct position.
- CSO/CSC valves are secured with cable-ties.
- LO/LC valves are secured with locks.
- Identification tags are securely in place and are legible.

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7.5 ISOLATING SYSTEMS AND EQUIPMENT

PURPOSE

This section provides guidelines to protect personnel and equipment when isolating equipment or systems for work. The responsibilities and protective measures to be taken when isolating and working on systems are also described.

SCOPE

Isolation activities include:

- Energy isolations (process, mechanical, and electrical)
- Draining and depressurizing
- Hydrocarbon freeing
- Breaking containment
- Blinding
- Cleaning

Each job must be evaluated for complexity and risks to determine isolation requirements and whether a job-specific isolation plan is required. This plan should include, as a minimum:

- Detailed activity instructions
- The order in which activities are to be performed
- Any special precautions that should be taken during each step

7.5.1 ENERGY ISOLATION AND ISOLATION CONTROL CERTIFICATE

7.5.1.1 INTRODUCTION

Energy isolations are intended to remove all sources of energy from equipment to enable them to be safely removed from service, and keep them out of service for the duration of the work.

Energy isolations are managed by an Isolation Control Certificate (ICC).

Reference: [GOTO/WMSEFORMS](#), "ICC & TD," Isolation Control Certificate

7.5.1.2 ENERGY ISOLATION PRACTICE

The company's energy isolation practice is to achieve "zero energy" and demonstrate zero energy to the Permit Holder. This ensures that work on facilities and equipment does not cause any unexpected or uncontrolled release of energy that can cause injury to personnel or damage to the facilities or the environment.

The goal of energy isolation is to allow the work to proceed safely by preventing:

- #### 7.5.1.4 ISOLATION FLOWCHART

Figure 14: Isolation Flow Chart

The diagram compares two isolation processes: WMS Isolation Process Flow and ePTWS ICC Life Cycle.

WMS Isolation Process Flow:

- Establish isolation requirements
- Isolation plan preparation (linked to: Isolation points, Isolation type)
- Isolation plan verification ★
- Isolation(s) installed ★ (linked to: IA installs isolation)
- Isolation(s) verification ★ (linked to: Second person verifies/ confirms isolation)
- Work (indicated by a dashed blue line)
- Isolation removed ★
- De-isolation Integrity Verified ★ (linked to: AO signs de-isolation verification)
- IC Closed ★

ePTWS ICC Life Cycle:

- Requested Incomplete
- Requested (linked to: Isolation points, Isolation type)
- Verified (AA)
- Isolation in Progress (AA/AO)
- Update Point Positions (IA)
- In Place (AO)
- De-isolation in Progress (AA/AO) (linked to: 2nd Verifier Signature)
- Update Point Positions (IA)
- Completed (AO)
- Archived (AA/AAF)

Integration:

- A red box labeled "PTW must be job complete or job incomplete (integrity restored)" is connected to the "Work" step in the WMS flow and the "De-isolation in Progress" step in the ePTWS cycle.
- A blue box labeled "De-isolation Verifier Signature" is connected to the "De-isolation in Progress" step in the ePTWS cycle.

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7.5.1.5 GENERAL ENERGY ISOLATION PROCESS

Use the following process for preparing and executing energy isolations to confirm that appropriate steps have been taken to isolate energy sources. In addition, a job-specific isolation plan by which mechanical isolation is undertaken may be required, depending on the complexity of the job. Refer to

Table 16: Energy Isolation Process.

Table 16: Energy Isolation Process

Task	Action
1.	Plan and prepare the isolation by completing an Isolation Control Certificate (ICC). An ICC is not required for "Personal Isolations" that meet the criteria described later in this Section 7.0: Energy Isolation and Temporary Defeat .
2.	For complex electrical isolations, review the appropriate electrical diagrams to identify the isolation, grounding, and testing points. A copy of these marked-up electrical diagrams should be included in the Permit Pack.
3.	For complex process isolations or when lines are not visibly traceable in the field, review the Piping and Instrumentation Drawings (P&IDs) and mark up the energy points that need to be isolated, flushed, or purged. A copy of these marked-up P&IDs should be included with the ICC.
4.	For complex mechanical isolations, review the equipment drawings and identify any moving parts that need to be stopped, wedged, removed, or uncoupled.
5.	If necessary, compare the P&ID for the system to be isolated to the actual system (also known as "walking the lines") to confirm that the P&ID is accurate and to identify: ▪ Low points where liquids may not be drained ▪ Drain, vent, and test points in piping or vessels ▪ Instrumentation that needs to be protected or defeated ▪ Bleed points, vents, gauges, or break methods where zero energy can be confirmed. ▪ Other energy sources. ▪ How zero energy will be demonstrated to the Permit Holder.
6.	Depending on complexity, prepare a job-specific energy isolation plan.
7.	If necessary, perform an engineering review to confirm that the design limits are not exceeded: ▪ If filling vessels and pipework with water, which would not normally be exposed to full water volumes (static head and specific gravity) ▪ When pressure testing the isolations. Confirm that all specification changes are accounted for in the isolation plan.
8.	Depending on complexity, energy sources, and magnitude of the isolations, consider performing a Risk Assessment on the proposed energy isolation plan (consult a Risk Screener).

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Task	Action
9.	[CriticalE] The AA must verify the Isolation Control Certificate and any specific isolation plan.

Table 16: Energy Isolation Process, continued

Task	Action
10.	Perform isolations, including Lock-out, Tag-out.
11.	Update the status of all isolation points on the ICC.
12.	[CriticalE] Verify (using a second independent, competent person) that the isolations are in place and effective. The verifier signs the field copy of the ICC.
13.	[CriticalE] The Area Operator demonstrates isolation integrity (zero energy) to the Permit Holder, who accepts the permit, if satisfied with the level of energy isolation for the proposed work.

7.5.1.6 REMOTE FIELD OPERATIONS AND NON CENTRALIZED REMOTE FIELD

[CriticalE] For Remote Field Operations, when an Area Operator is not present, the isolations shall be implemented by a competent Permit Holder, and updated and signed on the hardcopy confirming that the isolations are in place. If the isolation plan needs to be changed, the AA will be consulted, verbal approval obtained and the changes will be made on the hardcopy form.

Second verifier requirements for isolations that do not meet the “personal Isolation” criteria shall be conducted by the same individual, if working on their own.

7.5.1.7 ISOLATION PLANNING

Isolation of equipment is planned so that the risks associated with removing from service and reinstating on completion of work are minimized.

The following key points should be considered:

- Facility system isolation requirements should be identified early in the work planning cycle referencing [Table 14: Minimum Isolation Standard Table](#), so that adequate time is available for the risk assessment process, if required.
- Job-specific plan should be prepared for complex energy isolation. These plans should include (as applicable):
 - Marked-up P&IDs showing the isolation points and the drain, fill, purge, and pressure-testing points.

NOTE: Lines should be walked to confirm the accuracy of the P&IDs and to identify hazards such as low points where liquids can collect and check valves that can trap pressure.

 - Marked-up electrical diagram showing isolation and test points.
 - Engineering evaluation, when required.
 - Step-by-step detail of individual isolation points and any specific sequence for isolating or de-isolating required.

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- Emergency response (such as rescue plan and rescue equipment).
 - Provision for the protection of sensitive equipment.
 - Isolation testing plans.
 - Boundary isolations may be put in place to cover several planned activities in an area of the facility.
- Reference:** See "Boundary isolations" later in this section.
- [CriticalE] The equipment or facility system to be isolated must be rendered to a "zero energy" state.
 - Valve and blind isolations should be located as close as possible to the equipment or system to be isolated.
 - Inability to implement isolations:
 - [CriticalE] Changes to the ICC may be made by the Isolating Authority (IA) to improve the integrity of the isolation [such as wrong valve, leaking valve, seized valve or otherwise unable to isolate (UTI)] with the approval of the Area Authority. [CriticalE] The isolation changes **must** be recorded on the ICC and other relevant documentation (P&ID, written isolation and reinstatement plans updated).
 - Changes can be hand written on the printed ICC and isolation tags by the IA.
 - Lessons learned should be captured for future ICCs.
 - Changes should not be made to isolations that are "In place" other than to allow testing of the equipment using the Sanction to Test (STT) process.

7.5.1.8 TYPICAL ISOLATIONS

Facility systems to be isolated can have many sources of energy that need to be removed or isolated. These include:

- Liquid and gas lines
 - Analyzer lines
 - Drain and vent connections
 - Flare lines
 - Pressure safety valves (PSVs), vacuum breakers, and thermal relief devices
- ▲ CAUTION:** [CriticalE] Do not isolate pressure, vacuum or thermal relief safety valves until the equipment is isolated and adequate vents are open to prevent overpressure or vacuum from occurring.
- Temporary hoses and small-bore pipes
 - Recycle lines
 - External seal flush lines on pumps
 - Fuel lines to engines and turbines
 - Jacket water to pumps or compressors
 - Seal or buffer gas on compressors

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- Seal or lube oil systems on pumps, turbines, or compressors
- Stored energy (e.g. gravity, wind, spring tension and buoyancy)

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7.5.1.9 ISOLATION CONTROL CERTIFICATE

[CriticalE] The Isolation Control Certificate (ICC) **must** be used to document the isolation and approval of equipment or systems and will be used to track the status of all isolations that do not meet personal isolation requirements.

[CriticalE] The isolation plan **must** be Verified by the Area Authority (AA) or designate before it can be applied. [CriticalE] Isolation points **must** be signed off by the Isolating Authority (IA) to confirm they have been put in place. [CriticalE] Associated work permit(s) can only be issued once the AA or designate confirms that all isolations are In Place. [CriticalE] The ICC **must** remain in force until all work permits associated with the ICC have been completed.

The ICC should be cross-referenced to all relevant work permits and is the principal control once isolations are in place. The ICC performs the following functions:

- Lists the location of the isolation points and their normal status.
- It is a record of any fittings (such as analyzer points, sample points and plugs) that have been removed or moved from their normal state to ensure that they are reinstated.
- It is a record of all drain, vent, and bleed points for valve integrity tests. (Operation of these valves does not need to be recorded as an amendment on the ICC but would be checked when returning equipment to service.)
- Authorizes any temporary de-isolation for equipment testing (Sanction to Test).
- Authorizes and records each de-isolated point.
- As necessary, includes attachments of marked-up P&IDs, electrical diagrams, and isolation plans.

7.5.1.10 ICC REFERENCE

The Isolation Control Certificate (ICC) can be identified using a unique number or title. The certificate should be reference on related work permits and any relevant lock-out tags.

FORM LOCATION

Refer to GOTO/WMSFORMS, "ICC & TD," Isolation Control Certificate and in the electronic Permit to Work System.

7.5.1.11 BOUNDARY ISOLATIONS

A *boundary* isolation involves an area of the facility that is to be isolated so that several activities can safely take place within a common set of strategically placed isolations. Thorough planning and preparation is essential when planning boundary isolations. A planning team, including the AA, IA, and additional technical support may be required. Consideration should be given to equipment that may need to be returned to service or test run before other activities are finished when determining whether boundary or individual permit isolations are more appropriate.

The team should meet and thoroughly review and agree on the scope of the boundary isolation and on the associated tasks. This review should include the identification of potential areas of trapped inventory that normally may not be vented and drained (for example, check valves or non-return valves).

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[CriticalE] An ICC **must** be developed and identified as a boundary isolation that clearly identifies where isolation points are appropriate for all the individual tasks to be carried out within its scope of work. [E] Where multiple work permits are carried out against a boundary isolation, each work permit **must** be cross-referenced to the ICC. [CriticalE] All the cross-referenced work permits **must** be completed before the boundary ICC can be closed.

[CriticalE] If any of the proposed tasks require a level of isolation or implementation of control measures greater than that covered by the boundary isolation, then a separate ICC **must** be developed.

7.5.1.12 BOUNDARY ISOLATIONS AFTER WORK ACTIVITIES START

Once work is in progress, a subsequent work permit may be identified and may be applied within the recognized framework of existing boundary isolation. The subsequent work permit to be added must establish clearly that the existing isolations are appropriate for the scope of work.

The original boundary isolation must be reviewed for any subsequent work permits. [CriticalE] Entry into a confined space within a boundary isolation requires a specific ICC and Confined Space Entry Permit, including the implementation of additional security measures.

[CriticalE] Boundary isolations **must not** be removed until all cross-referenced work permits are complete.

If work on a particular area of the facility within the boundary isolation needs to be suspended due to scope change, then a new boundary isolation may need to be defined and documented before the existing boundary isolation can be altered. All affected permits will be suspended before the boundary isolations can be altered.

[CriticalE] When the de-isolation of the boundary isolation is proposed, an isolation review team that includes the AA and IA **must** confirm that all cross-referenced work permits are complete. [CriticalE] They **must** also confirm that equipment within the boundary isolation is in a safe condition.

7.5.1.13 ENERGY ISOLATION SAFETY PRECAUTIONS

Include the following safety precautions when performing energy isolations:

- When isolating equipment, be careful not to trap liquids in sections that do not have pressure protection to avoid pressure buildup due to increases in ambient temperature that could lead to failure of the equipment.
- Stand to the side (avoid line-of-fire) when operating valves to avoid being struck if the valve stem fails or if the packing dislodges or leaks.
- Do not stand below valves when operating chain wheels in case of mechanical failure.
- Confirm that stable access is available, and be aware of posture, obstructions, or pinch points.
- Valves that are difficult to operate should be lubricated and assistance (mechanical or people) should be obtained.
- Where practicable, use the valves closest to the equipment for isolation, while still allowing for depressurizing and draining.
- Immediately before approving permits to open equipment, open vents and confirm that pressure has not built up. Leave the vents open while the equipment is opened up to confirm that the pressure does not build up again. [CriticalE] Vent valves **must** be controlled by the ICC process.

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▲ **CAUTION:** [CriticalE] When venting hydrocarbons, flammable, or toxic substances to atmosphere as part of an energy isolation, all vent points **must** be directed to a safe location. For these purposes a safe location is defined as an area where venting can occur that minimizes exposure to personnel and potential ignition sources. Single valve isolation planning should consider potential system back pressure risk when leaving vent lines open to systems above ambient pressure.

Safe locations include but are not limited to;

- Flare systems
- Atmospheric vents and drains
- Sumps and tanks
- Outside of buildings, or congested / restricted areas

[CriticalE] A method **must** be in place to monitor vent points for the presence of liquids or vapors that could create a hazardous condition.

[CriticalE] Open-ended Hydrocarbon, Flammable, or Toxic Vents **must** be closed if unattended or not being monitored.

! **WARNING:** [CriticalE] Where hot work or confined space entry is performed in the proximity of a hydrocarbon vent (potential release point), appropriate control measures (such as gas testing and area monitoring) must be implemented.

- [E] Valve body cavity bleeds are not to be attempted if there is only a grease nipple fitting fitted to the valve bleed port without an isolation valve. If the valve body cannot be bled down, then assume that there are trapped vapors or liquids.
- [CriticalE] When working on reciprocating engines, such as diesel-driven fire pumps or standby generators, their starting medium (battery, air, or hydraulic) **must** be disconnected and tagged before the energy isolation is considered complete.
- [E] Off-Auto-Start switches **must** be placed in the off position.
- [CriticalE] Heat tracing **must** be isolated when there is the potential to damage the heat tracing and cause arcing or injury, when lagging or bridles are being removed or worked on, or when preparing for vessel entry.
- [CriticalE] Circuit breakers **must** be isolated and locked out. The circuit(s) must be proven to be dead before any electrical work can proceed.
- [CriticalE] When the electrical isolation is for mechanical work (for example, pump motor for pump repairs), the electrical isolation **must** be proven by confirming the right circuit breaker has been isolated, locked, and tagged and attempting to start the motor. Be mindful that process shutdowns may prevent the motor from starting. If in doubt, have an electrician test the circuit.

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7.5.1.14 PERSONAL ISOLATIONS

Personal Isolations can be performed for lower risk, short-duration tasks such as taking samples, isolating instruments, replacing pressure gauges, replacing small filter cartridges or small valves, working on small inhibitor pumps, small lighting circuit jobs and other similar jobs.

NOTE: Operating sites should establish a list of jobs that can be performed as Personal Isolations.

[CriticalE] The personal isolation **must** provide Care, Custody, and Control of the equipment to the person conducting the work (Permit Holder). Isolations will typically be visible to the PH (adjacent to the work face) and do not require isolation tags or break point tags. If a point is not visible, the isolation point will be locked and tagged with a Functional or Personal lock and the key will be retained by the PH.

[CriticalE] The criteria for personal isolation are as follows:

- Before approval, the AA or designate **must** be satisfied with the competence of the person conducting the work to execute the personal isolation (i.e. identify isolation & break points, isolation sequence, zero energy requirements, etc.) and to complete the task safely. The AA can request an AO to assist with the isolations, if required.
- De-isolation takes place immediately upon completion of the task.
- An ICC is not required if the task is completed before the end of shift. An ICC will be processed with lock-out/tag-out if not completed by end of shift.
- The person conducting the work may receive assistance from others but the person **must** be present at all times while work is in progress.
- The person conducting the work confirms zero energy and demonstrates zero energy to any person assisting.
- If the work site with a personal isolation is to be left unattended during the shift, the equipment will be isolated, locked, and tagged, including the capping or plugging of open ends.
- During permit closeout, the AO **must** confirm that the isolation points are returned to the normal state.

7.5.1.15 MULTIPLE ISOLATION POINT LOCKING PROCESS

[CriticalE] When more than one isolation point is required, the multiple isolation point locking process **must** be used: Refer to [Table 17: Multiple Isolation Points Locking Process](#).

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Table 17: Multiple Isolation Points Locking Process

Task	Activity
1	An Isolation Control Point (ICP) must be selected and be in a prominent and accessible location such as a Control Room, Permit Coordination Center, or a location on the equipment.
2	Each isolation point must be locked and tagged with a lock from the keyed alike lock set(s).
3	The key(s) from the lock set(s) used on each isolation point are placed in the ICP box and locked with a uniquely keyed Operations Functional or personal lock by the AO.
4	Each Permit Holder places their uniquely keyed Functional or personal lock onto the ICP box to have control of the isolations. The PH retains the key in their custody while working on the job. The key is handed to their relief or returned to their supervisor if they will leave the job or site.
5	As each job is completed, the PH <i>completes</i> their permit and removes their Functional lock from the ICP box.
6	When all associated permits are <i>complete</i> and all PH Functional locks have been removed, de-isolation of the equipment can begin. The Area Operator is then able to remove their Operations Functional lock, access the isolation key(s) from within the ICP box, unlock each isolation point, and return the equipment to service.

7.5.1.16 ELECTRICAL ISOLATIONS

[CriticalE] Competent personnel will be authorized by the Operating Site to conduct electrical isolations. Typically, an Electrical Technician is authorized to perform electrical isolations that require isolations beyond opening a breaker and locking it, for example, racking out breakers, disconnecting terminals, or applying grounds (earths).

7.5.1.17 ELECTRICAL ISOLATION PLANS

All circuit breakers, switches, and combined fuse-switch units for the equipment to be isolated should be locked out when possible.

[CriticalE] All electrical isolations **must** be tested and shown to be effective before exposing personnel to electrical equipment. Job-specific isolation plans **must** specify the methods by which the electrical isolations will be tested.

[CriticalE] The following process **must** be followed when isolating electrical equipment to be worked on, or reinstating it afterwards:

- Starting and stopping of high voltage motors should be controlled when personnel are working in high voltage switchgear rooms.
- The Electrical Technician **must** confirm that the equipment is switched off. An isolation lock **must** be attached to the appropriate isolation switch on the switchboard.
- **! WARNING:** If no locking facility exists, the Electrical Technician **must** disconnect supply cables or rack out the circuit breaker to achieve a positive electrical isolation. In this case, the cabinet **must** be secured and tagged to prevent reconnection of the cables, or racking the

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breaker closed, until the job is completed. There must be an attempt to energize or start the equipment to confirm de-energization.

- When the power cable is disconnected from the equipment, the Isolating Authority (Electrical Technician or Area Operator) will use a lock from the lock set to isolate the electrical supply.
- The Electrical Technician can then complete any grounding (earthing).
- The electrical isolation integrity **must** be confirmed and zero energy demonstrated by the Isolating Authority using appropriate methods.
- For electrical work, the three-step process (Ex. Live-dead-live) to test and confirm equipment has been de-energized **MUST** be conducted as close as possible to the point of work before disconnection. If workers leave the equipment prior to disconnection (ex. Breaks, lunch) the three-step process must be performed again before returning to the task.

▲ CAUTION: When verifying that electric circuits or electrically operated equipment is de-energized, consider that a process control such as a pump low-level shutdown may be preventing the equipment from starting when the start button is pressed. If possible, temporarily defeat those functions when attempting to check electrical isolations.

- If more than one permitted job is being conducted on isolated equipment, each Permit Holder **must** put a Functional lock on the ICP that prevents re-energizing before the last Functional lock is removed.
- As each permitted job is completed, the uniquely keyed Functional locks are removed from the ICP.

! WARNING: The Upstream Safety Manual Electrical SWP **must** be consulted for further precautions to be taken when isolating high voltage or high energy electrical equipment.

Reference: Upstream Safety Manual Electrical SWP "Electrical Safe Work Practices".

7.5.1.18 CREW CHANGES, SHIFT CHANGES, AND CALLOUTS

[CriticalE] The integrity of the energy isolation system **must** be maintained during crew changes, shift changes, and callouts. [CriticalE] For callouts or crew or shift changes ensure the following:

- Personnel leaving the facility before the work is completed inform their Functional Supervisor of the status of the work and turn over their Functional lock key(s) to their supervisor or the oncoming Permit Holder.
- Personnel coming on shift or callout to work on locked-out equipment actively participate in the pre-task briefing, including a review of the Permit Pack (including the ICC).
- The oncoming Permit Holder confirms that a Functional lock/Personal Lock is on the Isolation Control Point for the permitted work being performed and take control of the key from the off-going PH or their Supervisor before signing acceptance of the permit.

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7.5.1.19 REMOVAL OF FUNCTIONAL AND PERSONAL LOCKS BY THE AA

[E] If a job is complete but an individual leaves the site without removing their Functional/Personal lock or turning over their isolation key(s) to the Functional Supervisor, the only person authorized to remove the lock is the Area Authority. The AA should be accompanied by the Functional Supervisor when removing the lock (lock removal may be delegated to a competent person after consultation with the AA in RFO & NC-RFOs where traveling to location is impractical).

PIC endorsement is obtained prior to lock removal.

The AA verifies that the lock can be removed and all isolating devices may be returned to the normal operating position. The Functional Supervisor will notify the individual who initially installed the lock of its removal when the individual reports back to work.

The Functional/Personal Lock Removal form at [GOTO/WMS FORMS "ICC & TD"](#) section can be used to facilitate this process.

7.5.1.20 MULTIPLE ISOLATION REQUESTS

Instances can arise when several jobs, each with its own work permit, require the same isolation points. In these cases, the following precautions built into the energy isolation process will enable this situation to be safely managed:

- [E] The ICC **must** be cross-referenced to each permit.
- Work permits are cross-referenced to the ICCs.
- [CriticalE] If a point is part of two separate isolations, the point **must** have two locks and tags (one from each keyed alike lock set) that are secured in their respective ICP boxes. This prevents the point from being de-isolated until the jobs associated with both ICPs are completed.
- Each energy isolation job will be completed as a stand-alone job.
- Protection from de-isolation before all work is finished is achieved by:
 - Ensuring that all Permit Holders apply their Functional lock and tag to the ICP. This results in more than one Functional lock being in place on the ICP.
 - The tag on each lock identifies the job that the lock is protecting.
 - A Permit Holder can only remove the Functional lock associated with their permit (custodians of the key).
 - All affected Permit Holders who are still working under the isolation will have their Functional locks and tags on the ICP until their permitted work is completed.
 - As work is completed, each Permit Holder will remove their Functional lock and tag from the ICP.
 - Once all Permit Holder Functional locks are removed, the AO Functional lock can be removed and the ICP opened to retrieve the lock-set key(s) and commence de-isolation.

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7.5.1.21 LONG-TERM ISOLATIONS

Long-term isolations (LTIs) are defined as isolations that do not have work being performed against them and remain in the isolated state until parts, or other services, are provided.

[E] A register of long-term isolations should be maintained and subject to two levels of review:

- A monthly review of the LTI register to confirm status and forward plans (e.g. repairs scheduled, parts have not arrived, equipment not required, physical disconnection or blinding required, etc.).
- A quarterly field assessment to physically check the integrity of all isolation points e.g. tags legible, securing devices in place, locks in good state, no leaks and no pressuring up of the system. The intent of this assessment is to ensure the integrity of the isolation is maintained over time.

LTIs are removed when either:

- Isolation is transferred back to ICC for the equipment to be worked on and restored. [CriticalE] Before any work is recommenced against an LTI, a full integrity field inspection of the isolations **must** be conducted and the LTI moved to an ICC.
- Equipment is physically disconnected or blinded for an interim period (temporary MOC) or permanently made out of service (OOS).
 - Need for physical disconnection or blinding should be considered for LTIs based on the potential for personnel or process safety events. Retaining as an LTI will require, per above, a regular field assessment to ensure integrity of isolation is maintained.

Equipment that is deemed "out of service" is evaluated to determine whether it should be permanently removed and/or physically disconnected or blinded using the MOC process. The MOC shall include an evaluation for continued operations of the facility during normal and abnormal operations without the OOS equipment. Equipment strategies and integrity programs shall be updated accordingly.

7.5.1.22 LTI EQUIPMENT RISKS

LTI equipment may have impacts to safe and reliable, continued operability of facilities. [CriticalE] The impacts and associated risk of continued operation with equipment in LTI **must** be considered by a multi-disciplined Operations, Maintenance and Technical (OMT) team on a periodic basis defined by the Site in their WMS [Section 12.0 Appendix A](#). Considerations should include hazards (i.e. dead legs, corrosion, loss of automation, etc.), impact to the operating envelope(s), interim measures and resilience during abnormal operations, if applicable. OMT evaluation should consider impacts of unavailable equipment on a system basis (e.g. power generation, flare system, water handling, emergency response, etc.).

[CriticalE] The OMT team **must** report LTIs that pose a risk to safe and reliable continued operations to the Operations Manager at a minimum on a **6-monthly basis**. The intent of the Operations Manager review is for consideration of overall system risk and as appropriate:

- Additional controls/mitigations for safe continued operations.
- Alignment on prioritization for repair and reinstatement.
- Alignment on need for temporary or permanent physical disconnection/removal or blinding (temporary or permanent MOC).

Tools: [LTI/OOS Health - "How do you Know Questions"](#)
[Risk Assessment and Management System](#)

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7.5.2 DRAINING AND DEPRESSURIZING

7.5.2.1 INTRODUCTION

This section provides guidelines to protect personnel and equipment when draining and depressurizing equipment or systems, particularly process or hydrocarbon-containing systems.

7.5.2.2 PREREQUISITES

[CriticalE] Prior to draining or depressurizing, the equipment or system **must** be fully isolated.

Reference: Section 7.5.1: "Energy Isolation and Isolation Control Certificate"

7.5.2.3 PLANNING DRAINING AND DEPRESSURIZING ACTIVITIES

The following guidelines should be used to plan draining and depressurization:

- Confirmation that the equipment or system is isolated.
- The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared.
- Job planning should consider the following:
 - Special conditions to be established before draining and depressurizing.
 - Step-by-step activities to be performed.
 - Facility critical operating parameters (pressures, temperatures, capacities) that must not be exceeded.
 - Whether the equipment being isolated relates to one of the Site's highest consequence scenario(s) and associated Critical Safeguard(s) as designated by Scenario Management
 - Required temporary defeats, and when critical devices may be reinstated.
 - Means to monitor the progress of the preparation and confirm completion.
 - The gases to be vented and liquids to be drained, reviewing the appropriate Safety Data Sheets (SDS).
 - The necessity for a Risk Assessment of the activities to identify risk mitigation measures.
 - Any additional personal protective equipment (PPE) required.
 - Emergency or abnormal operating procedures.
 - If a Critical Work Permit for "Breaking Containment" is required.

Draining:

The following additional guidelines should be used to plan draining activities:

- The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared.
- Light ends (such as, methane or ethane) containing liquids should be displaced into vessels of similar composition and pressure to avoid the potential for low temperature embrittlement by draining these liquids to the drain systems, unless the drain systems are designed for low temperatures.

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- Drain liquids slowly to the closed drain system, making sure not to exceed its capacity. When this is not possible, an estimate of the liquid quantity should be made and sufficient local containment should be provided.
 - Drain liquids from the lowest point(s) of the system.
 - Avoid over pressuring or overfilling the closed drain system.
 - Be aware of any sand carryover from the vessels, which could cause problems with the process equipment.
 - Provide spill containment and waste disposal.

! WARNING: Do not depressurize gas into the drain system.

! WARNING: Some vessels have standpipes or weirs that can trap liquids on the side away from the drain line.

▲ CAUTION: [CriticalE] Do not allow the temperature of the drain system or hoses to go below their minimum design temperature or brittle failure can occur.

▲ CAUTION: [E] If it is necessary to drain liquids through hoses to an open drain, follow these precautions:

- Use hoses that are compatible with the fluids to be drained.
- Include a valve that will enable control of the rate of draining.
- Secure the ends of the hose.
- Provide spill containment at all hose connections.
- The hose nozzle **must** be electrically grounded (earthed).

NOTE: A JSA should be considered when draining or venting to atmosphere when pressures are above drain or flare header residual pressure.

▲ CAUTION: Do not use excessive force on drain valves.

▲ CAUTION: Any oil-contaminated materials **must** be removed because they can be prone to spontaneous ignition.

Check that all low points are drained and that no pockets of liquid remain.

Depressurizing:

The following additional guidelines should be used to plan depressurizing activities:

- Depressurize the equipment slowly, through the closed flare or vent system, avoiding liquid carryover to the flare system.
 - ▲ CAUTION:** [CriticalE] If hazardous or flammable gas **must** be depressurized to the atmosphere, it **must** be performed under strictly controlled conditions, taking into account wind strength and direction. Personnel working downwind of any system being drained or vented that could be impacted **must** be instructed by the Area Operator to leave the area. The precautions listed on the Critical Work Permit and/or JSA for "Breaking Containment" should detail the appropriate controls and communications requirements.
- Depressurize all valves through the valve body drain, and contain fluids.
- Verify equipment depressurization with several bleed points and pressure gauges.

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7.5.2.4 SAFETY PRECAUTIONS

Use the following safety precautions when draining and depressurizing facility systems:

- [E] Appropriate PPE **must** be worn to protect against hazards specific to draining and depressurizing. These include:
 - Increased eye protection (for example, goggles or face mask) when bleeding down equipment.
 - **Breathing apparatus** when conditions (confined area, where asphyxiation could occur) and state of the equipment to be broken (i.e. whether the equipment may contain toxic substances such as H₂S, hydrocarbons or benzene).
- Do not use excessive force if the bleed or drain valves are tight, use lubricant to free up valve stems and bushings or have them replaced during the shutdown.
- Drain before depressurizing fully so that liquids do not continue to vapor off and prolong the depressurizing phase and to avoid potential brittle fracture concerns
- Drain and depressurize slowly. This helps prevent:
 - Hydrates
 - Erosion of equipment and piping if sand particles are present
 - Flashing or blowback from open drain systems

! IMPORTANT: Potential flashing or blowback from open drain systems must be considered when evaluating other work or activities at the facilities.
- Open more than one bleed valve to confirm that pressure does not remain trapped in the system due to a blocked valve or hydrates.
- Be careful not to exceed equipment temperature, pressure, flow, or metallurgic limits (including drain systems, flare systems, and sumps). If necessary, allow excessively hot or cold fluids to stabilize before draining.

! IMPORTANT: Consider that volatile liquids can become colder when depressurized. In these cases, consideration should be given to displacing as much process liquid as possible back into the system before depressurizing. Do not allow the temperature of the drain system or hoses to go below their minimum design temperature or brittle failure can occur.
- Open drain water seals (P-traps) may allow flammable or toxic vapors to release to atmosphere if the seal is not maintained. Flush open drain systems with water before and after draining to an open drain system to prevent blowback or flashing of flammable gases.
 - Open drain valves and use hoses, if necessary, to drain liquids to drain systems. Liquids can be trapped in weirs, standpipes, instrument bridles, or dead legs.
- Use double manual drain valves to depressurize systems when the potential for hydrates exists. The upstream valve must be opened fully first, then the downstream valve can be opened. This is to ensure that, if the downstream valve hydrates up due to throttling, the upstream valve is unaffected and can be closed. The downstream valve is closed before the upstream valve when depressurizing has been completed. Bleed off any trapped fluids between the two drain valves by cracking open the downstream valve, then close it when pressure is fully bled.

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▲ **CAUTION:** Only valves designed for throttling should be used as the downstream, throttling valve.

- [E] Do not attempt to clear atmospheric bleed valves that are blocked and under pressure. Connect pressure-rated hoses between the blocked valve and the closed drain system and attempt to use differential pressure to clear the blockage. If the blockage cannot be cleared using this method, or if this method could overpressure the closed drain system, the system will need to be isolated elsewhere to enable maintenance to clear the blockage or replace the problem valve.
- Hot work at the facility should be restricted based on the following considerations:
 - Venting flammable gases to atmosphere
 - Potential blowback or flashing of flammable gases from open drain systems
 - Potential for liquid overflow or carryover into drain or flare systems
 - Wind direction and strength
- All potential spills must be banded or otherwise contained. Identify the need for, and type, of spill response equipment. Any waste process or utility fluids must be disposed of in a proper manner. The SDS for any contaminants should be consulted.
- Pressure relief valves, thermal relief valves, and vacuum breakers should not be isolated until the equipment is depressurized and adequate vents are open to prevent overpressure or under pressure of the equipment.
- Draining and depressurizing to closed drain or active flare systems may not bleed the pressure completely to zero. When residual pressure has been reached, it may be necessary to vent the remaining pressure to atmosphere in a safe manner.
- Corrosion in drain systems (especially offshore) can result in the loss of containment of drained fluids. Drain systems should be monitored or inspected, if practicable, before draining large volumes of fluids, or where drain systems may be exposed to design pressures.

▲ **CAUTION:** [CriticalE] When venting hydrocarbons, flammable, or toxic substances to atmosphere as part of an energy isolation, all vent points **must** be directed to a safe location. For these purposes a safe location is defined as an area where venting can occur that minimizes exposure to personnel and potential ignition sources. Single valve isolation planning should consider potential system back pressure risk when leaving vent lines open to systems above ambient pressure.

Safe locations include but are not limited to;

- Flare systems
- Atmospheric vents and drains
- Sumps and tanks
- Away from buildings, or congested / restricted areas

[CriticalE] A method **must** be in place to monitor open ended vent points for the presence of liquids or vapors that could create a hazardous condition.

[CriticalE] Open-ended Hydrocarbon, Flammable, or Toxic Vents **must** be closed if unattended or not being monitored.

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7.5.3 BREAKING CONTAINMENT

7.5.3.1 INTRODUCTION

This section provides guidelines to protect personnel and equipment when breaking containment in systems such as hydrocarbons, chemicals, air, steam, fluids, and gases.

This section is focused on:

- Unbolting flanges and manways or removing screwed fittings on systems, such as pipes, vessels, exchangers, valves, turbines, compressors, and pumps
- Opening enclosures such as pig launchers or receivers and filter housings
- Orifice plate removal

At this stage of the facility isolation process, the equipment should be depressurized, drained, and at atmospheric or flare/drain residual pressure. Depending on the isolations achieved and the proposed work, purging of hazardous materials can be conducted before or after breaking containment. Purging requirements and method should be done at the planning phase.

In circumstances where the equipment cannot be hydrocarbon freed prior to breaking containment, a slight positive pressure should be established to ensure that air does not migrate into the system and form a combustible mixture (hydrocarbons and air) inside the equipment.

Breathing apparatus – the conditions (confined area, where asphyxiation could occur) and state of the equipment to be broken (i.e. whether the equipment may contain toxic substances such as H₂S, hydrocarbons or benzene) will determine if breathing apparatus will be required.

7.5.3.2 PREREQUISITES

[CriticalE] Before breaking containment, the equipment and piping involved **must** be isolated, drained, and depressurized.

References:

- Section 7.5.1: "Energy Isolation and Isolation Control Certificate"
- Section 7.5.2: "Draining and Depressurizing"
- Section 7.5.5: "Hydrocarbon Freeing"

7.5.3.3 PLANNING BREAKING CONTAINMENT ACTIVITIES

The following guidelines should be used to plan breaking containment activities.

- The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared.
- Job planning should consider the following:
 - Special conditions that should be established before breaking containment is commenced.
 - Identifications of the locations where containment is to be broken.
 - Step-by-step activities to be carried out.
 - Any required temporary defeats of critical devices, and when critical devices may be reinstated.
 - Substances that might be present and a review of the appropriate Safety Data Sheets (SDS).

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- The necessity for a Risk Assessment of the activities to identify risk mitigation measures.
- Consider whether the breaking containment activity is on equipment that relates to one of the Site's highest consequence scenario(s) and associated Critical Safeguard(s).
- Spill containment and waste disposal that might be required.
- Additional personal protective equipment (PPE) required.
- Emergency procedures/process.
- The Area Operator or a designated competent person should identify all break points with the Permit Holder, and agree on the location of first break (to minimize risk) and subsequent breaks.
- The Area Operator or designated competent person and Permit Holder should physically walk the line(s) to identify:
 - Hazards such as low points where liquids may collect and not be drained.
 - Access and egress problems.
 - Confirm that the vessel/system is isolated, drained and depressurized.
- [CriticalE] The Area Operator or designated competent person **must** be present for the first break or when there is any doubt regarding the correct flange to be split.

7.5.3.4 BREAK POINT TAGGING

[CriticalE] A Break Point Tag **must** be placed at all Break Points and signed by both the Area Operator and the Permit Holder (or Designated Competent Person) responsible for the execution of the Breaking Containment Work. Tags will be removed after the completion of the work.

[CriticalE] When managing break point tags:

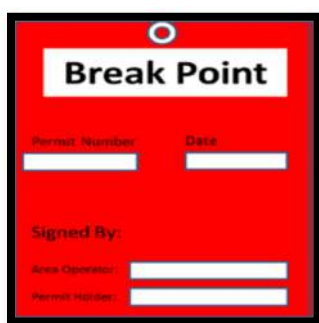
- The Area Operator (or Designated Competent Person) must apply the tag to the Break Point after verification of the energy isolation.
- Tag must be applied directly to break points (e.g. valves, flanges, manways, not adjacent equipment etc.).
- Tags must be highly visible
- The Area Operator (or Designated Competent Person) must demonstrate Zero Energy to the Permit Holder and then they both sign the tag to confirm the correct Break Point(s).
- The Area Operator (or Designated Competent Person) must be present and remain onsite during First Break.

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7.5.3.5 LINE CUT IDENTIFICATION

- [CriticalE] A line cut identification process **must** be used for demolition, cutting and removal of pipe; electrical conduits and other process or ancillary lines. [E] Line marking processes must ensure:
 - Line identification tape which can be applied and removed prior to and after job completion is preferred methodology. Use of paint and permanent marking **should** be avoided to eliminate risk of incorrect identification from previous jobs.
 - Tape **must** be highly visible and durable
 - **Must** be able to be applied to line cut points

Figure 15: Break Point Tagging and Line Cut Tape



Break Point Tag



Line Cut Identification Tape

7.5.3.6 CLAMP-LOCK FITTINGS

Particular care should be taken with a clamp-lock type of fitting. [E] Clamp-lock fittings **must** only be removed by personnel trained in accordance with the manufacturer's instruction manual. No bolts are to be fully removed until the clamp rotates freely.

7.5.3.7 OPENING LIVE/OPERATING SYSTEMS

Due to facility/equipment design, there are certain instances where Breaking Containment activities need to take place on equipment/operating systems that cannot be fully isolated. Examples:

- Removing analyzers or instruments from live low-pressure systems
- Sampling- production and regulatory
- Installing Blinds – small volume/low pressure where no isolation exists
- Removing PSVs where no isolation exists

[CriticalE] When opening live operating systems where there is no alternative isolation method(s) available, the following additional control measures **must** be employed:

- A Risk Assessment **must** be conducted and endorsed at the appropriate level (see the Risk Assessment & Management Manual).

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- Any Risk Assessment **must** consider the introduction of air into the system/equipment, duration of task, complexity, type of service etc.
- A procedure/process to control the activity as per outcomes of the Risk Assessment **must** be developed and followed.

▲ **CAUTION:** PSVs fitted with pilot valves can backflow if flare header pressure increases and the pilot operates while the vessel is depressurized. Ensure that the upstream valve (vessel side) is closed or pilot valve sensing line is isolated before breaking containment.

7.5.3.8 EMERGENCY PLANNING

The emergency plan should consider the following:

- How to respond to a leak
- Availability of spill containment or cleanup equipment
- Knowing the location of the next available isolation valve
- Planning escape routes when working in congested or confined locations

[CriticalE] Fire-fighting equipment **must** be readily available and the IA and the PH **must** be able to initiate the emergency alarm and contact the Central Control Room (CCR).

7.5.3.9 BREAKING CONTAINMENT SAFETY PRECAUTIONS

Use the following safety precautions when breaking containment:

- [E] Appropriate PPE **must** be worn to protect against hazards particular to breaking containment. These may include:
 - Increased eye protection (for example, goggles or face mask).
 - **Breathing apparatus** - the conditions (confined area, where asphyxiation could occur) and state of the equipment to be broken (i.e. whether the equipment may contain toxic substances such as H₂S, hydrocarbons or benzene) will determine if breathing apparatus will be required.
- [CriticalE] Hot work, **must** be restricted, in compliance with the SIMOPS Restrictions Table in [Section 9.0 Simultaneous Operations](#), when breaking containment.
- All potential spills **must** be contained. Identify the need for, and type of, spill response equipment.
- [CriticalE] Except for hot bolting under controlled conditions, do not remove any bolts until the line or equipment is completely depressurized and spill containment is in place.
- [CriticalE] Non-spark tools **must** be used when flanges are initially broken if there is a potential for hydrocarbons to be released.
- Do not place fingers between flanges or through bolt holes.

[CriticalE] When splitting insulated flanges (typically the first or last flange on a pipeline), cathodic protection units for each side of the flange **must** be turned off and the flanges electrically bonded before breaking.

▲ **CAUTION:** When breaking each flange, it must be treated as though it could have pressure or liquid present. The PH or work team member should always be positioned on the upwind side

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of the flange to be broken. They will first loosen the bolts on the lower and opposite side from them, keeping the bolts nearest to them under control and enabling the line to separate in a manner that causes any spillage/release to be away from them. The flange should then be spread apart to check for the presence of any gas or liquids.

- After isolating, draining, and depressurizing the system; if a valve bonnet is to be removed, drain the valve and place the valve in the open position before loosening the bonnet bolts.

▲ CAUTION: Ball and plug valves can have pressure in the cavity under the stem packing and bonnet, regardless of the position of the valve and the pressure in adjacent lines. [E] Ball and plug valves that are to be removed from the system **must** be opened, closed, and reopened to relieve pressure after the line is drained and depressurized.

! IMPORTANT: Consider hot bolting the flange(s) before breaking containment to reduce the potential for manual handling injuries and improved efficiency when executing the work.

7.5.4 BLINDING

7.5.4.1 INTRODUCTION

Blinding is a positive and safe means of preventing the migration of hydrocarbons or hazardous substances into equipment or systems that are isolated.

7.5.4.2 DEFINITION

- **Blind/Blank:** A pressure-rated solid-steel plate fitted to the end of an open spool, valve, vessel, or other piece of equipment.
- **Blinding:** The activity of isolating equipment or facility systems using blinds or spades.
- **Spade:** A pressure-rated solid-steel plate installed between two flanges.

7.5.4.3 GOAL

The goal of blinding is to establish a physical barrier between process fluids or vapors so that work can proceed safely.

7.5.4.4 WHEN TO BLIND

[CriticalE] Blinding **must** be conducted:

- For confined space entry into equipment.
- If the job site will not be attended at all times and a double block and bleed (DBB or DBV) cannot be established.
- Between existing production equipment and new equipment under construction.
- Between active production equipment and equipment that has been taken out-of-service (short or long term).
- Whenever segregation is required between systems.

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7.5.4.5 WHEN BLINDING IS NOT REQUIRED

Blinding is not required:

- If a double block and bleed (DBB or DBV) can be established and confined space entry is not involved.
- If immediate replacement of the item is to occur, and the job site will not be unattended.
- During short-term opening of the equipment or pipework (jobs completed within one shift), and the job site will not be unattended.
- On jobs where breaking containment to install blinds would involve greater exposure to risk (for example, time taken, location, size and number of blinds to be installed) than not blinding, provided that:
 - Not blinding is approved by the Person In Charge (PIC).
 - Approval not to blind is documented (for example, on the permit or JSA).
 - The work does not involve confined space entry.

7.5.4.6 PREREQUISITES

Blinding involves breaking containment. [CriticalE] Prior to blinding, the equipment and piping involved **must** be isolated, drained and depressurized. Install blinds and spades using the appropriate breaking containment and blinding precautions

7.5.4.7 PLANNING BLINDING ACTIVITIES

The following guidelines should be used to plan blinding activities.

- The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared.
- Job planning should consider the following:
 - Any special conditions that should exist before blinding is commenced.
 - Step-by-step blinding activities to be carried out. Attention should be paid to the order in which blinds are installed, and later removed (blinding priority) to ensure that the equipment is protected (pressure safety valves) while there is the potential for pressure to build up (for example, when *blinds or spades* are not in place).
 - Where practical, select isolation and blind positions to enable zero energy checks (drains or vents) before removing the blind.
- Locate blind and spade locations as follows:
 - Using P&IDs, identify all blind and spade locations.
 - Verify each location by field checking.
 - Record the blind locations on the ICC plan.
- Obtain a Critical Work Permit for Breaking Containment for the blinding to be done.
- Confirm that the vessel or facility system is isolated, drained, and depressurized.

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Install blinds and spades using the appropriate breaking containment and blinding precautions.

▲ CAUTION: When breaking flanges to install spades, treat each break as though pressurized fluids were present.

! WARNING: [CriticalE] The Permit Holder or person designated by the AA **must** confirm that:

- Blinds, spades and gaskets are design rated and are of the correct material specification (refer to piping/flange specification)
- New and correct gaskets are used.
- Correct bolts and nuts are used.
- Correct torque and pattern are used to tighten all bolts.

[CriticalE] After blinds and spades are installed, they **must** be recorded on the ICC.

7.5.4.8 BLINDING AND WORK PERMITS

[CriticalE] The following are requirements regarding work permits when blinding is executed:

- Hot work, **must** be restricted, in compliance with the SIMOPS Restrictions Table in [Section 9.0: Simultaneous Operations](#), when breaking containment.
- Work permits, if any, **must not** be issued or **must** be withdrawn if the work is in close proximity to a work site that has the potential to ignite hydrocarbons.

7.5.4.9 BLINDING PREPARATIONS

Blinding may be conducted as part of the isolation activity controlled by the isolation control certificate and Critical Breaking Containment Permit. All JSA and checklist requirements still apply in these cases. In most cases, the blinding process is as follows:

- An initial ICC₁ is processed by the Isolating Authority for the required isolations to enable draining and depressurizing of the equipment or system so that blinds can be safely installed.
- A separate ICC₂ for blinding is created. Future permitted activities relying on the installed blinds would be linked to ICC₂.
- [E] The Blinds List (ICC₂) and/or marked-up piping and instrumentation drawings (P&IDs) must be used when installing blinds or spades. The exact location for installation of a spade or blind needs to be identified by tagging each flange.

! IMPORTANT: [CriticalE] When preparing the blinding plan, existing blinds or spades **must** be field verified and included on the ICC.

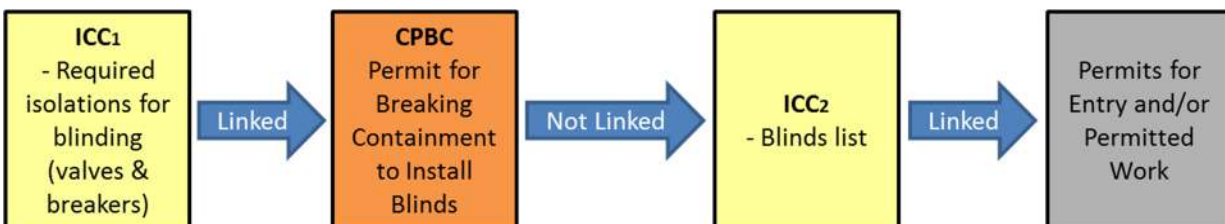
- Blinds or spades should be installed as close as practical to the equipment to be worked on.

▲ CAUTION: Splitting the flange on the wrong side of an isolation valve that has not been drained and depressurized can result in an uncontrolled hydrocarbon release.

- [CriticalE] The blinding plan **must** be approved by the Area Authority before execution.
- [CriticalE] The Area Authority **must** approve any changes to the Blinds ICC or the blinding plan after initial approval. The marked-up P&IDs, the Blinds ICC, and the blinding plan **must** be updated or replaced. Refer to [Figure 16: Blinds ICC](#).

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Figure 16: Blinds ICC



7.5.4.10 BLINDING PRIORITY

When blinds are installed, the order in which this is done is important. [E] Pressure safety valves (PSVs) are the last to be blinded in case of an unforeseen event or delay. Blinding priority should be listed in the blinding and de-blinding plan, and should be listed on the blinding tag.

The priority is as follows:

- **Major lines:** Blinding eliminates the risk of leakage from high-pressure or high-volume sources. Examples are inlet and outlet, suction, or discharge lines.
- **Secondary lines:** Blinding eliminates medium-pressure or medium-volume sources. Examples are recycling, bypass, purge, and drain lines.
- **Minor lines:** Includes instrument lines, sample points, and so forth.
- **Pressure safety valves:** Blinding PSVs last enables maximum vessel protection during other isolation activities.

▲ CAUTION: PSVs fitted with pilot valves can backflow if flare header pressure increases and the pilot operates while the vessel is depressurized. Ensure that the upstream valve (vessel side) is closed or pilot valve sensing line is isolated before breaking containment.

7.5.4.11 BLINDING PRECAUTIONS

The precautions regarding blinding are as follows:

- [CriticalE] All blinds that could be or will be exposed to system pressure **must** be full system-pressure rated.
- Always proceed as if the pipe were under pressure. Appropriate precautions **must** be taken, including:
 - When splitting flanges for blinding or de-blinding, the best practice for breaking containment **must** be followed (the bolts farthest away from the Technician should be loosened first and the flange parted to check for gas or liquids under pressure).

Reference: Section 0: "Breaking Containment" for additional precautions.

- Spill containment must be in place when there is the potential for liquids to be present.

! WARNING: Breaking containment can enable an oxygen atmosphere to coexist with hydrocarbons. Appropriate precautions must be taken, for example, by maintaining a slightly positive pressure in the system to prevent air entering. This may require breathing apparatus to be worn if working in a confined area where asphyxiation could occur.

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- [CriticalE] Any relevant heat tracing or cathodic protection **must** be electrically isolated to prevent spark potential.
- [CriticalE] The Area Operator or designated competent person **must** be present, when the first flange in a system is to be unbolted, to monitor and confirm that the system is drained and depressurized. [CriticalE] The Area Operator **must** be consulted whenever there is any doubt (identification is not visible) as to which flange to break to insert a spade or blind.
- The Area Operator identifies and remains in attendance for breaking of any flanges that are suspected of containing any fluids due to insufficient or inadequate drain valves.
- [CriticalE] All valves and flanges to be blinded **must** be identified with tags. Sequence should be considered in the de-blinding process, when the sequence is important e.g. PSVs de-blinded first.
- [CriticalE] The Area Operator **must** confirm that all blinds have been installed, bolts are tensioned (flange complete tape), and checked off on the ICC.
- **▲ CAUTION:** The Area Operator **must not** Issue permits for work or entry, relying on the blinds for isolation, until all blinds are verified as *in place*.
- [CriticalE] If there is suspicion or indication of any gas or liquid pressure in the pipe, bolts **must** be retightened immediately to seal the flange. The Area Operator **must** investigate the reason for the pressure or liquids and ensure that the system is not under pressure before work recommences.
- If there is no bleed valve between the blind or spade and the isolation valve, then a vented spade, blind, or venting ring should be installed to allow for depressurizing the spool and performing blind removal later, if the isolated valve leaks.
- If the ability to vent cannot be provided, a plan for the safe removal of the specific blind/spade is to be documented to address the potential hazards between the valve and blind/spade (for example, liquid, gas, pressure, temperature, H₂S, or other hazards).
- [CriticalE] All blinds and spades installed **must** be cleaned and fitted with new gaskets on the pressure side to ensure effective sealing, should pressure build up behind it due to valve leakage.
- **NOTE:** Old gaskets in good condition may be used on the non-pressure side to protect the flange faces.
- [CriticalE] All flange bolts **must** be fitted and tightened to the appropriate torque.
- The person in attendance is required to know the correct action to take if a leak occurs.

7.5.4.12 BLIND FABRICATION

[CriticalE] Blinds and spades that may be subjected to process pressure (due to leaking valves or inadvertent opening of a valve) **must** be certified and compatible with the process fluids and with a design pressure and temperature at least equal to that of the vessel or piping in which they are installed.

7.5.4.13 PERMANENT BLINDS

When blinds or spades are installed to segregate sections of plant no longer in use, or not being used in the original configuration, they are fitted for long or indefinite periods and their position must be adequately documented and identified.

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7.5.4.14 PLUGS

[CriticalE] The following precautions **must** be taken during the use of plugs (inflatable or mechanical expansion type):

- The plug **must** be checked before use.
- Adequate venting of the system **must** be confirmed to avoid pressure buildup behind the plug.
- During hot work, there **must** be a continuous flow of inert gas from the plug to the relief point, which **must** be located at a safe distance from the hot work location.
- The pressure behind the plug **must** be continuously monitored.
- Personnel **must not** be located in front of the pipe end where the plug is installed, in case it is ejected (line of fire).
- A barrier provided by a plug is not as secure as a blind and **must** be risk assessed to ensure that preventative and mitigative steps are identified and implemented to provide an acceptable hydrocarbon barrier and protect against ejection or leakage.

[CriticalE] All permanent blinds and spades **must** be captured as a long-term isolation until permanently marked on the P&IDs.

7.5.5 HYDROCARBON FREEING

7.5.5.1 INTRODUCTION

This section provides guidelines to protect personnel and equipment when performing hydrocarbon freeing.

At this stage of the facility isolation process, hydrocarbon freeing is intended to eliminate as much hydrocarbon liquid and vapor from the isolated facility system as can be done before breaking containment.

! WARNING: [CriticalE] Facility systems, vessels, tanks, or piping that could contain iron sulfide (FeS), which is a pyrophoric substance, **must** be kept damp at all times, even when equipment is hydrocarbon freed. These deposits are a possible source of auto-ignition and capable of producing sulfur dioxide (SO₂).

[CriticalE] **Hydrocarbon Freeing Required:**

- Hot Work or Confined Space Entry activities
- Breaking Containment of facility systems and equipment that have large volumes and/or complex components e.g. exchangers, compressors, tanks, vessels, turbines and pipelines
- Breaking Containment of facility systems and equipment that have higher risk of ignition or release of hazardous substances e.g. iron sulfide, acid gas, toxic H₂S levels, etc.

Hydrocarbon Freeing Discretionary:

- Breaking containment activities on “small volume process components” when the risk presented by an inadvertent hydrocarbon release to atmosphere (with or without ignition) or ignition of the combustible mixture within the equipment can be reduced to ALARP.

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Example: “Small volume process components:” Small filters and pumps, flame arrestors, meters, instrument piping/tubing, small bore pig launcher/receiver and short runs of process piping when replacing spools, valves, corrosion probes, coupons or orifice plates.

- Where hydrocarbon freeing activities would present additional hazards and increased risk.

[CriticalE] Additional Precautions When NOT Hydrocarbon Freeing:

- SIMOPS restrictions for hydrocarbon source activities
- No flame or spark generating tools or equipment will be used at, or near the work site
- Gas testing for toxic and flammable conditions is conducted and required PPE is worn
- Fire and gas detection equipment in the vicinity **must** be in service or alternate mitigations are in place

7.5.5.2 GOAL

[CriticalE] When hydrocarbon freeing prior to breaking containment, the goal is to attain a hydrocarbon concentration of zero or, if this is not possible, less than 80% of LEL in an inert atmosphere (e.g. N₂).

! IMPORTANT: [CriticalE] A gas detector capable of detecting hydrocarbon in an inert atmosphere **must** be used when hydrocarbon freeing with an inert medium. [E] Gas detection devices should be equipped with a sensor for the type of hydrocarbons being measured (i.e. methane, propane, etc.) and **must** be calibrated using the manufacturer recommended calgas for the sensor.

! IMPORTANT: Atmospheric readings must be taken as close to the source as practicable to achieve the most accurate readings (i.e. Probe inside the vessel, direct from bleeder opening, etc.)

! IMPORTANT: [CriticalE] Hot Work activities **must** achieve zero percent hydrocarbons (0% of LEL).

Hydrocarbon freeing activities are conducted with inert mediums such as nitrogen, water or steam to avoid a flammable gas-air mixture when passing through the LEL-UEL range.

7.5.5.3 PREREQUISITES

[CriticalE] Prior to hydrocarbon freeing, the equipment and piping involved **must** be successfully isolated, drained, and depressurized.

7.5.5.4 METHODS

There are various methods that can be used to free equipment of hydrocarbons. They are as follows:

- Water flushing or flooding
- Inert gas purging with nitrogen (N₂) or carbon dioxide (CO₂)
- Steaming
- Hot water circulation with filtering

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7.5.5.5 PLANNING HYDROCARBON-FREEING ACTIVITIES

The following guidelines should be used to plan hydrocarbon-freeing activities:

- The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared. Job planning should consider the following:
 - The method/technique to be used for hydrocarbon freeing the equipment.
 - Special conditions that should be established before the hydrocarbon-freeing is commenced.
 - Step-by-step activities to be performed.
 - Facility critical operating parameters (pressures, temperatures, capacities) that should not be exceeded.
 - Any required temporary defeats of critical devices, and when critical devices may be reinstated.
 - The means to monitor the progress of the preparation and confirm its successful completion.
 - The fluids to be drained and a review of the appropriate Safety Data Sheets (SDS).
 - A hazard assessment of the activities and risk mitigation measures.
 - Spill containment and waste disposal that might be required.
 - Additional personal protective equipment (PPE) required.
 - Emergency or abnormal operating procedures.
- Obtain a Critical Work Permit for Breaking Containment for any activities that involve venting hydrocarbons to the atmosphere other than through approved drain, vent, or flare systems. A permit is not required for an AO conducting these activities as part of their normal duties.
- Confirm that the vessel or facility system is isolated, depressurized, and drained of process fluids.

7.5.5.6 HYDROCARBON-FREEING SAFETY PRECAUTIONS

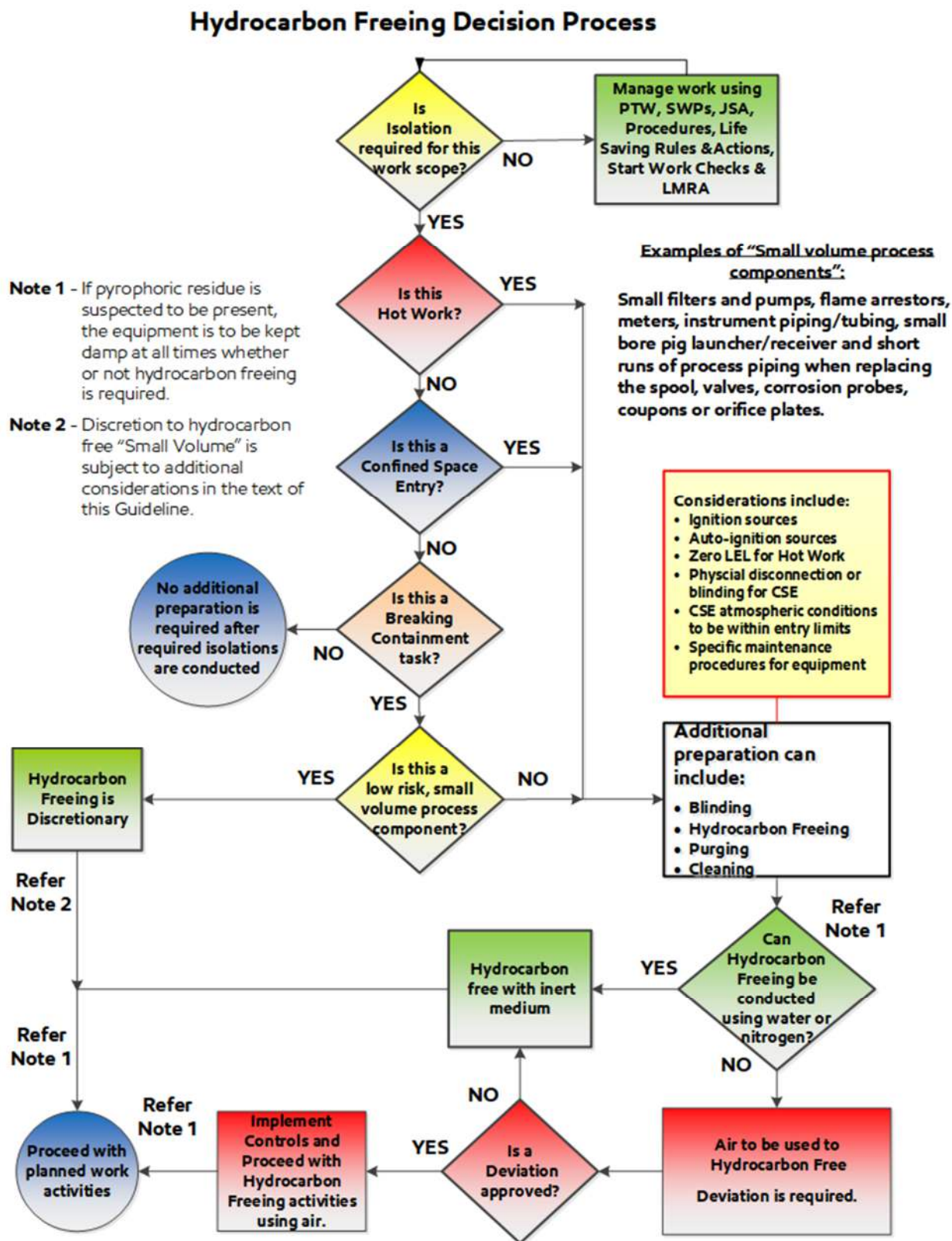
Use the following safety precautions when hydrocarbon-freeing systems:

- [E] Appropriate PPE **must** be worn to protect against hazards specific to flushing and purging (Refer to [Figure 17: Hydrocarbon Freeing Decision Process](#)). These include:
 - Increased eye protection (such as goggles or face mask) when bleeding down equipment.
 - **Breathing apparatus** - the conditions (confined area, where asphyxiation could occur) and state of the equipment to be broken (i.e. whether the equipment may contain toxic substances such as H₂S, hydrocarbons or benzene) will determine if breathing apparatus will be required.

! WARNING: Inert gas used for purging may not be toxic; however, appropriate precautions must be taken when bleeding down large volumes of inert gas, to prevent asphyxiation due to displacement of the oxygen.
- [CriticalE] In order to prevent auto-ignition, vessel internals **must** be kept wet if pyrophoric iron sulfides are suspected to be present and air will be introduced.
- The introduction of water into a higher-pressure or volatile hydrocarbon system can result in the formation of hydrates and should be avoided. The use of a hydrate inhibitor should be considered if water is to be used.
- Do not use excessive force if purge or drain valves are tight. Lubricant often frees up valve stems and bushings.
- Any water introduced to vessels or systems should be treated with corrosion inhibitor to prevent future corrosion of the system.

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Figure 17: Hydrocarbon Freeing Decision Process



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7.5.5.7 WATER FLUSHING PROCESS

Certain considerations should be followed when flushing vessels or facility systems with water before breaking containment. Refer to [Table 18: General Flushing Process](#).

Table 18: General Flushing Process

Task	Action
1	Before starting to fill with water, a review of the process should be made to determine: - Whether water can be used (for example, any seals in the system) - The level to which water can be filled: - Confirm that the structural design of the process equipment and its foundation are able to withstand the extra weight of the water column. - Confirm that the system to be flushed is not over pressured due to the height of the water column. - Any high point vents that need to be opened or any drain systems to be used when draining the system to avoid a vacuum.
2	Water and detergent should be added to the base of the vessel or tank to be flushed. The preferred method for displacing hydrocarbons with water is to use a portable or fixed pump fitted with a break tank. [CriticalE] If a hose is used, the following precautions must be taken: - A break tank must be used to prevent contaminated water backing up into the water supply system. - The hose must be secured to the fill valve to restrain the hose if the connection were to fail. - Hoses must be electrically bonded. ▲ CAUTION: [CriticalE] Never use hoses to connect utility or fire systems directly to vessels or pipework, to avoid the potential of hydrocarbons entering these safety systems.
3	The level in the vessel is monitored with a level or pressure gauge.
4	Pressure is vented from the top of the vessel to a flare or vent system.
5	After the equipment is filled, drain the water to the closed drain system or other appropriate location. ▲ CAUTION: Water used to flush a vessel or tank will be contaminated. Waste collection and disposal methods should be considered before commencement of the draining phase of the process. ▲ CAUTION: Avoid overfilling the closed drain system. ▲ CAUTION: [E] Ensure there is sufficient venting of the system to avoid under pressure or a vacuum while draining. This venting should not be allowed to pull air into the system unless all hydrocarbons have been previously displaced. [CriticalE] Nitrogen (N₂) or another inert gas must be used to displace any hydrocarbon vapors near the top of the vessel. Nitrogen should also be used to maintain positive pressure while the system or vessel is being drained. ! WARNING: Water flooding may not remove sludge or oils trapped in pipes or vessel internals. [CriticalE] Flooding with water must not be relied upon to remove all hydrocarbon traces.

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7.5.5.8 INERT GAS PURGING

Purging replaces hydrocarbon gases with an inert gas such as nitrogen (N₂) or carbon dioxide (CO₂), or a combination of both.

! WARNING: [CriticalE] When purging with inert gas, provision **must** be made to prevent the maximum allowable working pressure (MAWP) of the equipment or system from being exceeded. [CriticalE] Relief capability **must** be in place in case the nitrogen pressure regulator or compressor controls fail. [E] Temporary hoses **must** be tethered and rated for pressures that they could be exposed to if purging equipment controls fail (relief pressure).

There are two basic methods of purging with inert gases, as follows:

- **Purging by dilution:** The system or vessel is pressurized with inert gas. The inert gas and hydrocarbon mixture are then vented to a safe location. This process may be repeated several times, diluting the mixture further each time, until the vented gas is zero or, if this is not possible, less than 80% of LEL in an inert atmosphere.
- **Purging by displacement:** The system or vessel is purged by a flow through of nitrogen or other inert gas. High and low point vent and drain valves are opened to confirm effective purging, especially of dead legs.

The inert gas should be injected in the top of the system when it is lighter than the hydrocarbons contained in the system and it should be injected in the bottom of the system when the inert gas is heavier than the hydrocarbons contained in the system.

! WARNING: Purging by displacement may allow pockets of hydrocarbons to remain in vessels or pipework although the vented gas may indicate it is less than 80% of LEL volume in an inert atmosphere.

Nitrogen or inert gas purging can be effective for hydrocarbon freeing vessels and tanks in clean service (for example, LPG, LNG, light ends, fuel gas, or condensate).

The following general process should be followed when purging vessels or facility systems with nitrogen (N₂) or other inert gases before breaking containment or when performing hot work on pipelines or process piping.

! WARNING: [CriticalE] Hydrocarbon freeing with air **must not** be conducted because a gas-air mixture is flammable when transitioning through the LEL–UEL range. An incident could occur if there is a spark during this period due to static (purge velocity), movement of internal components, faulty internal electrical equipment, or if FeS deposits auto-ignite.

- **Purging by dilution method**

This method is preferred when purging large systems or vessels because it requires the usage of less inert gas and is more efficient at removing hydrocarbons, especially when the vessel being purged contains baffles, dead legs, or level bridles. Refer to [Table 19: Purge by Dilution Process](#).

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Table 19: Purge by Dilution Process

Task	Action
1	[E] The N ₂ or inert gas hose must not be connected to the system before draining and de-pressuring the vessel or system.
2	The system or vessel to be purged is properly isolated, drained, and depressurized. Reference: "General flushing process" described previously in this section.
3	Connect the inert gas hose to a low point drain.
4	Pressurize the system or vessel but do not exceed the maximum allowable working pressure (MAWP). ▲ CAUTION: The MAWP of any system or vessel should be no higher than 95% of the set point pressure of the PSVs.
5	Vent the inert gas and hydrocarbon vapors to a safe location as quickly as possible; that is, through the largest vent valve available. ! WARNING: If venting to atmosphere, consider restricting access to the area where inert gas is being vented to prevent asphyxiation. ▲ CAUTION: [CriticalE] When venting hydrocarbons, flammable, or toxic substances to atmosphere as part of an energy isolation, all vent points must be directed to a safe location. For these purposes a safe location is defined as an area where venting can occur that minimizes exposure to personnel and potential ignition sources. Single valve isolation planning should consider potential system back pressure risk when leaving vent lines open to systems above ambient pressure. Safe locations include but are not limited to; ▪ Flare systems ▪ Atmospheric vents and drains ▪ Sumps and tanks ▪ Locations away from buildings, or congested / restricted areas [CriticalE] A method must be in place to monitor open ended vent points for the presence of liquids or vapors that could create a hazardous condition. [CriticalE] Open-ended Hydrocarbon, Flammable, or Toxic Vents must be closed if unattended or not being monitored.
6	Test the hydrocarbon (HC) content of the gases as they are being vented. [CriticalE] The pressure and pop (pressure/de-pressure dilution) method must be repeated until the HC concentration is less than 80% of LEL in the inert atmosphere. ▲ CAUTION: Some HC monitoring devices do not measure accurately in an oxygen-deficient atmosphere; specialized equipment may be required.
7	After purging is complete, it is important to confirm that hydrocarbons or other combustible materials are prevented from re-entering the purged equipment.

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▪ Purging by displacement method

This method may be preferred in some situations when the desired pressure cannot be attained with the inert gas or when performing hot work. For example, a slow continuous nitrogen purge can be advantageous when welding on a pipeline. Refer to [Table 20: Purge by Displacement Process](#)

Table 20: Purge by Displacement Process

Task	Action
1	[E] The N ₂ or inert gas hose must not be connected to the system before draining and depressuring the vessel, facility system, or pipeline.
2	Connect the inert gas to a low drain point and begin N₂ injection. ! WARNING: Purging by displacement may allow pockets of hydrocarbons to remain in the vessels or pipework although the vented gas may show less than 80% of LEL in an inert atmosphere. ! WARNING: Consider restricting access to the area where inert gas is being vented to prevent asphyxiation. When flushing a large system, connect a vent hose to a high-point vent or open vent valve to flare in the isolated system as far away as possible from the inlet injection point.
3	Excessive purge rates should be avoided to prevent localized freezing of valves, gauges, and other equipment. ! IMPORTANT: Monitor the pressure to avoid unintended pressure buildup due to leaks from associated live systems. ! WARNING: [CriticalE] Where entry into a confined space is planned, the inert gas must be subsequently purged from the space with air to meet the confined space entry atmospheric limits/thresholds, even if breathing apparatus is to be used. Reference: Section 6.4: "Confined Space Entry"
4	Test vented gas with a certified instrument designed to read HC content in an oxygen-deficient atmosphere, to confirm that the content of HC is less than 80% of LEL. ▲ CAUTION: Some flammable vapor monitoring devices do not measure accurately in an oxygen-deficient atmosphere; specialized equipment can be required.
5	When purging is complete, confirm that hydrocarbons and other combustible materials are prevented from re-entering the purged equipment.

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7.5.5.9 STEAMING PROCESS

Steaming is the most effective means of removing residual hydrocarbons from vessels in crude or other heavy hydrocarbon service.

The facility system or vessel is purged with steam from the facility's own steam heat medium system or from a portable steam generator. High and low point vent and drain valves are opened to confirm effective purging of residual hydrocarbons, especially from dead legs and level bridles. Refer to [Table 21: Steaming Process](#).

Table 21: Steaming Process

Task	Action
1	[CriticalE] The nozzle of the steam hose must be bonded to the facility system or vessel and grounded (earthed) to decrease or prevent the generation of static electricity.
2	[E] The equipment operator must be in attendance at the steam generator at all times. ! WARNING: During steaming, the equipment operator must attend the steam generator at all times. In the event of the steam generator shutting down, adequate vents should be opened in the vessel immediately to avoid forming a vacuum as the system cools.
3	[CriticalE] If oxygen is present in the facility system or vessel, steam temperatures must not exceed the auto-ignition temperature of the previously stored hydrocarbons.
4	▲ CAUTION: Consider coating and insulation properties when steaming vessels and systems—for example, where the vessel is insulated with polyurethane foam, the temperature must not exceed 176°F (80°C) to prevent damage to the insulation.
5	When a vessel is being steamed for confined space entry, the temperature of the confined space must be cooled to an acceptable thermal environment prior to entry. Reference: Section 6.4: "Confined Space Entry"
6	[E] After steaming, process equipment must be adequately vented during cool-down to prevent the formation of a vacuum on the vessel being steamed.

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7.5.5.10 STEAMING STRIPS

To confirm adequate purging or steaming of equipment, blinds or spades are sometimes vented on the isolated equipment side to confirm that dead legs, created by blinding, are adequately purged or steamed. Steaming strips (standoffs) or grooved blinds are used to achieve this.

Steaming strips should not be used at locations where pressure can build up behind the blinds, because they may not allow the replacement of all flange stud bolts and even tensioning of stud bolts. The preferred option in these situations is to fit line-rated, grooved blinds or spades, specifically designed and fabricated for this purpose.

The locations of the steaming strips should be recorded on an ICC.

Reference: [Section 7.5.4: "Blinding"](#)

NOTE: Steaming strips should only be made of soft metal such as aluminum to avoid damage to the gasket surfaces of flange faces.

! WARNING: [CriticalE] Steaming strips **must** be removed, gaskets installed, and bolts tensioned before the start of confined space entry, to exclude the potential for toxic vapors to enter the space through these gaps.

▲ CAUTION: When using this method for cleaning a vessel or system, a vent of adequate size must remain open on the vessel to prevent the formation of a vacuum, if the steaming unit shuts down and the system cools.

7.5.5.11 HOT WATER CIRCULATION PROCESS

The facility system or vessel is filled with a mixture of water and detergent. The unit takes suction from the vessel/system and discharges back into the same vessel/system. Refer to [Table 22: Hot Water Circulation Process](#).

The soapy water mixture is continually heated and filtered to remove the hydrocarbon solids until the system or vessel is clean.

This method is preferred when preparing a vessel for entry that is known to have considerable quantities of solids mixed with hydrocarbons (for example, a three-phase production separator in a field that produces from a sandy formation or containing heavy oil).

NOTE: The unit typically requires dual filter installations to allow for filter cleaning due to the large amounts of hydrocarbon solids being removed from the vessel.

Reference: [Section 6.4: "Confined Space Entry"](#)

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Table 22: Hot Water Circulation Process

Task	Action
1	[CriticalE] The suction and discharge hoses of the hot water unit must be bonded to the facility system or vessel and grounded (earthed) to decrease or prevent the generation of static electricity.
2	[CriticalE] If oxygen is present in the facility system or vessel, water temperatures must not exceed the auto-ignition temperature of previously stored hydrocarbons. ▲ CAUTION: [E] When using this method for cleaning a vessel or system, a vent of adequate size must remain open on the vessel to prevent the formation of a vacuum, if the hot water unit shuts down and the system cools. ▲ CAUTION: Consider coating and insulation properties when heating vessels and systems with hot water—for example, where the vessel is insulated with polyurethane foam, the temperature must not exceed 176°F (80°C) to prevent damage to the insulation.
3	When a vessel is being cleaned for confined space entry, the temperature of the confined space must be cooled to an acceptable thermal environment prior to entry. Reference: Section 6.4: "Confined Space Entry"
4	Waste generated as a result of cleaning is collected, labeled, and disposed of as required by the Operating Site's <i>Waste Management Practices</i> .

7.5.6 CLEANING

7.5.6.1 INTRODUCTION

This section provides guidelines to protect personnel and equipment when cleaning hydrocarbon-containing equipment.

Some flushing and purging may have been performed before breaking containment. However, residual hydrocarbons can remain that must be removed by cleaning.

7.5.6.2 METHODS OF HYDROCARBON CLEANING

Methods of cleaning residual hydrocarbons are:

- Hydro blasting (normally requires vessel entry)
- Mechanical cleaning (normally requires vessel entry)

The exact method used depends on:

- The configuration of the equipment
- The nature of the fluids

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7.5.6.3 GOAL

The goal of cleaning is to achieve a hydrocarbon concentration of zero or, if this is not possible, less than 80% of LEL.

! WARNING: [CriticalE] If the cleaning requires vessel entry, confined space entry requirements regarding atmospheric limits **must** be followed.

Reference: Section 6.4: "Confined Space Entry"

7.5.6.4 PREREQUISITES

[CriticalE] Prior to cleaning, the equipment and piping involved **must** be successfully isolated, depressurized, drained, and hydrocarbon freed (to the extent possible).

References:

- Section 7.5.1: "Energy Isolation and Isolation Control Certificate"
- Section 7.5.2: "Draining and Depressurizing"
- Section 7.5.5: "Hydrocarbon Freeing"

! WARNING: Cleaning occurs after breaking containment when an oxygen atmosphere could coexist with hydrocarbons. Appropriate precautions must be taken.

7.5.6.5 PLANNING CLEANING ACTIVITIES

The following guidelines should be used to plan cleaning activities.

- The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared.
- Job planning should consider the following:
 - Special conditions that should be established before the process is commenced.
 - Step-by-step activities to be carried out.
 - Facility critical operating parameters (pressures, temperatures, capacities) that should not be exceeded.
 - Any required temporary defeats of critical devices, and when critical devices may be reinstated.
 - The means to monitor the progress of the preparation and confirm its successful completion.
 - Fluids to be vented and a review of the appropriate Safety Data Sheets (SDS).
 - A hazard assessment of the activities to identify risk mitigation measures.
 - Spill containment or waste disposal techniques that might be required.
 - Additional personal protective equipment (PPE) required.
 - Emergency or abnormal operating procedures.
- Obtain the required work permit for the cleaning work to be done.
- Hydrocarbons can then be removed from vessels and process systems by any one, or by a combination, of the following techniques:
 - Hydroblasting
 - Mechanical cleaning

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7.5.6.6 DEAD LEGS

[CriticalE] Dead legs where liquid hydrocarbons can collect **must** be considered when cleaning facility systems. Dead legs are created by:

- Weirs and standpipes in vessels
- Instrument bridles and sight glasses
- Low points in pipework

Dead legs that cannot be drained or cleaned should be isolated, if practicable.

7.5.6.7 HYDROBLASTING

Hydroblasting is required for cases where hydrocarbon deposits need mechanical force to be removed.

[E] The following steps should be followed when hydroblasting:

- The system **must** be isolated, drained, and depressurized.
- The high-pressure water cleaning system **must** be identified (contractor or facility equipment).
- During hydroblasting, the vessel should be continuously drained. The hydroblasting equipment **must** be staged in a manner that will not interfere with the draining process.

! WARNING: Hydroblasting of hydrocarbon deposits can create a toxic level of hydrocarbon vapors.

7.5.6.8 CLEANING SAFETY PRECAUTIONS FOR HYDROBLASTING

[E] Use the following safety precautions when hydroblasting equipment:

- Appropriate PPE **must** be worn to protect against hazards specific to hydroblasting.
- Hydroblasting lines and fittings **must** be properly grounded (earthed) to prevent static electricity generation.
- All potential spills **must** be banded or otherwise contained.
- Care **must** be taken not to exceed the maximum temperature of vessels. Temperature limits may exist because of:
 - Nature of the internals or any catalyst materials
 - Expansion of pipework and resultant stresses
 - Insulation materials
 - Vessel metallurgy
- [CriticalE] Steam temperatures **must not** exceed the auto-ignition temperature of process fluids, if oxygen is present.

7.5.6.9 MECHANICAL CLEANING

It is sometimes necessary to enter process equipment to remove sludge or scale to completely free the equipment of hydrocarbons. Mechanical cleaning methods include scraping, brushing, vacuuming, and so on.

! IMPORTANT: Entering a vessel is not the preferred method of cleaning. It should only be done when other methods are impractical or ineffective.

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The following steps should be followed when mechanically cleaning:

- Plan the method for removing the cleaning medium and cleaning debris from the vessel. Do not allow excessive amounts of debris to accumulate within the vessel.
- Plan and follow all of the precautions required for confined space entry.

Reference: Section 6.4: "Confined Space Entry"

! WARNING: When cleaning hydrocarbons from a confined space, continuous hydrocarbon gas monitoring is essential.

! WARNING: Hydrocarbon vapors can be released when sludge is disturbed or heated. Sludge containing heavy hydrocarbons (heavier than kerosene) can release ignitable vapors that may not be detected by gas detectors.

- Plan and follow all precautions required for the cleaning method(s) to be used.
- **! WARNING:** [CriticalE] Static electricity hazards **must** be eliminated when mechanically cleaning vessels.
- Plan and follow all precautions for naturally occurring radioactive material (NORM) and pyrophoric scale.
- The vessel must be continuously drained and ventilated.

7.6 REINSTATING FACILITY SYSTEMS AND EQUIPMENT

7.6.1 INTRODUCTION

This section provides guidelines to be followed when reinstating equipment and facilities and defines personnel responsibilities. The general sequence of events, when reinstating equipment and facilities, is as follows:

- Recommissioning

NOTE: Recommissioning is an activity that takes place throughout the reinstatement. [CriticalE] Mechanical completion checks **must** be made before and after equipment testing, air freeing, leak testing, and de-isolating.
- Equipment testing
- Air freeing
- Leak and pressure testing
- Removing energy isolations (mechanical, electrical, instrument)
- Removing temporary defeats
- Acceptance testing

! IMPORTANT: Each job should have its own reinstatement plan that includes detailed activity instructions, the order in which activities are to be performed, and any special precautions that should be taken during each step.

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7.6.1.1 PLANNING REINSTATEMENT ACTIVITIES

The following guidelines should be used to plan reinstatement activities.

- The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared. Job planning should consider the following:
 - Risk Assessment
 - Emergency procedures/process
 - Barrier requirements
 - Protection of sensitive equipment
 - Order of the steps required to reinstate equipment
 - Appropriate PPE for each step
 - Checklist or procedure for equipment status at each step
 - Marked-up piping and instrumentation drawing (P&IDs) showing existing isolations, drain, fill, and pressure testing points.
- Identify any required equipment testing.
 - ! IMPORTANT:** This can involve temporarily removing some isolation points for the test. After the test, a checklist or ICC should be used to re-isolate as required. Purge air from hydrocarbon systems with water or an inert gas such as nitrogen (N₂) or carbon dioxide (CO₂).
- Leak and pressure test equipment or facilities where pressure integrity was broken.
 - ! IMPORTANT:** Recommissioning checks should take place again before leak and pressure testing the facility system.
- Remove mechanical isolations and the locks and tags put in place for this work only.
 - ! IMPORTANT:** Recommissioning checks should take place again before restarting the facility system.
- Perform any required acceptance test of new or repaired equipment.
- Remove the temporary defeats of critical devices.

7.6.2 RECOMMISSIONING

7.6.2.1 INTRODUCTION

Recommissioning is the process of confirming the fit-for-purpose state of the system before it is air freed, leak tested, and put back into service.

7.6.2.2 PLANNING COMMISSIONING ACTIVITIES

The following guidelines should be used to plan recommissioning activities.

- Job planning should consider the following:
 - Scope and nature of the facility system worked on
 - The P&IDs (that were confirmed to be accurate by walking the lines) and the electrical diagrams

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- The need for communications to be established with other activities, or any other activity interfaces
- The de-isolation and reinstatement of equipment
- Step-by-step activities to be carried out
- A mechanical completion checklist

! IMPORTANT: Separate plans for performing recommissioning checks should be prepared as follows:

- Before purging
- Before leak and pressure testing
- Before putting the facility or equipment back into service.

7.6.2.3 POTENTIAL RECOMMISSIONING HAZARDS

Time should be taken to inspect the facilities and equipment. The following hazards may be present:

- Faulty valves and gauges (that were worked on and not function-tested)
- Valves that are in the wrong position
- Blocked bleed lines or drains
- Faulty gauges.
- Missing or improperly torqued flange bolts.
- Missing gaskets or plugs.
- Valves or drains open to atmosphere or drain.
- Small bore piping or analyzer inlet/outlet that have not reconnected.
- Instrument sensing lines that are not recommissioned.
- Control or safety systems not put back in service.
- Level glasses not being returned to service with the valves fully open (so that the blowout preventers can work if the glass breaks).
- Electrical connections that are not reestablished.
- Physical guards for rotating equipment that are not replaced.
- Incorrect, incompatible, or improperly pressure-rated materials used.

7.6.2.4 INTRODUCING PRESSURE

Take the following precautions when introducing pressure back into the facility system:

- Introduce pressure slowly to prevent pressure or temperature shocks to the system.
- Check all flanges that have been disturbed, manways, pump or compressor seals, or other fittings for leaks as pressure is increased, or as temperature is raised or lowered.
- Install a plug or blind on valves that are single valve isolations between the hydrocarbon process and atmosphere.

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- [E] Compressor and pump seal oil systems **must** be operating when any pressure is introduced. The discharge pressures should be increased slowly once the machine is running and leak checks conducted on the discharge piping.
- Exchangers and reboilers do not reach operating temperatures until they are in service. The temperature should be increased slowly while checking for leaks.

7.6.2.5 CRITICAL DEVICES

[CriticalE] If work was performed on any associated critical devices or systems while equipment was out of service, the critical devices or systems **must** be function-tested when the equipment is returned to service to confirm correct operation.

7.6.2.6 REMOVING BLINDS AND SPADES SAFETY PRECAUTIONS

Blinds and spades may have pressurized fluids that have built up behind them due to leaking isolation valves. [CriticalE] When removing blinds, the equipment and piping involved **must** be isolated, drained and depressurized. Remove blinds and spades using the appropriate breaking containment and blinding precautions. Consider the following safety precautions when removing blinds and spades:

- If possible, bleed any potential pressure through drain valves or vented spades.
- [E] Appropriate PPE **must** be worn to protect against hazards specific to draining and depressurizing. These include:
 - Increased eye protection (for example, goggles or face mask) when breaking containment.
 - The conditions (confined area, where asphyxiation could occur) and state of the equipment to be broken (i.e. whether the equipment may contain toxic substances such as H₂S, hydrocarbons or benzene) will determine if breathing apparatus will be required.
- Generally, de-blinding should be done in the following order:
 - Vessel or tank manways to be restored
 - Pressure safety valve (PSV) blinds
 - Minor line blinds
 - Secondary line blinds
 - Major line blinds
- [CriticalE] The Area Operator **must** confirm (witness) that all blinds are removed and check them off on the Blinds List.
- All potential spills must be banded or otherwise contained.
- [CriticalE] When blinds or spades are removed and the new joint is being reinstated:
 - New gaskets **must** be installed.
 - All flange bolts **must** be installed and tightened to the correct torque.
 - Joint complete tape **must** be wrapped around the flange by the person who completes the flange makeup.

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! IMPORTANT: To minimize future corrosion issues, the joint complete tape must be removed after leak testing the flange.

7.6.3 EQUIPMENT TESTING

7.6.3.1 INTRODUCTION

Sometimes equipment requires a test run to confirm the correct rotation of motors, seal containment, or engine operation before permits can be signed off as complete. This can require a full or partial removal of energy isolations.

NOTE: [CriticalE] When equipment testing requires the introduction of hydrocarbons, equipment testing **must** be performed after air freeing.

7.6.3.2 PLANNING EQUIPMENT TESTING ACTIVITIES

The following guidelines should be used to plan equipment testing activities.

The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared.

Job planning should consider the following:

- Step-by-step activities to be performed, including:
 - Required de-isolations (process, mechanical, and electrical) and their order of execution.
 - Determination of how a temporary partial or full de-isolation could affect other work groups working in the area or system (Sanction to Test).
 - Required suspension of temporary defeats of critical devices, and when critical devices may be reinstated.
 - Requirements for reinstating utility systems required for the test.
 - Means of introducing energy (process fluids and energy sources to prime movers).
 - How the equipment is controlled.
 - Test goals and parameters.
 - Means of re-isolating the system after the test.
- Facility critical operating parameters (pressures, temperatures, capacities) that should not be exceeded.
- The means to monitor the progress of the operations and confirm its successful completion.
- Identification of any fluids involved and a review of the appropriate Safety Data Sheets (SDS).
- A Job Safety Analysis if required or Risk Assessment for the de-isolations and for the equipment testing.
- Any personal protective equipment (PPE) required.
- Emergency or abnormal operating procedures.

7.6.3.3 SANCTION TO TEST

The following general practice regarding equipment testing should be followed:

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- The Isolating Authority (IA) should request the Sanction to Test (STT) and develop a plan for the points to be de-isolated during testing. [CriticalE] The Area Authority (AA) or delegate suspends all work permits affected by the STT (permits cross-referenced to the ICC).
- The AA reviews the plan and de-isolation points, and authorizes the sanction to test to progress. The IA performs the de-isolations, and reinstates or removes the required temporary defeats.
- The IA updates the isolation certificate to reflect the de-isolated points.
- The AA authorizes the permit for testing.
- The AO issues the permit for testing.
- The PH accepts the permit for testing. The permit is then **Live for Sanction To Test**.
- The equipment testing may then begin, including the introduction of energy, according to the written equipment test, if available.
- Upon completion of the test, the equipment is then re-isolated if further work is required, or if other permits require the isolations to be reinstated.
- Remaining isolated points can be de-isolated if the test was successful and there are no other permits requiring the isolations.
- **! IMPORTANT:** If the test was unsuccessful and further work on the equipment is required, isolation activities (for example, hydrocarbon freeing) may be required before the equipment can be worked on.
- The Area Operator then updates the ICC.
- The Area Operator may then reissue any work permits suspended for the equipment testing when it is safe to do so.

7.6.4 LEAK AND PRESSURE TESTING

7.6.4.1 INTRODUCTION

This section provides guidelines to protect personnel and equipment when performing leak testing of facility systems before putting them back into service.

7.6.4.2 GOAL

Leak and pressure testing confirm that the integrity of flanges, seals, glands etc., is restored after intrusive work. The goal is the prevention of leaks where flammable or toxic fluids can affect personnel, create an ignitable atmosphere or cause environmental damage.

Leak and pressure testing prior to return to service is not required for low risk or small volume systems and equipment:

- a. Low risk – non-hazardous, non-toxic services other than hydrocarbon service, where an inadvertent release presents no credible SHE risks e.g. potable water service, instrument air service etc.
- b. Small volume – the amount of hydrocarbon or toxic product that could be released when returning to service presents minimal risk of asphyxiation, toxicity or ignition. Additional mitigating measures must be implemented to reduce risk to ALARP (refer to decision process diagram).

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Example: “Small volume process components”: Small filters and pumps, flame arrestors, meters, instrument piping/tubing, small bore pig launcher/receiver and short runs of small-bore process piping when replacing the spool, valves, corrosion probes, coupons or orifice plates.

7.6.4.3 PRE-REQUISITES

Prior to testing, all applicable recommissioning checks must be completed, e.g. mechanical completions check, joint complete tape etc.

Reference: Section 7.6.2: "Recommissioning"

Prior to testing, appropriate test pressures and durations need to be determined.

7.6.4.4 TESTING GUIDANCE

▪ Leak test:

A leak test is an initial low-pressure test to identify obvious system integrity deficiencies e.g. missing gaskets or plugs, open vents or drains, missing or loose flange bolts, disconnected analyzer or instrument tapping points, etc.

Leak testing should be conducted at a low pressure, establish a leak test duration appropriate to the volume/complexity of the system/equipment being tested.

Pressure should be introduced slowly to avoid a large release of the pressurizing medium. Hold at test pressure and monitor for any fall in pressure, check all disturbed flanges and fittings.

▪ Pressure test (tightness test):

Pressure testing establishes that the system integrity is restored to normal operating pressure and any deficiencies are identified e.g. leakage at flanges and joints, valve packing, screwed connections and instruments etc.

Pressure testing is conducted to the normal operating pressure anticipated in that part of the system and should not exceed 90% of the pressure relief valve set point or 90% of the MAWP of the system.

Establish a pressure test duration appropriate to the volume/complexity of the system/equipment being tested. Pressure should be introduced slowly to avoid a large release of the pressurizing medium. Hold test pressure and monitor for any decrease in pressure, check all disturbed flanges and fittings.

NOTE: [CriticalE] In instances where the potential for equipment damage during pressure testing exists (e.g. seal damage/failure on a compressor or pump), the test pressure **must not** exceed equipment tolerances (e.g. suction pressure).

▪ Hydrostatic pressure test (proof test):

Conducted to establish system integrity during construction or re-establish integrity after facility modifications. Tested up to and exceeding MAWP (1.1 to 1.5 times) using water.

7.6.4.5 TEST MEDIUM

In all cases an inert medium (e.g. Nitrogen or water) is the preferred test medium. Where an inert medium is not available/practicable, hydrocarbons may be used for testing. [CriticalE] When conducting

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leak or pressure tests with hydrocarbon, additional mitigating measures **must** be implemented to reduce risk to ALARP as outlined in the decision process diagram.

- Factors that may influence “availability/practicality” of using an inert medium are:
 - Excessive delays to restart of facilities (typically unplanned circumstances).
 - Excessive cost due to remote location or volume required (with contingency for repeat tests)
 - Portable facilities not readily available for large volume.
 - Increased or additional hazards introduced by transporting or storing liquid nitrogen on a facility (especially on offshore steel structures).
 - Pressures required cannot be provided by medium

7.6.4.6 TESTING WITH LIQUIDS

The preferred liquid medium is water; for example, treated water. However, the effect of water on the metallurgy of the system or the effect of any residual water or additives on the facility equipment must be considered.

[E] At an ambient temperature below 41°F (5°C), antifreeze **must** be added when water is being used.

[E] The system to be tested with liquid **must** be equipped with valves to vent air pockets.

The selection of liquids other than water should take into account the following considerations:

- The boiling point relative to the test temperature.
- The flammability of the liquid, the flash point of which should be no less than 150°F (65°C) and at least 18°F (10°C) above the maximum test temperature.

[CriticalE] Use of hydrocarbon, flammable or toxic liquids **must** be Risk Assessed, endorsed at the appropriate level and managed with a procedure.

7.6.4.7 TESTING WITH AIR OR NITROGEN

Equipment may be tested with air or nitrogen for the following reasons:

- The equipment support structures cannot support the full weight of a liquid test (liquid may be used with an air or inert gas blanket on top of the partially filled equipment).
- The facility system to be tested normally contains gas, so a gas test is desired.

NOTE: When testing with nitrogen, a trace gas may be added which is better able than pressure monitoring to detect small leaks in large systems.

! WARNING: [CriticalE] Air shall not be used if hydrocarbons could be present in the system to be tested.

▲ CAUTION: High pressure testing with air or nitrogen adds a larger amount of total energy to the system.

▲ CAUTION: Nitrogen-testing large systems usually requires a large amount of liquefied nitrogen to be brought to the facility. Appropriate precautions for the handling of cryogenic liquids should be taken.

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7.6.4.8 SIMULTANEOUS LEAK AND PRESSURE TESTING

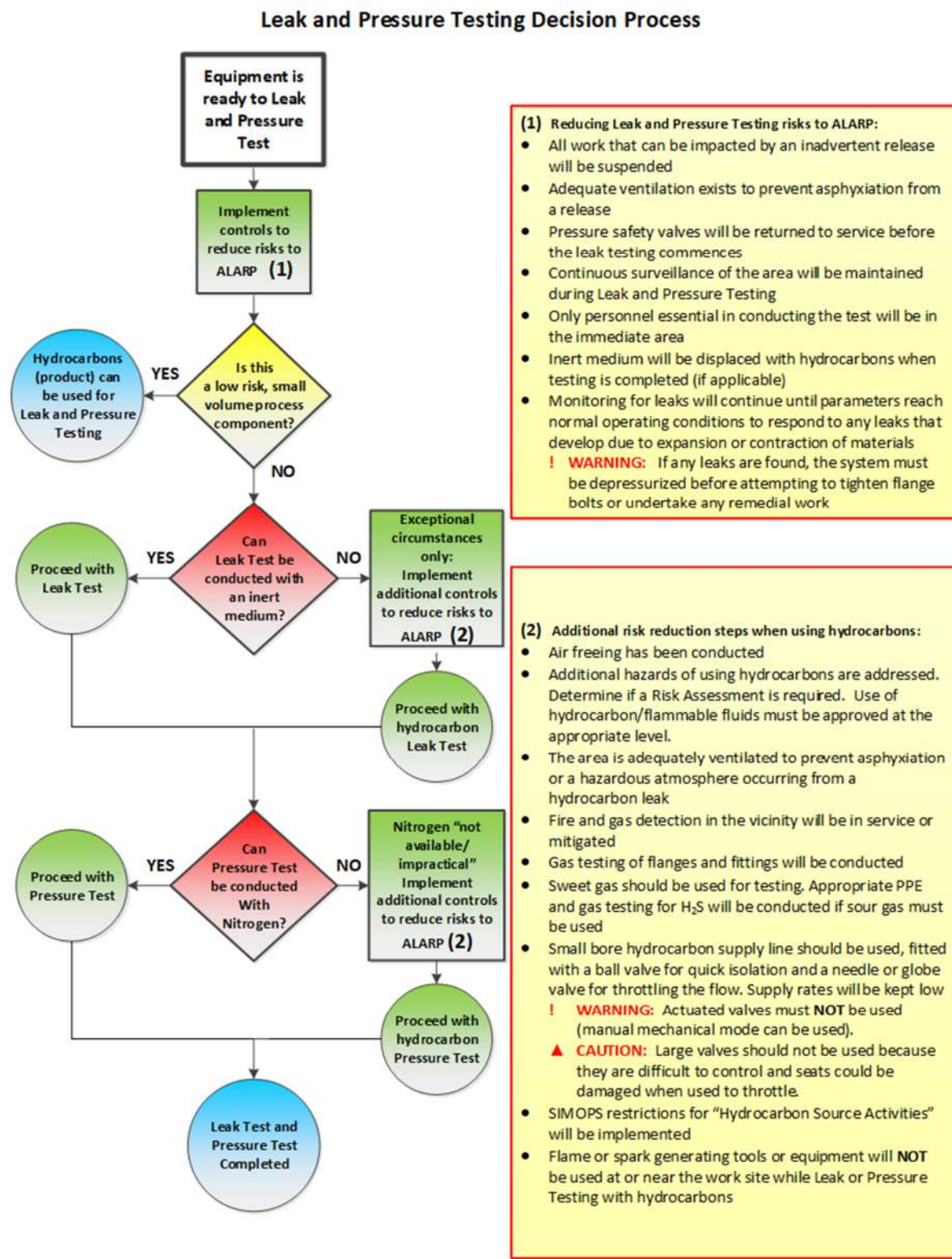
[E] Testing **must** be carried out only on a single, continuous pressure system. [E] Where it is necessary to simultaneously test two independent and non-connected systems adjacent to each other, careful consideration **must** be given to the effects of failure that one pressure system may have on the other. Refer to **Figure 18**: Leak and Pressure Testing Decision Process.

The following should be part of the pressure testing process:

- Any part that may be adversely affected by the test pressure inside the proposed system.
- The stability of components such as expansion joints and spring hangers.
- The ability to control the rate of pressure rise and the ability to depressurize quickly.
- Any interfaces with lower pressure systems or equipment (including exchangers). [CriticalE]
Provisions **must** be made to confirm that adjacent systems cannot be over pressured.

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Figure 18: Leak and Pressure Testing Decision Process



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7.6.4.9 PROCESS

The following general process should be used to determine requirements when leak and pressure testing facility systems after recommissioning. Refer to [Table 23: Testing Process](#).

Table 23: Testing Process

Task	Action
1	The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared. Job planning should consider the following: ▪ Marked-up P&IDs showing: – System boundaries – Specification change locations – Locations where the testing medium is to be introduced – Locations where the test pressure is to be monitored – Locations where the test pressure is to be bled off ▪ Marked-up plot plans showing: – Locations of equipment and pipework where leak/pressure testing will occur – Affected access and escape routes and alternate routes to be established – Location of testing equipment – Location of other unrelated work ▪ Any special conditions that should exist before leak/pressure testing is commenced. ▪ Confirmation that the vessel or equipment support structure is capable of supporting the full weight of the test medium. ▪ Step-by-step activities to be performed. ▪ Determine the appropriate Pressure Test duration. ▪ Any required temporary defeats of critical devices, and when critical devices may be reinstated. ▪ A hazard assessment of the activities, and risk mitigation measures. ▪ Any special spill containment and waste disposal equipment that could be required should leaks occur. ▪ Any personal protective equipment (PPE) required. ▪ Emergency procedures/process.

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Table 23: Testing Process, Continued

Task	Action
2	Instruct all personnel in their normal and emergency duties regarding the leak/pressure test.
3	[CriticalE] Prepare the system for leak/pressure testing by confirming the following: - Locations where containment was broken are wrapped with joint complete tape. - is above its minimum design metal temperature (MDMT) (cold-service equipment). - No cross-referenced work permits are live. - All bleed, drain, and vent valves have been closed prior to introducing pressure. ! IMPORTANT: All single valve isolations to the atmosphere must be plugged when not in use. - All sight glasses are returned to service with the valves reset fully open so that the blowout preventers can work if the sight glass fails. - Heat trace systems are recommissioned. - Instrumentation is returned to service. - Shutdown and protective equipment (for example, fire and gas detectors, high and low-pressure protection) are recommissioned. - Any lower-pressure lines or systems are isolated. - Required spill containment equipment is in place, especially if hydrocarbons will be introduced to the facility system during the leak/pressure test. - PSVs are recommissioned. If PSVs were replaced, confirm they are the correct size and pressure setting. ! IMPORTANT: Always open the downstream (flare/vent side) PSV valve first, due to the lower flange rating, then open the upstream (process side) valve to prevent PSV failure if the valve relieves immediately.
4	Establish site precautions as follows: - Cordon off the area (with caution or danger tape) at a safe distance from the equipment to be tested. - Post standby personnel to monitor the leak test from a safe distance as required. [CriticalE] All personnel not associated with the leak testing must stay outside of the cordoned-off area.

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Table 23: Testing Process, Continued

Task	Action
5	Whenever the test system extends beyond one physical area, establish a normal and emergency communication system for personnel involved in the test.
6	Leak testing: ▪ Notify all affected personnel of the leak testing before and after the test. ▪ Increase pressure up to specified leak test pressure. ▪ When test pressure is reached, the stabilized test pressure must be maintained for the determined period of time. ▪ Check all system bleed and vent valves, disturbed flanges and fittings and other potential leak points.
7	Pressure testing: ▪ Notify all affected personnel of the pressure testing before and after the test. ▪ Pressurize the system slowly to the test pressure. ▪ When the test pressure is reached, the stabilized test pressure must be maintained for the determined period of time. ▪ Check all disturbed flanges, fittings and other potential leak points. ▲ CAUTION: If any leaks are found, the system should be depressurized before attempting to tighten flange bolts or undertake any other remedial work. During a hydrostatic pressure test, the system must be depressurized before attempting to tighten flange bolts or leaks. ! IMPORTANT: [CriticalE] The pressurizing equipment or the equipment under test must never be left unattended at any time during the test. ! IMPORTANT: [E] The pressurizing equipment must be isolated or disconnected from the pressure system when the test pressure is reached. ▲ CAUTION: The possibility of a pressure increase due to thermal expansion should be considered in the case of an extended test. ! WARNING: [CriticalE] The equipment under test must not be subjected to any form of pressure shock (hammer testing, for example).

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Table 23: Testing Process, Continued

Task	Action
8	After the leak/pressure testing, remove pressure from the facility system as follows: - Slowly depressurize and drain the test medium from the facility system. [E] The pressure must be vented or drained gradually under controlled conditions until atmospheric pressure (or the desired operational pressure) is reached. Use the same plan and precautions as for depressurizing and draining hydrocarbons. ! WARNING: [E] Pressure and volume limitations of the drain or vent system must be evaluated and must not be exceeded. [E] Vents must be open at the high points before draining liquids to confirm that a vacuum is not created. [CriticalE] Inert gases must be vented to a safe location. NOTE: The pressurizing medium may be transferred into another facility system for testing in order to conserve the pressuring medium. - Checks should be made to confirm that no part of the tested facility system retains any trapped residual pressure; for example, behind non-return (check) valves. - Prior to returning the equipment to service, the risk of any residual pressuring medium must be assessed and actions taken to eliminate them, such as water and its effect on the process product, high or low temperatures, or service that may require the use of special drying methods such as hot dry air.
9	Upon successful completion of the leak/pressure test, remove the joint complete tape from tested flanges.
10	Return to service – Special precautions must be taken when pressuring up with a volatile or flashing liquid. Vessels and pipework can become brittle and fail due to cooling below their MDMT, if pressured up from ambient conditions with a volatile or flashing liquid. Equipment and facility systems handling such liquids may be pressurized with any suitable gas (for example, nitrogen, inert gas, methane, fuel gas, and so forth) before returning the equipment to service. However, [CriticalE] before filling any equipment with a volatile liquid, the gas must be first purged out with the <i>equilibrium vapor</i> (vapor similar in composition to the volatile liquid used to fill the equipment). [E] The equipment must then be pressurized to the normal operating pressure with the equilibrium vapor before liquid is added. This is to prevent the liquid flashing down to low temperatures, which can occur when volatile liquid is added to equipment pressurized with a non-equilibrium vapor.

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7.6.5 AIR FREEING

7.6.5.1 INTRODUCTION

This section provides guidelines to protect personnel and equipment when performing air freeing (purging air from a hydrocarbon system) activities. Air freeing should take place as soon as possible after a vessel or system is recommissioned or leak tested. Refer to [Figure 19: Equipment Re-instatement Air Freeing Decision Tree](#)

[CriticalE] The air-freeing activity is performed with all isolation valves that could introduce hydrocarbons in the isolated state, and **must not** be moved to the de-isolated state until an inert atmosphere is obtained.

7.6.5.2 GOAL

The goal of **air freeing** is to avoid a combustible mixture (hydrocarbons and oxygen) existing inside systems/equipment by removing oxygen prior to the introduction of hydrocarbons. Air freeing is achieved by purging or displacing air with an inert medium such as water or nitrogen before introducing hydrocarbons.

[CriticalE] Air may be displaced by the re-introduction of hydrocarbons only when Risk Assessed and approved, with additional mitigations in place to reduce risk to ALARP, except for air freeing discretionary scenarios described in this section.

! IMPORTANT: Displacing air with hydrocarbons will result in combustible mixtures being present within the system/equipment.

[CriticalE] **Air Freeing required:**

- Reinstating facility systems and equipment that contain large volumes and/or complex components that have been opened to the atmosphere e.g. exchangers, compressors, tanks, vessels, turbines and pipelines.
- Reinstatement of facility systems and equipment that have higher risk of ignition such as gas/air mixtures with a wide LEL-UEL range, equipment containing pyrophoric scale or internals that could cause a spark (moving parts or static).
- Reinstatement of systems and equipment where the gas/air mixture in the system could potentially reach known ignition sources such as the flare system or fired heater burners.

Air Freeing discretionary:

Small volume process components when the risk presented by an inadvertent hydrocarbon release to atmosphere (with or without ignition) or ignition of the combustible mixture within the equipment can be reduced to ALARP.

[CriticalE] **Considerations when NOT Air Freeing before introducing Hydrocarbons:**

- SIMOPS Restrictions for “Hydrocarbon Source Activities” **must** be implemented.
- All work in proximity to the activity that could be impacted by an inadvertent release **must** be suspended.
- Continuous surveillance of the area **must** be maintained while equipment is being reinstated.
- Fire and gas detection equipment in the vicinity **must** be in service or mitigated.

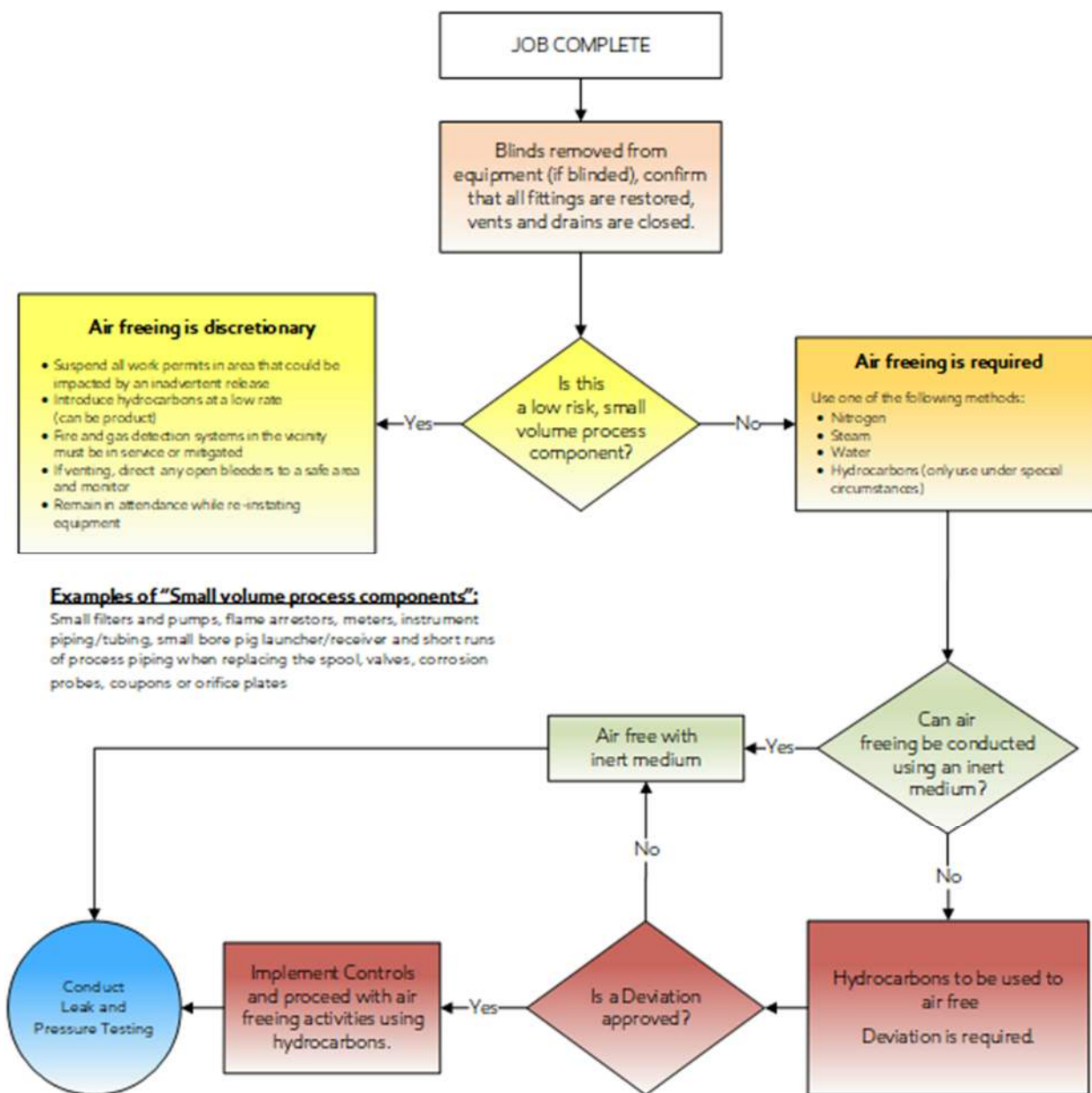
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- Process valves can be cracked open to introduce hydrocarbons to small volume process components. For larger volumes, small bore piping with a ball valve for quick isolation and a needle or globe valve for throttling should be used when introducing hydrocarbons. Keep rates low to minimize velocities and mixing (smaller area/front of combustible mixture).

! WARNING: [E] Actuated valves **must NOT** be used (unless manual mechanical mode can be used).

▲ CAUTION: Large valves should not be used because they are difficult to control and seats can be damaged when used to throttle flow.

Figure 19: Equipment Re-instatement Air Freeing Decision Tree



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7.6.5.3 AIR-FREEING METHODS

Three methods can be used for air freeing prior to introducing hydrocarbons:

- Displacing air with water
- Purging air with nitrogen, or other inert gas
- Purging with hydrocarbons (risk assessment required)

There are two methods of purging air with inert gas:

- Purging by displacement
- Purging by dilution

7.6.5.4 DISPLACING AIR WITH WATER

For process vessels and systems where moisture does not cause problems, or where an inert gas supply is unavailable, fresh or seawater can be used for displacement of air.

[E] When displacing with water, sufficient high point vents **must** be opened and the fill rate adjusted to suit venting capacity. Overflow hoses should be connected, properly secured, and directed to the drain system to confirm that any residual hydrocarbons do not contaminate the environment.

▲ **CAUTION:** If water is to be drained after displacing air, gas is allowed to enter the system by connecting fuel gas to vents. Care must be taken to avoid creating a vacuum or over pressuring equipment. PSVs and vacuum breakers should be commissioned prior to this.

NOTE: If water (fresh or sea) is used, a combined biocide and oxygen scavenger should be used.

7.6.5.5 USE OF BREAK TANKS

[CriticalE] When using fire or utility water to fill vessels, tanks, and facility systems, a break tank is required to confirm that the water system is not directly coupled to the process equipment. The break tank may be directly coupled to the fire or utility water system, with a pump used to pump from the break tank into the process equipment.

▲ **CAUTION:** [CriticalE] A non-return valve (check valve) at the fill point to a hydrocarbon system **must not** be used as a substitute for the break tank, because there is still a possibility of contamination or over pressuring the water system.

NOTE: As an alternative to a break tank and pump, a handheld or adequately secured fill hose may be used to fill a vessel from the top, provided that there is no direct coupling to the process equipment. [E] The hose **must** be electrically bonded to prevent static buildup.

[E] If the use of a break tank and pump is not practicable then a Risk Assessment is conducted as outlined in the Risk Assessment Management System manual and approved at the appropriate level.

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7.6.5.6 PURGING AIR WITH INERT GAS

Inert gas purging is the preferred method for removing air from process equipment in cold service or where moisture may cause problems, such as hydrate formation. Inert gas may be supplied from an inert gas generator or nitrogen supply. There are two methods of purging with inert gas:

- **Purging by displacement:** Small volume vessels and pipe spools are normally purged using displacement, because inert gas can be introduced quickly enough to displace the air with minimal mixing. Larger vessels can also be air purged using the displacement method, or a combination of displacement and dilution. For displacement, the inert gas is usually introduced at the bottom of the vessel or tank, and the air displaced through open vents at the top. This method is very effective if an inert gas that is heavier than air is used (for example, carbon dioxide).
- **Purging by dilution:** For larger process vessels and facility systems, purging by dilution is more commonly used. The interface is larger and more mixing occurs due to turbulence and convection. The vessel or facility system is pressurized with inert gas, taking care not to exceed the maximum working pressure. Pressure is held for approximately 15 minutes and then the vessel is depressurized from the end opposite to the one where the inert gas was introduced. Any dead legs must be thoroughly purged by opening bleeders or cracking flanges to confirm a satisfactory inert gas flow. This process is repeated until oxygen testing indicates less than 4% oxygen in an inert atmosphere. It is then safe to introduce hydrocarbons.

7.6.5.7 PLANNING AIR-FREEING ACTIVITIES

The following guidelines should be used to plan air-freeing activities.

- The complexity and risks associated with the activities should be used to determine if a job-specific plan should be prepared. Job planning should consider the following:
 - Any special conditions that should exist before the plan is commenced.
 - A marked-up P&ID showing system isolation points and purge and vent points.
 - Step-by-step activities to be performed.
 - Facility critical operating parameters (pressures, temperatures, capacities) that should not be exceeded.
 - Any required temporary defeats of critical devices, and when critical devices may be reinstated.
 - The means to monitor the progress of the operations and confirm its successful completion.
 - The means for collecting and disposing of the leak test medium.
 - A Risk Assessment of the activities and risk mitigation measures.
 - Any special spill containment or waste disposal equipment that could be required.
 - Any personal protective equipment (PPE) required.
 - Emergency or abnormal operating procedures.
 - Any restrictions on the use of air or water. For example, liquids may not be acceptable for gas compression systems because they can contaminate the seal oil and cannot always be drained from all low points. Liquid in compressor suction lines can cause serious damage to compressors.
- [CriticalE] All purge and vent points **must** be identified.

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7.6.5.8 WATER FLUSHING

The following guidelines should be followed when flushing with water:

- Before starting to fill with water, verify that the process equipment and its foundation are constructed to withstand the extra weight of the water.
- Water should be added to the base of the vessel or tank to be flushed. The preferred method for displacing air with water is to use a portable or fixed pump fitted with a break tank. [CriticalE] If a hose is used, the following precautions **must** be taken:
 - Hoses **must** be electrically bonded to prevent static electricity buildup.
 - Hoses **must** be secured to the fill valve with a clip.
 - A break tank **must** be used to confirm that no contaminated water backs up into the water system. A check valve alone is not acceptable.
- ▲ **CAUTION:** Never connect utility or fire hoses directly to vessels or pipe work, to prevent the ingress of hydrocarbons into these systems.
- An overflow hose must be connected to the high point in the vessel (if applicable) and be run to a closed drain.
- Close the overflow valve and remove the hoses when the vessel is full.
- Bolt up the flange on the fill point (as applicable).

! **IMPORTANT:** Some equipment can require vent valves to be opened.

7.6.5.9 PURGING

The following guidelines should be followed when purging air out of the system with an inert gas such as nitrogen (N₂):

- [CriticalE] Confirm that the following isolation conditions exist:
 - PSVs and vacuum breakers on process equipment **must** be recommissioned.
 - Blinds and valves isolating sources of hydrocarbons **must** remain in their isolated state.
- ! **IMPORTANT:** Only open valves or remove blinds that are required for the purging activity.
- Connect the inert gas to a low drain point.
- Purging with inert gas may be performed by:
 - Pressuring the system with inert gas, and bleeding it down through a vent system or another safe location until the vented gas is less than 4% oxygen by volume in inert atmosphere
 - Purging with inert gas to a vent system, open or closed drain, or to another safe location until the vented gas is less than 4% oxygen by volume in an inert atmosphere.
 - If using a flare system, consider type and gas mixture in the flare system

! **WARNING:** [CriticalE] If purging to a flare or any other system that contains hydrocarbons you **must** consider mitigations to keep the concentration of oxygen in the air gas mixture below 4% oxygen by volume in inert atmosphere.

! **WARNING:** Consider the need to restrict access to the area where inert gas is being vented to prevent asphyxiation.

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- When flushing a large system, connect to a high vent point in the isolated system as far away as possible from the inlet point.
- Excessive purge rates should be avoided to prevent localized freezing of valves, gauges, or other equipment.

! IMPORTANT: Monitor the pressure to avoid an unintended pressure buildup.

7.6.5.10 PURGING (DISPLACING WATER OR NITROGEN) WITH HYDROCARBONS

Purging with hydrocarbons is sometimes required to replace water or nitrogen (or a combination of both). Examples are:

- Diesel can be required for oil pumps to prevent contamination of seal or lube oil systems.
- Hydrocarbon gas can be required for compressors to prevent contamination of seal or lube oil systems.

Hydrocarbons can be used safely, provided that the appropriate controls and methods are used.

▲ CAUTION: [CriticalE] When hydrocarbon gas is used to purge the equipment, it **must** be performed in a manner that eliminates any risk of a major hydrocarbon release if there is an opening to the atmosphere.

7.6.5.11 PRECAUTIONS WHEN USING HYDROCARBON

[CriticalE] The following precautions **must** be observed when purging with hydrocarbons:

- All Work Permits of all types in the area **must** be withdrawn.
- A continuous surveillance of the area during gas purging **must** be maintained.
- Small bore piping **must** be used for the source of hydrocarbons.
- The purge line **must** be fitted with a ball valve for quick isolation and a needle or globe valve for throttling the flow rate. Large valves should not be used because they are difficult to control and seats could be damaged when cracking them open.

▲ CAUTION: [E] Actuated valves **must NOT** be used (unless manual mechanical mode can be used).

- Purge rates should be kept low to minimize velocities. Manual throttling valves should be used to control the purge rate.
- Fire and gas detection equipment **must not** be isolated during purging.
- Any bleed valves that are open to facilitate purging **must** be continuously monitored and **must** be directed to a safe area. The need for spill containment should be evaluated.

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7.6.6 REMOVING ENERGY ISOLATIONS

7.6.6.1 GOAL

The goal of removing energy isolations is to enable the introduction of process fluids and the re-energization of facility equipment so that the facility can be started up.

7.6.6.2 REMOVING ENERGY ISOLATIONS

When removing energy (mechanical, process, and electrical) isolations, the guidelines under the following margin headings should be followed:

- Planning energy isolation removals
- ICC requirements
- Additional precautions

7.6.6.3 PLANNING ENERGY ISOLATION REMOVALS

The following guidelines should be used to plan the removal of energy isolations.

The complexity and risks associated with the activities should be used to determine whether a job-specific plan should be prepared. Job planning should consider the following:

- Any special conditions that should exist before the plan is commenced.
- A marked-up P&ID clearly identifying where the isolations are. A cross-check should be made to the blind list and the valve isolation list to confirm completeness and correctness of the plan.
- Step-by-step activities to be carried out, including the sequence of isolations removal. De-isolate in the following order:
 - PSVs (to allow maximum vessel protection)
 - Minor facilities
 - Secondary facilities
 - Major facilities

▲ CAUTION: When isolations are removed, the order in which this is done is important. [E] PSVs should be the first to be de-isolated, in case of an unforeseen event or delay. The sequence confirms that the equipment is protected as it is returned to service. [E] When returning PSVs to service, always open the downstream valve first, so that the downstream flanges will not be overstressed if the PSV relieves.

- Special precautions to be taken for each isolation removal.
- Facility critical operating parameters (pressures, temperatures, capacities) that should not be exceeded.
- The original work pack should have attached copies of the blinding list, valve isolation list, marked-up P&IDs from the original isolation, and the job-specific plan for de-isolating the facility to be used in the de-isolation process.
- [CriticalE] All work permits for work that has the potential to ignite hydrocarbons or high-pressure release **must** be withdrawn.

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- Any required temporary defeats of critical devices, and when critical devices may be reinstated.
- A Risk Assessment of the activities and risk mitigation measures.
- Any special spill containment and waste disposal equipment that could be required.
- Any personal protective equipment (PPE) required.
- Emergency or abnormal operating procedures.

7.6.6.4 ENERGY ISOLATION REMOVAL PRACTICE

The following outlines the general practice regarding energy isolation removal:

- Upon completion of all recommissioning, leak testing, and air freeing, the Permit Holder notifies Operations that the system is ready to be de-isolated.
- [CriticalE] The Isolating Authority and Area Authority **must** confirm that it is safe to perform the requested de-isolations.
- [CriticalE] The Area Operator **must** confirm that the isolated system has been recommissioned, leak tested, and air freed, if required:
 - The Area Operator **must** confirm that no other planned or ongoing work requires any of the existing isolation points to remain isolated.
- The Area Operator collects the Isolation Control Certificate and requests the Area Authority to approve for reinstatement. The Area Authority approves the de-isolation activities to commence.
- For remote locations where the Area Authority is not present, the Area Operator or Permit Holder acts as the Area Authority and approves the de-isolation activities to commence.
- Once the Area Authority or designate has approved the de-isolation:
 - The Permit Holders remove their Functional lock(s) from the isolation control point (ICP).
 - The Area Operator removes their Functional lock from the ICP.
 - The Isolating Authority obtains the key(s) from the ICP and removes the locks and tags from the isolation points.
- The Area Operator is responsible for coordinating the de-isolating of the equipment.
 - The Isolation Control Certificate (ICC) is closed out.

7.6.6.5 ADDITIONAL PRECAUTIONS

The following additional precautions should be followed when de-isolating facility systems and reintroducing hydrocarbons:

- [E] The Area Operator **must** always confirm, if possible, that no pressure or liquids have accumulated behind blinds before their removal.
- [CriticalE] All flange and connections **must** be fitted and tightened to the appropriate torque.
- [E] Regardless of any examination showing no pressure or liquids in the flange areas, the Permit Holder and workers **must** always approach debinding as if there were pressure or liquids in isolated spools, valves, and similar devices.

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- [E] When splitting flanges for debinding, the best practice for breaking containment **must** be followed (bolts farthest away from the workers should be loosened first and the flange parted to check for gas or liquids under pressure).

Reference: Section 7.5.3: "Breaking Containment"

- [CriticalE] Non-spark hand tools **must** be used for the breaking flanges when there is a potential for hydrocarbons to be released.

▲ CAUTION: [CriticalE] During first break, when a flange is to be unbolted, the Area Operator **must** be present to confirm that the correct flange is being worked and confirm absence of pressure prior to breaking containment.

- [E] All blinds to be removed from valves and flanges **must** be identified with tags. Sequence should be considered in the de-blinding process, when the sequence is important e.g. PSVs de-blinded first.
- [CriticalE] The Area Operator **must** confirm all de-isolation activities and check them off the blinds list(s) and the valve isolation list(s).
- [CriticalE] A second independent, competent person **must** verify all isolation points have been removed and equipment has been returned to the de-isolated state. The second verifier then **must** sign the ICC.
- Actuated valves should not be de-isolated until the reinstatement of their operating medium, and the removal of any temporary defeats of their control system.

7.6.7 ACCEPTANCE TESTING

7.6.7.1 GENERAL

[E] The acceptance test of equipment **must** be performed when equipment has been out of service due to minor maintenance or change of parts.

Acceptance tests of equipment after major maintenance or modification to the equipment require specific performance or test practices.

7.6.7.2 PRACTICES

The Area Operator should perform the acceptance test in cooperation with the Area Authority.

The following requirements apply to the acceptance test:

- Check the equipment for leaks.
- Function test the equipment, including:
 - Start and stop of pumps
 - Correct settings for level or pressure on instruments
 - Alarm checks of instruments
 - Open or closure test of valves
 - Time to reach closed position for emergency shutdown (ESD) valves
 - Correct signals and alarms, both local and in the CCR
 - Equipment capacity verification test, when necessary

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When the acceptance test has been approved by the Area Operator and the Area Authority, the equipment can be put back into service.

7.6.7.3 POST WORK SURVEILLANCE

Close observation of the equipment and facility system should be maintained after it is returned to service regardless of the leak test results. Thermal expansion or contraction and increased pressures can result in a leak developing and quick intervention being required. Continuous surveillance should be maintained until the equipment has reached operating pressures and temperatures.

7.7 TOOLS

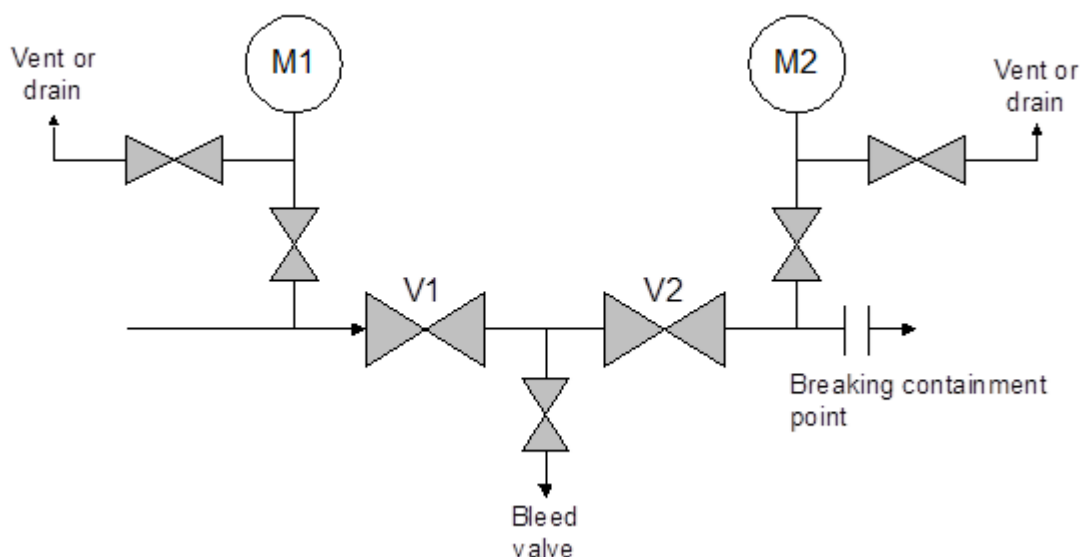
7.7.1.1 INTEGRITY TEST DIAGRAMS

The following tests are referenced in [Section 7.4.2: "Isolation Integrity"](#) under "Process isolation integrity tests."

- Refer to Figure 20: Double Isolation and Bleed (DIB) Isolation Integrity Test
- Refer to Figure 21: Double Isolation and Bleed Valve (DIBV) Isolation Integrity Test
- Refer to Figure 22: Single Valve Isolation (SVI) Isolation Integrity Test

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Figure 20: Double Isolation and Bleed (DIB) Isolation Integrity Test



LEGEND

- V1: First upstream isolation from live system
- M1: Live side monitoring point (pressure gauge, vent, or drain)
- V2: Second downstream isolation valve from live system
- M2: Monitoring point between valves and the breaking containment point

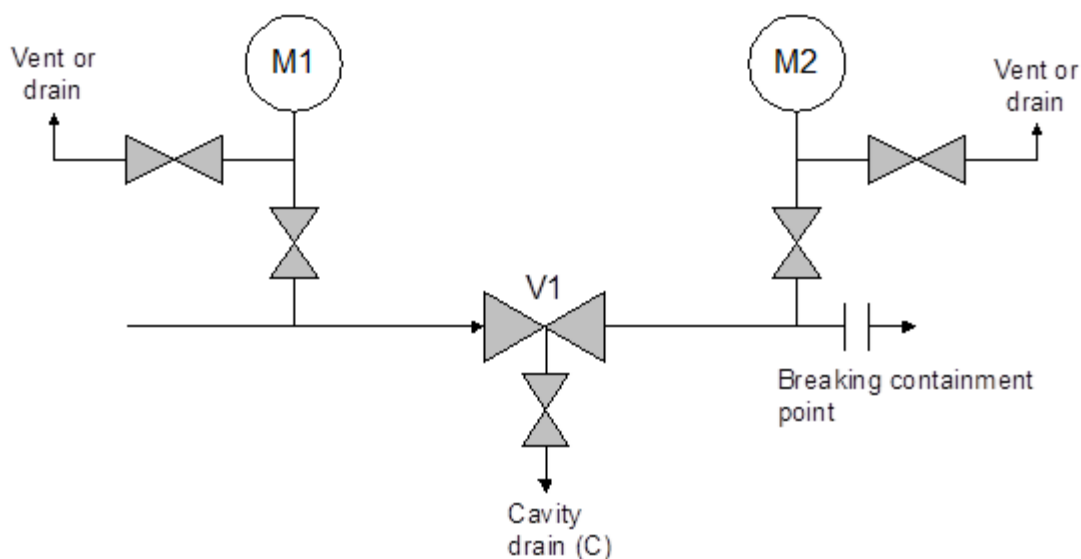
TEST

1. Ensure that M1, M2, and the bleed valve is not obstructed and the pressure gauges are reading properly.
2. Close the downstream valve V2.
3. Note pressure at monitoring points M1 and M2.
4. Vent or drain the section of line to be broken and monitor the pressure at M2 until the pressure is zero.
5. Close the vent or drain at the break point and monitor M2. No pressure buildup at M2 verifies the integrity of V2.
6. Close the upstream valve V1.
7. Note the pressure M1.
8. Open the bleed valve and bleed between V1 and V2. Monitor the bleed valve until the pressure is zero.
9. Leave the bleed valve in the open position to allow for further monitoring.

▲ CAUTION: Do not leave process vents to the atmosphere open unless they are directed to a safe location and secured. All vents and bleeds should be closed when not actively working on or monitored.

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Figure 21: Double Isolation and Bleed Valve (DIBV) Isolation Integrity Test



LEGEND

- V1: Double Isolation and Bleed valve (DIBV) isolation from the live system
- M1: Live side monitoring point (pressure gauge, vent, or drain)
- M2: Monitoring point between valves and the breaking containment point

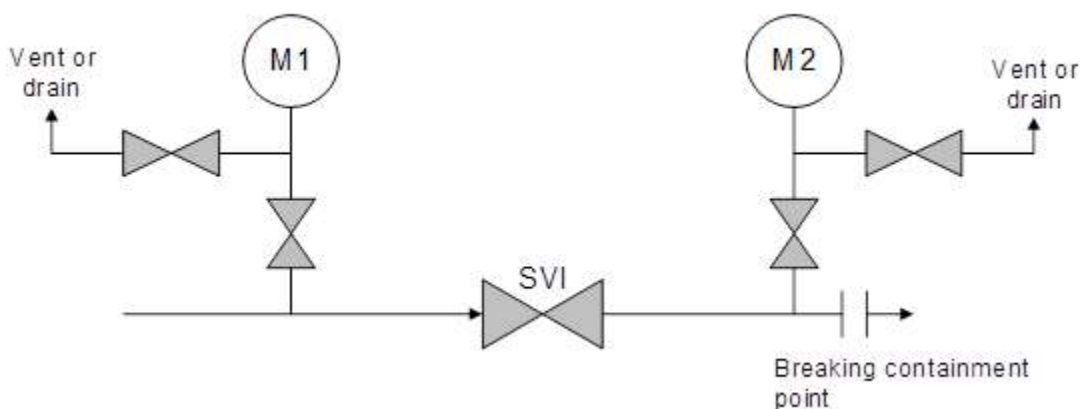
TEST

1. Ensure that M1, M2, and C are not blocked and are reading properly.
2. Close V1 isolation valve.
3. Note pressure at M1, C, and M2.
4. Vent or drain the downstream section of line to be broken. Monitor the pressure at M2 vent until pressure reaches zero.
5. Close the vent or drain at breakpoint and monitor M2.
6. Open C and bleed the cavity between seals until pressure is zero. Zero pressure buildup at C verifies the integrity of the upstream seal.
7. Leave C in the open position to allow for further monitoring.

▲ CAUTION: Do not leave process vents to the atmosphere open unless they are directed to a safe location and secured. All vents and bleeds should be closed when not actively

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Figure 22: Single Valve Isolation (SVI) Isolation Integrity Test



LEGEND

- SVI: Single valve isolation from live system
- M1: Live side monitoring point (pressure gauge, vent, or drain)
- M2: Monitoring point between valves and the breaking containment point

TEST

1. Ensure that M1 and M2 are free of obstructions and are indicating correctly.
2. Close the isolation valve.
3. Note pressure at M1 and M2.
4. Vent or drain downstream section of the line that is to be disconnected until M2 has zero pressure.
5. Close downstream vent/drain valve and monitor M2. No pressure buildup proves the integrity of the SVI.
6. Leave downstream vent or drain at the break point in the open position to allow for further monitoring.

▲ CAUTION: Do not leave process vents to the atmosphere open unless they are directed to a safe location and secured. All vents and bleeds should be closed when not actively working on or monitored.

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8.0 ATMOSPHERIC TESTING

8.1 INTRODUCTION

Atmospheric testing is performed to confirm a safe working environment by recognizing, assessing, eliminating, or reducing to acceptable levels of flammable or toxic gas or atmospheric hazards before work commences, and by monitoring and maintaining these conditions throughout the activity, or terminating the activity if adverse conditions develop.

REFERENCES:

- Section 6.2: "Work Permitting Process", "Work Site Verification"
- Section 6.4: "Confined Space Entry Precautions", "Atmospheric testing"
- [GOTO/WMSFORMS](#), "Atmospheric Test Record," for a supplementary Atmospheric Test Record for situations where work is conducted without a permit or additional space is required for recording values

8.2 ATMOSPHERIC TYPES, DETECTORS, TESTING, AND PROPERTIES

8.2.1 TYPES OF ATMOSPHERICS:

[CriticalE] Portable atmospheric detectors **must** be available to check for the following types of gas and atmospheric condition (where required):

- Oxygen (O₂) content (%) in air
- Hydrocarbon or flammable vapor (% of lower explosive limit [LEL] in air)
- Volatile Organic Compounds (VOC) (ppm) in air
- Hydrogen sulfide (H₂S) content (ppm) in air
- O₂ content (%) in inert atmosphere
- Hydrocarbon (% LEL) in inert atmosphere
- H₂S content (ppm) in inert atmosphere
- Other Toxic gases, such as:
 - Sulfur dioxide (SO₂)
 - Carbon monoxide (CO)
 - Benzene
 - Carbon dioxide (CO₂)

NOTE: The Operating Site in consultation with their MOH Department should determine any additional Atmospheric Test requirements related to their operations.

8.2.2 TYPES OF ATMOSPHERIC DETECTORS

In addition to permanently installed atmospheric monitoring systems which are tied into the facility alarm systems, portable atmospheric detectors are used to evaluate specific work site conditions.

There are two types of acceptable portable detectors that can be used for atmospheric gas testing:

- Electronic detection devices which operate via diffusion (passive) or by use of a pump (i.e., aspirated) to draw the atmospheric sample across the sensor
- Detector (e.g. Dräger) tubes

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8.2.3 FUNCTIONAL TEST/CHECK DEFINITIONS

Bump Test: A qualitative function check where a challenge gas is passed over the sensor(s) at a concentration and exposure time sufficient to activate all alarm indicators. This check confirms the gas can get to the sensor(s) and that the alarms are functional. This test does not provide a measure of calibration accuracy.

Calibration Check: A quantitative response test utilizing a known traceable concentration of test gas to demonstrate that the sensor(s) and alarms respond to the gas within the manufacturer's acceptable limits.

Full Calibration: A quantitative test and adjustment of the sensor(s) response to match the desired value compared to a known traceable concentration of test gas. The response value and acceptable range are set by the manufacturer.

8.2.4 ATMOSPHERIC DETECTOR APPLICATION

Diffusion (Passive) devices can be used for conducting atmospheric testing in general worksite locations, continuous area monitoring, etc.

Aspirated (Pump Type) should be used for conducting atmospheric testing using pump extraction capability (e.g. confined space entry, flange leak testing, inaccessible areas, etc.)

NOTE: It is important to utilize the appropriate instrument (passive or aspirated) for the specific location and atmosphere to be tested. It is also important to know and use the correct Atmospheric Detector when sampling in Inert Atmospheres.

Detector (e.g. Dräger) tubes are used for a variety of functions concerned with the measurement of the atmosphere. The chief function is to provide a rapid guide to air purity and, in certain cases such as leak detection, for production purposes.

8.2.5 ATMOSPHERIC TESTING AND MONITORING

Atmospheric testing is the act of sampling a defined area for specific atmospheric contaminants prior to work activities (where required) and intermittently using a portable detector.

Monitoring is the act of continuously using a portable detector to ensure air quality is safe at the work location for personal protection.

! IMPORTANT: [E] Sampling devices designed to measure average concentrations of specific contaminants over a measured time interval are used for Industrial Hygiene exposure monitoring and **must** not be used as atmospheric detection devices (e.g. passive exposure badges).

NOTE: "Monitoring" for real time determination of air quality should not be confused with IH Exposure Monitoring.

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8.2.6 TYPES OF ATMOSPHERIC TESTS

Atmospheric testing requirements are recorded as a control measure on the Atmospheric Test Record of the work permit. There are four different requirements for atmospheric testing as detailed in the following. [CriticalE] With the exception of continuous testing & monitoring of low risk, short duration activities described in this section, all atmospheric testing **must** be performed by an Authorized Gas Tester (AGT).

- **Initial oxygen test:** An initial test for oxygen performed before certain work can commence, such as confined space entry.
- **Initial atmospheric test:** An initial test for combustible or toxic atmospherics performed and recorded to confirm the safety of the work site before work can commence, such as confined space entry, hot work, and spark potential.
- **Periodic atmospheric test:** Repeat atmospheric tests carried out at specified intervals and recorded, typically after shift change, meal breaks or any other extended breaks (>30 minutes) from the work site where conditions may have changed.
- **Continuous atmospheric monitoring:** [CriticalE] When atmospheric testing is required, the work site is continuously monitored using a portable atmospheric detector for any changing conditions.

! IMPORTANT: [CriticalE] The results of initial and periodic atmospheric testing **must** be recorded on the Atmospheric Test Record and be available at the worksite.

8.2.7 HAZARDS

The reasons for and the potential hazards of improper atmospheric testing include:

- Use of atmospheric detector that was not properly calibrated and/or bump/functional tested per manufacturer guidance
- Use of incorrect or faulty electronic atmospheric detector.

! IMPORTANT: There are a number of reasons why a direct-reading device may fail to provide a proper response:

1. The sensor's protective membrane may be clogged with dirt or oil which may block the flow to the sensor.
 2. A broken electrode or contact, which prevents instrument function.
 3. Chemical degradation or sensor drift within electronic components, which occurs naturally over time.
 4. Exposure to high (over-range) concentrations of target gases or vapors, which can degrade the sensor.
 5. Exposure to chemicals, which can poison or inhibit the electrochemical or catalytic response of the sensor.
 6. Exposure or use in extreme environmental conditions such as high/low temperature, humidity, or high levels of airborne particulate, which can interfere with the electrochemical or catalytic response of the sensor.
 7. Harsh storage or operating conditions such as dropping the instrument onto a hard surface, submerging it in liquid or exposing it to vibration or shock, which can damage the instrument.
 8. Improper calibration of the device, which leads to incorrect measurement.
- Use of incorrect detector for testing in inert atmospheres
 - Use of incorrect detector tube, or faulty aspirator.

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- Checking in the wrong location.
 - Example:** Checking up high for a gas that is heavier than air.
- Presence of heavy liquids or sludge that yield no vapors until disturbed or heated.
- Checking for the wrong gases.
- Gases being released after checking due to the activity being performed. Activities that can cause gases to be released include:
 - Hot work
 - Steaming
 - Use of chemicals
 - Abrasive blasting
- The atmospheric tester is not competent.
- The potential consequences of improper atmospheric testing include:
 - Personnel injury or death.
 - Exposure of personnel to dangerous substances.
 - Inaccurate atmospheric readings.
 - Accumulation of dangerous gases (hydrocarbon rich or oxygen deficient).
 - Hot work in a flammable atmospheric environment.
 - Fire or explosion.
 - Damage to facilities or equipment.
- Potential risk mitigation steps for atmospheric testing include:
 - Properly trained personnel.
 - Following manufacture guidance on proper equipment calibration and bump/functional test.
 - Following atmospheric testing protocols
 - Understanding the limitations of the atmospheric testing device being used and potential sensor interferences (i.e. Hydrogen can yield false CO readings for some sensors)
 - Atmospheric detector use and maintenance logging.

8.3 GENERAL ATMOSPHERIC TESTING PRACTICES

8.3.1 INTRODUCTION

This section describes general precautions and practices pertaining to atmospheric testing.

8.3.2 ACTIVITIES REQUIRING ATMOSPHERIC TESTING

[CriticalE] Atmospheric testing **must** be part of the work planning and execution for the following broad categories of work:

- Confined space entry
- Hot work
- Spark potential work
- Application of coatings inside confined spaces
- Breaking containment, especially when toxic gases or hydrocarbons may be present
- Any work where hydrocarbon or toxic gases may be present
- Operation of a vehicle or internal combustion engine in the process/hydrocarbon area
- Work on live electrical equipment in the process/hydrocarbon area
- Use of non-hazardous area approved equipment in the process/hydrocarbon area

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[CriticalE] Once the work activity has been identified and that Atmospheric Testing is required, an Authorized Gas Tester (AGT) is assigned to both develop the Atmospheric Testing Plan and perform the required Atmospheric Tests to allow work to safely proceed.

The AGT is responsible for completing all necessary Tests and documenting the results on the worksite copy of the Atmospheric Test Record. The AGT ensures results are shared with the Supervisor and Work Team and that all necessary controls are in place.

8.3.3 ATMOSPHERIC TESTING FOR HOT WORK

[CriticalE] When hot work is to commence, atmospheric tests **must** be performed at the work site before and during the work according to the requirements of the work permit.

Some examples of where to perform atmospheric tests are:

- Flanges, valve packing, seals, vents, and drains near the hot work site
- Inside and outside a hot work habitat
- Continuous testing to ensure no hydrocarbons are migrating into the area

Reference: Section 6.3: "Hot Work"

8.3.4 ATMOSPHERIC TESTING IN CONFINED SPACES

[CriticalE] Initial atmospheric testing of confined spaces **must** be performed with the appropriate type of atmospheric testing device from a safe position outside of the confined space, or if a representative atmospheric test cannot be made from the outside, the following requirements apply:

- The person performing the atmospheric test **must** be equipped with the appropriate personal protective equipment (PPE), including:
 - Breathing apparatus (BA)
 - Eye protection
 - Clothing sufficient to protect the skin from exposure to potential toxic substances
- A Confined Space Entry Attendant **must** be present and monitor confined space for changing conditions.

Reference: Section 6.4: "Confined Space Entry Precautions," under headers "Atmospheric testing" and "Communications" which contain further requirements related to atmospheric testing and critical entry into a confined space.

8.3.5 PERFORMANCE ATMOSPHERIC TESTS

The following factors should be considered when performing atmospheric tests. These include:

- Order of atmospheric testing. [CriticalE] Atmospheric tests **must** be performed in the following order:
 1. Oxygen
 2. Flammability/Combustibility (LEL)
 3. Toxic agents i.e. H₂S, Benzene, SO₂, CO, etc.

! IMPORTANT: Oxygen is measured first. The most common type of LEL sensor (i.e., catalytic sensor), requires oxygen to function properly. At least 10% oxygen is necessary for the LEL sensor to function properly.

! IMPORTANT: LEL Sensors Measure Explosivity, Not Toxicity – Many VOCs are potentially toxic at levels that are well below their explosive levels and below the sensitivity of the LEL sensors. A low LEL

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reading does not indicate absence of VOC. [E] VOC readings **must** be taken using a device that measures VOC.

- Correct detector and scale/range for the purpose
- Chemical exposure limits
- Meter alarm set points
- Wind strength and direction
- Ambient temperature
- Equipment or building configuration
- Location of engine exhausts or other potential contaminants
- Potential soil contamination
- Natural or forced ventilation
- Low points, drains, or culverts
- Density of the atmosphere to be tested (heavier or lighter than air)
- Types of gases that may be present that may interfere with sensors or tubes being utilized to conduct the atmospheric test

8.3.6 WHERE TO PERFORM ATMOSPHERIC TESTS

The following guidelines should be considered when performing atmospheric tests:

- **Hydrocarbons (HC):** Test vents, open drains, flanges, bleeder valves, valve packing glands, screwed fittings, or other potential sources of hydrocarbon leaks (liquids or vapors).
NOTE: Most hydrocarbon vapors are heavier than air and can accumulate in low points, such as drains, culverts, basements, and so forth. However, some light ends can accumulate in high spots.
- **Oxygen (O₂):** Test for oxygen deficiency in vessels, tanks, pits, basements, and so forth prior to confined space entry.
▲ CAUTION: Oxygen deficiency can be caused by ongoing activities in a confined space (such as breathing, hot work, chemical reactions). Continuous monitoring of oxygen is necessary to evaluate the effectiveness of ventilation and the safety of continued presence of personnel in the confined space. If testing indicates, personnel may require breathing apparatus (BA).
- **Hydrogen sulfide (H₂S):** Hydrogen sulfide is slightly heavier than air, toxic, smells like rotten eggs, but cannot be recognized at higher concentrations due to loss of smell, and is flammable. The product of burning hydrogen sulfide, sulfur dioxide (SO₂), is also toxic. Hydrogen sulfide can be present in hydrocarbon production, or it can be formed over time from decaying drilling mud components or water. Where fixed hydrogen sulfide monitors and alarms do not exist or do not effectively protect the work area, portable continuous-monitoring devices are used.
- **Benzene (Bz):** is a chemical that is a colorless or light-yellow liquid at room temperature. It has a sweet odor and is highly flammable. Benzene evaporates into the air very quickly. Its vapor is heavier than air and may sink into low-lying areas. Benzene dissolves only slightly in water and will float on top of water. Benzene is also a natural part of crude oil and gasoline. Test for benzene when breaking containment, vents, open drains, flanges, bleeder valves, valve packing glands, screwed fittings, or other potential sources of hydrocarbon leaks (liquids or vapors).
- **Carbon monoxide (CO):** Carbon monoxide is produced by internal combustion engines and by inert gas generators. It is toxic, colorless, and odorless. Leaking exhaust systems in enclosed areas (such

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as generator buildings) can cause carbon monoxide to accumulate. Engine exhausts placed too close to, or upwind of, confined spaces (or confined space ventilation intakes) may lead to a dangerous concentration in a confined space. Carbon monoxide is continuously tested when there is the potential for these situations to exist.

- **Carbon dioxide (CO₂):** Carbon dioxide can occur naturally in production. It may also be used in some fire extinguishing systems, and may be used as the inert gas for process system purging. Although nontoxic, it can cause death by asphyxiation. Carbon dioxide is heavier than air, and can accumulate in basements, pits, or other low points. If its presence is suspected, atmospheric testing and thorough ventilation is required prior to confined space entry.
- **Nitrogen (N₂):** Nitrogen is used extensively to preserve or purge equipment. Although nontoxic, it can cause death by asphyxiation. Nitrogen is slightly lighter than air. If its presence is suspected, testing for O₂ ≥ 19.5% to ≤ 23.5% and thorough ventilation is required prior to confined space entry.

8.3.7 ATMOSPHERIC TESTER QUALIFICATIONS

Atmospheric testing is performed by an Authorized Gas Tester. Continuous monitoring can be assigned to a member of the work team.

[CriticalE] The Authorized Gas Tester **must**:

- Have received atmospheric testing training meeting OPITO Authorized Gas Tester Training (OPITO standard codes: 9240 or 9242) requirements and be deemed competent in atmospheric test requirements and sensor interferences.
- Be trained and competent in the use and functionality of the atmospheric detection equipment.
- Be knowledgeable of the potential atmospheric hazards and conditions.
- Be knowledgeable in exposure limits.
- Be knowledgeable of the testing requirements for the testing plan to include the equipment, area, and specific test locations (e.g. flanges, bleeds, vents, low points).
- Be competent on how to interpret and make informed decisions based on the atmospheric test results.

In some instances, atmospheric testing and monitoring can be performed by the Permit Holder for lower risk activities provided the following conditions are adhered to:

- **Short-term, lower-risk activities:** This includes use of a camera, opening low voltage junction boxes, and use of non-intrinsically safe tools and electronics (ie. Laptops, drills, etc.). Work scopes must be limited to those that require only general area monitoring for LEL. If any specific toxic substances may be present, an AGT is required for testing. A continuous atmospheric testing device will be provided to the individual only after the individual:
 - Has been briefed on the testing & monitoring requirements by the AGT
 - Receives a demonstration on the correct operation of the atmospheric test device by the AGT (alarms, not to cover the sensing point, care of the unit, and so forth).
 - Can demonstrate the correct operation of the device back to the AGT
 - Is aware or briefed on what hazards to look for.
 - Can explain what actions are to be taken if the atmospheric detector alarms.
- **Short-or long-term, higher-risk activities:** [CriticalE] An Authorized Atmospheric Tester (AGT) must conduct the test.

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8.4 USING ELECTRONIC ATMOSPHERIC TESTERS

8.4.1 INTRODUCTION

This section describes the general practices to be reviewed when using portable electronic atmospheric detectors. Model-specific operation instructions should be reviewed by the users.

8.4.2 OPERATIONAL CHECKS

[CriticalE] Users are to perform a functional check in a "clean" atmosphere to ensure cells are not contaminated, followed by a bump test against a known gas concentration at least once per shift, before use.

8.4.3 GENERAL PRACTICE

Follow this general practice when using portable electronic atmospheric detectors:

- Check the battery level.
- Check alarm settings
- Calibrate the instrument according to manufacturer's instructions.
- If an extension tube or hose is used, allow enough time for the sample gas to reach the sensor. Establish a time by testing and timing on a known source of gas.
- Begin aspirating on approach, not after the probe or tester is inserted into the atmosphere. Watch the meter at all times.
- On manually aspirated models, confirm that check valves are functioning properly.

8.4.3.1 CALIBRATION

[CriticalE] Calibration of the portable electronic atmospheric detectors **must** be performed either by the supplier or by competent personnel in accordance with the supplier's calibration instruction.

[E] The calibration process **must** include the following activities:

- Conduct an inspection and calibration of all portable electronic atmospheric testers as per manufacturer's recommendations to include:
 - Function control of the instruments
 - Calibration of the instruments
 - Cleaning the instruments
 - A test against a known concentration of gas
 - Approval of the instruments for use
- Records of inspection, calibration and maintenance **must** be maintained for each piece of atmospheric detection equipment.
- Portable electronic atmospheric detectors **must** have unique identification numbers.
- The last calibration date for each device **must** be readily available and detectors not serviced within the period **must not** be used.

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8.4.3.2 MAINTENANCE

[CriticalE] Atmospheric detector maintenance **must** be performed by qualified personnel using the manufacturer's recommendations.

8.4.3.3 PRECAUTIONS

When using portable electronic atmospheric detectors, follow these precautions:

- Do not immerse the probe or detector into liquid.
- Use of the atmospheric detector for long periods of time in high gas concentration areas may damage the filament. In this case the instrument should be calibrated against a known gas source more frequently to confirm it is working properly.
- [E] The atmosphere to be tested **must** be at atmospheric pressure when aspirated through the tester. Do not force the mixture through the detector.
- Do not use portable electronic atmospheric detectors near raw (high concentration) oxygen, hydrogen, or acetylene as the instrument may ignite the atmosphere.
- Be aware of the accuracy range of detectors and use the results accordingly.
- [CriticalE] If it is suspected that the detector controls are damaged or not operating properly, the unit **must not** be used.
- Do not place the unit in your pocket or otherwise cover the atmospheric detector inlet.

8.5 USING TUBE DETECTORS

8.5.1 INTRODUCTION

This topic describes the general practices to be followed when using tube-type atmospheric detectors. Operation and storage instructions supplied from the manufacturer should be reviewed by the users.

8.5.2 STORING TUBE DETECTORS

Adhere to the following precautions regarding the storage of tube detectors:

- Tube detectors have a limited shelf life. Check the expiration date of all packets and return or discard expired packets.
- Store tubes per manufacturer's recommendations

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8.5.3 GENERAL PRACTICE

[E] When using tube-type atmospheric detectors:

- Confirm the tube is within its expiration date.
- Select the appropriate tube for the job, based on the expected conditions and the atmosphere to be tested.
- Read the manufacturer's instruction sheet and verify it is the correct instruction for the tube model selected.

! IMPORTANT: Consider what other contaminants may be present that can affect the results. Read the manufacturer's instruction sheet on interferences (cross-sensitivity). Choose the correct tube or tubes.

- Verify that the pump or bellows does not leak before each use by holding a vacuum on an unbroken tube. Refer to the manufacturer's Operating Instructions Manual.
- Break both ends of the tube and insert it into the pump with the arrowhead mark towards the pump. Be careful not to allow liquids to enter the tube. Do not break tube tips until immediately before using.

! IMPORTANT: Some tubes can be used again after a negative test, others cannot. Check the manufacturer's instruction sheet for specific details.

- Check the atmosphere at the operator breathing zone first, actuating the pump the correct number of full strokes. Failure to wait until the chain tightens will cause a low reading and, hence, a possible false sense of security. If in doubt, perform another test.
- Do not mix different manufacturers' pumps and tubes. The volume of a gas tested can differ and affect the reading.
- A Dräger carbon pretube should be used if checking for carbon monoxide where fairly high concentrations of hydrocarbon gases may be present. The charcoal in the pretube will absorb gases such as propane and butane that would otherwise interfere with the reading.

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9.0 SIMULTANEOUS OPERATIONS

9.1 INTRODUCTION

Many different operations are periodically conducted on offshore production platforms and onshore plants, facilities, and fields. Wherever possible, operations should be managed to avoid situations where simultaneous operations (SIMOPS) are required, however at times, it may be necessary to perform critical operations simultaneously. This section provides information on how to safely manage routine and non-routine SIMOPS. It provides the following:

- Basis for safe performance of these operations
- Common approach to routine SIMOPS on Company facilities
- Consistent approach to the planning and execution of non-routine SIMOPS on Company facilities
- Identifies work that cannot be performed at the same time

Adherence to the requirements set forth in the Permit to Work System in this manual, as well as Company and regulatory requirements, is intended to reduce the risks associated with the performance of SIMOPS to as low as reasonably practicable (ALARP). The safety requirements for activities associated with all SIMOPS operations are reviewed by the Person in Charge (PIC) and adhered to during execution.

9.2 SIMOPS DEFINITIONS

The following definitions are used for simultaneous operations:

- **Simultaneous operations:** Are separate Job Scopes of Work that take place at the same time and meet both of the following conditions:
 - Are in the same area.
 - Directly or indirectly could potentially affect the safe performance of any other activity at the facility.

SIMOPS can be *routine* or *non-routine*.

- **Routine SIMOPS:** Simultaneous operations that meet both of the following requirements. All activities are:
 - Identified on the SIMOPS Matrix.
 - Covered by existing, approved written instructions or procedures (or Procedure Checklist).
- **Non-routine SIMOPS:** [E] SIMOPS not covered by the matrix or detailed work execution plan **must** be reviewed by appropriate subject matter experts and approved by management.
 - Well Control Operations are non-routine and are not listed explicitly in the SIMOPS matrix or table. [E] Work execution unrelated to the well control operation that is concurrent, **must** be reviewed and approved by site management in charge of well control operations before the unrelated work is allowed to continue.

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9.3 APPLICABILITY

These simultaneous operations guidelines were prepared to comply with Operations Integrity Management System (OIMS) System 6-4 Work Management and are consistent with the requirements of the *Upstream Safety Manual*.

9.4 SIMOPS PROCESS

9.4.1 INTRODUCTION

The number of operations that can be conducted simultaneously on the same structure depends on various factors such as:

- Type and size of the intended activity
- Location of the proposed work in relation to other activities being performed
- Duration of the work effort
- Impact of the activities on the overall operations

These considerations must be thoroughly discussed and a decision made on the number of operations that can be performed simultaneously without jeopardizing safety of personnel, the environment, and the facility.

In addition to evaluating the risk associated with each operation, compliance with related Company policies, safety rules, regulatory requirements and other safe work guidelines is mandatory. In some instances, these requirements may prevent certain operations from being performed simultaneously.

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9.4.2 SIMOPS ACTIVITIES

9.4.2.1 SIMOPS ACTIVITIES

The activities that comprise SIMOPS operations are listed in the following table. Refer to

Table 24: SIMOPS Activities

SIMOPS Activities	
Operations	Painting and Coating Operations
	Confined Space Entry
	Crane Activities or Complex Lifts (or both)
	Engine-Driven Equipment and Vehicles
	Fire System Inoperable
	Hot Work
	Hydrocarbon Source Activity
	Pigging Operations
Marine	Boat Operations
	Derrick Barge
	Diving Operations
	Remotely Operated Vehicle (ROV) Operations
Wellwork	Blowout Preventer (BOP) Movement
	Coiled Tubing Unit (CTU)
	Drilling or Workover
	Electrically Fired Explosives Operations
	Rig Skid
	Snubbing Operations
	Well Control
	Well Stimulation
	Wireline Operations
Other Potential Adds	Critical Casing Pressure
	Specified Night Operations
	Flammable Product Transfer
	Hull Maintenance/Inspection
	Tanker Loading or Unloading
	Helicopter Operations
	Lifeboat testing unavailable
	Others as required by each Operating Site

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9.4.3 SIMOPS DEFINITIONS

9.4.3.1 TERMS AND DEFINITIONS

This topic presents definitions of the terms used in the SIMOPS Matrix and the SIMOPS Restrictions Tables.

Table 25: SIMOPS Definitions

Term	Description
abrasive blasting	High-velocity air that is carrying sand or other abrasive particles to clean metal surfaces.
adjacent wells	Those wells that are located in any of the four quadrants bordering the well being worked on.
boat operations	Any activity involving marine vessel operations. NOTE: The loading and unloading of supply vessels are considered to be a Crane Activity.
BOP stack	Abbreviation for blowout preventer (BOP) stack or an assembly of heavy-duty valves attached to the top of the wellhead or tree to control well pressure.
BOP stack movement	Any time the BOP stack is moved from the storage area or wellhead, including rigging up or rigging down.
Coating Operations (activities)	Any activities to remove coatings from, or place coatings onto equipment. Coating activities include: ▪ Spray painting ▪ Abrasive blasting ▪ Installing insulation, such as fiberglass
crane activities	Any activities where a crane is used to raise, move, or lower a load. NOTE: Loads moved by a derrick barge are considered to be a <i>derrick barge</i> activity. Reference: Crane activities must be supervised in compliance with the Rigging and Lifting SWP.
critical casing pressure	Refer to the Well Integrity Manual "Maximum Allowable and Operating Annulus Surface Pressures" Reference: Section 12.0: Appendix A.4 "Annulus Trigger Pressures" .
CTU	Coiled tubing unit
designated welding area designated hot work area	An area at the facility, pre-approved by the Company, that does not require special preparation to provide isolation either horizontally or vertically from hazardous areas to allow hot work.

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Table 25: SIMOPS Definitions, Continued

Term	Description
drilling	The actual work process, performed by the drilling rig that involves circulation of fluid or movement of tubulars in the wellbore.
engine-driven equipment and vehicles	Any equipment powered by an internal combustion engine including vehicles. ! WARNING: These items are a potential source of ignition and exhaust gas emissions contain carbon monoxide that is hazardous for a confined space atmosphere.
electrically fired explosives operations	Includes any operation that sets off an explosive charge by electrical signal, such as perforating, chemical cutters, tubing punches, setting bridge plugs, or permanent packers. Usually, but not limited to wellwork activities. ! IMPORTANT: A radio-safe detonating system uses a radio-safe blasting cap or electro-explosive that does not require radio silence. If radio-safe detonation systems are used, the PIC must determine if this mitigates the SIMOPS identified hazard(s), and renders the SIMOPS situation "not applicable." Reference: Upstream WellWork Execution Manual
electric line	Electric or conductor line protected by one or more layers of steel braid or armor. It is generally utilized for conveyance of electrically fired explosives operations, well logging operations, or mechanical runs.
emergency shutdown (ESD)	A system of manual stations located throughout the platform or facility, which, when activated, initiate a shutdown of the production processes.
firewater system inoperable	Whenever the firewater system is unable to deliver the rated capacity of a single pump, or the system is out of service in the area of the work.
fire protection system inoperable	Whenever the fire protection system is out of service in the area of the work. Fire protection system can be firewater system, deluge, dry chemical, and so forth that is determined to be safe and adequate for the foreseeable risk of the activity or the area.
flammable product transfer	Transfer of flammable products using hose lines such as between facility and boat, and refueling.

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Table 25: SIMOPS Definitions, Continued

Term	Description
flowline	Piping that directs the well stream from the wellhead to the first downstream process component.
Gemoco® unit	An automatic wireline unit used to cut paraffin in a wellbore.
hull maintenance and inspection	Regulatory or integrity required inspection of vessel hull.
lifeboat testing	Regulatory or integrity required testing of lifeboat or key components.
major activity	A significant work effort that may have an impact on simultaneous operations, such as welding while drilling and producing.
MIRU	Abbreviation for move in and rig up—usually associated with an onshore modular drilling rig, a coiled tubing unit, or a snubbing unit.
MOB/DEMOB	mobilization/demobilization Complex Lift activities associated with the movement of equipment during rig up or rig down operations—usually associated with a major piece of offshore equipment, such as a modular drilling rig, a coiled tubing unit, or a snubbing unit.
Non-rig wireline work	Wireline work performed on any well that is not rig-related and positioned directly under the rotary of the rig.
operation	Any activity that directly or indirectly affects the safe performance of any other activity at the location.
paraffin cutter	Sometimes referred to as Gemoco® unit or automatic paraffin cutting unit. The unit is an automatic wireline unit which is manually started by an operator and automatically cuts paraffin to a preset depth, spools back up, and turns off.
partial operating facility	When the production operations involve continuous production separation or pumping, but is less complex than full facilities.
perforating	The process of arming a perforating gun on the surface, lowering it into the wellbore, and detonating charges into the formation at the desired location. Perforating operations are considered to be ongoing until the assembly is safely retrieved at the surface.

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Table 25: SIMOPS Definitions, Continued

Term	Description
pigging	Using pressure and flow to push a device (for cleaning, dewatering, separating, or inline inspection) through a pipeline or flowline. <i>For SIMOPS, a critical operation is opening the pig trap door to the atmosphere to insert or remove a pig.</i>
pipeline	Piping that directs hydrocarbon fluids or gases between platforms or to a shore facility.
potential source of hydrocarbon leak	Equipment that either contains or has contained hydrocarbons and the potential for discharge (flanges, connections, relief or vent points, valves, and so forth) is present.
producing	The phase of operations where well fluids or gases are brought to the surface or received via pipelines from another facility to be separated, processed, and delivered into departing pipelines.
prohibited	Prohibited is defined for two SIMOPS uses: • SIMOPS Matrix: A combination of activities that are not allowed to be conducted concurrently. No deviations are allowed. • SIMOPS Restrictions Tables: A specified condition under which a combination of activities is not allowed to be conducted concurrently. No deviations are allowed.
PSHH or PSLL	Pressure switch high-high or pressure switch low-low. A safety device that detects abnormal pressure (high or low) and transmits a signal to perform a shut-in function.
quick-release fusible lock-out device	Fusible lock-out for a surface safety valve (SSV) that features a releasing mechanism that can be remotely activated to enable the SSV to close upon loss of pneumatic ESD system pressure. See SSV.
radio-safe detonating system	A radio-safe detonating system uses a radio-safe blasting cap or electro-explosive that does not require radio silence. If radio-safe detonation systems are used in electrically fired explosives operations, the PIC must determine if this mitigates the SIMOPS identified hazard(s), and renders the SIMOPS situation not applicable.
rig skid	To move the rig from one well location to another with minimal or no dismantling of equipment.
rig-related	Wellwork performed in a well that is rig-related and has equipment positioned on the drilling rig (directly under or over the rotary).

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Table 25: SIMOPS Definitions, Continued

Term	Description
ROV (remotely operated vehicle)	A submersible vehicle controlled from the surface by a tether that is able to send pictures and data to the surface and perform limited activities underwater.
Safety device	A device that detects an abnormal operating condition and transmits a signal to perform a specific shutdown function.
SCSSV (surface-controlled subsurface safety valve)	A device installed in a well below the wellhead, SCSSVs are controlled from the surface by hydraulic, electric, mechanical, or other means, with the design function to prevent uncontrolled well flow when actuated.
SSV (surface safety valve)	An automatic wellhead valve assembly that closes upon loss of power supply caused by, for example, a PSHH, PSLL, or ESD).
shut in	Shut down or stop an operation.
site-specific station bill	A supplemental or amended <i>station bill</i> that describes emergency response responsibilities for all personnel at a location where simultaneous operations are to be conducted.
snubbing	A wellwork process by which concentric pipe is inserted into a well under pressure. Snubbing can be utilized as part of a routine well workover procedure or non-routine as needed during well control response.
subsea maintenance	Subsea maintenance is diving-based or ROV-based inspection of the structural integrity of underwater or splash zone structural components.
tanker loading or unloading	Tanker transfer operations generally for product sales, specifically, crude oil or liquefied natural gas (LNG).
TCP	tubing conveyed perforating
TIW valve	Brand name for a quarter-turn, full-bore ball valve that can be installed on drill pipe or a workstring.
tree-on BOP stack	The BOP stack used when the Christmas tree is installed on the tubing head.
tree-removed BOP stack	The BOP stack used when the Christmas tree is not installed on the tubing head or the wellhead.

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Table 25: SIMOPS Definitions, Continued

Term	Description
unprotected lubricators	When the wireline lubricator is located at or above the top (drill) deck of an offshore platform.
unprotected process equipment	Process equipment not protected by a solid steel deck.
well control problem (drilling or workover)	Unplanned insufficient hydrostatic weight to overbalance the formation pressure resulting in pressure or flow on the surface, and operations to correct the situation. Generally, refers to a situation where there is a tree-removed BOP stack on the well rather than a Christmas tree.
well control problem (producing)	Loss of surface or subsurface pressure containment on a well with a Christmas tree installed.
well control operations	Operations to correct <i>well control problems</i> . These operations can include pumping fluids, snubbing operations, coiled tubing operations, or wireline operations and are considered non-routine.
well stimulation	The pressure phase of stimulating a well.
wireline operations	For the purposes of utilizing this SIMOPS matrix and tables, <i>wireline operations</i> include downhole work involving wireline, slickline, or braided line. <i>Does not include</i> electrically fired explosive operations.
workover	The actual work process performed by the workover rig that involves circulating fluid downhole, or any movement of tubulars in the wellbore. Can also include routine snubbing or coiled tubing operations when not involved in well control response.

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9.4.4 SIMOPS MATRIX

9.4.4.1 SIMOPS MATRIX

The Simultaneous Operations Decision Matrix (SIMOPS Matrix) is a quick reference tool to indicate whether activities can be performed simultaneously. A number of activities are identified that require additional consideration when conducted simultaneously. This matrix is not a stand-alone document and is used in conjunction with the associated tables, potential differences in facilities or operations, and other work management control tools including the judgment of the operations team. Each Operating Site should develop specific matrices and associated tools as necessary to meet their specific regulatory requirements or unique operational aspects not addressed in this matrix.

[CriticalE] SIMOPS **must** be managed in accordance with the SIMOPS Matrix, associated restriction tables, required deviations and defined approval levels. These restrictions and approval levels are listed in the SIMOPS Restrictions Tables presented in this section.

NOTE: Sites with extra management layer, Department Head (DH, Ops TLS) and Operations Manager (Ops 4th Line), Operating Sites have flexibility to assign approval authority to persons deemed competent to fulfill the role and are capable of executing their responsibilities outlined in WMS.

References:

- Refer to Figure 23: Example SIMOPS Matrix
- [GOTO/WMSEFORMS](#), "SIMOPS," SIMOPS Matrix (Operating Site specific)
- Section 9.4.5: "SIMOPS Restrictions Tables"
- Upstream WellWork Execution Manual
- Rigging and Lifting SWP

9.4.4.2 SIMOPS MATRIX CLASSES

The SIMOPS Matrix is a chart with predefined activities showing the class of the SIMOPS to which the combination of the two activities is assigned. These classes are:



Simultaneous activities are *prohibited*. They will *not* be performed. No deviations are allowed.



Simultaneous activities may be *restricted*. The restrictions in the SIMOPS Restrictions Tables **must** be met where applicable. Written authorization from the PIC (Level 1), Operations Manager (Level 2), or Site Manager (Level 3) **must** be obtained to deviate from the restrictions.



Simultaneous activities are allowed using normal procedures and instructions.



Simultaneous activity combinations are not applicable.

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9.4.4.3 ASSESSMENT CONSIDERATIONS

The SIMOPS matrix is based on Risk Assessments of routine SIMOPS. These assessments consider:

- Potential conflicts with one another in normal or emergency situations
- Nature of potential risks and hazards
- Causal factors and consequences from simultaneous operations

9.4.4.4 USING THE SIMOPS MATRIX

Use the following steps for the SIMOPS Matrix:

- Identify the activities to be performed. The activities are based on the Job Scope and not all tasks within the scope.
- Enter both operations on the SIMOPS Matrix and locate the:
 - Activity that is closest to the bottom on the alphabetical axis.
 - Other activity on the numeric axis.
- Arrive at a SIMOPS classification:
 - Prohibited
 - Restricted
 - Allowed
 - Not applicable
- The color of the square where they intersect indicates whether the two operations can be conducted simultaneously, or if a conflict in activities exists requiring additional approval.
- **! IMPORTANT:** When evaluating the simultaneous execution of more than two activities, every combination of two activities **must** be evaluated on the SIMOPS Matrix.
- The combination that yields the most restrictive SIMOPS classification is the classification for the planned SIMOPS.

The SIMOPS Matrix must be used in conjunction with the following:

- The written procedures, practices and instructions for each activity
- The precautions listed on the Work Permits and the Permit Pack
- SIMOPS Restrictions Tables

! IMPORTANT: The SIMOPS Matrix is designed as a reference tool, not as a stand-alone tool. It **must** be used in conjunction with the SIMOPS Restrictions Tables.

- The following approval levels **must** be obtained to deviate from the restrictions and perform the activities simultaneously:
 - Level 1 - PIC
 - Level 2 - Operations Manager
 - Level 3 - Site Manager

NOTE: Refer to [Table 26: SIMOPS Restrictions Tables](#), for Level 1, 2 and 3 Restriction designations

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9.4.4.5 EXAMPLE SIMOPS MATRIX

Use the following SIMOPS Matrix as a reference to the text in this section.

Each Operating Site may have its unique SIMOPS Matrix which will be located as a separate file in the local Addendum. Refer to [Figure 23: Example SIMOPS Matrix](#)

Figure 23: Example SIMOPS Matrix

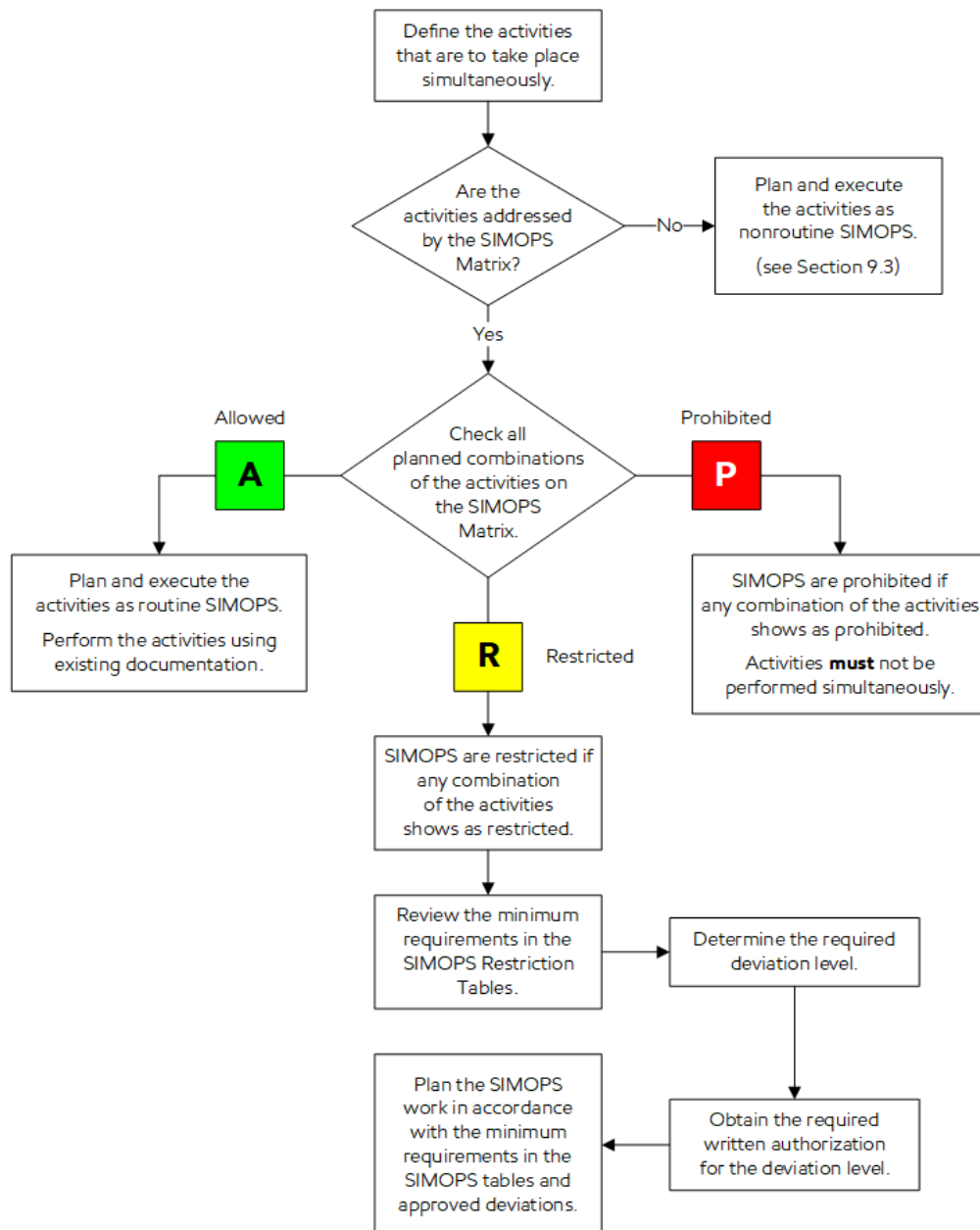
Simultaneous Operations Decision Matrix																						
- Normal operations assumed -																						
NOTE! Well Control Operations is non-routine SIMOPS																						
This Matrix must be used in conjunction with the SIMOPS Restriction Tables																						
			Coating Operations	Confined Space Entry	Crane Activities	Engine-Driven Equipment/Vehicles	Fire Protection System Inoperable	Hot Work	Hydrocarbon Source Activity	Pigging Operations	Boat Operations	Derrick Barge	Diving Operations	ROV Operations	BOP Movement	Coiled Tubing Unit	Drilling or Workover	Electrical Fired Explosives Operations	Rig Skid	Well Stimulation	Wireline Operations	
Operations	Coating Operations	A	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
	Confined Space Entry	B	R	2											R	SIMOPS Restricted						
	Crane Activities	C	R	R	3										P	SIMOPS Prohibited						
	Engine-Driven Equipment and Vehicles	D	R	R	A	4									A	SIMOPS						
	Fire Protection System Inoperable	E	R	R	R	R	5								X	N/A						
	Hot Work	F	R	R	R	A	P	6														
	Hydrocarbon Source Activity	G	R	R	R	R	R	R	7													
	Pigging Operations	H	R	R	R	R	R	R	A	8												
Marine	Boat Operations	I	A	A	A	A	R	A	A	A	9											
	Derrick Barge	J	R	A	R	A	R	A	R	R	A	10										
	Diving Operations	K	R	R	P	R	R	R	R	R	R	R	11									
	ROV Operations	L	A	R	R	A	A	A	A	A	A	R	R	12								
WellWork	BOP Movement	M	R	R	R	R	R	R	A	A	A	R	R	R	13							
	Coiled Tubing Unit	N	R	R	R	R	R	R	R	R	A	P	R	R	R	14						
	Drilling or Workover	O	R	R	R	R	R	R	R	R	A	R	R	R	R	X	A	15				
	Electrically Fired Explosives Operations	P	R	P	P	R	P	P	P	P	P	P	P	P	P	P	P	R	16			
	Rig Skid	Q	R	P	R	A	R	A	P	P	A	P	P	R	R	X	P	X	P	17		
	Well Stimulation	R	R	R	R	R	R	R	A	A	A	P	P	R	R	R	R	R	R	18		
	Wireline Operations	S	R	R	R	R	P	R	R	R	A	R	P	R	R	R	R	R	R	R	19	
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		

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9.4.4.6 FLOW CHART

Use this flowchart for classifying two simultaneous activities on the SIMOPS Matrix. Refer to [Figure 24: Classification of Simultaneous Activities on the SIMOPS Matrix](#)

Figure 24: Classification of Simultaneous Activities on the SIMOPS Matrix



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9.4.4.7 MORE THAN TWO SIMOPS ACTIVITIES

[CriticalE] In cases where more than two SIMOPS activities are being considered at the same time, all activities **must** be plotted on the matrix for each simultaneous combination. [CriticalE] Restriction tables are applied for every matrix point and all controls outlined **must** be applied for application. [CriticalE] Any restriction that cannot be met **must** be managed through the SIMOPS Deviation Request Form.

Reference: [GOTO/WMSFORMS](#), "SIMOPS," SIMOPS Deviation Request and Approval Form

9.4.5 SIMOPS RESTRICTIONS TABLES

9.4.5.1 SIMOPS RESTRICTIONS TABLES: DEFINITIONS AND DEVIATIONS

The SIMOPS Restrictions Tables list the activity restrictions for managing conflicts between activities identified on the SIMOPS Matrix as "Restricted," and act as a supplement to the matrix. When a combination of activities is classified as restricted and depicted by a yellow box, the SIMOPS Restrictions Tables are to be consulted for guidance related to that particular combination of activities to be conducted.

Restriction Table criteria must be assessed and increased if circumstances are not favorable and would increase the risk (for example, wind direction could extend the distance vapor travels).

Should any deviation be required from the rules contained in the SIMOPS Restrictions Tables, the PIC, with the input of the Operations Management Team, determines the mitigating steps to be taken and confirms that these steps are documented on the SIMOPS Deviation Request Form or equivalent process such as management of change (MOC). This confirms that the mitigation steps are "fit for purpose" for the activities to be conducted at the facility.

! IMPORTANT: The SIMOPS Restrictions Tables and associated activity restrictions are not necessarily all inclusive. Supervisory judgment and a Risk Assessment may be required to confirm safe operations.

! IMPORTANT: Some restrictions involve situations that are *prohibited*. No deviations are allowed from *prohibited* restrictions.

Reference: [GOTO/WMSFORMS](#), "SIMOPS," SIMOPS Deviation Request and Approval Form.

9.4.5.2 SIMOPS RESTRICTION TABLES

The SIMOPS Restrictions Tables are included on the following pages.

Reference: [GOTO/WMSFORMS](#), "SIMOPS," for the SIMOPS Matrix (Operating Site specific). Refer to Table 26: SIMOPS Restrictions Tables

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Table 26: SIMOPS Restrictions Tables

Matrix	Requirement
B1	Abrasive blasting conducted concurrently with confined space entry (CSE) on an offshore installation, or within 82 ft. (25 m) of each other onshore, is restricted and requires a Level 1 approval. Spray painting performed within 49 ft. (15 m) of openings or ventilation air intakes of any CSE operation, is restricted and requires a Level 1 approval. Fiber-glassing conducted concurrently with CSE on an offshore installation, or within 82 ft. (25 m) of each other onshore, is restricted and requires a Level 1 approval. Hazards – Spark potential, fire or explosion, air quality
C1	Coating operations conducted concurrently with crane activities on an offshore installation or within 82 ft. (25 m) of each other onshore, are <i>restricted</i> and require a Level 1 approval. Abrasive blasting conducted concurrently while a lift is taking place on an offshore installation, or within 82 ft. (25 m) of each other onshore, is <i>restricted</i> and requires a Level 1 approval. Hazards – Spark potential, fire or explosion, obstructed operator vision, dropped object ! IMPORTANT: Crane Activities must be in accordance with the <i>Rigging and Lifting SWP</i>.
C2	CSE conducted concurrently with complex lift crane activities within 85 ft. (26 m) is <i>restricted</i> and requires a Level 1 approval. Hazards – Dropped object
D1	Engine-driven equipment and vehicles operated concurrently within 49 ft. (15 m) of large volume spray coating operations is <i>restricted</i> and requires a Level 3 approval. Hazards – Spark potential, fire or explosion
D2	Engine-driven equipment and vehicles operated concurrently with CSE on an offshore installation, or within 49 ft. (15 m) of the confined space onshore, are <i>restricted</i> and require a Level 1 approval. Hazards – Engine-driven equipment and vehicles are a source of ignition that could ignite hydrocarbons and give off toxic exhaust fumes that create a hazard in a confined space
E1	Major coating operations conducted concurrently with the fire protection system being inoperable on an offshore installation or in the same area onshore are <i>restricted</i> and require a Level 3 approval. Hazards – Fire or explosion without fire response

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Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
E2	CSE conducted concurrently with the fire protection system (entire) being inoperable on an offshore installation or in the same area onshore is <i>restricted</i> and requires a Level 2 approval. Hazard – Safe and timely evacuation in the event of a fire
E3	Complex Lifts by stationary or mobile cranes performed concurrently with the fire protection system being inoperable are <i>restricted</i> and require a Level 1 approval. Hazard – Dropped object impact with no fire response
E4	Engine-driven equipment and vehicles operated concurrently with the fire protection system being inoperable on an offshore installation or in a hazardous location onshore are <i>restricted</i> and require a Level 1 approval. NOTE: This is generally covered by Permit to Work controls. Hazard – Spark potential with no fire protection response
F1	Large volume spray coating operations conducted concurrently with hot work activities on an offshore installation, or within 85 ft. (26 m) of each other onshore, are <i>restricted</i> and require a Level 3 approval. Hazards – Fire or explosion
F2	Unrelated hot work conducted concurrently with CSE activities within 85 ft. (26 m) is <i>restricted</i> and requires a Level 2 approval. ! IMPORTANT: All Hot Work rules apply for work within a confined space. Hazard – Air quality contamination
F3	Complex lifts performed concurrently with hot work within the process area are <i>restricted</i> and require a Level 2 approval. Hazard – Dropping the load and causing a hydrocarbon release
F5	Hot work is <i>prohibited</i> while the fire protection system in the affected area is inoperable or there is no alternate fire protection system available. NOTE: Work can only be performed when additional fire protection is readily available. Fire protection must be sufficient for the risk exposure as determined by a formal Risk Assessment. Approval is at a minimum Level 3 or higher in accordance with the unmitigated risk. Hazards – Fire or explosion
G1	Abrasive blasting conducted concurrently with a hydrocarbon source activity on an offshore installation, or within 82 ft. (25 m) of each other onshore, is <i>restricted</i> and requires a Level 1 approval. Hazards – Spark potential, and fire or explosion

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Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
G2	CSE activities conducted concurrently with hydrocarbon source activities within 85 ft. (26 m) are <i>restricted</i> and require a Level 2 approval. Hazard – Hydrocarbon entering confined space
G3	Crane activities conducted concurrently with hydrocarbon source activity within 85 ft. (26 m) are <i>restricted</i> and require a Level 1 approval. Hazards – Spark potential from internal combustion engines and non-intrinsically safe electric equipment
G4	Engine-driven equipment and vehicles operated concurrently with hydrocarbon source activity within 85 ft. (26 m) are <i>restricted</i> and require a Level 1 approval. Hazard – Engine-driven equipment and vehicles are a source of ignition that could ignite hydrocarbons.
G5	Hydrocarbon source activity is <i>restricted</i> while the fire protection system is inoperable and requires Level 2 approval. Hazards – Fire or explosion due to spark potential (for example, hot work, engine-driven equipment, and so on)
G6	Hot work is <i>restricted</i> from occurring concurrently with hydrocarbon source activity within 85 ft (26 m), and requires a Level 3 approval. Hazards – Fire or explosion
H1	Abrasive blasting conducted concurrently with pigging operations on an offshore installation, or within 82 ft. (25 m) of each other onshore, is <i>restricted</i> and requires a Level 3 approval. Hazards – Spark potential, fire, or explosion
H2	CSE activities conducted concurrently with pigging operations within 85 ft. (26 m) are <i>restricted</i> and require a Level 2 approval. Hazards – Air quality, hydrocarbon entering confined space
H3	Crane activities conducted concurrently with pigging operations within 85 ft (26 m) are <i>restricted</i> and require a Level 1 approval. Hazards – Spark potential from internal combustion engines and non-intrinsically safe electric equipment
H4	Engine-driven equipment and vehicles operated concurrently with pigging operations within 85 ft (26 m) are <i>restricted</i> and require a Level 1 approval. Hazard – Engine-driven equipment and vehicles are a source of ignition that could ignite hydrocarbons

Work Management System (WMS)

Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
H5	Pigging operations are <i>restricted</i> while the fire protection system is inoperable and require Level 2 approval. Hazard – Hydrocarbon release, potential fire or explosion with no fire protection response
H6	Hot work is <i>restricted</i> from occurring concurrently with pigging operations within 85 ft. (26 m) and requires a Level 3 approval. Hazard – Hydrocarbon release with open flame resulting in fire or explosion
I5	Boat operations carried out in close proximity to a manned offshore installation when the installation's fire protection system is inoperable, are <i>restricted</i> and require a Level 1 approval. Hazards – Collision or contact with offshore risers, fire or explosion potential
J1	Coating operations on an offshore installation or on the derrick barge performed concurrently with derrick barge lifts are <i>restricted</i> and require a Level 1 approval. Hazard – Dropped object impact
J3	Complex lift – Derrick barge activities within 85 ft. (26 m) are restricted and require a Level 1 approval. Hazard – Dropped object impact
J5	Derrick barge lifts carried out in close proximity to a manned offshore installation are <i>restricted</i> if the installation's fire protection system is inoperable and require a Level 2 approval. Hazards – Fire and explosion due to contact with offshore risers
J7	Hydrocarbon source activity conducted when the derrick barge boom is directed toward the platform, and the derrick barge is within 500 ft. (150 m) of an offshore installation, is <i>restricted</i> and requires a Level 2 approval. Hazards – Fire or explosion
J8	Pigging operations conducted when the derrick barge boom is directed toward the platform, and the derrick barge is within 500 ft. (150 m) of an offshore installation, are <i>restricted</i> and require a Level 2 approval. Hazards – Fire or explosion
K1	Abrasive blasting conducted concurrently with diving operations is <i>restricted</i> and requires a Level 3 approval. Hazard – Contaminated air supply
K2	CSE conducted concurrently with diving operations is <i>restricted</i> and requires a Level 2 approval. Hazard – Assumes fire protection system inoperable (firewater pumps) due to diving operations.

Work Management System (WMS)

Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
K3	Any lifts unrelated to the diving activities <i>that could impact divers in the water or their support equipment on the facility (while in use)</i> conducted concurrently with diving operations are <i>prohibited</i>. Hazards – - Objects dropped into the sea can travel great distances <i>in unpredictable directions and could strike the divers</i>. - Dropped object causing damage to diver's support equipment onboard the facility while in use (divers in the water or decompression) or damage to production facilities causing a release and potential fire.
K4	Engine-driven equipment operating in close proximity to the diving operations breathing air compressor intake is <i>restricted</i> and requires Level 1 approval. Hazard – Contaminated air supply
K5	Diving operations requiring bypassing of the automatic start function of the firewater system, conducted with all operations except production, are <i>restricted</i> and require a Level 2 approval. In order for producing operations to continue, a knowledgeable production person in direct communications with the Diving Supervisor must stand vigil over the firewater pump so he or she can alert the Diving Supervisor to get the diver away and, once clear, start the firewater pump. ! IMPORTANT: The temporary defeat process must be used to control bypassing the firewater system.
K6	Hot work conducted concurrently with diving operations assumes fire protection is operable, that is, not dependent on the firewater system. If this is not the case, go to F5 and apply SIMOPS for fire protection system. Hazard – Fire pump may be inoperable due to proximity of divers to pump intake
K7	Hydrocarbon source activity conducted concurrently with diving operations is <i>restricted</i> and requires a Level 1 approval. Hazard – Contamination of diver's air or fire protection system inoperable (or both)
K8	Pigging operations activity conducted concurrently with diving operations are <i>restricted</i> and require a Level 1 approval. Hazard – Contamination of diver's air or fire protection system inoperable (or both)
K9	Unrelated boat operations in close proximity conducted concurrently with diving operations are <i>restricted</i> and require a Level 1 approval. Hazards – Divers or equipment being struck or impacted by the boat

Work Management System (WMS)

Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
K10	Unrelated derrick barge operations in close proximity conducted concurrently with diving operations are <i>restricted</i> and require a Level 1 approval. Hazards – Divers being struck by the boat or impacted by derrick barge operations
L2	CSE conducted concurrently with ROV operations is <i>restricted</i> and requires a Level 2 approval. Hazard – Assumes fire protection (firewater) system inoperable due to ROV operations
L3	Any lifts unrelated to the ROV activities conducted concurrently in the vicinity of ROV operations are <i>restricted</i> and require Level 2 approval. Hazard – Dropped object impact—equipment dropped into the sea can travel great distances before hitting bottom.
L10	Derrick barge lifts made in the vicinity of ROV operations are <i>restricted</i> and require a Level 1 approval. Hazard – Potential interference with derrick barge or dropped load
L11	Unrelated diving operations in the vicinity of ROV operations and both taking place within facility-controlled zone (for example, 500-meter zone) are restricted and require Level 1 approval. Hazard – Potential interference between ROV operations and diver support equipment
M1	Abrasive blasting conducted concurrently with BOP movement operations on an offshore installation, or within 82 ft. (25 m) of each other onshore, is <i>restricted</i> and requires a Level 1 approval. Hazard – Spark potential in potentially explosive atmosphere
M2	CSE during BOP movement operations and within 85 ft (26 m) is <i>restricted</i> , and requires a Level 1 approval. Hazard – Potential interference and dropped object impact to production, drilling, or workover package
M3	Complex lifts within 85 ft (26 m) during BOP movement operations are <i>restricted</i> and require a Level 1 approval. Hazard – Potential interference and dropped object impact to production, drilling, or workover package
M4	Engine-driven equipment and vehicles operating concurrently during BOP movement without the wellhead and within 85 ft (26 m), are <i>restricted</i> and require a Level 1 approval. Hazards – Spark potential, hydrocarbon release
M5	BOP movement operations conducted in an offshore facility when the fire protection system inoperable are <i>restricted</i> and require a Level 3 approval. Hazards – Fire or explosion and open wellbore

Work Management System (WMS)

Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
M6	Hot work conducted concurrently with tree-removed BOP stack movement within 85 ft. (26 m), is <i>restricted</i> and requires a Level 1 approval. Hazards – Fire or explosion, open wellbore, potential hydrocarbon source
M10	Derrick barge lifts conducted concurrently with BOP movements are <i>restricted</i> and require a Level 2 approval. Hazard – Dual complex lifts, increases supervisory oversight requirements
M11	Diving operations in close proximity of an offshore installation conducted concurrently with BOP movement are <i>restricted</i> and require Level 1 approval. Hazards – Dropped object, diver being struck if BOP dropped
M12	ROV operations in close proximity of a fire pump intake while conducted concurrently with BOP movement are <i>restricted</i> and require Level 1 approval. Hazards – ROV could obstruct the intake of the fire pump or the fire pump is inoperable. A dropped BOP could contact the ROV
N1	Abrasive blasting conducted concurrently with coiled tubing unit (CTU) work within 82 ft. (25 m) is <i>restricted</i> and requires a Level 1 approval. Hazard – Spark potential in potentially explosive atmosphere
N2	CSE is <i>restricted</i> during CTU operations when using a lubricator within 85 ft. (26 m) and requires a Level 2 approval. Hazard – Evacuation of confined space in the event of a hydrocarbon or toxic vapor release NOTE: Distances may be greater for gas toxicity (such as, H ₂ S).
N3	Complex lifts conducted concurrently with CTU operations when the path of the complex lift is over the coiled tubing or any of the CTU equipment, are <i>restricted</i> and require a Level 1 approval. Hazards – Dropped object impact or equipment collision
N4	Unrelated engine-driven equipment and vehicles operated concurrently with CTU work within 85 ft. (26 m), are <i>restricted</i> and require a Level 1 approval. Hazards – Hydrocarbon release with spark potential and equipment collision
N5	CTU operations conducted in an offshore facility when the fire protection system is inoperable are <i>restricted</i> and require a Level 3 approval. Hazards – Gas release and vapor cloud explosion potential
N6	Hot work conducted concurrently with CTU operations within 85 ft. (26 m) is <i>restricted</i> and requires a Level 2 approval. Hazards – Gas release and vapor cloud explosion potential

Work Management System (WMS)

Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
N7	Hydrocarbon source activity conducted concurrently with CTU operations within 85 ft. (26 m) is <i>restricted</i> and requires a Level 1 approval. Hazards – Gas release and vapor cloud explosion potential
N8	Pigging operations conducted concurrently with CTU operations within 85 ft. (26 m) are <i>restricted</i> and require a Level 1 approval. Hazards – Gas release and vapor cloud explosion potential
N10	Derrick barge lifts are <i>prohibited</i> from being performed concurrently with CTU operations. Hazards – Collision of derrick barge crane boom with CTU operations or with platform impacting CTU operations
N11	Diving operations conducted concurrently with CTU operations are <i>restricted</i> and require a Level 1 approval. Hazard – Tools or other equipment falling and striking divers
N12	ROV operations in close proximity of a fire pump intake conducted concurrently with CTU operations are <i>restricted</i> and require a Level 1 approval. Hazards – ROV could obstruct the intake of the fire pump or the fire pump is inoperable. Tools or other equipment falling and striking the ROV
N13	Unrelated BOP movement conducted concurrently with CTU. Operations are <i>restricted</i> and require a Level 2 approval. Hazards – Crane collision, gas release, and vapor cloud explosion potential
O1	Major coating operations conducted concurrently with drilling or workover activities on an offshore platform, or within 82 ft. (25 m) of an onshore well site, are <i>restricted</i> and require a Level 1 approval. Abrasive blasting conducted concurrently with drilling or workover activities on an offshore installation, or within 82 ft. (25 m) of an onshore well site, is <i>restricted</i> and requires a Level 1 approval. Hazards – Spark potential with hydrocarbon release potential; explosion or fire

Work Management System (WMS)

Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
O2	CSE conducted concurrently with drilling or workover operations on an offshore installation, or within 164 ft. (50 m) of an onshore drilling rig, during any of the following conditions is <i>restricted</i> and requires a Level 1 approval: ▪ Workover or live wireline work with open perforations to a hydrocarbon zone (unless the well is secured while the CSE Permit is <i>live</i>) ▪ Drilling with a hydrocarbon zone open to the wellbore ▪ Drilling surface hole Hazards – Gas release and vapor cloud explosion potential with personnel in confined space and restricted escape
O3	<u>Unassociated</u> complex lifts conducted concurrently with drilling or workover operations are <i>restricted</i> and require Level 1 approval. Hazards – Complex Lift over process, drilling, or workover equipment and potential for impact to concurrent operations (for example, dropped object on compressor package) ! IMPORTANT: Shift communications processes are key for crane operations concurrent with drilling or workover operations to ensure SIMOPS issues are identified and mitigated. <i>Rigging and Lifting SWP.</i>
O4	Unrelated engine-driven equipment and vehicles operated concurrently with drilling or workover work within 85 ft. (26 m) are <i>restricted</i> and require Level 1 approval. Hazards – Hydrocarbon release with spark potential and equipment collision
O5	Drilling or workover activities conducted in an offshore facility when the fire protection system is inoperable are <i>restricted</i> and require a Level 3 approval. Hazards – Fire or explosion
O6	Hot work conducted concurrently within 85 ft. (26 m) of drilling or workover operations is <i>restricted</i> and requires a Level 1 approval. ! IMPORTANT: On active drilling rig operations, the signature of the Drilling Supervisor and the PIC must be obtained on Hot Work Permits. Hazards – Hydrocarbon release and potential fire or explosion
O7	Hydrocarbon source activity conducted concurrently with drilling or workover activity within 85 ft. (26 m) is <i>restricted</i> and requires a Level 1 approval. Hazards – Gas release and vapor cloud explosion potential
O8	Pigging operations conducted concurrently with drilling or workover activity within 85 ft. (26 m) are <i>restricted</i> and require a Level 1 approval. Hazards – Gas release and vapor cloud explosion potential

Work Management System (WMS)

Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
O10	Derrick barge lifts conducted concurrently with drilling or workover activities are <i>restricted</i> and require a Level 2 approval. Hazard – Collision of derrick barge crane boom with drilling or workover activities
O11	Diving operations conducted concurrently with drilling or workover activities are <i>restricted</i> and require a Level 1 approval. Hazard – Tools or other equipment falling and striking the diver(s)
O12	ROV operations in close proximity of a fire pump intake conducted concurrently with drilling or workover activities are <i>restricted</i> and require a Level 1 approval. Hazards – ROV could obstruct the intake of the fire pump or the fire pump is inoperable. Tools or other equipment falling and striking the ROV
P1	Coating operations conducted concurrently with electrically fired explosives operations on an offshore installation, or within 82 ft (25 m) onshore, are <i>restricted</i> and require a Level 1 approval. ! WARNING: Static electricity precautions must be taken during the coating activities.
P2	CSE activities are <i>prohibited</i> from being conducted concurrently with electrically fired explosives operations requiring radio silence. Hazard – Potential lack of communication due to radio silence
P3	Unassociated crane activities are <i>prohibited</i> from being conducted concurrently with electrically fired explosives operations requiring radio silence. Hazards – Impact of crane operations into lubricator, e-line, and slick line and lack of communication due to radio silence
P4	Engine-driven and vehicle equipment unassociated with electrically fired explosives operations on an offshore installation, or within 164 ft (50 m) onshore, are <i>restricted</i> and require a Level 1 approval. Hazards – Impact to workover equipment, explosion hazard with vapor release, possible electrical interference caused by electric starts, and radio control ! IMPORTANT: Before using electro-explosive detonators, all radio, radar, cellular, and other signal transmitting equipment within 500 ft (150 m) must be identified, located, and shut off. Reference: WellWork Execution Manual NOTE: Unassociated engine driven equipment should be equipped with intrinsically safe components if within designated hazardous areas.
P5	Electrically fired explosives operations are prohibited at offshore facilities when the fire protection system is inoperable. Hazards – Fire or explosion

Work Management System (WMS)

Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
P6	Electric welding is <i>prohibited</i> from being conducted while running, handling, or pulling electrically fired explosives on surface to the first 300 ft. (100 m) below mudline at an offshore location or the first 300 ft. (100 m) below the wellhead (surface) at an onshore location (see following reference for additional guidance). Hazards – Welding machine generated electric current could set off electrically fired explosives. ! IMPORTANT: Before using electro-explosive detonators, all radio, radar, cellular, and other signal transmitting equipment within 500 ft. (150 m) must be identified, located, and shut off. Reference: <i>WellWork Execution Manual</i> Hot work is <i>prohibited</i> from being conducted concurrently with electrically fired Explosives operations offshore or within 85 ft. (26 m) onshore. Hot work conducted concurrently in proximity with electrically fired explosives operations, beyond 85 ft. (26 m), and outside designated hot work area, is <i>restricted</i> and requires a Level 1 approval. Hazards – Spark travel, explosives, and potential of hydrocarbons
P7	Hydrocarbon source activity is <i>prohibited</i> while the electrically fired explosive is on surface and the activity is within 164 ft. (50 m) of electrically fired explosives operations. Hazards – Noise distraction, radio interference, or vapor cloud resulting in a fire or explosion
P8	Pigging operations are <i>prohibited</i> while the electrically fired explosive is on surface and the activity is within 164 ft. (50 m) of electrically fired explosives operations. Hazards – Noise distraction, radio interference, or vapor cloud resulting in a fire or explosion
P9	Boat operations within offshore installation safety zone (500 meters) are <i>prohibited</i> to be conducted concurrently with armed electrically fired explosives not 300 ft. (100 m) or more below the mudline. Hazard – Radio interference with electrically fired equipment
P10	Derrick barge operations within offshore installation safety zone (500 m) are <i>prohibited</i> to be conducted concurrently with electrically fired explosives operations. Hazards – Noise distraction or radio interference resulting in electrically fired explosive inadvertently discharging; potential collision with platform
P11	Diving operations are prohibited to be conducted concurrently with electrically fired explosives operations (includes pressure detonation) within facility safety zone. Hazards – Radio interference resulting in explosives inadvertently discharging. Fire protection system may be inoperable while divers are in the water. Exposure of divers to premature detonation in less than 300 foot (100 m) below mudline.

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Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
P12	ROV operations in close proximity to an offshore installation are prohibited to be conducted concurrently with electrically fired explosives operations (includes pressure detonation). Hazards – Radio interference resulting in electrically fired explosive inadvertently discharging. Fire protection system may be inoperable while ROV is in the water.
P13	BOP movement (open hole) is <i>prohibited</i> to take place concurrently with electrically fired explosives operations (includes pressure detonation). Hazards – Radio interference resulting in electrically fired explosive inadvertently discharging. Open hole potential for gas release and vapor cloud resulting in a fire or explosion.
P14	CTU operations are <i>prohibited</i> to take place concurrently with unrelated electrically fired explosives operations until electrically fired explosive is 300 ft. (100 m) below wellhead onshore and below mudline offshore. Hazard – Radio interference resulting in electrically fired explosive inadvertently discharging
P15	Unrelated electrically fired explosives operations conducted concurrently with drilling or workover activities are <i>restricted</i> and require a Level 2 approval. The following precautions must be taken regarding electrically fired perforating equipment (with electrically fired detonators, normally run on wireline): - Only one electrically fired perforating assembly is allowed on an offshore installation or onshore well site at a time. - While explosives are being handled, the SIMOPS requirements regarding perforating and hot work (especially electric welding) must be followed. - Before electrically fired explosives are armed, the most current facility Radio Silence Checklist must be completed. ! WARNING: Radio silence must be maintained from the arming of the charges until they are run to a depth of 300 ft. (100 m) and, on pulling out of the hole, from a depth of 300 ft. (100 m) until the charges are safe (verified as discharged or disarmed). ! WARNING: Electric welding machines, cathodic protection, rig top-drive systems and other devices that can produce induced voltages or stray currents must be isolated. Hazard – Radio interference resulting in electrically fired explosive inadvertently discharging
Q1	Coating operations conducted concurrently with rig skids on an offshore installation, or within 82 ft. (25 m) of an onshore rig, are <i>restricted</i> and require a Level 1 approval. Hazards – Vision, noise obstructions, flammable vapors with spark potential
Q2	CSE is <i>prohibited</i> from being conducted concurrently with rig skids. Hazards – Spark potential, potential gas release

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Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
Q3	Complex Lifts conducted concurrently with rig skid operations on an offshore installation, or within 82 ft. (25 m) of an onshore drilling rig, are <i>restricted</i> and require a Level 1 approval. Hazards – Congested or conflicted work areas or increase in supervisory oversight requirements
Q5	Rig skid operations conducted over live wellheads when the fire protection system is inoperable are <i>restricted</i> and require Level 3 approval. Hazards – Fire or explosion from impact on live wellhead
Q7	Hydrocarbon source activity is <i>prohibited</i> from being conducted concurrently with rig skid operations within 85 ft. (26 m). Hazards – Spark potential, fire or explosion
Q8	Pigging operations are <i>prohibited</i> from being conducted concurrently with rig skid operations within 85 ft. (26 m). Hazards – Spark potential, fire or explosion
Q10	Derrick barge lifts are <i>prohibited</i> during rig skid operations. Hazards – Interferences with rig skid or derrick barge operations
Q11	Diving operations during rig skid operations are restricted and require Level 1 approval. Hazards – Dropped object into water potential; if fire protection system inoperable with divers in the water, other SIMOPS apply
Q12	ROV operations in close proximity of a fire pump intake conducted concurrently with rig skid operations are <i>restricted</i> and require a Level 1 approval. Hazards – ROV could obstruct the intake of the fire pump or the fire pump is inoperable. Tools or other equipment falling and striking the ROV
Q14	CTU operation is <i>prohibited</i> during rig skid operations. Hazards – Spark potential, fire or explosion; equipment collision impacts
Q16	Electrically fired explosives operations on the surface and within 300 ft (100 m) below wellhead and rig skid operations are <i>prohibited</i> from occurring concurrently. Hazards – Radio interference resulting in electrically fired explosive inadvertently discharging
R1	Coating operations conducted concurrently with well stimulation operations on an offshore installation, or within 82 ft. (25 m) of onshore well stimulation operations, are <i>restricted</i> and require a Level 1 approval. Abrasive blasting conducted concurrently with well stimulation operations on an offshore installation, or within 164 ft. (50 m) of onshore well stimulation operations, requires a Level 1 approval. Hazard – Spark potential in proximity to potential hydrocarbon presence

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Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
R2	CSE activities conducted concurrently with well stimulation operations are <i>restricted</i> within 85 ft. (26 m) and require a Level 2 approval. Hazard – Uncontrolled release
R3	Crane activities conducted concurrently with well stimulation operations within 85 ft. (26 m) are <i>restricted</i> and require Level 1 approval. Hazard – Hydrocarbon release
R4	Unrelated engine-driven equipment and vehicles operated concurrently with well stimulation within 85 ft. (26 m), are <i>restricted</i> and require a Level 1 approval. Hazard – Spark potential, impact to well stimulation operations, hydrocarbon release
R5	Well stimulation operations when the fire protection system is inoperable, are <i>restricted</i> and require Level 3 approval. Hazards – Hydrocarbon, well control potential issues—fire or explosion
R6	Hot work outside designated hot work area, conducted concurrently with well stimulation operations within 85 feet (26 m), is <i>restricted</i> and requires a Level 1 approval. Hazards – Hydrocarbon release—fire or explosion
R10	Derrick barge lifts are <i>prohibited</i> from being performed concurrently with well stimulation operations. Hazards – Interfaces between the two operations and potential interference from derrick barge boom, impact to risers, well control, hydrocarbon release, and fire or explosion
R11	Diving operations with the fire pumps inoperable are <i>prohibited</i> from being conducted concurrently with well stimulation operations. Hazard – Inoperable fire protection system
R12	ROV operations in close proximity of a fire pump intake conducted concurrently with well stimulation operations are <i>restricted</i> and require Level 1 approval. Hazards – ROV could obstruct the intake of the fire pump. If fire pump and protection system is inoperable, other SIMOPS apply (R11 – prohibited). Tools or other equipment falling and striking the ROV.
R13	Tree-removed blowout preventer (BOP) stack movement conducted concurrently with well stimulation operations is <i>restricted</i> and requires a Level 2 approval. Hazard – Hydrocarbon release—fire or explosion
R14	Unrelated CTU operations conducted concurrently with well stimulation operations are <i>restricted</i> and require a Level 1 approval. Hazard – Interfaces between the operations (such as, pumping, pressures)

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Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
R15	Unrelated drilling or workover operations conducted during well stimulation operations are <i>restricted</i> and require a Level 2 approval. Hazards – Congested work area, equipment spacing, dropped objects, well under pressure, hydrocarbon release potential
R16	Electrically fired explosives operations conducted concurrently with unrelated well stimulation operations are <i>restricted</i> and require a Level 2 approval. Hazards – Radio interference resulting in electrically fired explosive inadvertently discharging; potential gas release
R17	Rig skidding conducted concurrently with well stimulation operations is <i>restricted</i> and requires Level 1 approval. Hazards – Spark potential with flammable vapor potential, congested work area, and equipment spacing issues
S1	Abrasive blasting conducted concurrently with wireline operations (with lubricator) within 85 ft. (26 m) of a well site, is <i>restricted</i> and requires a Level 1 approval. Hazard – Spark potential with potential for hydrocarbons
S2	CSE activities conducted concurrently with wireline operations are <i>restricted</i> within 85 ft. (26 m) and require Level 1 approval. Hazards – Potential hydrocarbon release, fire or explosion
S3	Crane activities conducted that are simultaneous with wireline operations are <i>restricted</i> and require Level 1 approval. This does not include the use of a crane to support the wireline operation and within the work scope. Hazards – Impact to the wireline or lubricator by crane or crane load, dropped object impact
S4	Unassociated engine-driven equipment and vehicles operated concurrently with wireline operations within 85 ft. (26 m), are <i>restricted</i> and require a Level 1 approval. Hazards – Impact to workover equipment, explosion hazard with vapor release NOTE: Unassociated engine driven equipment should be equipped <i>with</i> intrinsically safe components if within <i>designated hazardous areas</i>.
S5	Wireline operations conducted on a manned offshore facility when the fire protection system is inoperable are <i>prohibited</i>. Hazards – Gas release and potential fire or explosion

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Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
S6	Hot work outside the designated hot work area conducted concurrently with offshore wireline operations within 85 ft. (26 m) is <i>restricted</i> and requires a Level 2 approval. Hot work conducted concurrently with wireline operations within 85 ft. (26 m) on a single well site onshore is <i>restricted</i> and requires a Level 1 approval. Hazards – Gas release and potential fire or explosion
S7	Hydrocarbon source activity conducted concurrently with wireline operations within 85 ft. (26 m) is <i>restricted</i> and requires a Level 1 approval. Hazards – Hydrocarbon release and engine-driven equipment
S8	Pigging operations conducted concurrently with wireline operations within 85 ft. (26 m) are <i>restricted</i> and require Level 1 approval. Hazard – Hydrocarbon release
S10	Unassociated derrick barge operations are <i>restricted</i> in the vicinity of a producing platform during wireline operations and require Level 2 approval. Hazards – Potential interference and impact from derrick barge boom with wireline operations. Potential well control, hydrocarbon release, fire or explosion. NOTE: Include Risk Assessment (RA) considerations for large lift boat or higher risk scenarios.
S11	Diving operations with the fire protection system inoperable are <i>prohibited</i> from being conducted concurrently with wireline operations. Hazard – Inoperable fire protection system (such as, disabled fire water pumps due to divers in the water)
S12	ROV operations in close proximity of a fire pump intake conducted concurrently with wireline operations are <i>restricted</i> and require Level 1 approval. Hazard – ROV could obstruct the intake of the fire pump or the fire pump is inoperable.
S13	BOP movement conducted concurrently with wireline operations is <i>restricted</i> and requires a Level 1 approval. Hazards – Lift impacts resulting in hydrocarbon release - fire or explosion
S14	Unrelated CTU operations conducted concurrently with wireline operations are <i>restricted</i> and require Level 1 approval. Hazard – Interfaces or impact between the two operations (such as, pumping, pressures)
S15	Unrelated drilling or workover operations conducted during wireline operations are <i>restricted</i> and require a Level 1 approval. Hazards – Interfaces or impact between the two operations, congested equipment spacing, dropped objects, well under pressure, hydrocarbon release

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Table 26: SIMOPS Restrictions Tables, Continued

Matrix	Requirement
S16	Electrically fired explosives operations conducted concurrently with unrelated wireline operations are <i>restricted</i> and require a Level 2 approval. Hazards – Radio interference resulting in electrically fired explosive inadvertently discharging. Potential gas release from interfacing operations.
S17	Rig skidding conducted concurrently with wireline operations is <i>restricted</i> and requires Level 1 approval. Hazards – Spark potential, gas release, fire or explosion, congested equipment spacing
S18	Well stimulation operations conducted concurrently with unrelated wireline operations are <i>restricted</i> and require Level 1 approval. Hazards – Interfaces or impact between the two operations, congested equipment spacing, and hydrocarbon release with potential fire or explosion

9.4.6 ROUTINE SIMOPS JOB PLANNING

9.4.6.1 INTRODUCTION

This section describes the guidelines for the planning and preparation of routine SIMOPS

9.4.6.2 SIMOPS PLANNING ACTIVITIES

[CriticalE] The PIC for the SIMOPS **must**:

- Identify the restrictions and conditions under which the SIMOPS can be performed.
- Coordinate the planned activities with other line supervisors and designate an Area Interface Authority / Site Coordinator if required.
- Agree on communication lines with other line supervisors.

9.4.6.3 DEVIATION REQUEST APPROVAL

Follow these steps to initiate a deviation request from the activity restrictions and obtain appropriate approval to carry out SIMOPS. Refer to [Table 27: Deviation Request](#).

Work Management System (WMS)

Table 27: Deviation Request

Task	Activity
1	Obtain a copy of the SIMOPS Deviation Request and Approval Form. Reference: GOTO/WMSEFORMS , "SIMOPS" for the SIMOPS Deviation Request and Approval Form
2	Complete the form and include pertinent information. Attach any proposed procedures and relevant documents.
3	Submit the SIMOPS Deviation Request and Approval Form for approval at the appropriate level (1, 2, or 3).
4	A copy of the <i>approved</i> SIMOPS Deviation Request and Approval Form should be returned to the originator and kept per unit's Management and Protection of Information (MPI) guidelines.

9.4.7 NON-ROUTINE SIMOPS JOB PLANNING

9.4.7.1 INTRODUCTION

This section provides guidelines for planning and preparation for non-routine simultaneous operations.

9.4.7.2 RESPONSIBILITY FOR PLANNING

The Operations Superintendent (Drilling or Workover Superintendent for drilling or workover operations) has the overall responsibility and accountability for planning and approving the SIMOPS plan. The Operations Superintendent appoints the supervisor responsible for planning SIMOPS.

The Responsible Supervisor should be appointed early enough to allow him or her the necessary time for preparing and planning the non-routine SIMOPS.

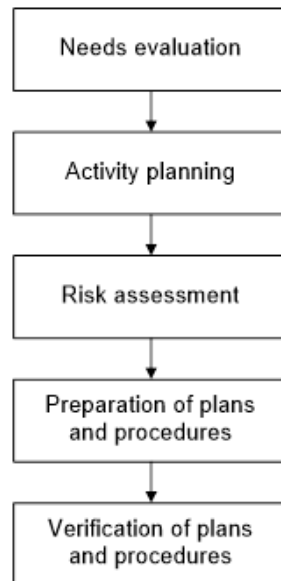
The Site Manager (if drilling is involved) may be responsible for evaluating and approving non-routine simultaneous operations.

9.4.7.3 PLANNING PROCESS FLOWCHART

The following figure shows the planning process for non-routine SIMOPS. Refer to [Figure 25: Planning Process for Non-routine SIMOPS](#).

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Figure 25: Planning Process for Non-routine SIMOPS



9.4.8 EVALUATION OF NEEDS

Evaluation of non-routine SIMOPS includes:

- Providing an overview of the:
 - Background for performing the planned operation
 - Purpose of the planned operation
- Identifying the following:
 - Other activities planned during the same period that may introduce a higher risk
 - Other limitations or restrictions that may be imposed on the planned operations
 - Circumstances (for example, weather) that may affect planned operations

9.4.9 ACTIVITY PLANNING

The activity planning phase should address the following:

- Description of the operation, work to be performed and timing aspects.
- Identification of additional supervision and technical support required during the operation, including coordinating responsibility and allocating responsibilities.
- Safety, health, environmental, security, and regulatory requirements.
- Risk Assessment of the planned operation.
- Description of requirements for equipment, tools, materials, communication, and safety equipment to be used.
- Interfaces with other operations including commercial aspects.
- Requirements, if any, for specific qualifications and training of personnel involved and identification of any extra personnel.
- Emergency preparedness measures and alternative plans.

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- Requirement for work permits and Isolation Certificates including the signature or countersignature of an Area Interface Authority (AIA) / Site Coordinator (SC).

References:

- Section 6.0: "Permit To Work"
- Section 7.0: "Equipment Isolation and Reinstatement (Lock-out/Tag-out)"

9.4.10 SIMOPS KEYWORD TOOL

The following SIMOPS keyword tool may be used for creating non-routine SIMOPS procedures.

Table 28: SIMOPS Keyword Tool

Risks and Hazards	Causal Factors	Consequence Drivers
▪ Fire or explosion ▪ Hydrocarbon release ▪ Impact risk: ○ Dropped or swinging loads or tools ○ Collision ▪ Rotating equipment ▪ Wires or cables snapping ▪ Entanglement ▪ Structural failure ▪ Ballast failure ▪ Loss of heading control; reduced power for heading control ▪ Man overboard ▪ Personnel exposure to air or water intakes, outlets, vents and so forth ▪ Working environment threats: ○ Dust ○ Chemical exposure ○ Noise ○ Gases and fumes ○ Migration of sparks or chemicals into adjacent areas, including above and below	▪ Safety systems affected ▪ Safety barriers impaired ▪ Monitoring, control, or detection systems affected ▪ Overload of utility or safety systems ▪ Distraction of personnel ▪ Overburdening personnel ▪ Key personnel tied up ▪ Wrong timing or coordination ▪ Communication or information disturbed ▪ Work environment affected ▪ Not covered by procedure ▪ Availability of emergency response or rescue resources (personnel and equipment)	▪ Ignition source ▪ Potential hydrocarbon releases ▪ Effect on personnel, environment or equipment ▪ Effect on escape ways ▪ Effect on evacuation means ▪ Effect on safety systems (emergency shutdown [ESD], fire water, fire, and gas) ▪ More personnel present in area ▪ Personnel working in exposed positions ▪ Critical equipment in exposed position

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9.4.11 JOB EXECUTION

9.4.11.1 INTRODUCTION

This section describes guidelines for the execution of non-routine simultaneous operations, including prestart verification and preparation for execution.

9.4.11.2 PREPARATION ACTIVITIES

[E] Prior to starting the non-routine SIMOPS, the Area Interface Authority (AIA) / Site Coordinator (SC)

must:

- Ensure work teams complete a Permit/Work Pack.
- Hold a safety discussion and briefing.
- Familiarize the personnel involved in the operation with the tasks to be performed and with the communication lines. Confirm that tasks are understood.
- Communicate any safety and emergency measures to the personnel involved in the operation.

9.4.11.3 SIMOPS SAFETY DISCUSSION AND BRIEFING

The Area Interface Authority (AIA) / Site Coordinator (SC) responsible for the operation may call for a safety discussion prior to starting the SIMOPS operation.

The SIMOPS safety discussion should be conducted in accordance with the following requirements:

- The safety discussion is arranged in advance of the start-up of operations to allow sufficient time for personnel to become familiar with procedures and to carry out any remedial actions if required.
- Involved personnel attend the SIMOPS safety discussion, and discuss the following items:
 - Scope, purpose, and objectives
 - Review of operational details
 - Life Saving Rules & Actions
 - Permit/Work Packs, Procedures and instructions
 - Emergency preparedness and contingency plans
 - Utilization of safety equipment and protective clothing
 - Hazardous chemicals
 - Individual responsibilities
 - Management of the operations
 - Communication lines
 - Any other item important for the operation

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10.0 SAFE WORK PRACTICES

10.1 INTRODUCTION

This section contains links to globally required Upstream Work Management Safe Work Practices. [CriticalE] Sites **must** meet or exceed the minimum expectations defined in the Upstream Work Management Safe Work Practices and associated checklists when executing applicable work activities.

10.2 DOCUMENTATION AND CHANGE MANAGEMENT

The Upstream Work Management Safe Work Practices documents will be managed as stand-alone tools that support the Work Management UBP. Review frequency will be determined by the respective subject matter contacts.

Change requests for any of the Upstream WMS Safe Work Practices can be requested through the change request process outlined in WMS [Section 4.3: Deviation, Exceptions, and Changes](#)

All Upstream SWPs can be accessed at [GOTO/UPSTREAMSWPs](#)

UPSTREAM SAFE WORK PRACTICES - LINKS AND CONTACTS

Title	Subject Matter Contact
7.1 Rigging & Lifting SWP	Upstream R&L Maintenance Advisor
7.2 Forklift Operations SWP	Upstream R&L Maintenance Advisor
7.3 Working at Heights SWP	Upstream WMS Advisor
7.4 Upstream Electrical SWP	Upstream I&C Engineering Advisor

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11.0 GLOSSARY

11.1 TERMS, ABBREVIATIONS, AND ACRONYMS

The following terms, abbreviations, and acronyms are used throughout this document

Table 29: Glossary

Term	Description
AA	Area Authority
AAF	Area Authority Functional
AGT	Authorized Gas Tester
AIA / SC	Area Interface Authority / Site Coordinator
ALARP	as low as reasonably practicable
AO	Area Operator
API	American Petroleum Institute
AS	Australian Standard
BA	breathing apparatus
Barg	bar gauge (pressure)
BOP	blowout preventer
°C	degrees Celsius
CAS	Competency Assurance System
CCP	central control point
CCR	Central Control Room
CIMS	Controls Integrity Management System
CO	carbon monoxide
CO ₂	carbon dioxide
COP	code of practice
COP	critical operating parameter
CPR	cardiopulmonary resuscitation
CSC	car sealed closed
CSE	confined space entry
CSO	car sealed open
CTU	coiled tubing unit

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Table 29: Glossary, Continued

Term	Description
CWP	critical work permit
DIB	double isolation and bleed
DIBV	double isolation and bleed valves
DCS	distributed control system
DEMOB	demobilization
DWCM	Daily Work Coordination Meeting
EM	ExxonMobil
EMOG	ExxonMobil Oil and Gas
ePTWS	electronic Permit to Work System
ESD	emergency shutdown
°F	degrees Fahrenheit
FeS	iron sulfide
FIMS	Facilities Integrity Management System
FPSO	floating production, storage, and offloading (vessel)
Ft	feet
GRES	Global Real Estate and Facilities
HAZOP	Hazard and Operability Review
HC	hydrocarbon
H ₂ S	hydrogen sulfide
IA	Isolating Authority
IC	Integrity Critical
IC	Isolation Certificate
ICC	Isolation Control Certificate
ICP	Integrity Critical Procedure
ICP	isolation control point
ID	identification
IDLH	immediately dangerous to life and health
JSA	Job Safety Analysis

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Table 29: Glossary, Continued

Term	Description
Kg	kilogram
kPa	kilopascal
LAN	local area network
Lb	pound
LC	locked closed
LEL	lower explosive limit
LMRA	last minute risk assessment
LNG	liquefied natural gas
LO	locked open
LO/TO	lock-out/tag-out
LPG	liquefied petroleum gas
LSA	low specific activity
LSHH	level shutdown high-high
LSLL	level shutdown low-low
LSRA	Life Saving Rule & Action
LTI	long term isolation
M	meter
m ³	cubic meter
MAWP	maximum allowable working pressure
MDMT	minimum design metal temperature
Mg	milligram
MIRU	move in and rig up
MOB	mobilization
MOC	management of change
MODU	mobile offshore drilling unit
MOH	Medical and Occupational Health
MOV	motor operated valve
MPI	Management and Protection of Information

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Table 29: Glossary, Continued

Term	Description
N ₂	nitrogen
NC	normally closed
NIOSH	National Institute for Occupational Safety and Health (USA)
NO	normally open
NOHSC	National Occupational Health and Safety Commission (Australia)
NORM	naturally occurring radioactive material(s)
O ₂	oxygen
OBO	operated by others
OIM	Offshore Installation Manager
OIMS	Operations Integrity Management System
OS	Operations Support
P&ID	pipng and instrumentation diagram
PA	public address
PEL	permissible exposure limit
PFD	process flow diagram
PH	Permit Holder
PIC	Person in Charge
PLC	programmable logic controller
PM	preventive maintenance
PPD	Permit Pack Document
PPE	personal protective equipment
Ppm	parts per million
PSHH	pressure switch high-high
Psi	pounds per square inch
Psig	pounds per square inch gauge (gauge pressure)
PSLL	pressure switch low-low
PSMS	Personnel Safety Management System
PSV	pressure safety valve

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Table 29: Glossary, Continued

Term	Description
PTW	permit to work
PTWS	Permit to Work System
PU	Production Unit
RA	risk assessment
RAD	radiation
RAMS	Risk Assessment and Management System
RFIC	Remote Field Isolation Certificate
RFO	Remote Field Operations
ROV	remotely operated vehicle or valves
SVI	single valve isolation
SCBA	self-contained breathing apparatus
SCSSV	surface-controlled subsurface safety valve
SDS	Safety Data Sheet
SHE	safety, health, and environment
SHES	safety, health, environment, and security
SIMOPS	simultaneous operations
SME	subject matter expert
SO ₂	sulfur dioxide
SSV	surface safety valve
STEL	short term exposure limit
STOP	safety training observation program
STT	sanction to test
SWC	Start Work Check
SWP	Safe Work Practice
TCP	tubing conveyed perforating
TD	Temporary Defeat
TEMPSC	totally enclosed motor propelled survival craft

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Table 29: Glossary, Continued

Term	Description
THA	Task Hazard Assessment
TWA	time weighted average
UBP	Upstream Best Practice
UEL	upper explosive limit
UPS	uninterruptible power supply
VP	Vice President
WAH	work at heights
WBRM	wellbore risk management
WMS	Work Management System

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12.0 APPENDIX A OPERATING SITE ADDITIONAL INFORMATION

12.1 INTRODUCTION

The sections in this appendix are either required by this Work Management System or suggested Site inclusions. The individual Operating Site decides the location of the additional information to make it readily available to all personnel (employees or contractors) involved in conducting work. Reference to local regulatory requirements are included when they meet or exceed the requirements of this Work Management System.

12.2 CONTENT

This appendix may contain the following information (for example):

- Work Not Requiring a Permit
- Work That Can Be Performed by a Single Person in a Remote Field Operation
- Work That Can Be Performed Using a Personal Isolation
- Two and Three Permit Area Maps
- Day In the Life Of (DILOs)
- Local SIMOPS Matrix (If Required)
- Local Regulatory Requirements
- Additional Checklists